



RADIO TEST SET 2968 for TETRA Phase 3.2



Operating Manual

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RADIO TEST SET 2968 for TETRA Phase 3.2

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About this manual

This manual explains how to operate the TETRA test system (Phase 3.2) in the 2968 Radio Test Set. It describes the full TETRA functionality of the Test Set. Depending on which options you have purchased, certain functions may not be available.

This manual applies to test sets with versions 11.x of software fitted to A6/2.

Intended audience

People who need to test the performance of TETRA mobile radios and base stations. It is assumed that the reader will either have a working knowledge of the test set or access to its Operating Manual, and will be familiar with TETRA telecommunication terms.

Structure

Chapter 1

Introduction, and an overview of the test system

Chapter 2

An introduction to the features of the TETRA test system.

Chapter 3

How to select the required test system and customize it to your requirements.

Chapter 4

A detailed guide to using the TETRA Mobile manual test system.

Chapter 5

A detailed guide to using the TETRA Base station test system.

Chapter 6

A detailed guide to using the TETRA Direct Mode test system.

Chapter 7

A detailed guide to using the TETRA Mobile Automatic test system.

About this manual (continued)

Chapter 8

A detailed guide to using the TETRA Digital Duplex test facility.

Chapter 9

Information about using the data display facility.

Chapter 10

Information about using the Store and Recall facility.

Chapter 11

Listing of Performance Data and Test Parameters.

Appendix A

Glossary of terms and abbreviations used in this manual.

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Document conventions

The following conventions apply throughout this manual:-

AUTORUN	Screen text, such as menu titles and messages, is shown verbatim in this font.
[SYSTEMS]	Hard key titles are indicated by normal lettering in square brackets.
[SET-UP]	Soft key titles are shown in italic lettering in square brackets.

Associated publications

Other manuals that cover specific aspects of this test set are:-

- **Operating Manual** (46882/274) describes the test set
- **Programming Manual** (46882/280) provides programming information for those wishing to write their own test programs

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Chapter 1	GENERAL INFORMATION
Chapter 2	TESTING MOBILES AND BASE STATIONS
Chapter 3	SELECTING AND CUSTOMIZING THE SYSTEM
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See the Operating Manual, 46882/274.

Chapter 1

GENERAL INFORMATION

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Introduction

The Systems mode of the 2968 Radio Test Set enables you to test radio communication equipment such as mobile radios and base stations. The inbuilt software configures the Test Set to make measurements on signals produced by equipment operating on the selected system and produces signals structured to that system.

This manual refers to the use of the Test Set to test the performance of mobiles and base stations designed to operate to the TETRA specification.

The three modes of TETRA operation supported by the Test Set are:-

TETRA Mobile (MS)	The Test Set tests signals transmitted by a mobile operating in normal trunked mode.
TETRA Base Station (BS)	The Test Set tests signals transmitted by a base station.
TETRA Direct Mode (DM)	The Test Set tests signals transmitted by a mobile operating in direct mode.

Manual testing is provided for the TETRA Mobile (MS), TETRA Base Station (BS) and TETRA Direct Mode (DM) modes.

Automatic testing is provided for the TETRA Mobile (MS) mode.

Each of these TETRA modes is provided with an individual Systems identity within the software structure of the Test Set. The modes enabled on a specific Test Set will depend on the *Options*, or combination of *Options* pertinent to it.

The test facilities provided for the TETRA system are:-

- On-channel transmitter measurements using standard or user-configured channel plans.
- Comprehensive modulation analysis with power profile, constellation, phase trajectory and vector error vs. time diagrams.
- Display of parameters and decoded data received from mobiles and base stations to aid diagnosis of system problems.
- Base station control channel simulation and signaling to provide effective network simulation, allowing mobile registration and call set-up to Test Set.
- Mobile test mode control using RF loopback and T1 signaling; and audio loopback, test sound and silence modes for mobile audio system testing.
- Uplink T1 test signal generator synchronized to base station down link frame structure supporting conformance testing of base station receivers.

Capability

Transmitter tests

The Test Set is able to test the performance of transmitters in TETRA mobiles, TETRA base stations and TETRA direct mode terminals. The specific test capabilities are:-

- Tx burst power.
- Tx power level control (mobile only).
- Tx frequency error.
- Tx burst timing (frame alignment) (mobile only).
- Modulation accuracy for peak and RMS vector error and residual carrier.
- Burst profiles (transmitter output power versus time).
- Constellation, rotated vector and phase trajectory diagrams.
- Vector error, magnitude and phase versus. time displays.
- Display of parameters and data received from mobiles and base stations.

Measurements can be made on *on-channel* signals without signaling or synchronization, for manufacturing applications where radios can be operated in test modes. In addition the Test Set provides a base station simulation and TETRA signaling to allow a mobile to register and set up calls with the Test Set, allowing the transmitter of a mobile to be tested without the aid of special test modes.

Frequency setting

When the Test Set is operating in Systems mode, the RF signal generator and the receiver within the Test Set can be set as frequencies if NO PLAN is selected as the CHANNEL PLAN on the SYSTEM PARAMETERS MENU.

Receiver tests

The signal generator within the Test Set produces the following signals:-

- Simulated base station Main Control CHannel (MCCH).
- Simulated base station Traffic CHannel (TCH).
- T1 test signals types 1, 2 and 4 as defined in EN 300 394-1.
- RF loopback and T1 signaling control of mobile test mode.
- Synchronization to base station downlink frame structure for synchronized uplink T1 type 7 signal generation for base station test.

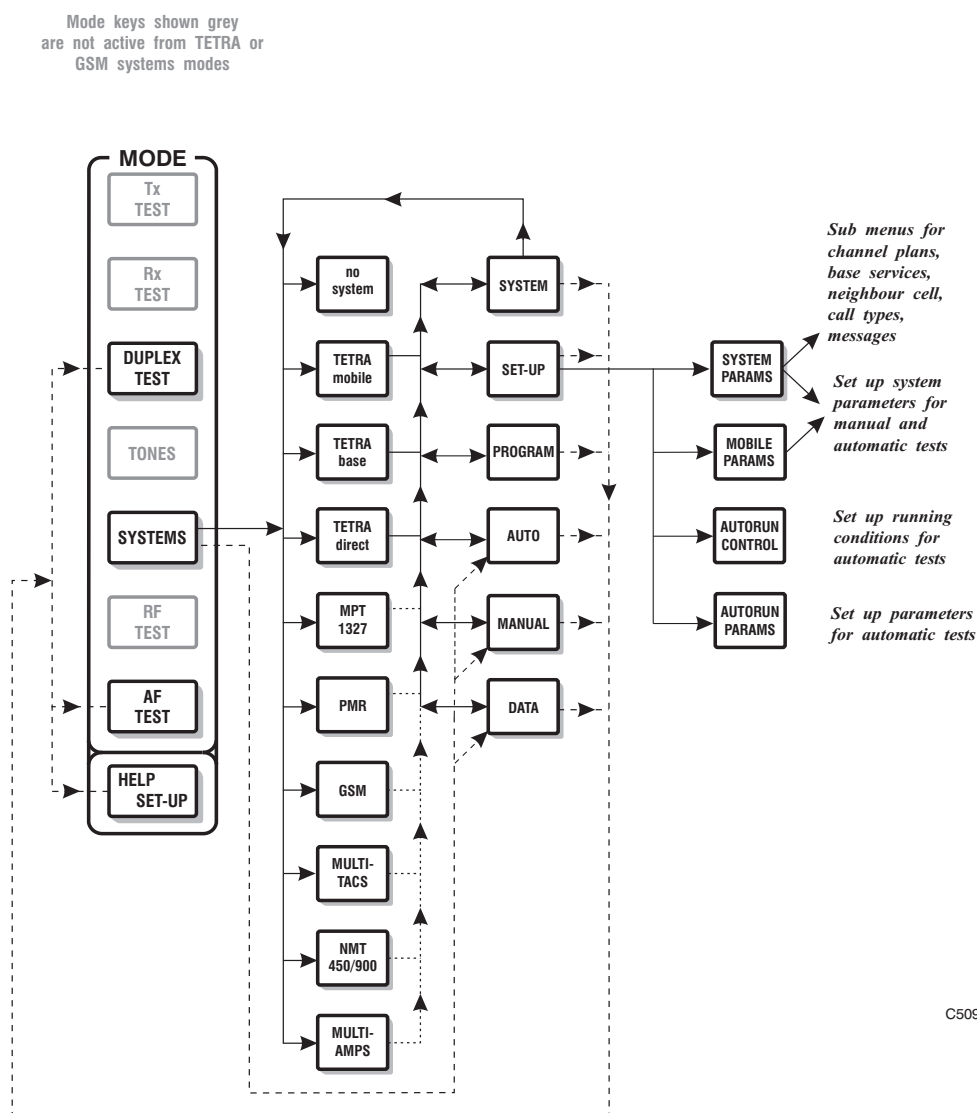
Frequency setting

See this heading in *Transmitter tests* above.

General

When the Systems mode is selected the Test Set is set up to produce signals and make measurements appropriate to the selected system. The non-system modes of the Test Set can be used to give real time readings of equipment performance.

Figures 1-1 and 1-2 show how the TETRA system interfaces with the other operating modes of the Test Set. The dashed lines indicate that you can enter one of the standard test modes by pressing the associated hard key - such as [AF TEST] or [DUPLEX TEST] - and then return directly to the TETRA system screens simply by pressing the [SYSTEMS] hard key.



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Fig. 1-1 TETRA Mobile Systems interface with main Test Set modes

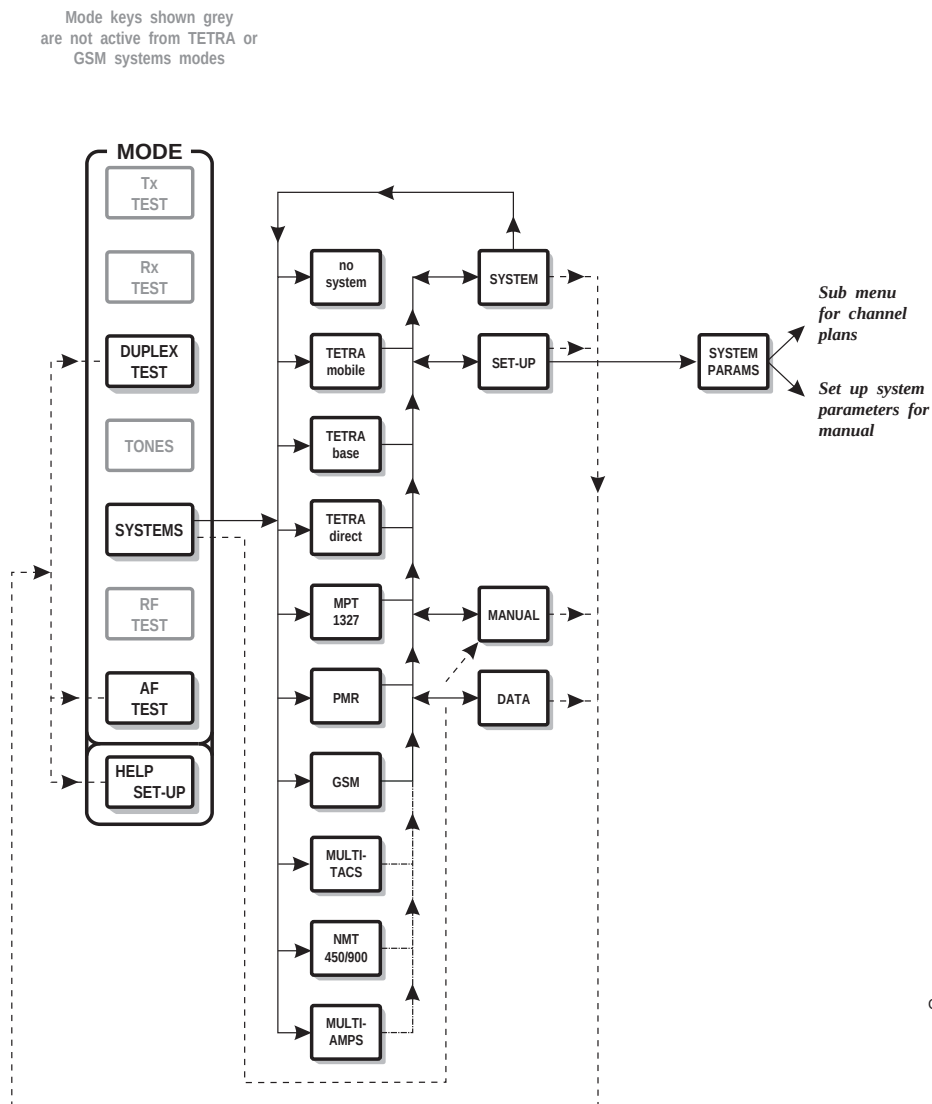


Fig. 1-2 TETRA Base Station and Direct Mode Systems interface with main Test Set modes

Manual and automatic testing

The Manual test mode provides user control of the signal parameters produced by the Test Set for TETRA Mobile (MS) or TETRA Base Station (BS) testing. This is useful for fault finding or adjustment of the mobile or base station under test.

The Auto test mode provided for the TETRA Mobile (MS) mode, allows you to run test programs that are resident within the Test Set to test the operational performance of the mobile. A user-defined program can be loaded into the Test Set and run in a similar way.

RF connections

Prior to testing, you must connect one of the RF connectors (N-type or TNC) on the Test Set to the antenna connector of the equipment under test using a suitable RF cable. See *Before starting*, in Chapter 2.

Manual testing

Manual testing enables you to test specific parts of the operation of the mobile or base station under test. The required signaling is generated by the Test Set under your control.

Automatic testing

The Auto test mode provides two built-in test programs, written in MI-BASIC (see below), which are run to test the performance of the mobile.

The test programs differ in content to give you a choice of tests to suit your particular requirements. You can control both the range of the tests and the test parameters by setting up the system and autorun parameters, as explained in Chapter 3, *Selecting and customizing the system*.

The progress of tests and the results obtained from them are displayed on the screen and stored in memory within the Test Set for later analysis. Results can be printed while the test program is running or at the end of the test program.

The set-up menus allow many combinations of data presentation, storage, retrieval, and printing. Memory cards and remote control facilities provide further flexibility.

The use of memory cards is explained in Chapter 10 *Store and recall facility*.

For information on remote control, see the Programming Manual.

MI-BASIC

The internal MI-BASIC interpreter allows you to write your own test programs using the built-in tests. It also allows you to control the Test Set's keyboard, display, and ports. You can save the current user-defined test program on a memory card or overwrite the current user-defined test program by recalling a program stored on a memory card, as explained in Chapter 10 *Store and recall facility*. For further information about writing test programs, see the Programming Manual.

Remote operation

The TETRA Systems modes can be operated using the GPIB or RS232 remote control facilities provided on the Test Set.

Chapter 2

TESTING MOBILES AND BASE STATIONS

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Introduction

The following chapters explain how to use the Test Set to carry out tests on TETRA mobiles and base stations. The instructions given assume that you are familiar with the basic operation of the Test Set in its various modes and that it has been prepared for use as described in the Test Set 2968 Operating Manual (46882/274) supplied with the Test Set.

If you are unfamiliar with the Test Set, it is suggested that you spend a short time exploring the various menus and displays before connecting it to a mobile or base station.

The operating details that you require when using the Test Set in the TETRA Systems mode fall into two groups:-

- Customization, where you change default test parameters for certain items to be tested or specific test requirements.
- Testing of individual mobiles or base stations. You decide on the procedure required for the particular unit or batch of units, connect the equipment under test to the Test Set, and carry out manual or automatic tests.

The set-up menus of the Systems mode give you control of the test system parameters. Once set to your requirements, the Test Set will retain the settings for both immediate and future use.

Different settings can be stored within the Test Set or on memory cards - refer to Chapter 10, *Store and recall facility*.

Before starting

Warm-up time

It is recommended, in line with best practice in electronic measurements, that the working temperature of the Test Set is allowed to stabilize before commencing tests. See *Performance data; Frequency standard; Warm-up time* in Chapter 1 of the Test Set 2968 Operating Manual (46882/274).

System configuration and default settings

The complexity of the TETRA system requires that key parameters are set to values specific to the requirement of the piece of equipment under test before testing is possible. The relevant parameters are identified and explained at the beginning of Chapters 4, 5, 6, and 7 under *Initial Configuration*.

The procedures for setting the parameters are given in Chapter 3, *Customizing the system*.

The default values of all the parameters for the current System can be recalled using the key sequence below. These values are listed in Chapter 11, *Performance data*.

[HELP SET-UP], [SET-UP], [MISC CONFIG], [default values], [init current].

Note

Ending the sequence with the [init systems] soft key (instead of the [init current] soft key), will recall the default values of **all** Systems within the Test Set.

RF connections

The mobile or base station to be tested must have an RF connection (or connections) to the Test Set.

1. Ensure that the mobile or base station to be connected is not powered up.
2. Use a suitable RF cable (and adapter if required) to connect the antenna connection on the mobile or base station to be tested, to the appropriate RF connector (N-type or TNC) on the Test Set. A two port set-up will be required for some base stations.
3. Use the [RF SELECT] hard key if necessary, to select the appropriate Test Set connector (but see note below). A typical Test Set-up is shown in Fig. 2-1.

Any loss in the RF cable can be compensated for by setting an RF generator level and a receiver level offset via the HELP AND SET-UP menus using the following key sequence:-

[HELP SET-UP], [SET-UP], [TEST OPTIONS], [rf port setup].

A *negative* value must be entered for a *loss*; a *positive* or *unsigned* value represents a *gain*.

CAUTION

You are advised to check that the maximum RF output from the unit under test will not exceed the permissible input of the RF connector used. See *Performance data; RF signal generator;; Reverse power protection*, and *Performance data; RF power meter;; Maximum continuous rating*, in Chapter 1 of the Test Set 2968 Operating Manual (46882/274).

Note

If you enter Digital DUPLEX TEST mode from Systems mode, then change the setting of the [RF SELECT] key, the original selection will be reset when you return to Systems mode.

RF Interference

Plastic-cased mobiles tend to be susceptible to interference and also may pick up transmissions from Base Stations. This can affect the recognition of the simulated control channel, and also mobile receiver tests when the Test Set RF generator output is low. If interference is a problem, you are advised to place the mobile in a screened enclosure where possible.

Note

When a mobile is tested in the presence of a strong local TETRA base station signal, it may register to the base station in preference to the Test Set. If it is not possible to screen the mobile from these transmissions, set the Test Set control channel to the same frequency as the base station and set the Test Set RF generator to its maximum level.

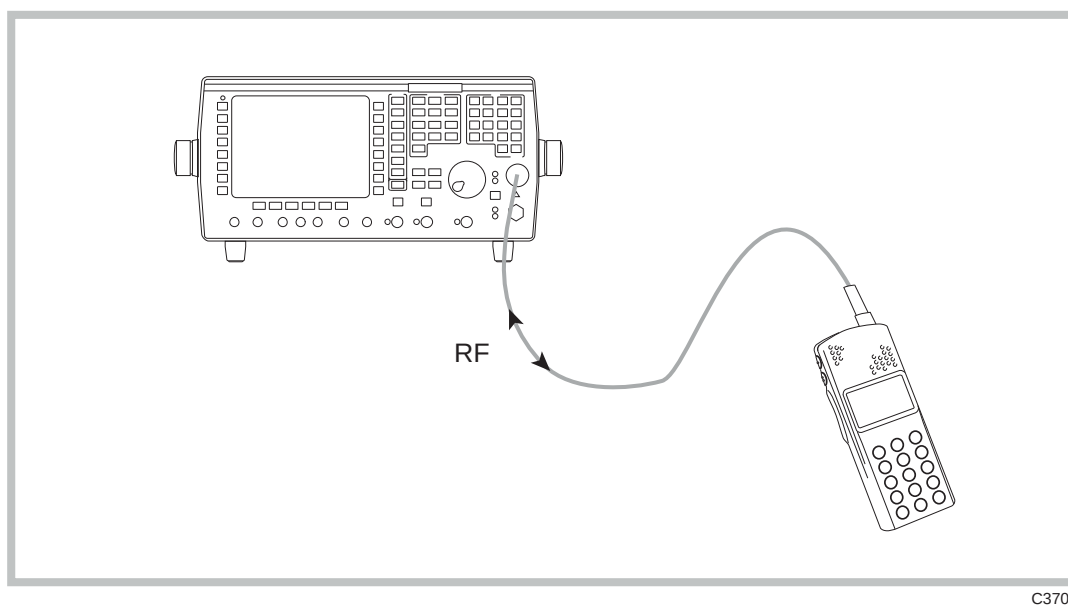


Fig. 2-1 A typical test set-up

Manual testing

This mode is provided to assist you in fault location and repair. You select the Manual test mode by pressing the *[MANUAL]* soft key.

The test procedures for manual testing vary for each of the TETRA systems and these are described in the appropriate chapters.

- TETRA Mobile in Chapter 4.
- TETRA Base Station in Chapter 5.
- TETRA Direct Mode in Chapter 6.

However, there are some common features that are described below.

Switching between test modes

When the Test Set is operating in Systems manual test mode within one of the TETRA systems, you can change to DIGITAL DUPLEX TEST by pressing the *[DUPLEX TEST]* key or to AF INPUT/OUTPUT TEST or MULTIMETER by pressing the *[AF TEST]* hard key. When you do this, the equipment under test remains active and all the settings of the test mode relate to the on-going manual test. This allows measurements to be made on the receiver or transmitter of the equipment under test. You can return directly to the Systems manual test by pressing the *[SYSTEMS]* hard key.

From DIGITAL DUPLEX TEST mode, you can access the spectrum analyzer, RF power analyzer, vector analyzer and constellation diagram displays.

Further information about duplex testing is given later in Chapter 8, Digital duplex testing.

Notes

1. All operations involving signalling between a mobile and the Test Set (registration, de-registration, call setup, call clear-down, transmit request) are only possible in Systems manual test mode or Systems auto test mode.
2. Parameters changed in the DIGITAL DUPLEX TEST mode of the Test Set (such as BURST TYPE, RF GEN LEVEL or RF SELECT) are reset to their Systems manual test mode values on returning to the Systems mode.

Selecting measurement options

Within the HELP SET-UP menus there are screens to allow you to select various options relating to the Rx and Tx measurement procedures for manual tests.

Tx measurement options

You can set the number of single burst samples that will be used to determine each of the measurements tabled below and whether the value obtained is the average of all readings or the worst case obtained. The number of samples can be set within the range 1 to 250. The measurements are:-

- Power.
- Frequency error.
- RMS vector error.
- Peak vector error.
- Residual carrier.
- Timing error.

The Tx settings are accessed via the SET-UP OPTIONS: DIGITAL SYSTEMS screen, that you display by the following key sequence:-

[HELP SET-UP], [SET-UP], [TEST OPTIONS], [digital systems].

Then press the [TETRA Tx meas] soft key to display the SET-UP OPTIONS: TETRA Tx SAMPLES screen.

To change the number of samples taken for a particular measurement, press the appropriate soft key to highlight the measurement, then key in the required number of samples and press the [ENTER] key.

To reset the number of samples taken for all measurements to the default values press the [default values] soft key.

To change the measurement type for a particular measurement, use the appropriate [meas type] soft key on the right of the display, corresponding to the measurement. Each press of the key will step through the options of WORST CASE and AVERAGE, or for the Power measurement, MAXIMUM, MINIMUM and AVERAGE.

Rx measurement options

For TETRA Mobile testing you can set the number of samples (bits or messages) that will be used to determine the value for the TT loopback BER, RBER and MER measurements, and for the T1 BER TCH/7.2. The number of bits for BER/RBER measurements can be set within the range 1000 to 10,000,000 and the number of messages for the MER measurement can be set within the range 10 to 1,000,000.

The Rx settings are accessed via the SET-UP OPTIONS: DIGITAL SYSTEMS screen, that you display by the following key sequence:-

[HELP SET-UP], [SET-UP], [TEST OPTIONS], [digital systems].

Then press the [TETRA Rx meas] soft key to display the SET-UP OPTIONS: TETRA Rx SAMPLES screen.

For example, to change the number of bits used for the BER TCH/7.2 measurement, press the [BER samples] soft key to highlight the measurement, then key in the number of bits to be used and press the [ENTER] key.

To reset the number of bits to be used to the default values press the [default values] soft key.

Highlighted measurements

Measurement inaccuracies

In particular cases where there is some doubt about the accuracy of a measurement, the displayed result is highlighted by appearing in inverse video. This may be only temporary, for example:-

While the Test Set is collecting sufficient samples to make the measurements statistically valid, the intermediate result is shown in inverse video. Once sufficient samples have been collected the displayed result returns to normal video.

Other examples are given where appropriate.

RF level offsets

An asterisk (*) in inverse video beside the RF generator level or power measurement indicates that an RF generator or receiver level offset has been set via the HELP SET-UP menu. See *RF connections* on page 2-2.

Data displays

The data contained in the signal bursts can be displayed as received on the air interface, and in some cases in decoded form, on the DATA DISPLAY screens. The use of this facility is described in Chapter 9.

Automatic testing

This mode provides automatic test programs that carry out a series of tests on TETRA mobiles with a minimum of operator intervention. Test results can be obtained as hard copy or stored electronically for future retrieval and analysis.

You select the Auto test mode by pressing the *[AUTO]* soft key. The test procedures for automatic testing are described in Chapter 7.

Switching between test modes

When the Test Set is operating in Systems auto test mode within the TETRA Mobile system, you can change to DIGITAL DUPLEX TEST by pressing the *[DUPLEX TEST]* key or to AF INPUT/OUTPUT TEST or MULTIMETER by pressing the *[AF TEST]* hard key. When you do this, the equipment under test remains active and all the settings of the test mode relate to the on-going automatic test. This allows measurements to be made on the receiver or transmitter of the equipment under test. You can return directly to the Systems auto test by pressing the *[SYSTEMS]* hard key.

From DIGITAL DUPLEX TEST mode, you can access the spectrum analyzer, RF power analyzer, vector analyzer and constellation diagram displays.

Further information about duplex testing is given later in Chapter 8, *Digital duplex testing*.

Notes

1. All operations involving signalling between a mobile and the Test Set (registration, de-registration, call setup, call clear-down, transmit request) are only possible in Systems manual test mode or Systems auto test mode.
2. Parameters changed in the DIGITAL DUPLEX TEST mode of the Test Set (such as BURST TYPE, RF GEN LEVEL or RF SELECT) are reset to their Systems auto test mode values on returning to the Systems mode.

Selecting measurement options

There are various options relating to the automatic test parameters. These are explained in Chapter 7, *TETRA Mobile automatic testing*.

Chapter 3

SELECTING AND CUSTOMIZING THE SYSTEM

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System selection

In addition to the TETRA Mobile, TETRA Base Station and TETRA Direct Mode systems described in the manual, other test systems are available with the Test Set. Initially¹ the [SYSTEMS] hard key displays a SYSTEM SELECTION menu showing the installed systems. (See Fig. 3-1.) The titles of systems that are not installed (but are available for the Test Set) are shown in dim video.

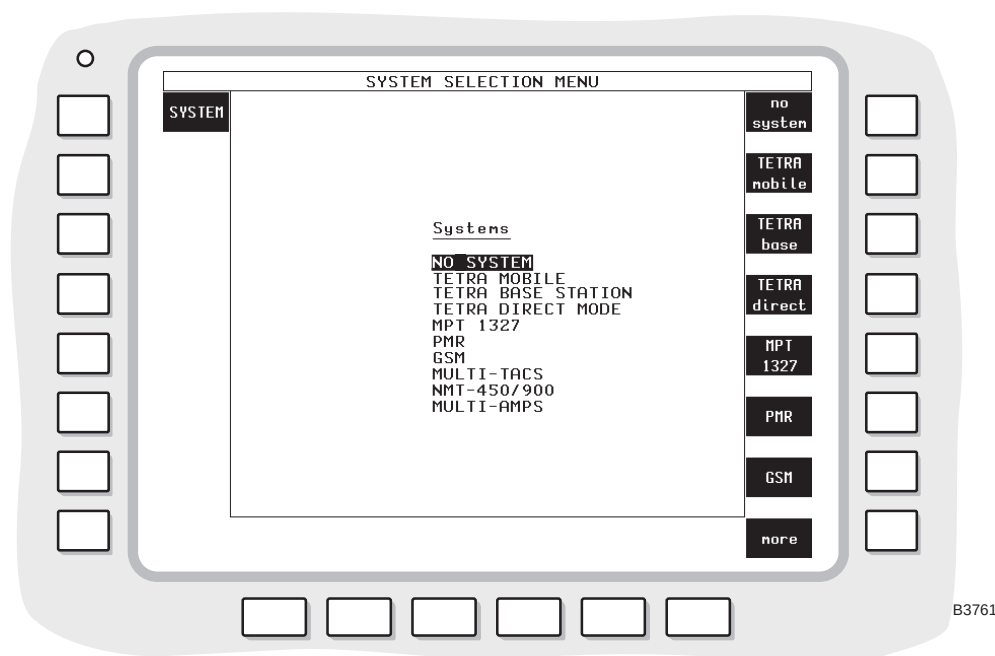


Fig. 3-1 System selection menu

Note

If a *HELP* or *SET-UP* screen is displayed the Test Set must be returned to a mode from which Systems mode can be selected. This can be done by pressing the *[return]* soft key on the display if one is present or by pressing the [HELP/SET-UP] key to return to the top level HELP AND SET-UP menu, then again to return to the screen displayed when HELP AND SET-UP was selected.

The SYSTEM SELECTION MENU assigns one soft key to each system. The three TETRA test modes are each programmed as individual Systems within the Test Set. To select the TETRA Mobile, TETRA Base Station or TETRA Direct Mode system, press the *[TETRA mobile]*, *[TETRA base]* or *[TETRA direct]* soft key as appropriate.

With a TETRA system selected, the soft keys at the left of the screen are allocated to explicit functions or user operations. This follows the general philosophy of the Test Set design. The keys and their functions are shown below. Also shown are references to sections of the manual that give further details.

¹ The first time that the [SYSTEMS] key is pressed after switch-on. Otherwise the latest Systems screen appears.

[SYSTEM]	System selection menu	Page 3-5; Selecting the system
[SET-UP]	System set-up menus	Page 3-5; Selecting the system
[PROGRAM]	Autorun test program selection menu	Chapter 7; Mobile automatic testing
[AUTO]	Auto test mode	Chapter 7; Mobile automatic testing
[MANUAL]	Manual test mode	Chapter 4; Mobile manual testing Chapter 5; Base station manual testing Chapter 6; Direct mode manual testing
[DATA]	Data display mode	Chapter 9; Data displays

Not all of the keys listed above are present for all TETRA systems.

Channel Plan Selection

Having selected the system, you can select alternative channel plans.

First press the [SET-UP] soft key to select the SYSTEM PARAMETERS menu (or if it is shown, press the [System Params] soft key). Next press the [channel plan] soft key.

This gives you access to the appropriate set of keys:-

NO PLAN	or	NO PLAN	or	NO PLAN
USER DEFINED		USER DEFINED		USER DEFINED
TETRA 380MS		TETRA 380BS		TETRA 380DM
TETRA 380+0MS		TETRA 380+0BS		TETRA 380+0DM
TETRA 410MS		TETRA 410BS		TETRA 410DM
TETRA 410+0MS		TETRA 410+0BS		TETRA 410+0DM
TETRA 410-6MS		TETRA 410-6BS		TETRA 410-6DM
TETRA 450MS		TETRA 450BS		TETRA 450DM
TETRA 450+0MS		TETRA 450+0BS		TETRA 450+0DM
TETRA 800MS		TETRA 800BS		TETRA 800DM
TETRA 800+0MS		TETRA 800+0BS		TETRA 800+0DM
TETRA 870MS		TETRA 870BS		TETRA 870DM
TETRA 870+0MS		TETRA 870+0BS		TETRA 870+0DM

The procedure is described in detail under *System Parameters* on pages 3-7 for TETRA Mobile (MS); 3-22 for TETRA Base Station (BS); and 3-31 for TETRA Direct (DM).

The title of the selected system channel plan is shown at the top left of the display. This continues to be displayed on all systems screens until the [no system] soft key on the SYSTEM SELECTION MENU is pressed or until a different system or channel plan is selected. If you open the DIGITAL DUPLEX TEST mode or the HELP facility, directly from the SYSTEMS mode, the title changes to TETRA MS, TETRA BS or TETRA DM as appropriate.

Customizing the system

The subjects covered by this chapter are:-

- Selecting the system
- Setting system parameters
- Setting mobile parameters (TETRA Mobile only)
- Setting autorun controls (TETRA Mobile only)
- Setting autorun parameters (TETRA Mobile only)
- Setting up a user-defined channel plan

Before you use the Test Set for automatic or manual testing, you may need to customize the system by changing some of the default test parameters. You can do this via the [SYSTEMS] hard key and [SYSTEM] and [SET-UP] soft keys, as explained in this section.

Note

If you attempt to enter a parameter value that is outside the permitted limits, the screen message **Entered value is out of range** appears and the Test Set waits for you to enter a permitted value.

Selecting the system

The [SYSTEMS] hard key initially² displays the SYSTEM SELECTION menu (Fig. 3-1 on page 3-3) showing all systems. To select the TETRA Mobile, TETRA Base Station or TETRA Direct Mode, press the [TETRA mobile], [TETRA base] or [TETRA direct] soft key on this menu.

Note

If the Test Set has additional Systems software installed (such as GSM) and you have used this prior to selecting a TETRA System, downloading the software within the Test Set can take in the order of two minutes.

With TETRA Mobile system selected, six of the soft keys at the left of the screen are allocated to explicit functions; the keys and their functions are shown below.

[SYSTEM]	System selection menu
[SET-UP]	System set-up menus
[PROGRAM]	Auto test program selection menu
[AUTO]	Auto test mode
[MANUAL]	Manual test mode
[DATA]	Data display

With TETRA Base Station or TETRA Direct Mode system selected, four of the soft keys at the left of the screen are allocated to explicit functions; the keys and their functions are shown below.

[SYSTEM]	System selection menu
[SET-UP]	System set-up menus
[MANUAL]	Manual test mode
[DATA]	Data display

[SYSTEM]

The [SYSTEM] soft key, which is always visible in Systems mode, takes you to the SYSTEM SELECTION menu, allowing access to all of the test systems that are installed in the Test Set and an option [no system] to disable the active system prior to leaving the Systems mode.

Selecting [no system] configures the 2968 as a conventional analogue radio Test Set.

² The first time that the [SYSTEMS] key is pressed after switch-on. Otherwise the latest Systems screen appears. If a digital System (TETRA MS, TETRA BS TETRA DM or GSM) was selected previously, an initialisation process will run for a few seconds.

[SET-UP]

With TETRA Mobile system selected, the [SET-UP] soft key allows access to the four SYSTEM SET-UP menus. These are: SYSTEM PARAMETERS, MOBILE PARAMETERS, AUTORUN CONTROL, and AUTORUN PARAMETERS. Repeatedly pressing [SET-UP] steps through all four menus.

Note

You can access any of the four menus immediately through the soft keys [System Params], [Mobile Params], [Autorun Control], and [Autorun Params] at the bottom of the screen.

With TETRA Base Station or TETRA Direct Mode systems selected, the [SET-UP] soft key allows access to the SYSTEM SET-UP SYSTEM PARAMETERS menu.

The SYSTEM SET-UP menus enable you to perform the following functions:-

SYSTEM PARAMETERS (All TETRA systems)	Change the values of system parameters such as the channel numbers, network identity numbers, etc.
MOBILE PARAMETERS (TETRA Mobile only)	Change the values of mobile parameters such as the Short Subscriber Identity (SSI), Power Class etc.
AUTORUN CONTROL (TETRA Mobile only)	Select the format of the displayed auto test results, choose the conditions that will cause the auto test program to pause for user attention between tests, and switch the print mode on or off.
AUTORUN PARAMETERS (TETRA Mobile only)	Change the parameters of each of the tests that are available in the auto test programs.

The rest of this section describes each of the SYSTEM SET-UP menus and the parameters on it.

System parameters

When you select the SYSTEM PARAMETERS menu via the [SET-UP] soft key, the system parameters are displayed, and each parameter has a soft key on the right of the screen dedicated to it. When TETRA MOBILE or TETRA BASE STATION is selected the bottom right key is a [more] soft key, that re-assigns the soft keys to allow access to other parameters. Pressing this key repeatedly will cycle you through all the available system parameter screens. (See Fig. 3-2.)

Changing system parameter values

To change a system parameter value, first press the soft key assigned to the parameter; the selected parameter will be highlighted. Then:-

- If the parameter is set by a choice of options press the appropriate secondary soft key.
- If the parameter requires a value, enter the required value using the data input keys and then press the selected parameter soft key or any of the ENTER data keys [MHz / V], [kHz / mV], [Hz / μ V], [dB / %], [dBm / rad].

The selected parameter will then be de-highlighted to indicate that the data entry is complete.

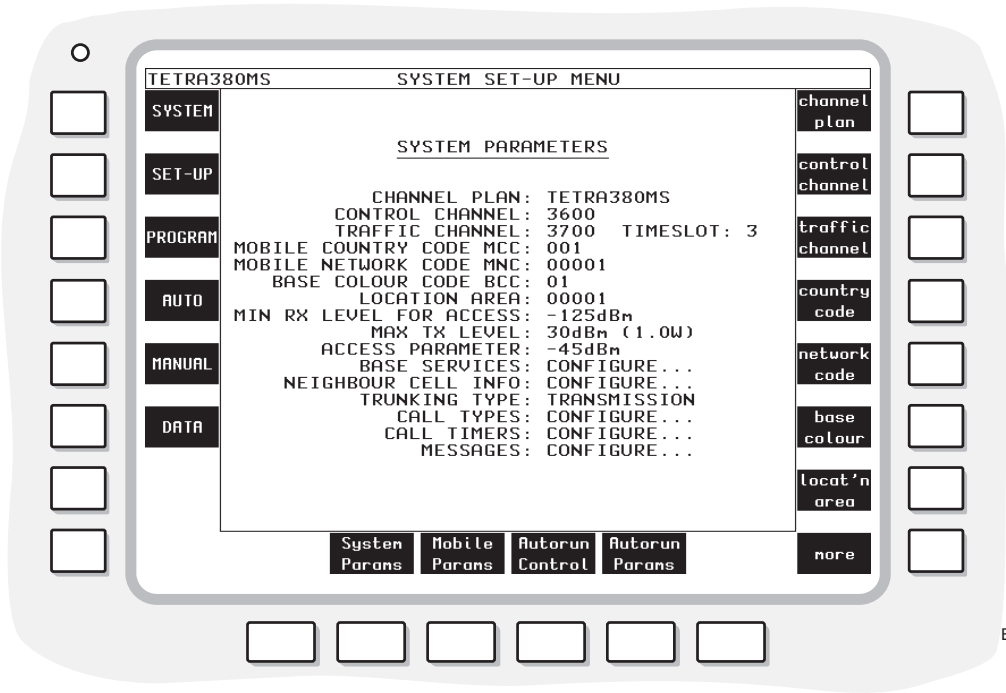


Fig. 3-2 System parameters menu TETRA Mobile

The system parameters that you can modify are described below. Default values are shown in Chapter 11, *Performance data*, under the heading *Test parameters*.

The parameters are different for each of the TETRA systems, therefore those for TETRA Mobile are described first followed by those for TETRA Base Station, then those for TETRA Direct Mode.

TETRA Mobile System parameters

Channel Plan

[channel plan]

The current channel plan is shown at the top left of the screen.

To select a channel plan first press [channel plan] on the SYSTEM PARAMETERS menu.

This gives you access to the appropriate set of keys:-

NO PLAN	or	TETRA 450MS
USER DEFINED		TETRA 450+0MS
TETRA 380MS		TETRA 800MS
TETRA 380+0MS		TETRA 800+0MS
TETRA 410MS		TETRA 870MS
TETRA 410+0MS		TETRA 870+0MS
TETRA 410-6MS		
more	↔	more

Note

You may find that it is necessary to use a USER DEFINED channel plan. See the note regarding channel numbers in chapter 11, *Performance data*.

If you select USER DEFINED as the channel plan, a *[user defined]* soft key becomes available on the SYSTEM PARAMETERS menu. Pressing this *[user defined]* soft key gives you access to the USER DEFINED CHANNEL PLAN menu, that allows you to enter the appropriate channel plan parameters. For further information see *Setting up a user-defined channel plan* starting on page 3-39.

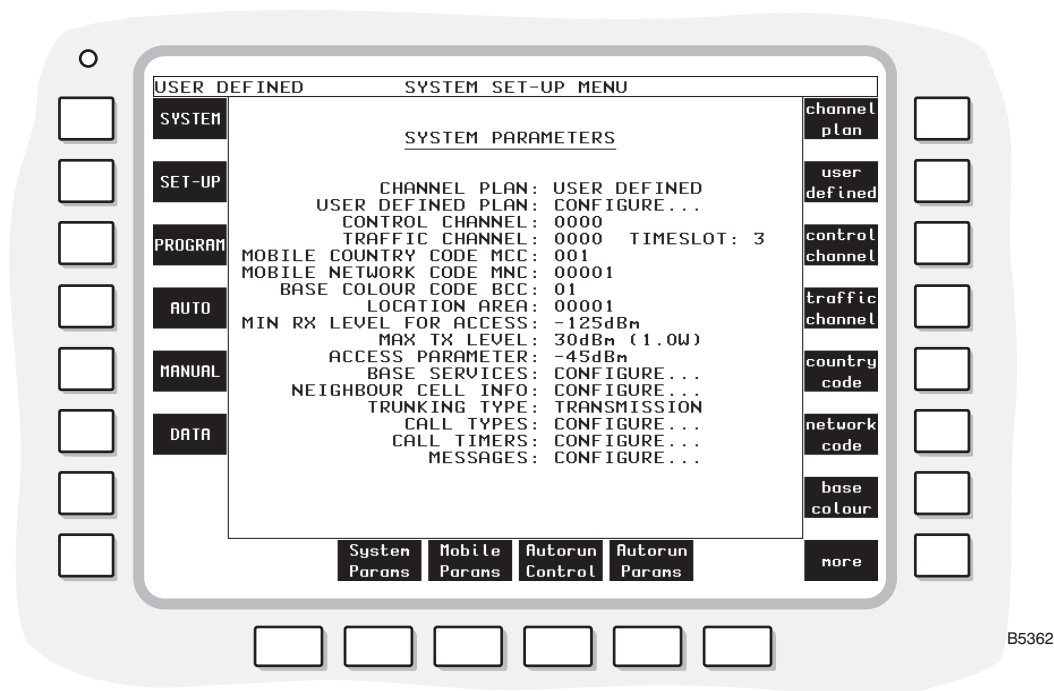


Fig. 3-3 System parameters menu TETRA Mobile USER DEFINED channel plan

If you select NO PLAN as the channel plan, the mobile's CONTROL CHANNEL Tx frequency can be set directly in MHz. An Rx OFFSET field and soft key become available to allow you to set the offset between the mobile's transmitter and receiver frequencies.

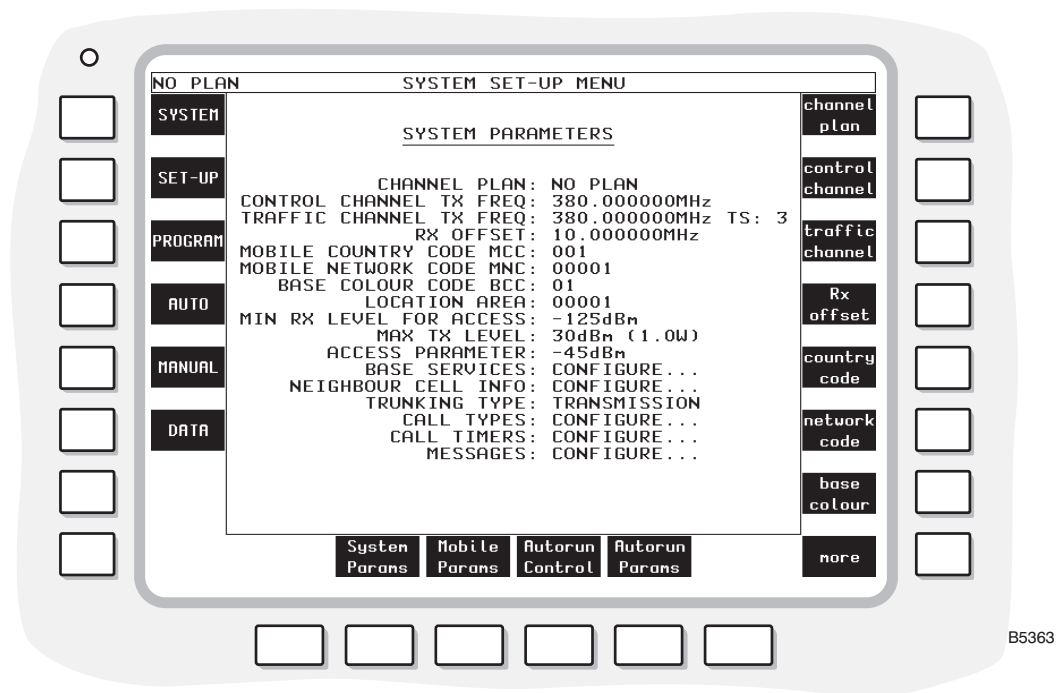


Fig. 3-4 System parameters menu; TETRA Mobile NO PLAN

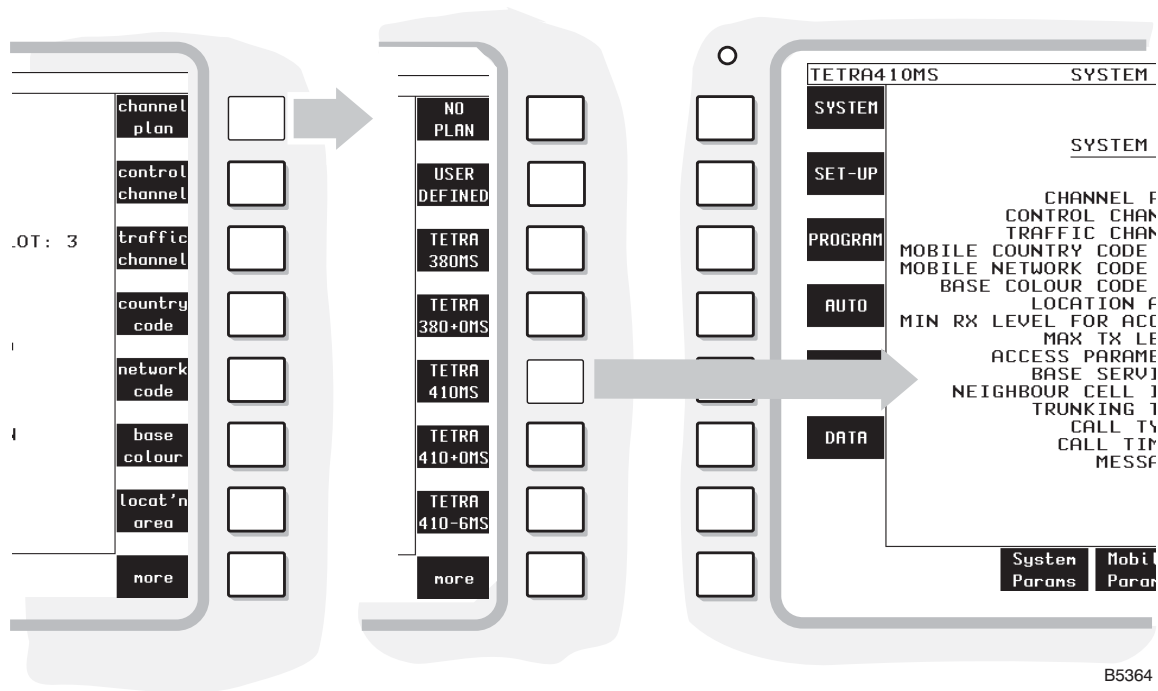


Fig. 3-5 Channel plan selection for TETRA Mobile (MS)

User Defined Plan

[user defined]

Note

The *[user defined]* soft key is only available if the USER DEFINED channel plan has been selected.

Pressing the *[user defined]* soft key gives you access to the USER DEFINED CHANNEL PLAN menu. (See *Setting up a user-defined channel plan* starting on page 3-39.)

Note

The following Control Channel, Traffic Channel and Timeslot parameters are used during Automatic testing and as the initial values for Manual testing.

Control Channel

or

Control Channel Tx Freq

(if channel plan is set to NO PLAN on the SYSTEM PARAMETERS menu)

[control channel]

[control channel]

The Control Channel parameter can be set to any value within the range permitted by the selected channel plan. (See the appropriate section in Chapter 11, *Performance data*.) To change the control channel, enter the required channel number (decimal) or frequency using the data keys. The timeslot for the control channel is always TN 1.

Note

The Control Channel Tx FREQ parameter sets the frequency of the Test Set receiver, i.e. the expected frequency of the mobile transmitter. The Test Set signal generator frequency is offset from this value by the value of the Rx offset parameter.

TrafficChannel

or

Traffic Channel Tx Freq

(if channel plan is set to NO PLAN on the SYSTEM PARAMETERS menu)

[traffic channel]

[traffic channel]

The Traffic Channel parameter can be set to any value within the range permitted by the selected channel plan. (See the appropriate section in Chapter 11, *Performance data*.) To change the traffic channel, enter the required channel number (decimal) or frequency using the data keys. Pressing the *[traffic channel]* soft key twice allows the Timeslot parameter to be changed. (See *Timeslot* below.)

Note

The Traffic Channel Tx FREQ parameter sets the frequency of the Test Set receiver, i.e. the expected frequency of the mobile transmitter. The Test Set signal generator frequency is offset from this value by the value of the Rx offset parameter.

Rx Offset

[Rx offset]

Note

The *[Rx offset]* soft key is only available if the NO PLAN channel plan has been selected.

This sets the offset between the mobile's transmitter frequency (the Test Set receiver frequency) and the mobile's receiver frequency (the Test Set signal generator frequency).

$$\text{MS RxFreq} = \text{MS TxFreq} + \text{Rx Offset}$$

Note

Reverse operation can be set by entering a negative value of Rx Offset.

Timeslot

[traffic channel] [traffic channel]

To change the TIMESLOT value, press the *[traffic channel]* soft key twice and enter the required timeslot number (1-4) using the data keys.

Note

You can set any or all of the following system parameters, MCC, MNC or BCC, to match a real Base Station. MCC, MNC and BCC are all parameters on the Broadcast Synchronisation Channel (BSCH) that is sent in MCCH, TCH and T1 signals. Changing any of these will change the Scrambling sequence that applies to all signals (MCCH, TCH and T1) generated and received by the Test Set. This does not apply to the BSCH itself, that substitutes zero values for BCC, MCC and MNC when generating the Scrambling sequence. If the mobile under test is in a test mode where it expects a certain fixed value of scrambling sequence, the MCC, MNC and BCC must be set to produce the scrambling sequence expected.

Mobile Country Code (MCC)

[country code]

This sets the Mobile Country Code (MCC) parameter sent to the mobile in the broadcast synchronisation information. The range is 0 to 999 (decimal).

Set the Mobile Country Code (MCC) to 001 if you are testing a mobile using TETRA Test mode (TT). Otherwise set the code that is appropriate to the mobile under test.

Mobile Network Code (MNC)

[network code]

This sets the Mobile Network Code (MNC) parameter sent to the mobile in the broadcast synchronisation information. The range is 0 to 16383 (decimal).

Set the Mobile Network Code (MNC) to 00001 if you are testing a mobile using TETRA Test mode (TT). Otherwise set the code that is appropriate to the mobile under test.

Base Colour Code (BCC)

[base colour]

This sets the Base (Station) Colour Code (BCC) parameter sent to the mobile in the broadcast synchronisation information. The range is 0 to 63 (decimal).

Note

Setting the BCC to 0 causes the scrambling sequence for all logical channel types to be generated with an all zeros scrambling seed, as for the BSCH, regardless of the values of MCC and MNC. Some TETRA terminals do not support BCC value 0

Location Area (LA)

[locat'n area]

This sets the Location Area (LA) parameter sent to the mobile in the system information. The range is 1 to 16383 (decimal).

Min Rx Level for Access

[min Rx level]

This parameter sets the minimum received level of a signal from a base station (i.e. the Test Set in this instance) that the mobile is permitted to accept. (This is the RXLEV_ACCESS_MIN parameter in system information.) The range is from -125 dBm to -50 dBm, in 5 dB steps.

Max Tx Level

[max Tx level]

This parameter sets the maximum power level at which a mobile is permitted to transmit when it is using MS Link Control (open loop power control). (This is the MS_TXPWR_MAX_CELL parameter in system information.) The range is 15 dBm to 45 dBm, in 5 dB steps.

Access Parameter

[access param]

This sets the ACCESS_PARAMETER value in system information, that controls MS Link Control Operation (open loop power control). The range is from –53 dBm to –23 dBm in 2 dB steps.

Base Services

[base service]

This gives access to all of the Base Services system information parameters. These are tabled below, with the default settings and available options:-

Parameter	Default	Options	
Power on registration	Required	Not required,	Required
Power off de-registration	Required	Not required,	Required
Priority cell	Yes	No,	Yes
Minimum mode service	Never used	Never used,	May be used
Migration	Supported	Not supported,	Supported
System wide services	Normal mode	Not supported,	Normal mode
TETRA voice service	Supported	Not supported,	Supported
Circuit mode data service	Not supported	Not supported,	Supported
(Reserved)	Not available	Not available,	Available
SNDCP service	Not available	Not available,	Available
Air interface encryption	Not available	Not available,	Available
Advanced link	Not supported	Not supported,	Supported

Power on registration

[registn]

This parameter indicates whether or not a mobile is required to perform ITSI Attach registration when it is powered on.

Normally this parameter should be set to ‘required’.

Power off de-registration

[de-registn]

This parameter indicates whether or not a mobile is required to perform ITSI detach de-registration when it is powered off.

Normally this parameter should be set to ‘required’.

Priority cell

[priority cell]

This parameter indicates whether or not a mobile should select this ‘cell’ in preference to non-priority cells.

Normally this parameter should be set to ‘yes’.

Minimum mode service

[minimum mode]

This parameter indicates whether or not there is a possibility of the base station main control channel (MCCH) being replaced by a traffic channel in frames 1-17 and a minimum mode control channel in frame 18.

Normally this parameter should be set to ‘never used’ as the Test Set does not use minimum mode.

Migration

[migratn]

This parameter indicates whether or not the base station supports registration attempts by a mobile belonging to a different network (different MCC or MNC values).

Normally this parameter should be set to 'supported'.

System wide services

[system wide]

This parameter indicates whether the base station is operating normally or whether it is temporarily in 'fallback mode' in which it is unable to communicate with the rest of the TETRA network.

Normally this parameter should be set to 'normal mode'.

Note

Setting this parameter to 'not supported' may cause the mobile to indicate e.g. 'Local Area Service' or 'Fallback'.
The mobile may not attempt operations requiring connections outside the local cell e.g. telephone calls.

TETRA voice service

[tetra voice]

This parameter indicates whether or not the base station supports speech calls using the TETRA codec.

Normally this parameter should be set to 'supported'.

Circuit mode data service

[circuit data]

This parameter indicates whether or not the base station supports circuit mode data calls.

Normally this parameter should be set to 'not supported' as the Test Set does not support circuit mode data calls.

(Reserved)

[reservd]

This parameter is reserved for future definition in the TETRA standard

Normally this parameter should be set to 'not available'.

SNDCP service

[SNDCP service]

This parameter indicates whether or not the base station supports the TETRA packet data service.

Normally this parameter should be set to 'not available' as the Test Set does not support packet data.

Air interface encryption

[air encrypt]

This parameter indicates whether or not the base station supports air interface encryption.

Normally this parameter should be set to 'not available' as the Test Set does not support air interface encryption.

Note

Setting this parameter to 'available' may cause the mobile to wait before registering in order to receive the associated security class and cipher key information broadcasts. Since the Test Set does not generate these information broadcasts, the mobile may fail to register.

Advanced link

[advance link]

This parameter indicates whether or not the base station supports the advanced link service which is required for packet data.

Normally this parameter should be set to 'not available' since the Test Set does not support the advanced link.

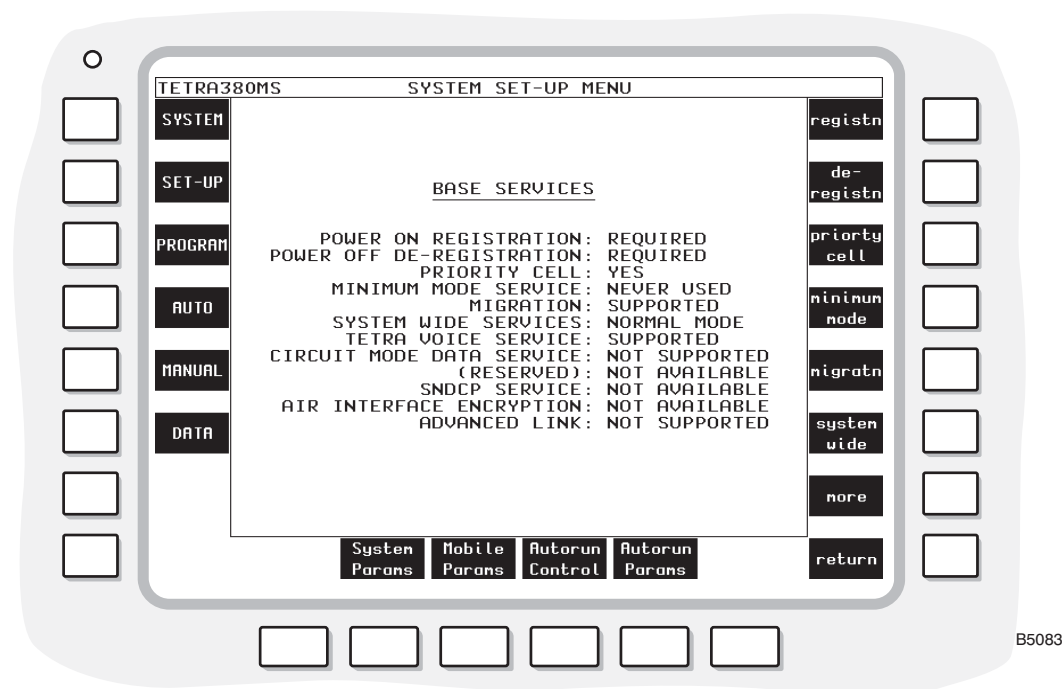


Fig. 3-6 System parameters, TETRA Mobile; Base Services menu

Neighbour Cell Info

[neighbr cell]

You can define the neighbour cell information sent to the mobile in the Neighbour Cell Broadcast message by pressing the [neighbr cell] soft key to display the NEIGHBOUR CELL INFO menu shown in Fig. 3-7. This information is used by the mobile during cell re-selection. (See *Neighbour Cell Info* in Chapter 4.)

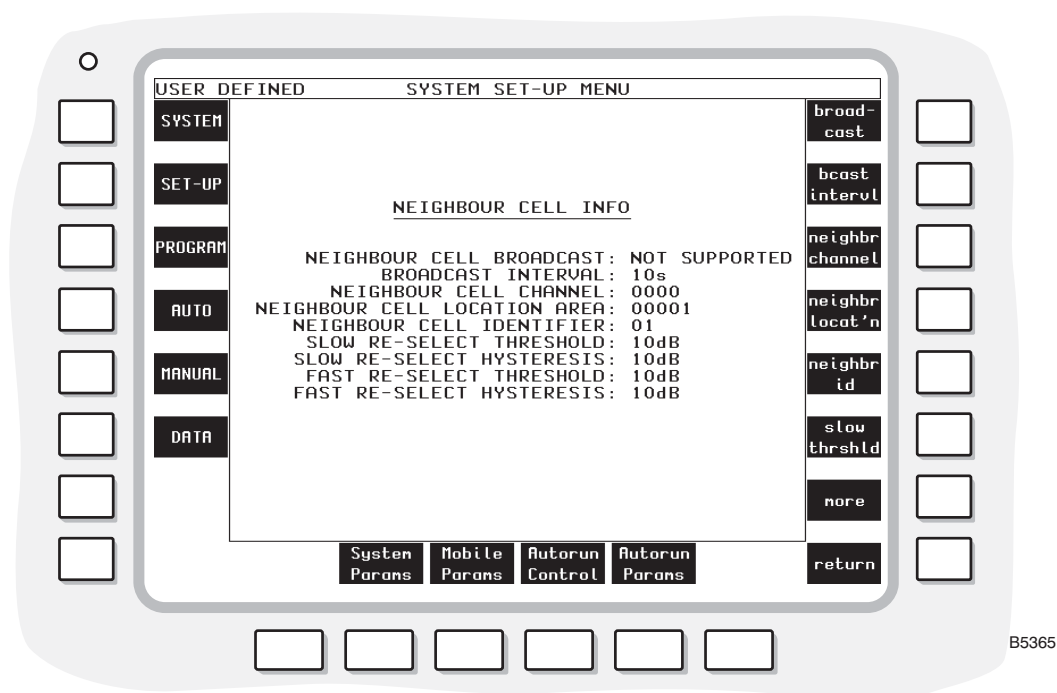


Fig. 3-7 System parameters, TETRA mobile Neighbour Cell Info menu

The options available from each key on the NEIGHBOUR CELL INFO menu are:-

Neighbour Cell Broadcast

[broad-cast]

Sets whether the Neighbour Cell Broadcast message is sent to the mobile (SUPPORTED) or not (NOT SUPPORTED).

Broadcast Interval

[bcast interval]

Sets the interval between sending neighbour cell broadcast messages. The range is 5 to 30 seconds.

Neighbour Cell Channel

[neighbr channel]

Sets the channel number (decimal) of the neighbour cell. The parameter will accept any value within the range 0 to 4095. However, only a value that is valid for the selected channel plan should be entered. (See the appropriate section in Chapter 11, *Performance data*.)

Neighbour Cell Location Area

[neighbr la]

Sets the Location Area (LA) of the neighbour cell. The range is 0 to 16383.

Neighbour Cell Identifier

[neighbr id]

Sets the Identifier number of the neighbour cell. The range is 1 to 31.

Slow Re-select Threshold (Above Fast Re-select Threshold)

[slow thrshld]

Sets the slow neighbour cell re-selection threshold. The range is 0 to 30 dB in steps of 2 dB, and is additional to the Fast Re-select Threshold.

Slow Re-select Hysteresis

[slow hysters]

Sets the slow neighbour cell re-selection hysteresis. The range is 0 to 30 dB in steps of 2 dB.

Fast Re-select Threshold

[fast thrshld]

Sets the fast neighbour cell re-selection threshold. The range is 0 to 30 dB in steps of 2 dB.

Fast Re-select Hysteresis

[fast hysters]

Sets the fast neighbour cell re-selection hysteresis. The range is 0 to 30 dB in steps of 2 dB.

Trunking Type

[trunk type]

MESSAGE trunked, TRANSMISSION trunked or QUASI TRANSMISSION trunked.

This applies only to simplex calls, for duplex calls this parameter can be set to any value. (See *Selection of Trunking Mode* in Chapter 4.)

Note

Quasi Transmission Trunking is not currently supported on the Test Set. If trunking type is set to Quasi Transmission Trunked, the Test Set behaviour will be the same as for message trunked.

Call Types

[call types]

You can configure parameters of each call type, but *some* parameters of *some* call types are pre-configured.

Note

Not all combinations of call type options are valid. If you set up an invalid combination, the call may fail.

Pressing the *[call types]* soft key displays a key for each of the call types. Then pressing the soft key for the specific call type changes the display to show the parameters that can be configured.

Pressing the soft key for the specific parameter either:-

- Changes the display to show soft keys for each option; or
- Highlights the parameter and soft key if a value is required.

To select an option press the appropriate soft key. (See Fig. 3-8.)

To change a value, key in the *new value*, and press the highlighted soft key. An error message is displayed if the entry is not valid or out of range. To correct a invalid or out of range value first press the [DELETE] data key to remove the incorrect digit(s), then enter the correct value.

The tables below and on the next page show the parameters for each call type, with the default values and settable ranges, or pre-configured values where appropriate.

Group call

Parameter	Defaults or pre-configured value	Range
Group/Individual	Group Call	Pre-defined value
Simplex/Duplex	Simplex Call	Pre-defined value
Signalling Type	Direct	Pre-defined value
Priority	00	00 to 15
Calling Party SSI	00000001	00000000 to 16777215
ESN Required	Not required	Pre-defined value

Private call

Parameter	Defaults or pre-configured value	Range
Group/Individual	Individual Call	Pre-defined value
Simplex/Duplex	Simplex Call	Simplex Call, Duplex Call
Signalling Type	Hook	Direct, Hook
Priority	00	00 to 15
Calling Party SSI	00000001	00000000 to 16777215
ESN Required	Not required	Pre-defined value

Phone call

Parameter	Defaults or pre-configured value	Range
Group/Individual	Individual Call	Pre-defined value
Simplex/Duplex	Duplex Call	Pre-defined value
Signalling Type	Hook	Pre-defined value
Priority	00	00 to 15
Calling Party SSI	16777184 (PSTN gateway)	00000000 to 16777215
Calling Party ESN	01438742200	0 to 99999999999999999999999999999999 Up to 24 digits
ESN Required	Required	Required, Not required

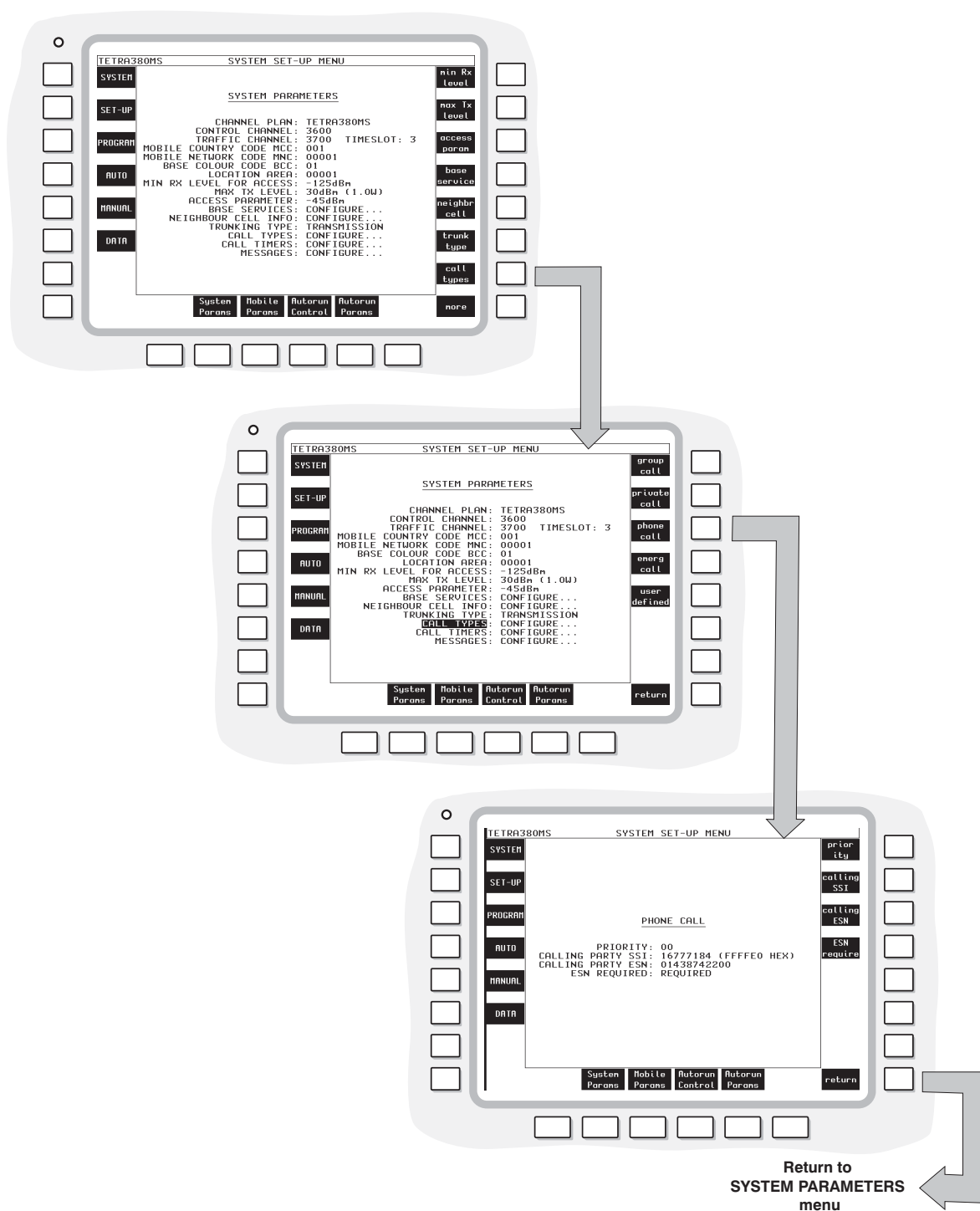
--	--	--

Parameter	Defaults or pre-configured value	Range
Group/Individual	Individual Call	Individual Call, Group Call
Simplex/Duplex	Simplex Call	Simplex Call, Duplex Call
Signalling Type	Direct	Direct, Hook
Priority	15	Pre-defined value
Calling Party SSI	00000001	00000000 to 16777215
ESN Required	Not required	Pre-defined value

User defined

The User Defined call type allows a call type to be set up for your specific requirements. The parameters that can be included are:-

Parameter	Defaults	Range
Group/Individual	Individual Call	Individual Call, Group Call
Simplex/Duplex	Duplex Call	Simplex Call, Duplex Call
Signalling Type	Hook	Direct, Hook
Priority	00	00 to 15
Calling Party SSI	16777186 (PABX gateway)	00000000 to 16777215
Calling Party ESN	742200	0 to 999999999999999999999999 Up to 24 digits
ESN Required	Required	Required, Not required



B5366

Fig. 3-8 System parameters TETRA Mobile Call Type set-up procedure

Call timers

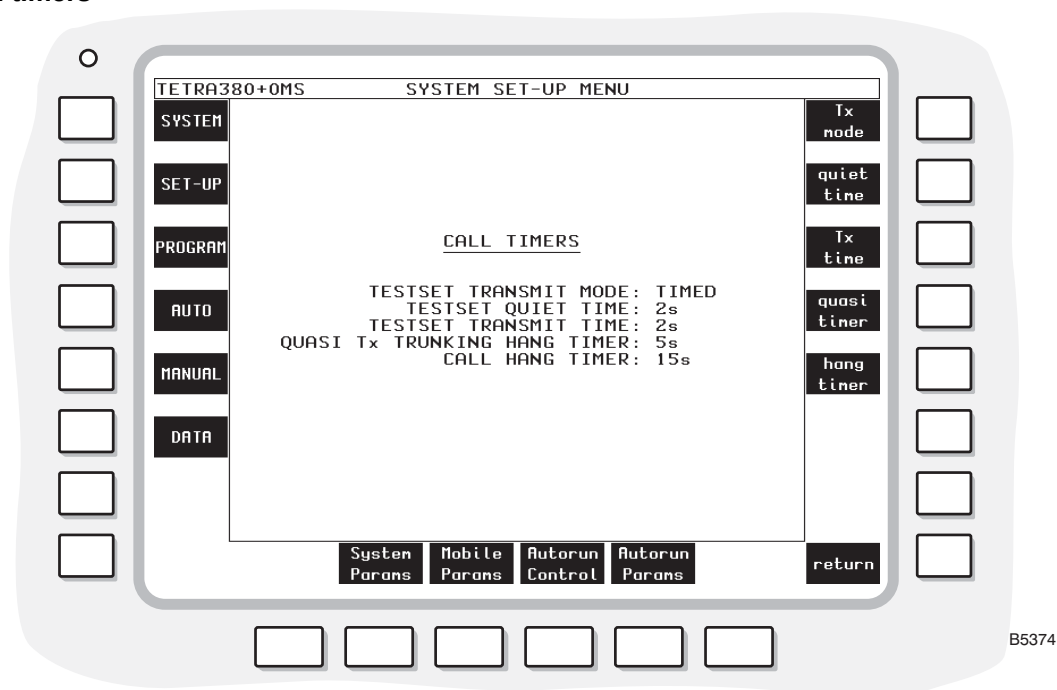


Fig. 3-9 System parameters TETRA Mobile Call Timers menu

The following soft keys are used to set the CALL TIMERS parameters. (See *Selection of Trunking Mode* in Chapter 4.) These parameters only apply to simplex calls.

Test Set transmit mode

[Tx mode]

This allows you to select the Test Set behaviour on a simplex call when the mobile is not transmitting. The options are NONE, TIMED or CONTINUOUS. The default is TIMED.

The options are as follows:-

- **NONE:** The Test Set will not simulate another user transmitting when the mobile under test is not transmitting, so you can press the PTT again at any time to request transmission. However group calls are subject to the call hang timer. (See below.)

Note

Transmission mode NONE is only available when the trunking type is set to transmission trunking.

- **TIMED:** When you release the PTT of the mobile under test, the Test Set waits for the TEST SET QUIET TIME period, during which you may press the PTT again to request transmission. If you do not press the PTT, the Test Set will simulate another user talking for the TEST SET TRANSMIT TIME period, after which it will revert to QUIET mode. During QUIET mode you may press the PTT again to request transmission. In a group call, the maximum duration of QUIET mode is determined by the CALL HANG TIMER period. On expiry of the call hang timer the Test Set automatically clears down a group call.

Note

Transmission mode TIMED is only available when the trunking type is set to transmission trunking.

- **CONTINUOUS:** When you release the PTT of the mobile under test, the Test Set immediately simulates another user talking indefinitely. Depending on the mobile, you may be able to interrupt this other user and request to transmit by pressing the PTT again. The timer values do not apply when CONTINUOUS is selected. Use CONTINUOUS mode if you need to keep the mobile receiver open in a simplex call.

Note

If you have selected TRUNKING TYPE to be Message or QUASI TRANSMISSION, the TESTSET TRANSMIT MODE will be CONTINUOUS regardless of the setting of the parameter.

Test Set quiet time

[quiet time]

This parameter allows you to define the period for which the Test Set waits after PTT release before simulating another user talking in TIMED mode. The range is 0 to 30 seconds and the default is 2 seconds.

Test Set transmit time

[Tx time]

This parameter allows you to define the period for which the Test Set simulates another user talking in TIMED mode. The range is 1 to 30 seconds and the default is 2 seconds.

Quasi Tx hang timer

[quasi timer]

Quasi transmission trunking is not currently supported on the Test Set, so this parameter currently has no effect. The range is 1 to 30 seconds and the default is 5 seconds.

Call hang timer

[hang timer]

This parameter allows you to define the maximum quiet period in group call, in TIMED mode or NONE mode, after which the Test Set will automatically clear down the group call. The range is 1 to 30 seconds and the default is 15 seconds.

Note

The hang timer value should normally be longer than the TEST SET QUIET TIME period, otherwise a group call will be cleared down before the Test Set has an opportunity to simulate another user talking.

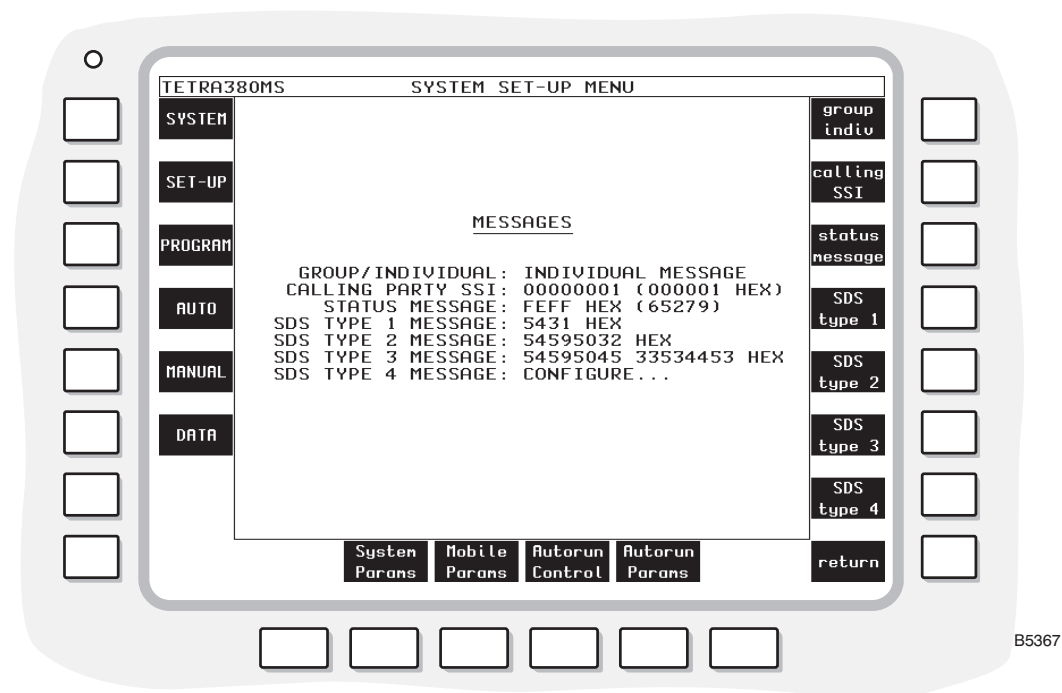


Fig. 3-10 System parameters, TETRA Mobile; Messages menu

Messages

The following keys are used to set the MESSAGES parameters. (See *Mobile Terminated Status and Short Data (SDS) Messages* starting in Chapter 4.)

Group/Individual

[group indiv]

This allows you to select the type of message sent for all SDS types and STATUS. The options are Group Message or Individual Message. The default is Individual Message.

Calling party SSI

[calling SSI]

Allows a Calling Party SSI to be entered within the range 00000000 to 16777215 for all SDS types and STATUS. The default is 00000001

Status message

[status message]

Allows a status message to be entered within the range 0000 to FFFF HEX. The default is FEFF HEX (CALLBACK).

SDS type 1 message

[SDS type 1]

This allows the SDS type 1 message to be set. The range is 0000 to FFFF HEX and the default 5431 HEX. (ASCII "T1")

SDS type 2 message

[SDS type 2]

This allows the SDS type 2 message to be set. The range is 00000000 to FFFFFFFF HEX and the default 54595032 HEX. (ASCII "TYP2")

SDS type 3 message

[SDS type 3]

This allows the SDS type 3 message to be set. The range is 00000000 00000000 to FFFFFFFF FFFFFFFF HEX and the default 54595045 33534453 HEX. (ASCII "TYPE3SDS")

Each word of the message is set individually. Pressing the [SDS type 3] soft key allows the digits of the first word to be keyed in. Press the [enter] key and the digits of the second word can be keyed in. Press the [enter] key when the second word is complete and the new SDS type 3 message will be entered.

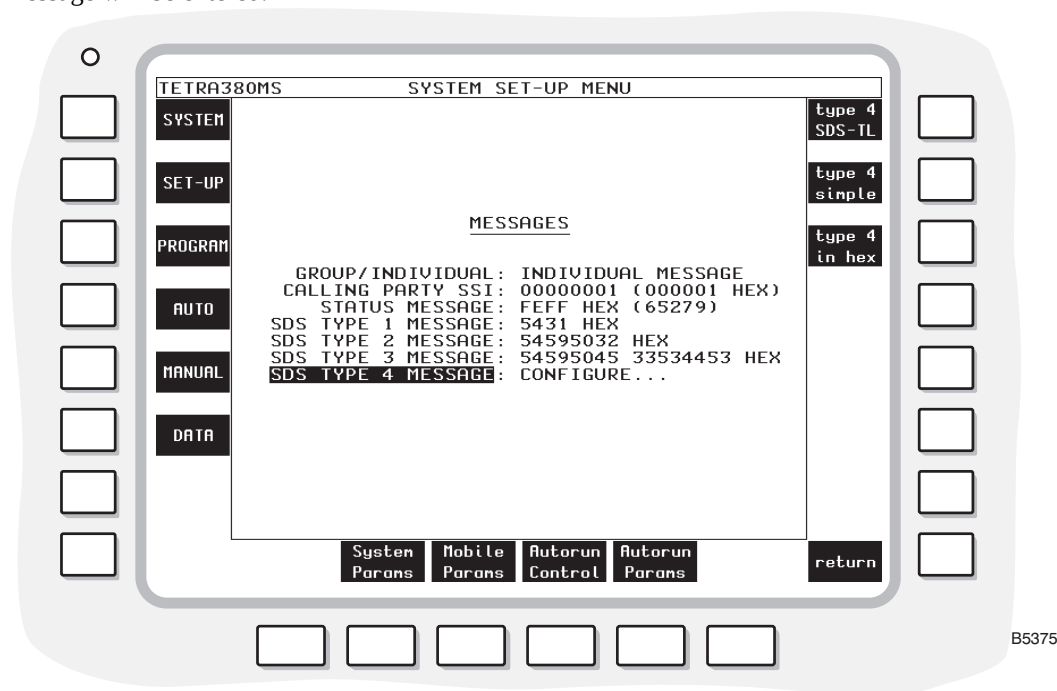


Fig. 3-11 System parameters, TETRA Mobile; SDS Type 4 Messages selection menu

SDS type 4 message

[SDS type 4]

You can configure parameters of different types of SDS type 4 message:

- Type 4 SDS-TL text message.
- Type 4 simple text message.
- Type 4 HEX data message.

(See *Mobile Terminated Short Data (SDS) Message Type 4* in Chapter 4.)

Type 4 SDS-TL text message.

[Type 4 SDS-TL]

The following keys are used to set the SDS-TL text message parameters.

Note

The message reference number is not user settable but is automatically incremented for each message sent by the Test Set.

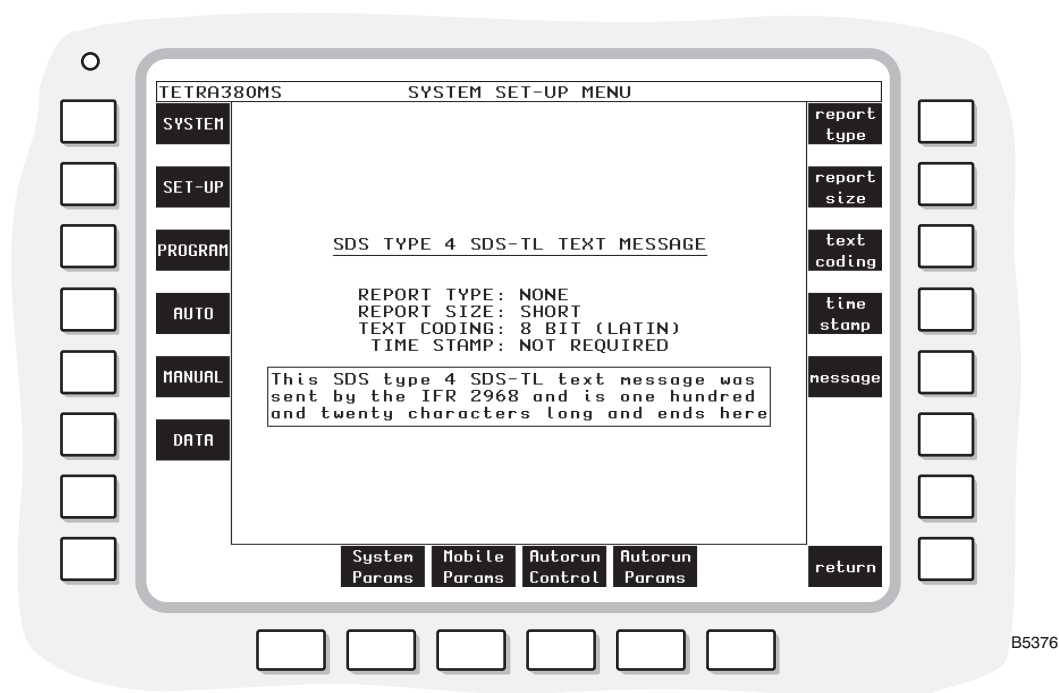


Fig. 3-12 TETRA Mobile; SDS Type 4 SDS-TL Text Message parameters menu

Report type

[report type]

This parameter allows you to request the type of report sent by the mobile in response to the SDS-TL text message. The options are: NONE; MESSAGE RECEIVED; MESSAGE CONSUMED; or MESSAGE RECEIVED AND CONSUMED. The default is NONE.

Note

The mobile may not necessarily send the type of report requested or any report. However, the mobile should not send a report if the requested report type is NONE.

Report size

[report size]

This parameter allows you to request either a SHORT report from the mobile (status message) or a FULL report (SDS-TL standard report). The default is SHORT.

Note

The mobile may not necessarily send the size of report requested or any report. If the requested report type is NONE, then no report should be sent.

Text coding

[text coding]

This parameter allows you to select whether the text message is sent to the mobile in standard 8-bit TETRA compatible text coding or in 7-bit GSM compatible text coding. The default is 8-BIT.

Note

Some mobiles may not be capable of receiving 7-bit coded text messages.

Time stamp

[time stamp]

This parameter allows you to select whether or not a time stamp is included in the text message sent to the mobile. The default is NOT REQUIRED.

Note

Time stamping information consists of month, day, hours, minutes, and is taken from the Test Set real time clock. The real time clock is set by pressing [HELP/SETUP], [SET_UP], [misc config], [time & date].

Text message

[message]

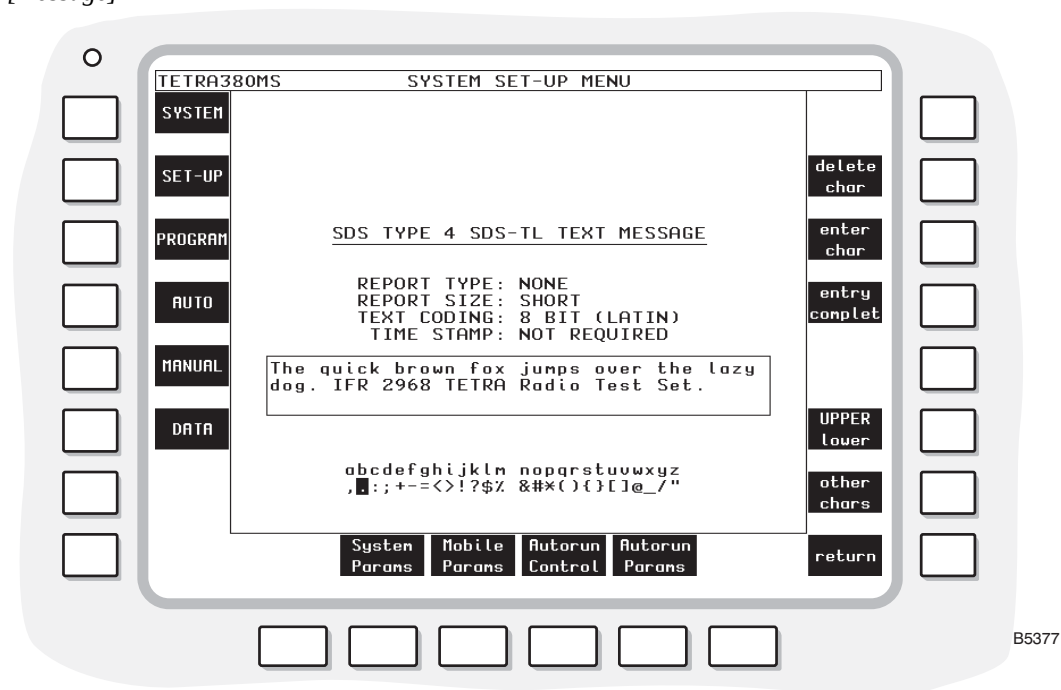


Fig. 3-13 TETRA Mobile; SDS Type 4 SDS-TL Text Message entry

The default text message is of length 120 characters. This key allows you to enter your own text message of up to 120 characters. Refer to *Channel Plan Title* on page 3-39 for details of text entry.

Note

If you wish to restore the default text message you will need to reset all TETRA MOBILE TEST system parameters to the factory default values. To do this, press [HELP/SETUP], [SET_UP], [misc config], [default values], [init current], [return], [return], [return], [RESET].

Type 4 simple text message.

[Type 4 simple]

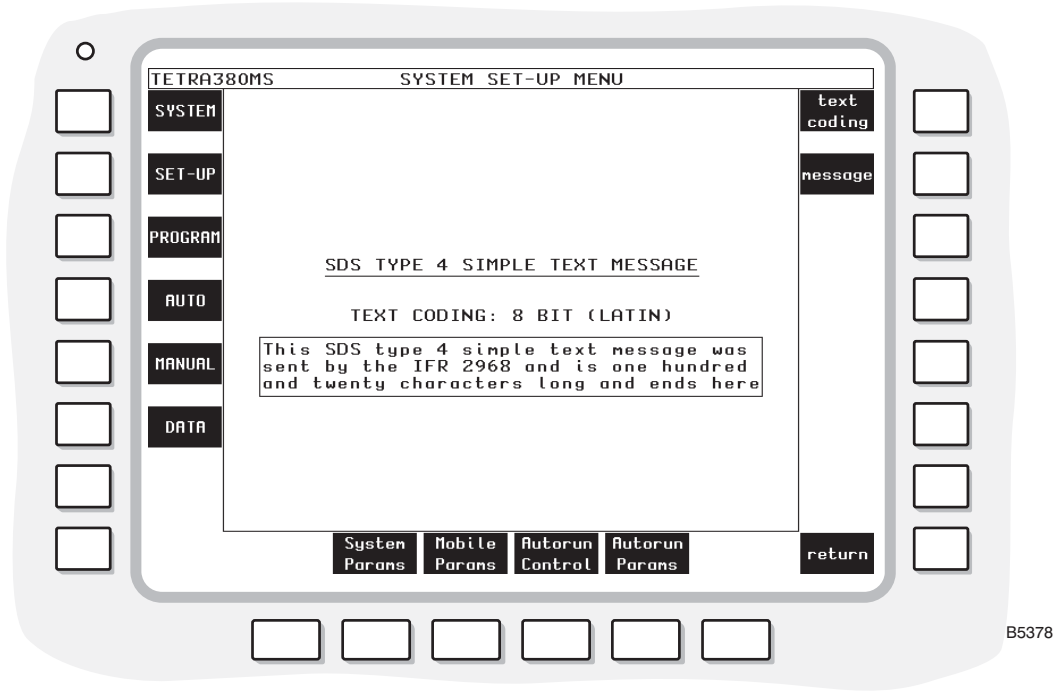


Fig. 3-14 TETRA Mobile; SDS Type 4 Simple Text Message

The following keys are used to set the simple text message parameters.

Text coding

[text coding]

This parameter allows you to select whether the text message is sent to the mobile in standard 8-bit TETRA compatible text coding or in 7-bit GSM compatible text coding. The default is 8-BIT.

Note

Some mobiles may not be capable of receiving 7-bit coded text messages.

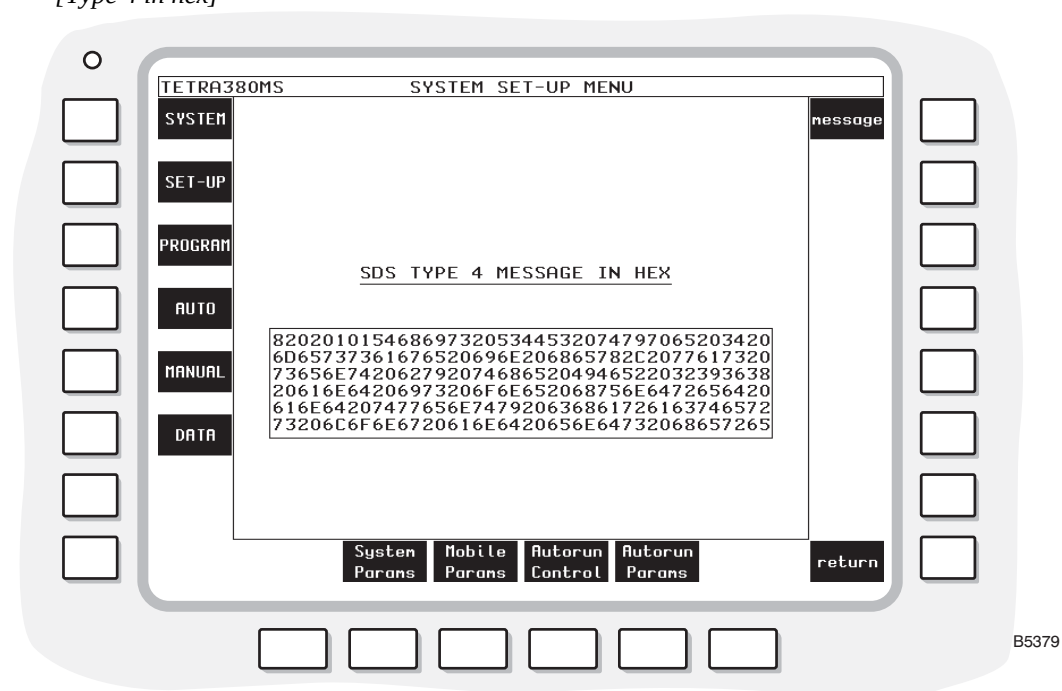
Text message

[message]

The default text message is of length 120 characters. This key allows you to enter your own text message of up to 120 characters. Refer to *Channel Plan Title* on page 3-39 for details of text entry.

Note

If you wish to restore the default text message you will need to reset all TETRA MOBILE TEST system parameters to the factory default values. To do this, press [HELP/SETUP], [SET_UP], [misc config], [default values], [reset current], [return], [return], [return], [RESET].

Type 4 message in HEX.*[Type 4 in hex]**Fig. 3-15 TETRA Mobile; SDS Type 4 Text Message in HEX*

The default HEX data message represents an SDS-TL text message. Use the *[message]* key to enter HEX data for your own SDS applications. The data is the SDS type 4 message content and should be an even number of HEX digits (max. 240 digits = 120 octets) and should start with the SDS-TL protocol identifier defined in ETSI EN 300 392-2 Clause 29.

TETRA Base Station System parameters

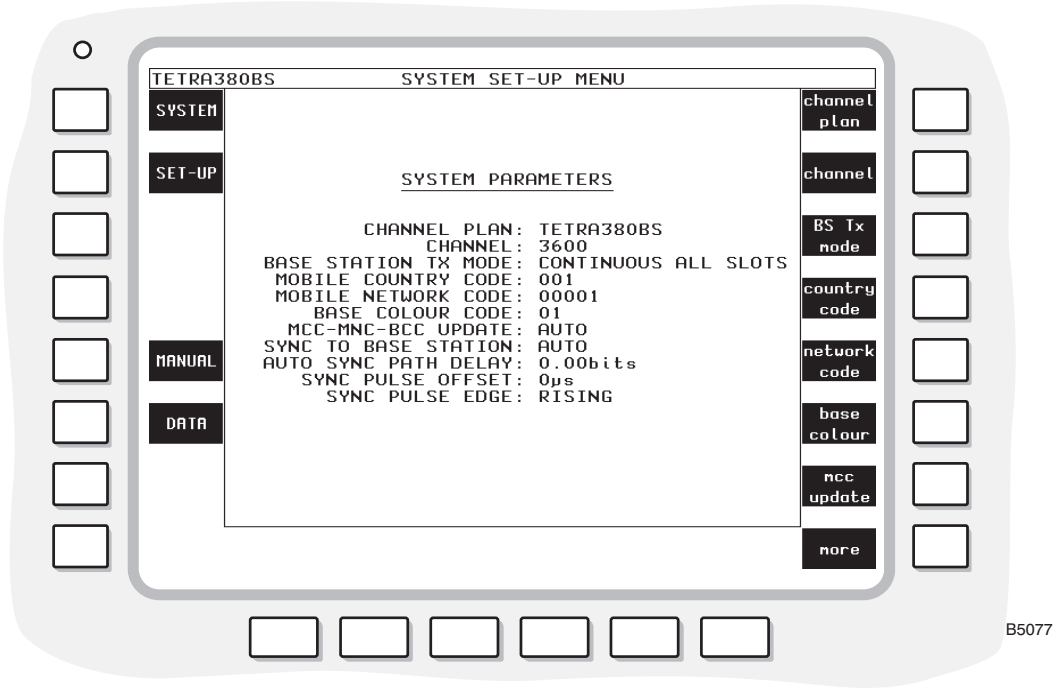


Fig. 3-16 System parameters menu TETRA Base Station

Channel Plan

[channel plan]

The current channel plan is shown at the top left of the screen.

To select a channel plan first press [channel plan] on the SYSTEM PARAMETERS menu.

This gives you access to the appropriate set of keys:-

NO PLAN	or	TETRA 450BS
USER DEFINED		TETRA 450+0BS
TETRA 380BS		TETRA 800BS
TETRA 380+0BS		TETRA 800+0BS
TETRA 410BS		TETRA 870BS
TETRA 410+0BS		TETRA 870+0BS
TETRA 410-6BS		
more	↔	more

Note

You may find that it is necessary to use a USER DEFINED channel plan. See the note regarding channel numbers in Chapter 11, *Performance data*.

If you select USER DEFINED as the channel plan, a *[user defined]* soft key becomes available on the SYSTEM PARAMETERS menu. Pressing this *[user defined]* soft key gives you access to the USER DEFINED CHANNEL PLAN menu, that allows you to enter the appropriate channel plan parameters. For further information see *Setting up a user-defined channel plan* starting on page 3-39.

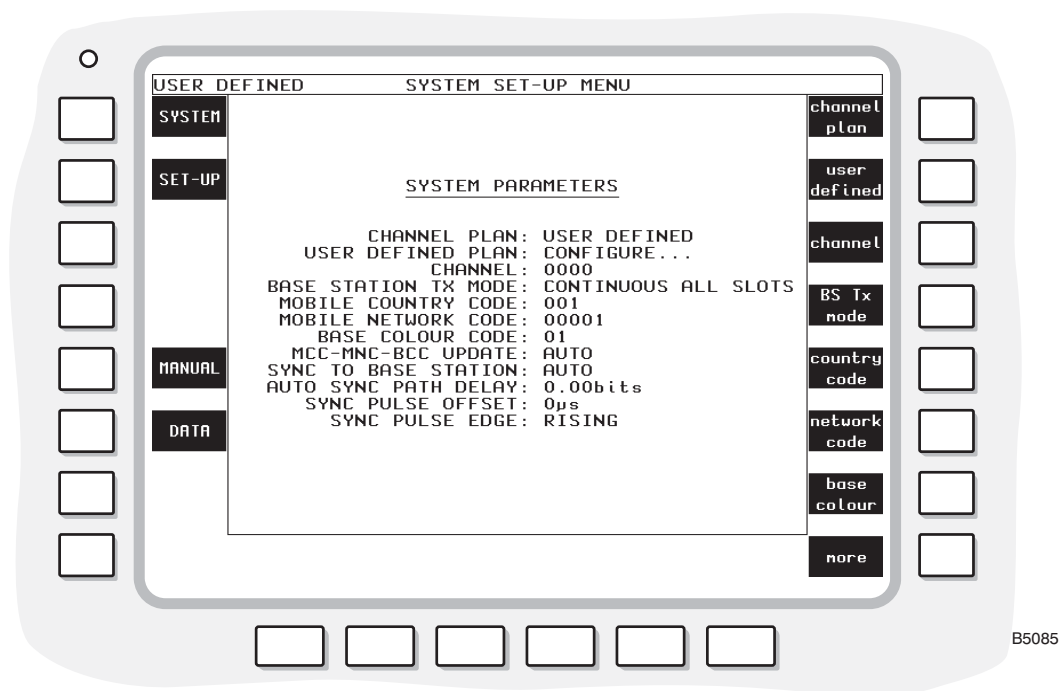


Fig. 3-17 System parameters menu; TETRA Base Station USER DEFINED channel plan

If you select NO PLAN as the channel plan, then the base station's Tx frequency can be set directly in MHz. An Rx Offset field and soft key become available to allow you to set the offset between the base station's transmitter and receiver frequencies.

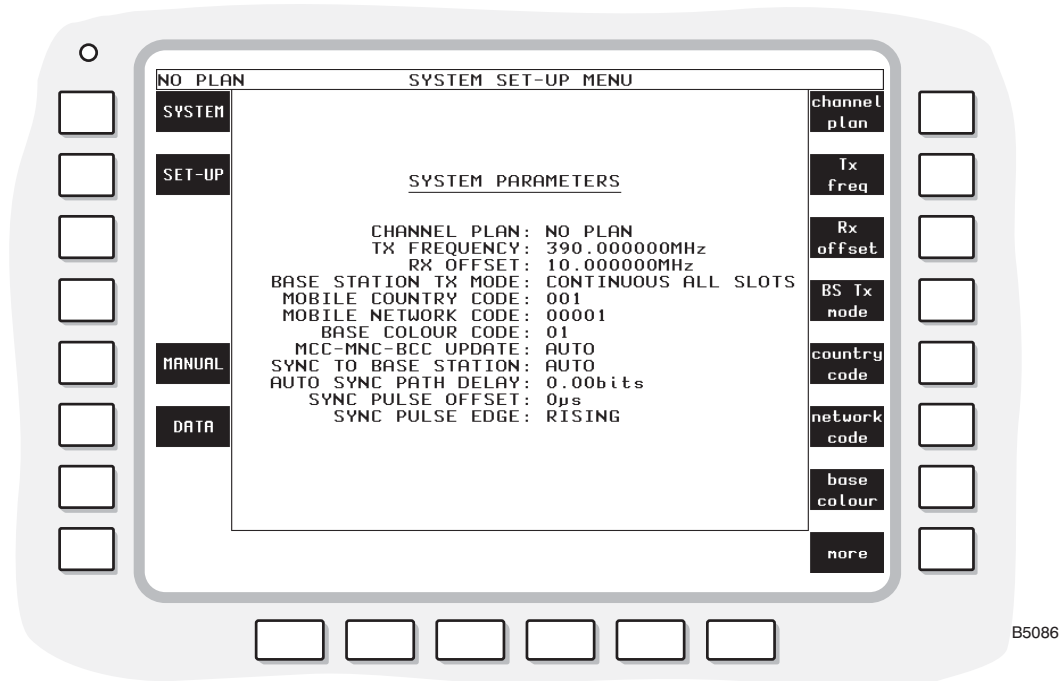


Fig. 3-18 System parameters menu; TETRA Base Station NO PLAN

User Defined Plan

[user defined]

Note

The *[user defined]* soft key is only available if the USER DEFINED channel plan has been selected.

Pressing the *[user defined]* soft key gives you access to the USER DEFINED CHANNEL PLAN menu. (See *Setting up a user-defined channel plan* starting on page 3-39.)

Channel or **Tx Freq** (if channel plan is set to NO PLAN
on the SYSTEM PARAMETERS menu)

[channel]

[Tx freq]

The Channel parameter is used to set the frequency of the Test Set's receiver and signal generator. It should be set to match the channel (frequency) on which the base station is operating.

The Channel parameter can be set to any value within the range permitted by the selected system type. (See the appropriate section in Chapter 11, *Performance data*.) To change the channel, enter the required channel number (or frequency if NO PLAN is set to on the SYSTEM PARAMETERS menu) using the data keys.

Rx Offset

[Rx offset]

This sets the offset between the base station's transmitter frequency (the Test Set receiver frequency) and the base station's receiver frequency (the Test Set signal generator frequency).

$$\text{BS RxFreq} = \text{BS TxFreq} - \text{Rx Offset}$$

Note

The *[Rx offset]* soft key is only available if the NO PLAN channel plan has been selected.

Base Station Tx Mode

[BS Tx mode]

Allows you to set the Test Set to receive CONTINUOUS ALL SLOTS transmission (base station transmits in all slots), CONTINUOUS 1 SLOT transmission (base station transmits in a single slot with ramps outside the slot) or DISCONTINUOUS transmissions (base station transmits in a single slot with ramps inside the slot).

Mobile Country Code (MCC)

[country code]

This parameter must be set to the MCC of the Base station under test for the Test Set to display decoded data or to perform Base Station receiver tests. (But see *MCC-MNC-BCC Update* below.) The range is 0 to 999. For making transmitter measurements, the value set is irrelevant.

Mobile Network Code (MNC)

[network code]

This parameter must be set to the MNC of the Base station under test for the Test Set to display decoded data or to perform Base Station receiver tests. (But see *MCC-MNC-BCC Update* below.) The range is 0 to 16383. For making transmitter measurements, the value set is irrelevant.

Base Colour Code (BCC)

[base colour]

This parameter must be set to the BCC of the Base station under test for the Test Set to display decoded data or to perform Base Station receiver tests. (But see *MCC-MNC-BCC Update* below.) The range is 0 to 63. For making transmitter measurements, the value set is irrelevant.

Note

The MCC, MNC and BCC parameters set the scrambling sequence used by the base station and Test Set transmissions. If the Test Set does not use the same scrambling sequence as the base station, it will not be able to decode the base station transmitted data (except the sync bursts) and it will not be able to generate the correct signal for testing the base station receiver; however the Test Set will perform transmitter measurements on the base station regardless of the MCC, MNC and BCC values.

Setting the BCC to 0 causes the scrambling sequence for all logical channel types to be generated with an all zeros scrambling seed, as for the BSCH, regardless of the values of MCC and MNC.

MCC-MNC-BCC Update

[mcc update]

You can select AUTO or MANUAL.

In AUTO mode the Test Set decodes the values of the MCC, MNC and BCC from the SYNC bursts received from the base station and automatically sets its MCC, MNC and BCC parameters to those values.

In MANUAL mode the Test Set uses the MCC, MNC and BCC parameter values set up in the Test Set and does not update them to the values received in the base station SYNC bursts.

Note

The following "Sync" parameters are associated with the T1 Test facility. Details relating to this will be found in Chapter 5. The operational information relating to each key is given below

Sync to Base Station

[sync to BS]

This allows you to set the T1 Test sync. to Base Station mode to AUTO or SYNC PULSE.

Auto Sync Path Delay

[path delay]

This allows you to set the T1 Test auto sync path delay. The range is -45.00 to +45.00 bits, (-22.5 to +22.5 symbols) in 0.01 bit increments.

Sync Pulse Offset

[pulse offset]

This allows you to set the T1 Test sync pulse offset. The range is 0 (zero) to 1.02 s (1020 ms, 1020000 μ s) in μ s increments. Use the [s] ([MHz]), [ms] ([kHz]) or [μ s] ([Hz]) key to terminate the entry. **Check the displayed value to confirm that you have entered the correct order of magnitude.**

Sync Pulse Edge

[pulse edge]

This allows you to set the T1 Test sync. pulse edge to be RISING or FALLING.

TETRA Direct Mode System parameters

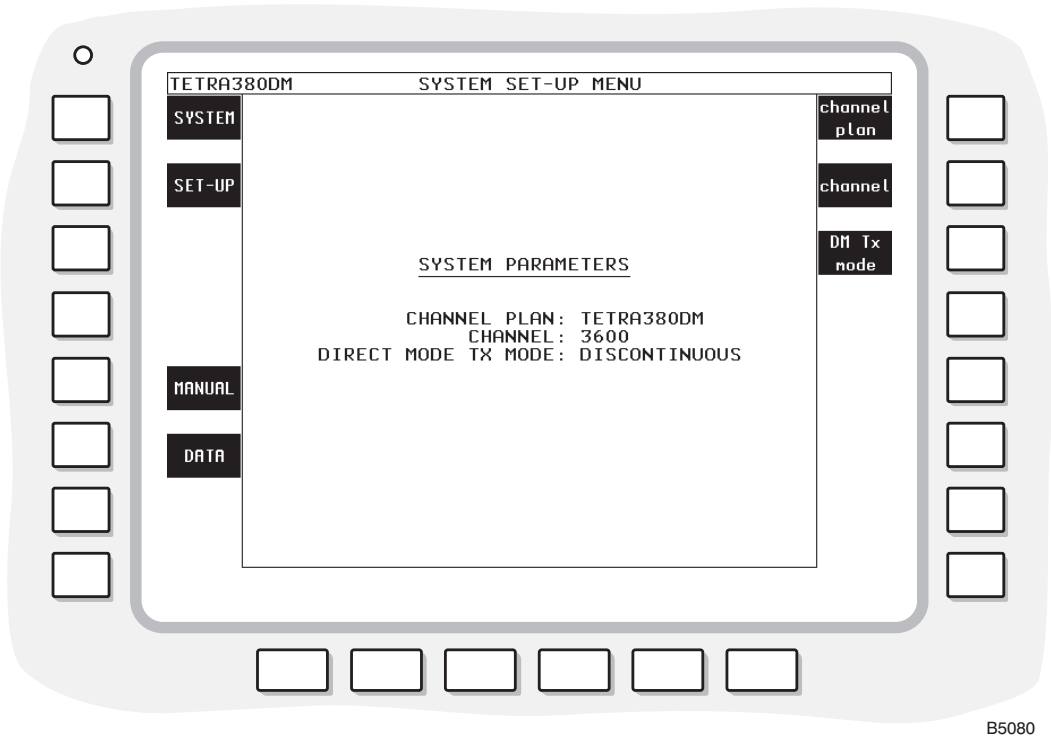


Fig. 3-19 System parameters menu TETRA Direct Mode

Channel Plan

[channel plan]

The current channel plan is shown at the top left of the screen.

To select a channel plan first press [channel plan] on the SYSTEM PARAMETERS menu.

This gives you access to the appropriate set of keys:-

NO PLAN	or	TETRA 450DM
USER DEFINED		TETRA 450+0DM
TETRA 380DM		TETRA 800DM
TETRA 380+0DM		TETRA 800+0DM
TETRA 410DM		TETRA 870DM
TETRA 410+0DM		TETRA 870+0DM
TETRA 410-6DM		
more	↔	more

Note

You may find that it is necessary to use a USER DEFINED channel plan. (See the note regarding channel numbers in chapter 11, *Performance data*.)

If you select USER DEFINED as the system type, a *[user defined]* soft key becomes available on the SYSTEM PARAMETERS menu. Pressing this *[user defined]* soft key gives you access to the USER DEFINED CHANNEL PLAN menu, that allows you to enter the appropriate channel plan parameters. For further information see *Setting up a user-defined channel plan* starting on page 3-39.

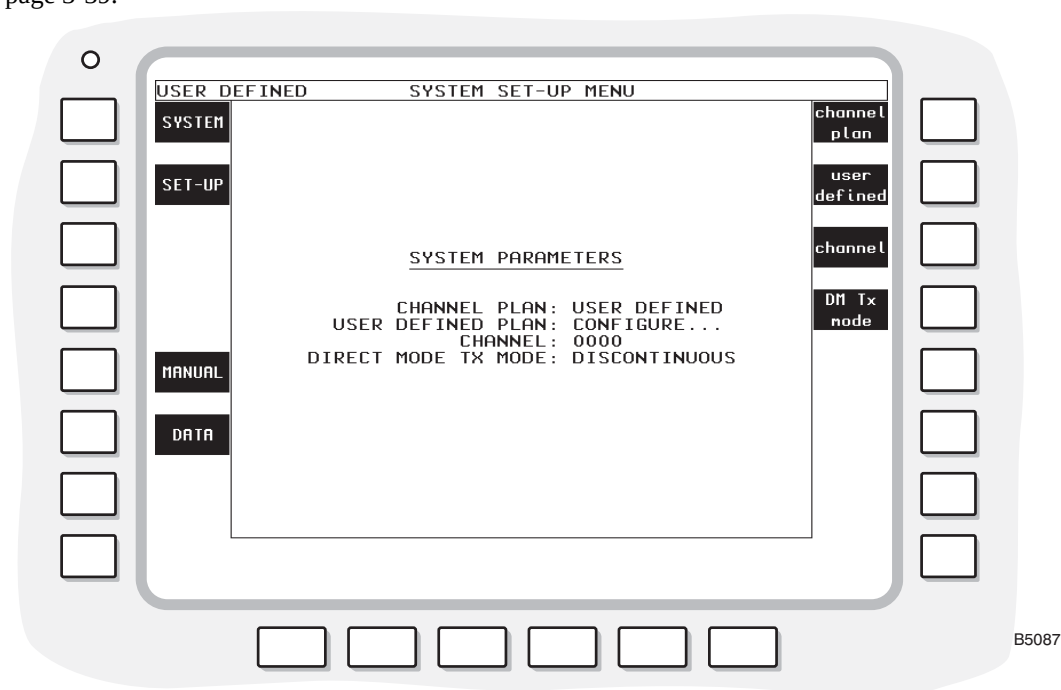


Fig. 3-20 System parameters menu; TETRA Direct Mode USER DEFINED channel plan

If you select NO PLAN as the channel plan, then the mobile's Tx frequency can be set directly in MHz.

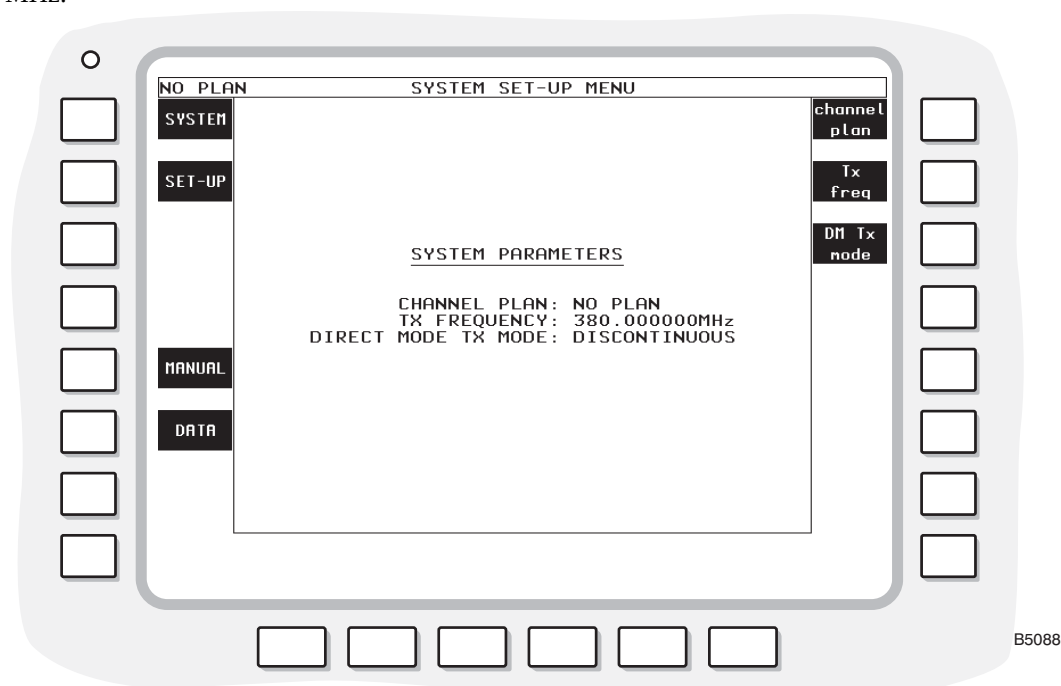


Fig. 3-21 System parameters menu; TETRA direct Mode NO PLAN

User Defined Plan

[user defined]

Note

The [user defined] soft key is only available if the USER DEFINED CHANNEL PLAN has been selected.

Pressing the [user defined] soft key gives you access to the USER DEFINED CHANNEL PLAN menu. (See *Setting up a user-defined channel plan* starting on page 3-39.)

Channel or **Tx Freq** (if channel plan is set to NO PLAN
[channel] [Tx freq] on the SYSTEM PARAMETERS menu)

The Channel (or frequency) can be set to any value within the range permitted by the selected system type. (See the appropriate section in Chapter 11, *Performance data*.) To change the channel, enter the required channel number (or frequency if NO PLAN is set to on the SYSTEM PARAMETERS menu) using the data keys.

Note

The Direct Mode Rx Frequency is the same as the Direct Mode Tx frequency, therefore there is no Rx Offset Setting.

Direct Mode Tx Mode

[DM Tx mode]

This parameter controls whether the Test Set measures continuous or discontinuous transmissions from the mobile. Set to CONTINUOUS when measuring initial continuous SYNC bursts from a direct mode mobile or to DISCONTINUOUS when measuring initial discontinuous SYNC bursts or the subsequent normal bursts from a direct mode mobile

Note

For normal measurements on a Direct Mode mobile, set this parameter to DISCONTINUOUS. If set to CONTINUOUS, the regular discontinuous bursts will not be correctly aligned for measurement and will appear to have large vector errors.

Mobile parameters

(Applicable only for TETRA Mobile system)

The MOBILE PARAMETERS menu, shown in Fig 3-22 below, enables you to make changes to the Short Subscriber Identity (SSI), the selected Group Short Subscriber Identity (GSSI), the Power Class and the Receiver Class.

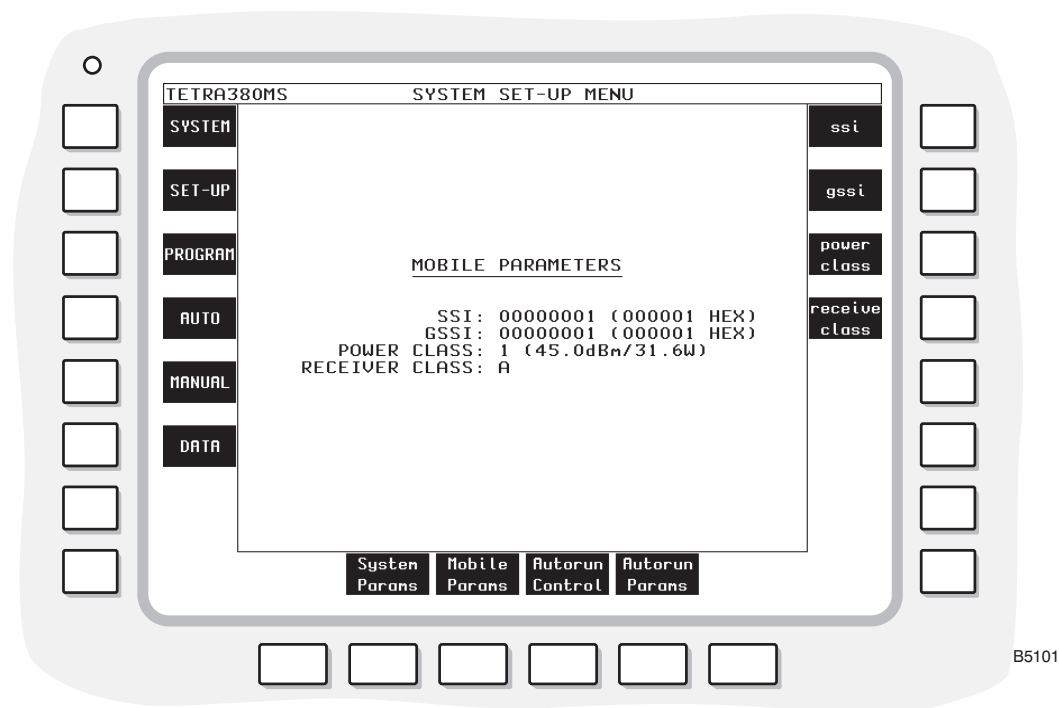


Fig. 3-22 Mobile parameters menu

Changing mobile parameter values

To change a mobile parameter value, first press the soft key assigned to the parameter; the selected parameter will be highlighted. Then as appropriate;-

Key in the required value using the data input keys and press the selected parameter soft key again or the ENTER data key.

Pressing the soft key relevant to the required setting.

Short Subscriber ID (SSI)

This is the Short Subscriber Identity used by the Test Set when making calls or sending messages to the mobile.

It is automatically updated when a mobile registers with the Test Set, therefore it is not normally necessary to enter this value manually.

Group Short Subscriber ID (GSSI)

This is the Group Short Subscriber Identity used by the Test Set when making a group call or sending a group addressed message to the mobile.

It is automatically updated when the mobile informs the Test Set it has selected a group. If the mobile selects "no group" this parameter is left at the last group selected.

It is not normally necessary to enter this value manually. However, if the mobile has attached multiple groups you may wish to make a call to a group ID other than the selected group.

Power Class

This is the Power Class of the mobile and is used by the Test Set to decide the maximum power level expected from the mobile. In Manual test it is used to indicate the accuracy of the Test Set's estimation of the mobile power level displayed.³ During an auto test program it is used to limit the maximum power level measured and decide the limits used for the Tx Power Level test.

If the mobile has completed an extended TT Test mode registration this parameter will be updated to the value received from the mobile. (See Chapter 4, *TETRA mobile manual testing*.)

Receiver Class

This is the receiver class of the mobile, indicating the mobile's receiver sensitivity. It is used by the Test Set to decide which limits to use for the BER/RBER/MER Autotests. (See Chapter 7, *TETRA Automatic Mobile Testing*.)

If the mobile has completed an extended TT Test mode registration this parameter will be updated to the value received from the mobile. (See Chapter 4 *TETRA mobile manual testing*.)

Autorun control

(Applicable only for TETRA Mobile system)

The AUTORUN CONTROL menu, shown in Fig. 3-23, allows you to select how the Autorun program will respond to the results of tests.

Changing autorun control values

To change an autorun control value, first press the soft key assigned to the parameter; the selected parameter will be highlighted. Then press the appropriate secondary soft key. The following subsections explain the significance of each secondary key.

Display Format

The Display Format soft keys *[full]* and *[summary]* give you some control over what data is displayed on the completion of each test in the program:-

- *[summary]* Causes one line containing the test title to be displayed for each test; this is followed by PASS or ★FAIL★, with a brief summary of the test result.
- *[full]* As for *[summary]*, but with additional test information (such as test limits) when running a Comprehensive testing or user-defined program.

³ During open loop power control, a TETRA MS has a permitted tolerance of ± 4 dB on its RSSI measurement which dictates the transmit power level step used. Therefore the MS may legitimately be transmitting on a higher or lower level than the power level step estimated by the Test Set. However, the MS should not transmit at a level higher than its power class or higher than the TxPWR_MAX_CELL parameter, nor lower than 15 dBm, hence it is possible for the Test Set power level estimation to be accurate for maximum and minimum levels.

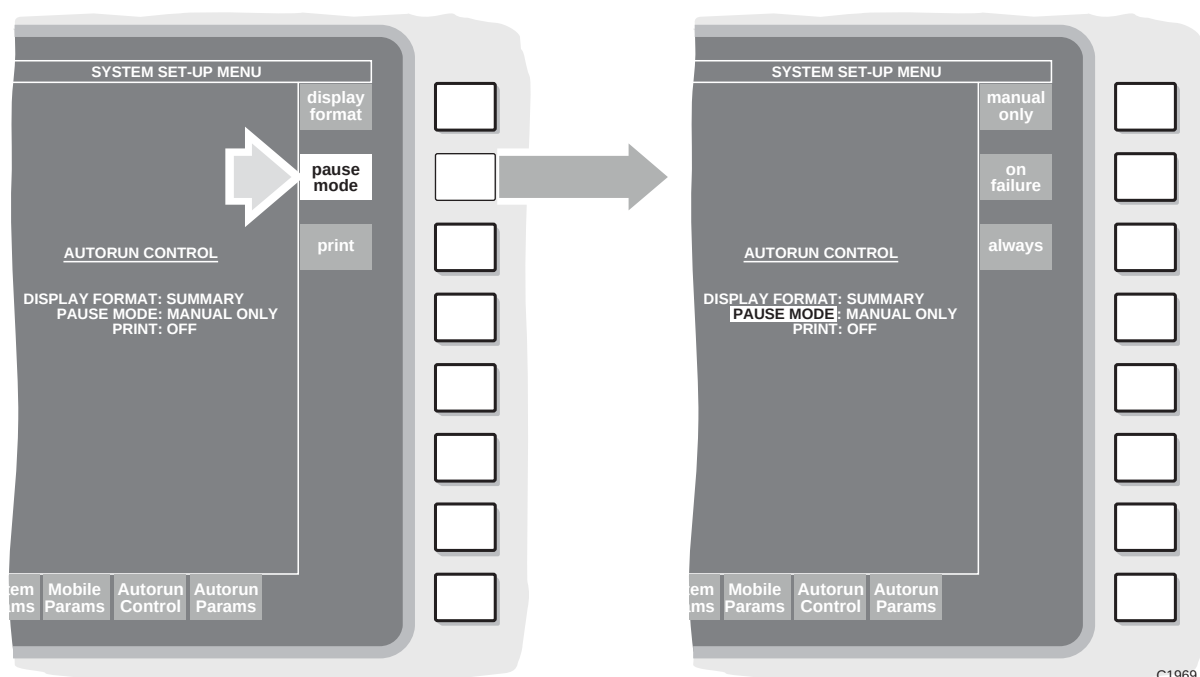


Fig. 3-23 Autorun control menu

Pause Mode

Pause Mode allows you to select the criteria that may cause the test program to pause and wait for intervention.

Three options, displayed on a soft key menu, are available to you after pressing the *[pause mode]* soft key. They are:-

- *[manual only]* The Autorun program will run through all tests. If the *[pause]* soft key is pressed during a test, the program will pause at the end of the current test.
- *[on failure]* The Autorun program will pause at the end of any test which fails.
- *[always]* The Autorun program will pause at the end of every test.

Once a program has paused, there are three options. You can:-

- Continue the program by pressing the *[cont]* soft key.
- Stop the program completely by pressing the *[stop]* soft key.
- Examine the operation of the mobile by selecting one of the other modes of the Test Set, such as Manual, DUPLEX TEST or AF TEST; the mobile remains active if the *[MANUAL]*, *[DUPLEX TEST]* or *[AF TEST]* key is pressed and all the settings of the basic test modes relate to the on-going system test. This allows you to carry out fault diagnosis or adjustment of the mobile. Information about duplex testing is given in Chapter 8.

You can return to the test program at any time by pressing the *[AUTO]* soft key (when in Manual test mode) or *[SYSTEMS]* hard key (when in DUPLEX TEST or AF TEST mode). The AUTORUN screen will be displayed, and if you press the *[cont]* soft key the program will continue from the point where it paused.

Print

When the Print parameter is set to On, all Autorun results are printed as they appear on the screen. The *[print]* soft key provides access to the *[on]* and *[off]* soft keys.

Note that it is possible to set up a Company name, user names, and the current user name, via the HELP AND SET-UP menu. (See *Help and setting up;.....; User names* in Chapter 3 of the Test Set 2968 Operating Manual, 46882/274.)

Autorun parameters

(Applicable only for TETRA Mobile system)

The AUTORUN PARAMETERS menu allows you access to the parameters of each of the tests used in the built-in test programs in the TETRA Mobile system. The parameters of each test are listed, with their ranges and default values, in Chapter 11, *Performance data*, in the same order as that in which they appear on the display.

Switching tests on or off

Any of the automatic tests can be disabled so that the test program does not run it. This gives you flexibility to meet specific requirements. When a test is disabled, this is indicated on the AUTORUN PARAMETERS menu by the word OFF beside the test name.

A quick method of switching tests on or off is to use the soft keys *[up]* and/or *[down]*, or the VARIABLE rotary control, to highlight the name of the required test, and then to press the *[test on/off]* soft key.

Changing autorun parameters

See Fig. 3-24 on page 3-38.

To gain access to the individual test parameters:-

1. Use the soft keys *[up]* and/or *[down]*, or the VARIABLE rotary control, to highlight the name of the required test Diagram A.
2. Press the *[select]* soft key; the test parameters are then displayed, as shown in Diagram B.
3. Now use the *[up]* and/or *[down]* soft keys, or the VARIABLE rotary control, to highlight the required test parameter and then press the *[edit]* soft key.
4. Select the required option (see Diagram C): or, for numerical parameters, by pressing the appropriate data keys or a mixture of data keys and the appropriate soft key terminator.
5. Press the *[return]* soft key to return to the AUTORUN PARAMETERS menu.

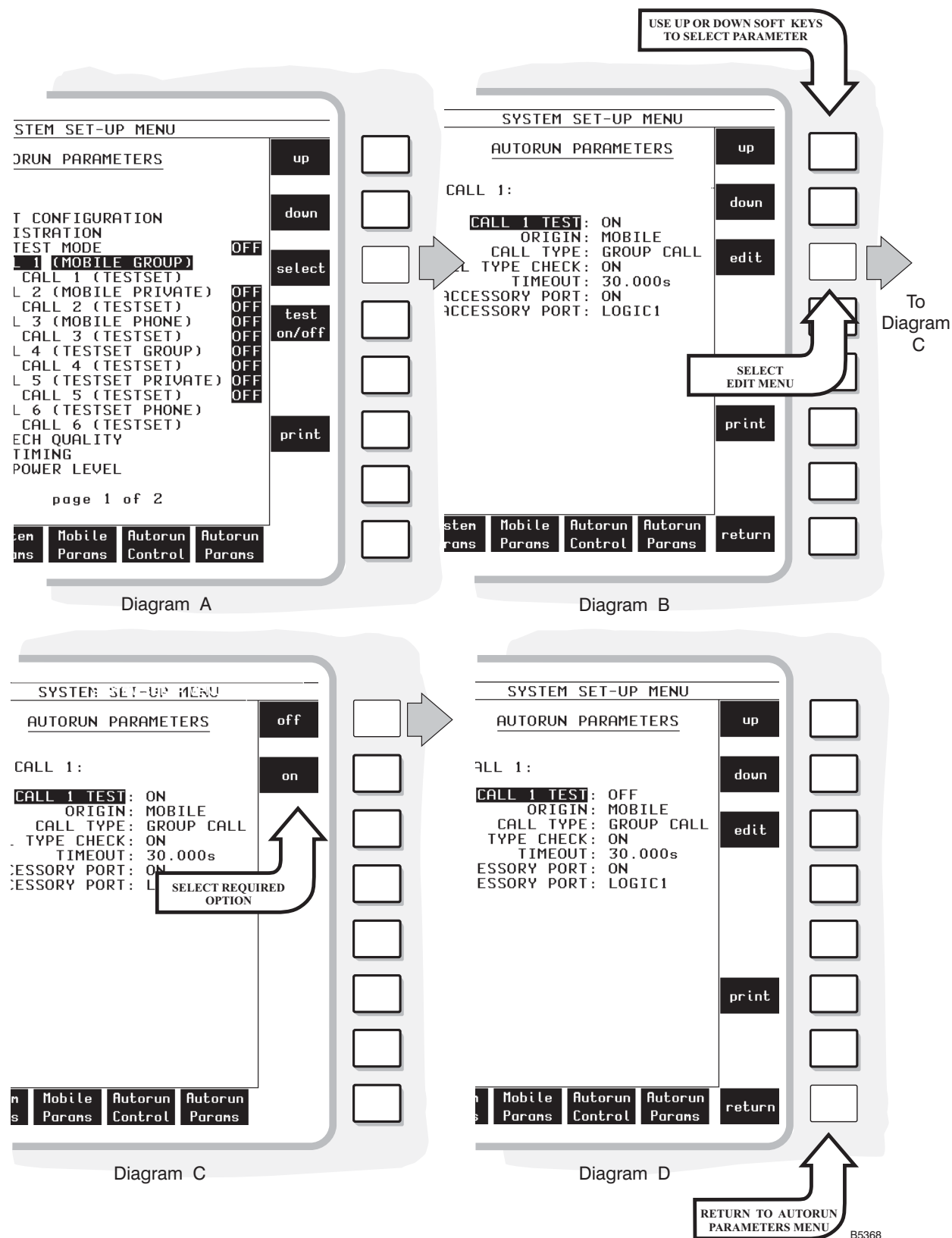


Fig. 3-24 Editing Autorun parameters

Setting up a user-defined channel plan

If you want to set up your own channel plan (or modify a previously defined user plan), you can do this as follows:-

1. Change the CHANNEL PLAN to USER DEFINED, by pressing the *[channel plan]* soft key on the SYSTEM PARAMETERS menu and then pressing the secondary soft key *[USER DEFINED]*.
2. Select the USER DEFINED CHANNEL PLAN menu by pressing the *[user defined]* soft key on the SYSTEM PARAMETERS MENU.
3. Set up the USER DEFINED CHANNEL PLAN parameters as required.

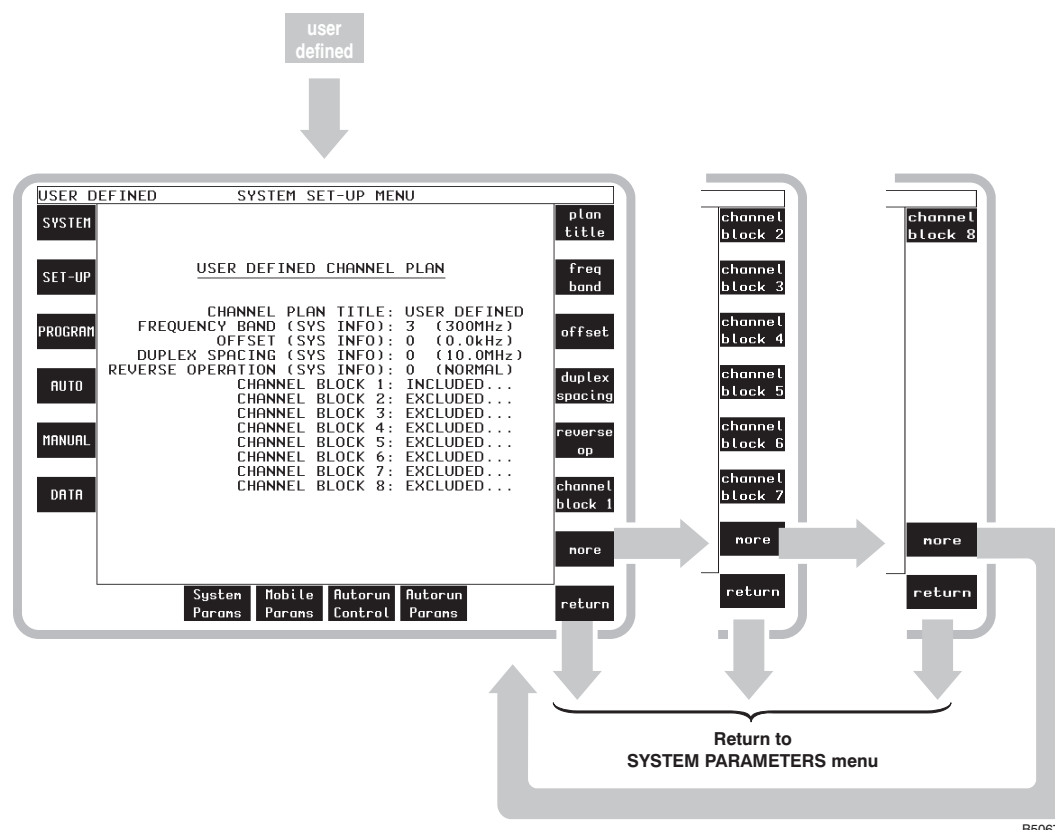


Fig. 3-25 Editing a user-defined channel plan .

The parameters on the USER DEFINED CHANNEL PLAN menu are described below.

Channel Plan Title

The user defined plan can be given any title of up to 12 characters.

To change the channel plan title, first press the *[plan title]* soft key. A screen similar to that shown in Fig. 3-26 will be displayed. The legend *AF?LLCK# NK?L# R&KCF* will be highlighted and an inverse video cursor will be present on the list of characters at the bottom of the screen.

Use the rotary control to place the cursor over the character to be the first character in the new title. Press the *[enter char]* soft key. The selected character will replace the previous title. Use the rotary control and *[enter char]* soft key to complete the title. The *[delete char]* soft key removes the latest character entered in the title. When the title is complete press the *[title complet]* soft key. Use the *[UPPER lower]* soft key to toggle upper case or lower case characters. Use the *[otherchars]* soft key to select non-alphabetic characters; press *[UPPER lower]* to continue entering alphabetic characters. Use the DATA keypad to enter numeric characters. The USER DEFINED CHANNEL PLAN screen will be displayed with the new title shown against the CHANNEL PLAN TITLE: legend.

The new channel plan title will also be shown at the top left of the display after you exit the USER

DEFINED CHANNEL PLAN menu.

The text on the [USER DEFINED] soft key for selecting the USER DEFINED CHANNEL PLAN will not change.

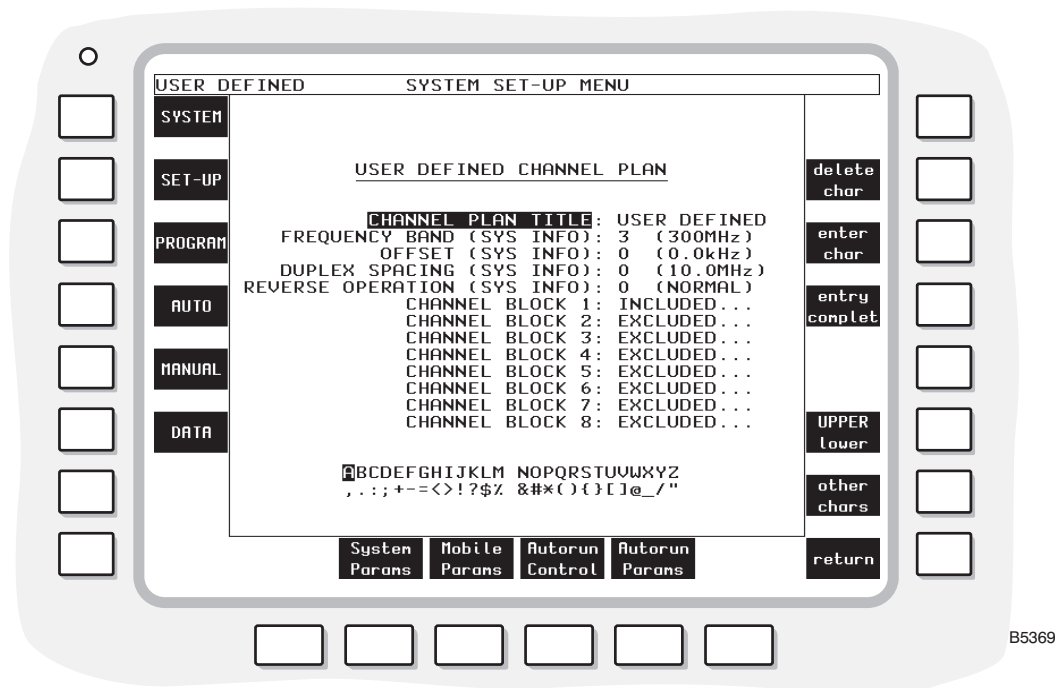


Fig. 3-26 USER CHANNEL PLAN TITLE editing screen

Frequency Band (SYSINFO)

(Applicable only for TETRA Mobile system)

Sets the Frequency Band parameter in the system information sent to the mobile. The range is 0 to 15. The equivalent frequency band in MHz is displayed where this has been defined in the TETRA specifications.

Offset (SYSINFO)

(Applicable only for TETRA Mobile system)

Sets the Offset parameter in the system information sent to the mobile. The range is 0 to 3. The equivalent offset in kHz is displayed.

Duplex Spacing (SYSINFO)

(Applicable only for TETRA Mobile system)

Sets the Duplex Spacing parameter in the system information sent to the mobile. The range is 0 to 7. The equivalent duplex spacing in MHz, related to the Frequency Band selected above, is displayed where this has been defined in the TETRA specifications.

Note

The interpretation of this parameter is dependent on the Frequency Band - ensure that the Frequency Band is set correctly first.

Reverse Operation (SYSINFO)

(Applicable only for TETRA Mobile system)

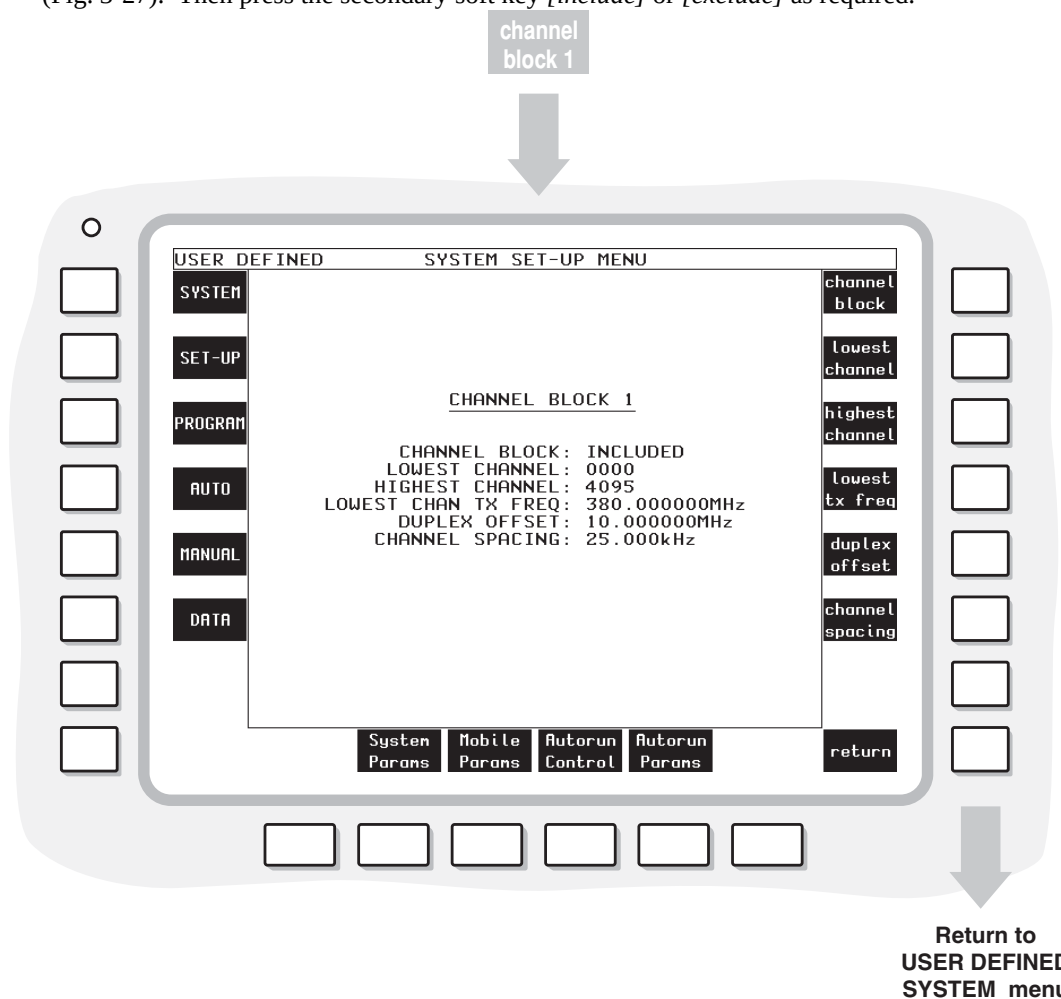
Sets the Reverse Operation parameter in the system information sent to the mobile. The choices are 0 (normal) or 1 (reverse).

Channel Blocks 1 to 8

The allocation of frequencies is not necessarily in continuous frequency bands. To make the maximum use of allocated frequencies, some frequency plans may be complex. Also, the duplex offset and channel spacing can be different for different channels within the same system. To enable the testing of mobiles or base stations programmed with such complex frequency plans, the Test Set has 8 channel blocks, each of which can hold a simple channel plan together with the appropriate duplex offset and channel spacing.

Channel Block

The frequency plan must include at least one channel block, but any channel block or blocks can be included or excluded. Select the appropriate channel block by pressing one of the soft keys *[channel block 1]* to *[channel block 8]* (Fig. 3-25), followed by the *[channel block]* soft key (Fig. 3-27). Then press the secondary soft key *[include]* or *[exclude]* as required.



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Fig. 3-27 Setting up channel blocks

The other channel block parameters are described below.

Lowest Channel

Press *[lowest channel]* and enter a number in the range 0 to 4095. The number must not exceed the Highest Channel number. The frequency entered as the Lowest Chan Tx Freq (see below) will be given to the channel number entered here.

Highest Channel

Press *[highest channel]* and enter a number in the range 0 to 4095 as for the Lowest Channel

(above). Only a number equal to or higher than that entered as the *Lowest Channel* will be accepted.

Lowest Chan Tx Freq

This is the lowest frequency of the transmitter frequency range of the device under test i.e.

- In MS test the MS Tx frequency (uplink);
- In BS test the BS Tx frequency (downlink);
- In DM test the DM Tx frequency.

Press [*lowest tx freq*] and enter a value in the range 10 MHz to 999.9999 MHz. The frequency entered here is the frequency used by the channel number set as the *Lowest Channel* (see above). e.g. If the *Lowest Channel* is set as 101 and the *Lowest Chan Tx Freq* is set as 380.025 MHz, then channel 101 will set 380.025 MHz, channel 102 will set 380.050 MHz and channel 100 cannot be selected.

Duplex Offset (not applicable for TETRA Direct Mode)

Press [*duplex offset*] and enter a value in the range 0 to 100 MHz. Negative values may be entered for reverse operation channel plans.

With TETRA Mobile system selected:-

Mobile receiver frequency = mobile transmitter frequency + duplex offset.

With TETRA Base Station system selected:-

Base station receiver frequency = base station transmitter frequency – duplex offset.

Channel Spacing

Press [*channel spacing*] and enter a value in the range 5 kHz to 500 kHz. The system rounds to the nearest 50 Hz. Negative values may be entered for systems where the frequency of channel 1 is higher than that of the maximum channel number.

Note

To check that you have correctly entered the channel block parameters for your user defined system, set the required channel number in Manual Test and use Digital Duplex Test to see the actual frequencies to which the Test Set GEN FREQ and mobile or base station Tx FREQ are set.

Chapter 4

TETRA MOBILE MANUAL TESTING

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Introduction

The TETRA Mobile test arrangement is shown below.

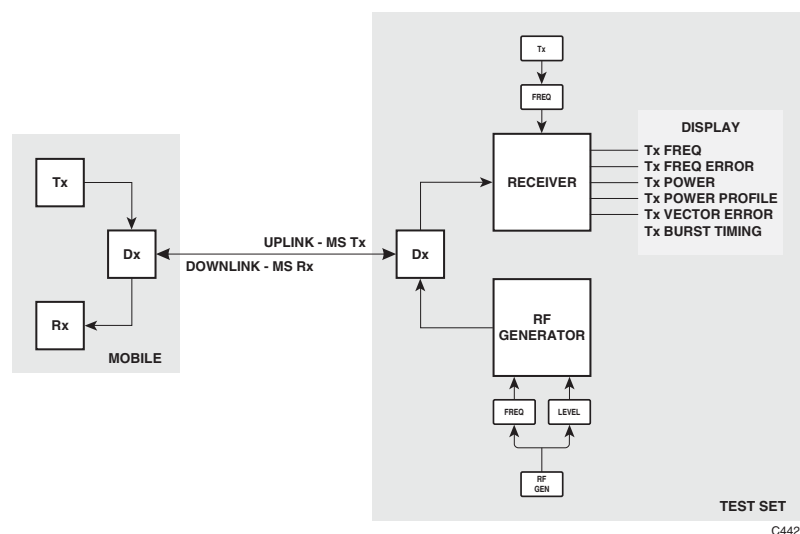


Fig. 4-1 TETRA Mobile test arrangement

Initial Configuration

The TETRA Mobile (MS) system is selected as described under *System selection* in Chapter 3. The parameters referred to below must be set correctly in the Test Set to allow a mobile to be tested.

They are set on the SYSTEM SET-UP SYSTEM PARAMETERS menu, accessed using the following key sequence:-

[SYSTEMS], [SET-UP], [System Params]

For further information see *Customizing the system* in Chapter 3.

Channel plans

The correct channel plan configuration must be set up such that the Rx and Tx frequencies and parameters of the Test Set correspond to those recognised by the mobile. You may need to contact the manufacturer or TETRA network operator supplying the mobile to obtain this information.

Predefined Channel plans

For a mobile which is designed to use one of the standard TETRA channel plans, set the CHANNEL PLAN parameter on the SYSTEM PARAMETERS menu to the corresponding channel plan. To do this, select the [channel plan] soft key, then the appropriate soft key. The frequency offset differs for different channel plans. The Channel Plans are defined in Chapter 11, *Performance data*.

The CONTROL CHANNEL, TRAFFIC CHANNEL and TIMESLOT parameter set on the SYSTEM PARAMETERS menu will be used as the default for the Manual test mode.

User Defined Channel plans

For a mobile which does not conform to one of the standard TETRA channel plans supported by the Test Set, precise channel plan information can be set up by setting the CHANNEL PLAN to USER DEFINED on the SYSTEM PARAMETERS menu. The parameters for the user defined system can then be set on the USER DEFINED CHANNEL PLAN menu accessed by the [user defined] soft key. This allows the TETRA system information parameters such as FREQUENCY BAND, OFFSET, DUPLEX SPACING and REVERSE OPERATION to be set.

Selecting the *[channel block 1]* sub-menu allows you to define the relationship between the TETRA system information channel number parameter and the mobile's uplink/transmitter frequency. The channel block must be set to INCLUDED for it to take effect. See *Channel Block* in Chapter 3.

For further information see *Setting up a user-defined channel plan*, also in Chapter 3.

No channel plan

For testing mobiles with unknown channel plans but where the frequency of operation is known, the mobile Tx FREQUENCY and Rx OFFSET can be set directly by setting the CHANNEL PLAN to NO PLAN. The Tx FREQUENCY and Rx OFFSET set on the SYSTEM PARAMETERS menu will be used as the default for the Manual Test mode. The Test Set signal generator frequency is set to (Tx FREQUENCY + Rx OFFSET). Negative values of Rx OFFSET correspond to reverse operation.

Note

When NO PLAN is selected, the parameters CONTROL CHANNEL, TRAFFIC CHANNEL, FREQUENCY BAND, OFFSET, DUPLEX SPACING and REVERSE OPERATION are all set to zero. Mobiles operating in normal trunked mode may not understand control channel and traffic channel assignments unless a channel plan is defined. Mobiles operating in T1 Test mode may ignore these parameters and work with NO PLAN.

System Identity

The correct MOBILE COUNTRY CODE (MCC) and MOBILE NETWORK CODE (MNC) must be set on the SYSTEM PARAMETERS menu to allow the mobile to recognise the downlink signalling from the Test Set. These values are entered and displayed in *decimal* not *hexadecimal*. You may need to contact the manufacturer or TETRA network operator supplying the mobile to obtain this information. Refer to ETSI EN 300 392-2 Annex K2 for Mobile Country Code values

RF Level Parameters

The following parameters on the SYSTEMS PARAMETERS menu define the way the mobile behaves with respect to the RF level it receives from the Test Set. The default values should be suitable for most mobiles.

MIN RX LEVEL FOR ACCESS defines the minimum RF generator level that the mobile will accept before it will attempt to access the system (register with the Test Set) and should therefore be set to a value lower than the RF generator level to ensure registration (**-125 dBm**). This parameter also affects the behaviour of the mobile when performing cell re-selection. See *Neighbour Cell Broadcast* etc. starting on page 4-27.

MAX TX LEVEL defines the maximum power level the mobile is permitted to transmit when using "open loop power control" (link control) and by default is set to 30 dBm.

ACCESS PARAMETER is used by the mobile together with the measurement of its received RF level (the Test Set's RF generator level) to decide the power level it will transmit when using "open loop power control" (link control). It is also used by the Test Set to adjust its input attenuation to suit the estimated mobile power level. The default value (**-45 dBm**) should cause the mobile to use a power level of 30 dBm with the default Manual test RF generator level of -75 dBm.

Selecting manual test modes

You select the Manual test modes using the soft keys at the bottom of the Manual test screen. The MODE field in the middle of the screen indicates the current mode.

MCCH (Main Control Channel) mode

[restart MCCH]

This is the normal mode for testing mobiles and is the default mode when the manual test mode is first selected.

In this mode the Test Set is generating a main control channel at a frequency determined by the CONTROL CHANNEL setting and at the level of the RF GEN LEVEL setting.

A TETRA mobile may not necessarily scan all of the channels in the band and may expect to find a MCCH at one or more pre-configured channels only. You may need to contact the manufacturer or TETRA network operator supplying the mobile to obtain this information.

The MCCH mode screen is shown in Fig. 4-2.

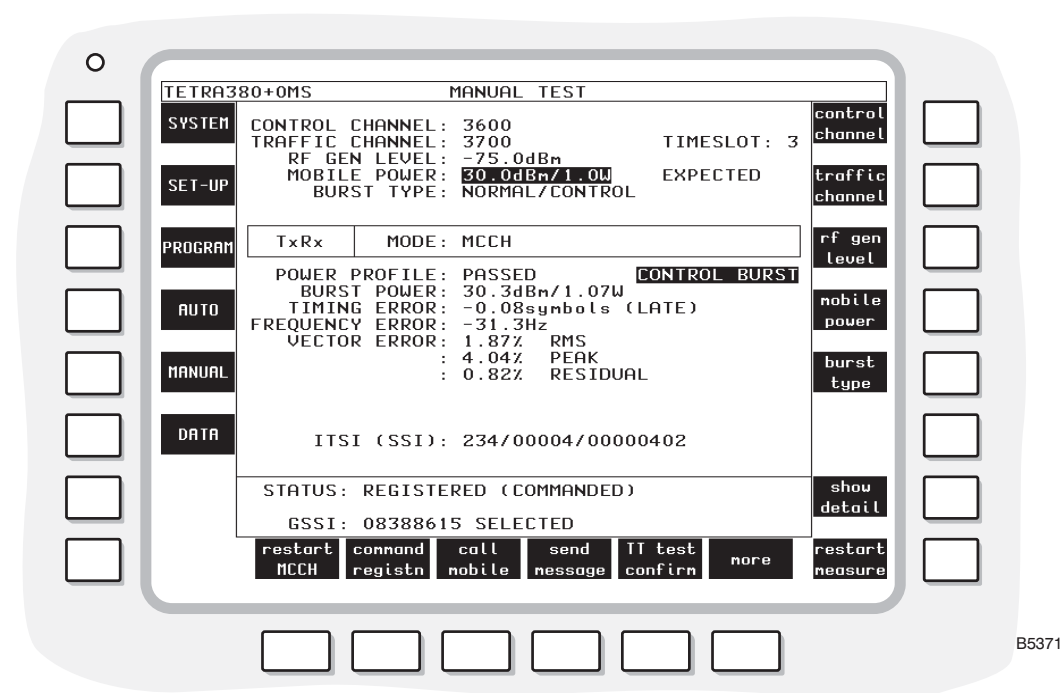


Fig. 4-2 TETRA Mobile Manual test screen (MCCH mode)

From the MCCH mode you can:-

- Change manual test settings to suit your requirements.
- Test mobile registration and de-registration.
- View the mobile's registration information.
- Obtain TEI, power class and receiver class from a mobile that supports TT test mode.
- Set up calls to and from the mobile.
- Send and receive status or Short Data Service (SDS) messages to and from the mobile.
- Receive group identity attachments from the mobile.
- Test mobile cell re-selection and call restoration. This requires two 2968 Test Sets. See *Neighbour Cell Broadcast* etc. starting on page 4-27.
- Select the special TCH or T1 Test modes.

When a call has been established to or from the mobile the Test Set enters the CONVERSATION mode.

CONVERSATION mode

In CONVERSATION mode the Test Set is generating a speech traffic channel at a frequency determined by the TRAFFIC CHANNEL setting, on the timeslot determined by the TIMESLOT setting and at the level of the RF GEN LEVEL setting.

The CONVERSATION mode screen is shown in Fig. 4-3.

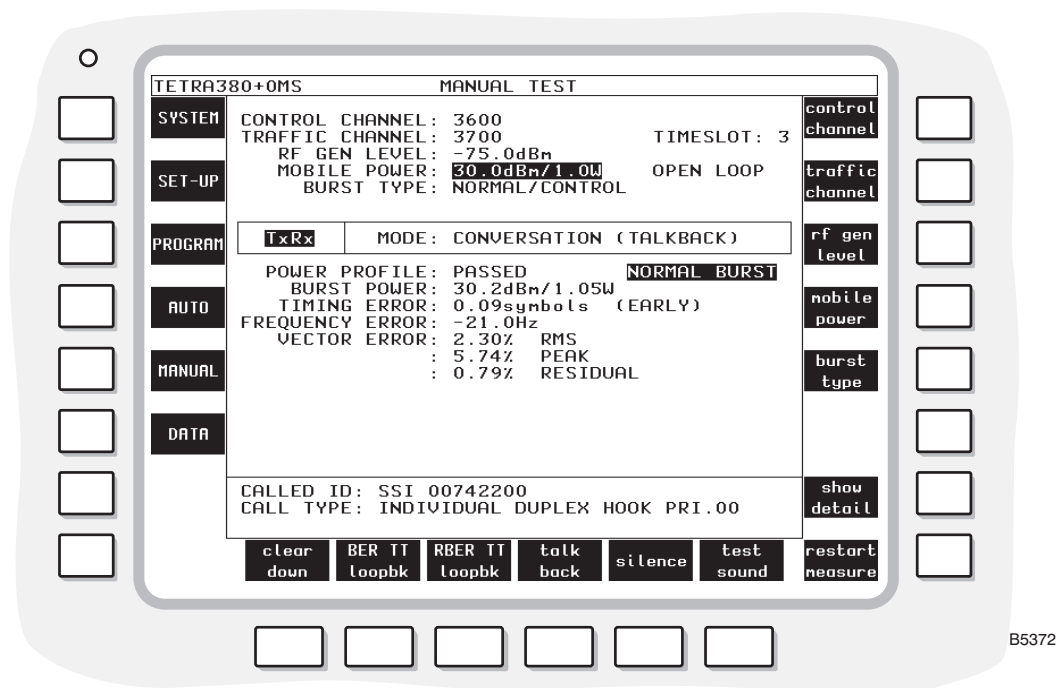


Fig. 4-3 TETRA Mobile Manual test screen (Conversation mode)

From this mode you can:-

- Change manual test settings to suit your requirements.
- View the results of mobile transmitter measurements.
- Test the mobile's audio performance.
- Test the mobile's DTMF signalling.
- Select the audio signal generated by the Test Set.
- Enter TT LOOPBACK modes
- Test mobile cell re-selection and call restoration. This requires two 2968 Test Sets. See *Neighbour Cell Broadcast* etc. starting on page 4-27.
- Cleardown calls from the mobile or from the Test Set.

TT LOOPBACK modes

The TT loopback protocol is implemented in the Test Set. To use these modes the TT loopback protocol (ETSI EN 300 394-1 Annex D) must also be implemented in the mobile. This provides facilities to put a mobile into loopback using commands sent via the RF connection so that it loops back transmitted bursts, allowing Bit Error Rate (BER) and Residual Bit Error Rate (RBER) measurements to be made.

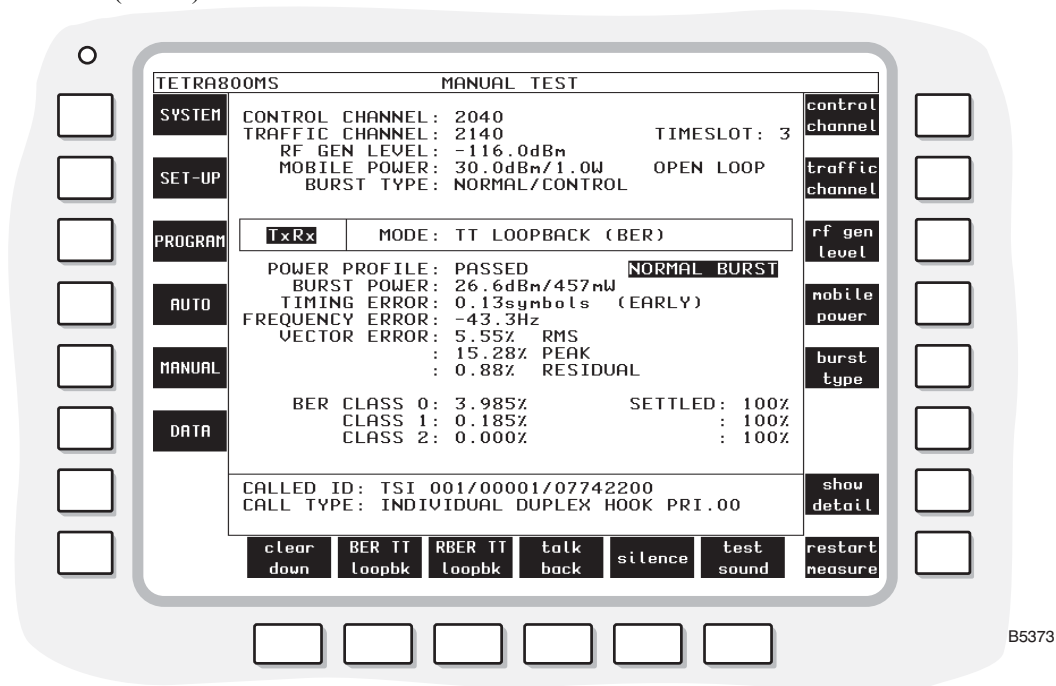


Fig. 4-4 TETRA Mobile Manual test screen (TT Loopback mode)

From these modes you can :-

- Change manual test settings to suit your requirements.
- View the results of the mobile transmitter measurements.
- View the results of the mobile receiver measurements (Bit error rate or Residual bit error rate).
- Return to the CONVERSATION mode.
- Cleardown calls from the mobile or from the Test Set.

TCH TEST mode

[more] [TCH test]

The TETRA Mobile test system has the capability to test mobiles with TCH test capability. This facility within some TETRA mobiles, allows the mobile to generate traffic channel signals when given relevant manufacturer specific instructions (via the keypad or test interface), rather than only after receiving instructions from a base station (or a Test Set) over the air interface.

The *TCH test mode* is described starting on page 4-40.

T1 TEST mode

[more] [T1 test]

A facility within some TETRA mobiles is a capability to produce specific test signals when so instructed over the air interface. The TETRA Mobile test system is able to produce the T1 Test signals required to access this facility, and to perform measurements on the resultant signals from the mobile.

The *T1 Test mode* is described starting on page 4-37.

Changing Manual test settings

To change a test setting, first press the soft key assigned to the test setting; the selected setting will be highlighted. Then:-

If it is set by a choice of options press the appropriate secondary soft key.

If it requires a value, enter the required value using the data input keys and then press any of the ENTER data keys or the selected parameter soft key.

The RF GEN LEVEL can also be adjusted using the VARIABLE control.

Control Channel

[control channel]

Sets the frequency of the RF generator and the receiver within the Test Set when on the control channel (in MCCH mode), by specifying a channel number.

Note

If channel plan NO PLAN is selected, CONTROL CHANNEL specifies the transmitter frequency of the mobile under test (the Test Set receiver frequency) in Hz. The Test Set RF generator is set to CONTROL CHANNEL + Rx OFFSET. However, it is not normally possible to perform call set-up with a mobile if NO PLAN is selected.

The CONTROL CHANNEL can be set to any decimal value permitted by the selected system type (specified under *Test parameters* in Chapter 11, *Performance data*). Initially the CONTROL CHANNEL defaults to that set on the SYSTEM PARAMETERS menu. See Chapter 3, *Selecting and customizing the system*.

The TIMESLOT for the CONTROL CHANNEL is always TN 1.

Ensure that the value of the CONTROL CHANNEL is in *decimal*, not *hexadecimal* and is within the range that the mobile can receive or is set to one of the specific values that the mobile recognizes if the mobile does not scan all channels of the band. You may need to contact the manufacturer or TETRA network operator supplying the mobile to obtain this information.

Traffic Channel and Timeslot

[traffic channel]

The TRAFFIC CHANNEL and TIMESLOT settings are both accessed using the [traffic channel] soft key. The first press highlights the TRAFFIC CHANNEL setting, allowing a new value to be keyed in; the second press highlights the TIMESLOT setting and allows it to be changed; the third press enters the new values (if changed) into the test settings.

TRAFFIC CHANNEL sets the frequency of the RF generator and receiver within the Test Set when on a traffic channel (a call has been established). It can be set to any decimal value permitted by the selected system type (specified under *Test parameters* in Chapter 11, *Performance Data*).

The TIMESLOT for the TRAFFIC CHANNEL can be set to a value between 1 and 4.

Initially, the TRAFFIC CHANNEL and TIMESLOT default to the settings on the SYSTEM PARAMETERS menu. See Chapter 3, *Selecting and customizing the system*.

Ensure that the value of the TRAFFIC CHANNEL is in *decimal*, not *hexadecimal* and that it is within the range that the mobile can use. You may need to contact the manufacturer or TETRA network operator supplying the mobile to obtain this information.

RF Generator Level

[rf gen level]

The RF GEN LEVEL parameter can be set to any value within the output range of the Digital Signal Generator. The default setting is -75 dBm. The range depends on the setting of the RF input/output connector selection as shown in the following table:-

RF output	RF input	RF generator level range
N type	N type	–50.0 to –135.0 dBm
N type	TNC	–40.0 to –135.0 dBm
TNC	TNC	–30.0 to –135.0 dBm
TNC	N type	–20.0 to –135.0 dBm

Notes

1. If an RF generator level offset has been set via the HELP AND SET-UP menu it will be included in the RF GEN LEVEL value displayed and may therefore restrict its range. (See *RF Connections* in Chapter 2.)
2. If the mobile is using open loop power control, changing the RF GEN LEVEL may change the power level that the mobile uses. (See *Mobile Power* below.)

Mobile Power

[mobile power]

This field displays the power level that the Test Set estimates the mobile will transmit at and is used to set the input sensitivity of the Test Set. The value is highlighted in inverse video if the estimate is not considered to be accurate.

The [mobile power] soft key allows control over the way this estimate is made as follows:-

Note

Some mobiles may not support the open and closed loop power control methods described below

When on the control channel (MCCH), T1 Test or TCH Test modes

When on the control channel or in T1 or TCH Test modes, the [mobile power] soft key allows the choice of [expected level] or [open loop].

If [expected level] is chosen, the input sensitivity can be adjusted, to the level expected from the mobile in 5 dB steps using the soft keys now available. The value is always highlighted to show that the estimate can not be considered accurate.

Note

The range over which the expected level can be adjusted is between –40 dBm and 70 dBm, but depends on the RF input port selected and any input offset introduced.

If [open loop] is chosen, the input sensitivity is set to the level it is estimated that the mobile will be using assuming it is using open loop power control (link control). The estimate therefore depends on the following parameters:-

- The Access Parameter as set on the SYSTEM PARAMETERS menu.
- The Max Tx Level as set on the SYSTEM PARAMETERS menu.
- The Power Class of the mobile as set on the MOBILE PARAMETERS menu or as received from the mobile if it is in the TETRA test mode (TT). See *TT Test mode* on page 4-41.
- The RF Generator Level as set on the Manual Test screen.

The value is highlighted, to show the estimate is not considered to be accurate due to permissible tolerances in open loop power control, unless the power level is estimated to be above the maximum or below the minimum power level assumed to be available to the mobile.

Note

Some mobiles may transmit initial control bursts at the maximum permitted by the Max Tx Level parameter, regardless of the RF Generator Level. In this case choosing [expected level] is advisable.

When on a traffic channel (Conversation mode)

When on a traffic channel, the [mobile power] soft key will not be available if the mobile is *not* transmitting. The input sensitivity is set to the level it is estimated that the mobile will be using assuming that it is using open loop power control, in the same way as when on the control channel described previously.

When the mobile is transmitting the [mobile power] soft key allows the choice of [closed loop] or [open loop].

If *[closed loop]* is chosen, the mobile is requested to use closed loop power control. The mobile can then be requested to increment or decrement its transmitter power, between its minimum and maximum power using the soft keys now available. The input sensitivity of the Test Set is therefore adjusted between the minimum and maximum power level assumed to be available to the mobile. The maximum power level available to the mobile will depend on its power class. The estimate will therefore depend on the Power Class parameter as set on the MOBILE PARAMETERS menu or as received from the mobile if it is in the TETRA test mode (TT). See *TT Test mode* on page 4-41.

If *[open loop]* is chosen, the input sensitivity is set as described previously.

The value is highlighted to show that the estimate can not be considered accurate unless the power level is estimated to be above the maximum or below the minimum power level available to the mobile.

Burst type

[burst type]

The burst type can be set to:-

- NORMAL
- CONTROL
- NORMAL/CONTROL.

The Test Set will measure the selected burst type and disregard bursts of the alternative type. If NORMAL/CONTROL is set, both types of burst will be measured.

Show detail

[show detail]

Pressing this key replaces the measurements display with additional information obtained from the mobile: group attachments, status messages, SDS messages and class mark information. Refer to *Registration information* on page 4-10 and *Mobile Originated Status and Short Data (SDS) Messages* on page 4-22 for information on these displays. Press the *[return]* soft key to return to the measurements display.

Restart measurements

[restart measure]

Pressing this key restarts the measurement averaging process. Use this facility to allow time for mobile adjustments to take effect before commencing a measurement over a number of bursts. See *Tx measurement options* or *Rx measurement options* in Chapter 2 for information on setting up measurement averaging.

Mobile Registration

Registering a mobile with the Test Set gives a good indication that the frequency and system parameters in the Test Set correspond to those with which the mobile will operate. In addition it allows the Test Set to capture the Short Subscriber Identity (SSI) and the selected Group Short Subscriber Identity (GSSI) of the mobile which are required to set up a call to the mobile. (These may also be set manually on the SYSTEM SET-UP MOBILE PARAMETERS menu).

In order to register the mobile the POWER ON REGISTRATION and POWER OFF DE-REGISTRATION parameter on the SYSTEM PARAMETERS, BASE SERVICES menu should be set to REQUIRED. This sets the relevant parameters in the system information to command the mobile to register at power-on and de-register at power-off.

The mobile should then register automatically once it has powered up and has found the simulated base station signal from the Test Set. This may take as long as 30 seconds from power on. On registration the SSI of the mobile and its selected GSSI are displayed. The SSI and GSSI parameters in the SYSTEM SET-UP MOBILE PARAMETERS menu are updated to the values received from the mobile.

The mobile may attach group identities during or after registration, and may subsequently change the selected GSSI. These can be viewed in detailed. See *Registration information* on page 4-10 for information.

If the mobile does not register with the Test Set and the initial configuration parameters described previously have been set correctly this may be because the RF generator level of the Test Set is insufficient for the mobile. You can try changing the RF GEN LEVEL setting, but be aware that this may affect the power level chosen by the mobile. See *Changing Manual Test settings* starting on page 4-7 and also the section *RF interference* in Chapter 2.

Failure to register may also be because the signal transmitted by the mobile is outside the capture range of the receiver in the Test Set. The Test Set is not able to predict the power level which the mobile will use, since this depends on the power class of the mobile and whether it transmits at its maximum power or uses open loop power control to decide the power level to use. You can try changing the MOBILE POWER setting to the level the mobile is expected to use. See *Changing Manual Test settings* starting on page 4-7.

After the mobile has registered, the *[command registn]* soft key can be used if required to send signalling to the mobile to force it to re-register with its full ITSI and attach groups.

The mobile should automatically de-register when it is powered down.

Note

The *[command registn]* function can only be used to request information from a mobile which is already registered to the Test Set. If the mobile does not register with the Test Set at power on, pressing this key will not make it do so.

Registration information

The following information, obtained during registration is displayed on the manual screen. Only the ITSI, GSSI and class mark information are displayed unless an extended (TT test mode) registration has taken place. See *TT Test mode* later in this section.

Class mark information

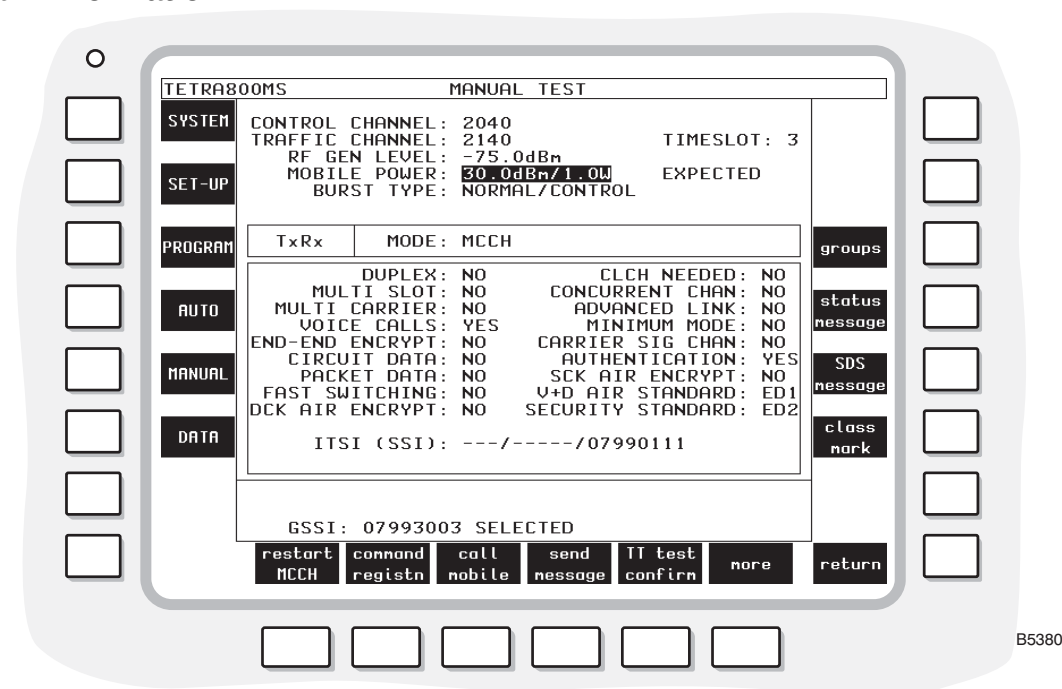


Fig. 4-5 TETRA Mobile Manual test screen from *[show detail]* soft key.

You can display the mobile's class mark information, obtained during registration, in place of the measurements by pressing the *[show detail]* then *[classmark]* soft keys.

The other information obtained at registration is also redisplayed.

To revert to displaying measurements press the *[return]* soft key.

ITSI

The Individual TETRA Subscriber Identity, displayed in *decimal* not *hexadecimal*, consisting of the Mobile Country Code (MCC), Mobile Network Code (MNC) and the Short Subscriber Identity (SSI).

The MCC and MNC are only sent during a commanded registration.

GSSI

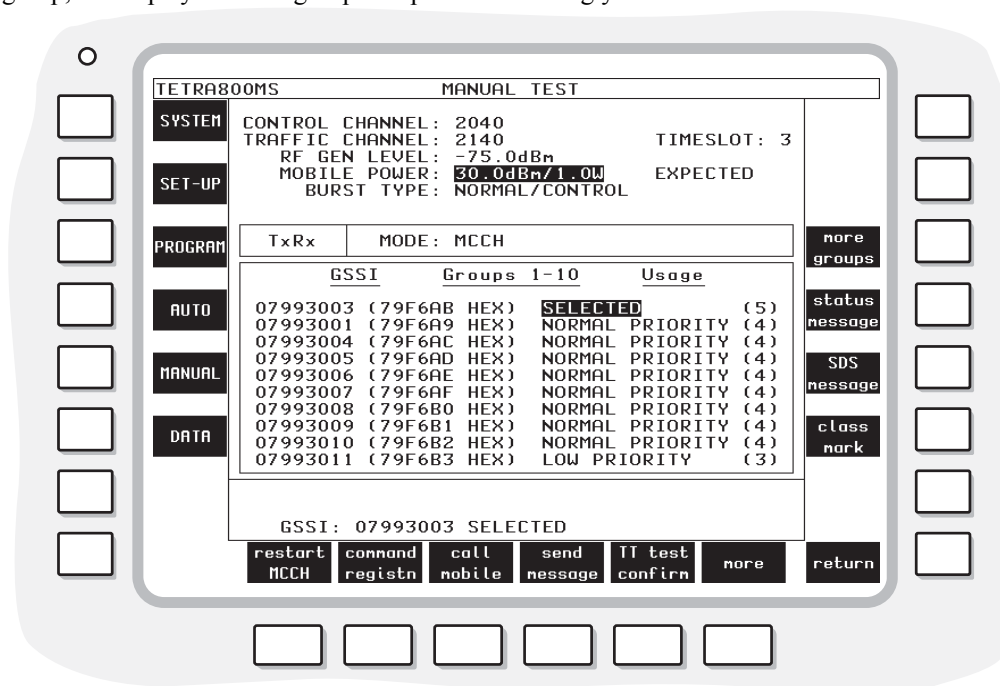
The Group Short Subscriber Identity for the selected group (Class of Usage 5) is displayed in *decimal* or as NO GROUP SELECTED. If an additional group is attached (Class of Usage ≠5) or a group is detached, it is displayed separately.

Some mobiles may attach multiple additional groups, in which case the Test Set displays only the most recent attached or detached groups beside the selected group. You can display all attached groups (up to 40) by pressing the *[show detail]* then *[groups]* soft keys. If there are more than 10 attached groups, use the *[more groups]* soft key to cycle the display through groups 1-10, 11-20, 21-30 and 31-40.

Each attached group is shown in decimal and HEX.

The class of usage of each group is also shown, e.g. SELECTED, NORMAL PRIORITY, LOW PRIORITY.

If you change the currently selected group, alter the priority of a group, detach a group or attach a group, the displayed list of groups is updated accordingly.



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Fig. 4-6 TETRA Mobile Manual test screen showing attached group details.

TEI

The TETRA Equipment Identity.

Only displayed if an extended (TT Test mode) registration has taken place.

Power class

Defines the mobile's maximum power level.

See the table *Mobile transmitter power levels* in Chapter 11, *Performance data*.

Only displayed if an extended (TT Test mode) registration has taken place.

Receiver class

Defines the receiver performance of the mobile. The classes are A, B or E.

Only displayed if an extended (TT Test mode) registration has taken place.

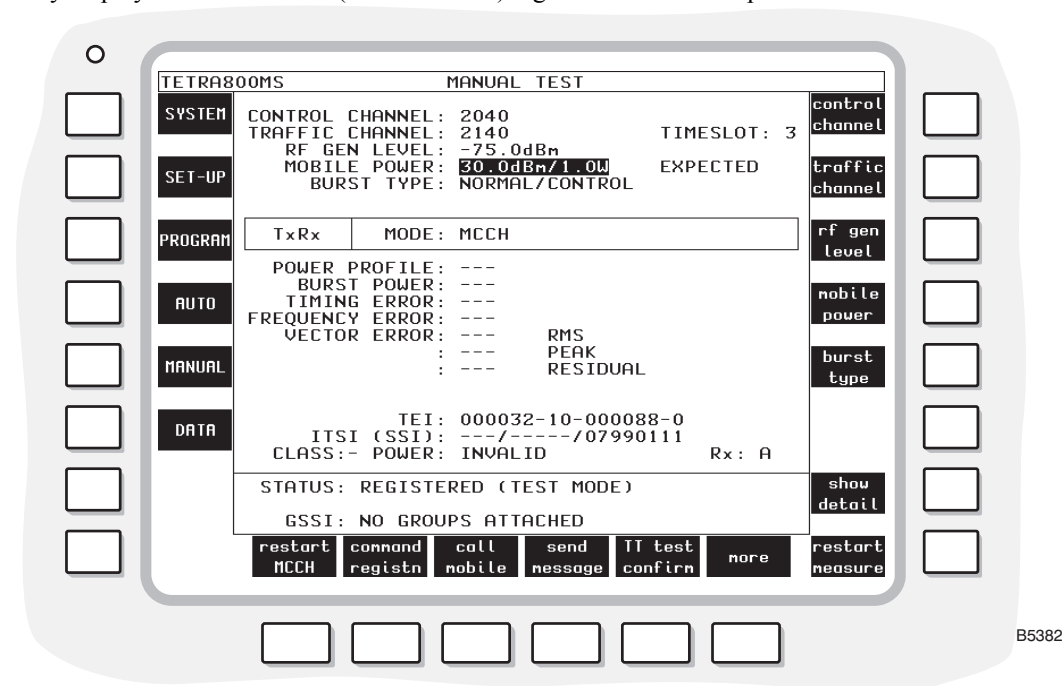


Fig. 4-7 TETRA Mobile Manual test screen, showing TEI, power class and receiver class

Setting up Calls to or from the mobile

Calls may be set up from a mobile to the Test Set (Mobile Originated or MO calls) and from the Test Set to a mobile (Mobile Terminated or MT calls) to allow the RF and audio performance of the mobile to be tested. The Test Set supports *group*, *private*, *phone* and *emergency* type calls, simplex and duplex.

Not all TETRA mobiles will support all of the above call types. TETRA mobiles from different manufacturers may differ in their implementation of certain call functions, and may not necessarily be fully compatible with the Test Set for all of these call types. The Test Set is compatible with mobiles conforming to version 3 of the TETRA MoU Interoperability Profile (TIP V3 core services).

Notes

1. All operations involving signalling between a mobile and the Test Set (registration, de-registration, call setup, call cleardown, transmit request) are only possible in Systems manual test mode or System auto test mode. If you switch to the [DUPLEX] mode during a call (for example to examine mobile transmitter performance using the vector diagrams or vector analyzer) you must return to the [SYSTEMS], [MANUAL] mode to cleardown the call.
2. Note also that parameters changed in the [DUPLEX] mode of the Test Set (such as BURST TYPE, RF GEN LEVEL or RF SELECT) are reset to their [SYSTEMS], [MANUAL] mode values on returning the Test Set to the [SYSTEMS] mode. This prevents any standard system settings being changed inadvertently

Mobile Originated Calls

For mobile-originated calls, the relevant call type is selected on the mobile; user operations on the mobile cause signalling to be sent to the Test Set to set up a call.

The following mobile-originated call types are supported and are described below:

- Group Call.
- Private Call (Individual Call).
- Telephone Call (PSTN Call or PABX Call or ISDN Call).
- Emergency Call.

Mobile Originated Group Call

This call type has the following characteristics:-

- Call is addressed to a group of TETRA users (point-to-multipoint).
- Simplex call only.
- Direct set-up only.
- Automatic cleardown by network.

The exact procedure will depend on the mobile under test, but normally the following operations are necessary to initiate a group call from the mobile to the Test Set:

- Ensure that the mobile has registered to the Test Set (SSI or full ITSI displayed).
- Ensure that the mobile has attached a selected Group ID (GSSI) to the Test Set.
- Ensure that the mobile's selected Group ID (GSSI) is not 'NO GROUP'.
- Ensure that the mobile is in group mode, if it has a mode selection.
- Ensure that the mobile's display shows no dialled digits.
- Press and hold the mobile's PTT (Push To Talk) switch.

The Test Set will indicate that a Group Simplex Direct Call has been initiated and that it is in conversation on the traffic channel (MODE: CONVERSATION). The called Group ID (GSSI) is shown. The mobile will initially be transmitting. See *PTT operation* on page 4-20. Depending on the mobile, it may be possible to cleardown the call from the mobile; otherwise the call can be cleared from the Test Set. See *Call Cleardown* on page 4-22.

The Test Set will cleardown the call automatically after expiry of the Hang Timer if neither the mobile nor the Test Set are transmitting. See *PTT operation* on page 4-20.

Note

TETRA mobiles set up group calls to a pre-selected group of TETRA users, hence there is no number to be 'dialled' when initiating a group call. Normally a TETRA mobile will have a rotary selector or up / down keys to select the required group, or to select NO GROUP. The selected group will normally be attached and indicated on the Test Set when registration is performed; if the group selection is subsequently altered the new selected group will normally be attached and indicated on the Test Set. TETRA mobiles can only set up group calls when a selected group has been attached to the Test Set.

Mobile Originated Private Call (Individual Call)

This call type has the following characteristics:

- Call is addressed to an individual TETRA user (point-to-point).
- Can be simplex or duplex call.
- Can be direct set-up or hook signalling.

The ability to originate a simplex or duplex private call may depend on the mobile under test and the particular configuration programmed into it. The use of direct set-up or hook signalling will normally depend on the mobile configuration, but may be affected by user selection of Loud Mode (speaker) or Discreet Mode (earpiece) on the mobile. Use of duplex may automatically switch the mobile to use the earpiece instead of the speaker. The exact procedure will depend on the mobile under test, but normally the following operations are necessary to initiate a private call from the mobile to the Test Set:

- Ensure that the mobile has registered to the Test Set (SSI or full ITSI displayed).
- Ensure that the mobile is in private mode, if it has a mode selection.
- Key in on the mobile a valid called party individual short subscriber ID (SSI) (see note below).
- Simplex call: if configured for direct set-up, press and hold the mobile's PTT switch.
- Simplex call: if configured for hook signalling, press and release the mobile's PTT switch.
- Duplex call: press the mobile's key marked "SEND" (handset up) or "Y" (yes) or "OK".

The Test Set will indicate the type of individual call that has been initiated and that it is in conversation on the traffic channel (MODE: CONVERSATION). The called ID (SSI) is shown. This may differ from the dialled number, e.g. the mobile may add the dialled number to a base address. For a simplex call, the mobile will initially be transmitting (direct set-up) or receiving (hook signalling). See *PTT operation* on page 4-20. For a duplex call, the mobile will be transmitting and receiving, and PTT operation is not required. The call can be cleared down from either the mobile or the Test Set. See *Call Cleardown* on page 4-22.

Note

The mobile may be configured to only accept numbers within a certain limited range as valid SSI values for a private call. If unsure of what is acceptable, try a number which is close to the mobile's own SSI; e.g. alter the least significant digit. For some mobiles it may be necessary to prefix the SSI or make a menu selection to indicate that the number is a TETRA individual subscriber ID - refer to the mobile's operating instructions

Mobile Originated Telephone Call (PSTN Call or PABX Call or ISDN Call)

This call type has the following characteristics:

- Call is addressed to an individual telephone user (point-to-point).
- Called user is accessed via the PSTN, PABX or ISDN gateway on the TETRA network.
- Normally duplex call for both users, but can be simplex call for the TETRA user.
- Normally hook signalling for both users, but can be direct set-up for the TETRA user.
- DTMF overdial for end to end signalling.

The ability to originate a telephone call may depend on the mobile under test and the particular configuration programmed into it. The use of direct set-up or hook signalling will normally depend on the mobile configuration, but may be affected by user selection of Loud Mode (speaker) or Discreet Mode (earpiece) on the mobile. Use of duplex may automatically switch the mobile to use the earpiece instead of the speaker. The exact procedure will depend on the mobile under test, but normally the following operations are necessary to initiate a telephone call from the mobile to the Test Set:

- Ensure that the mobile has registered to the Test Set (SSI or full ITSI displayed).
- Ensure that the mobile is in telephone mode, if it has a mode selection.

- Key in on the mobile a valid called party telephone number (see note below).
- Press the mobile's key marked "SEND" (handset up) or "Y" (yes) or "OK".

The Test Set will normally indicate that an individual duplex hook signalling call has been initiated and that it is in conversation on the traffic channel (MODE: CONVERSATION). The called ID (SSI) of the TETRA network's PSTN, PABX or ISDN gateway is shown, along with the telephone number of the called user on the PSTN, PABX or ISDN (the External Subscriber Number). For a duplex call, the mobile will be transmitting and receiving, and PTT operation is not required. It may be possible to send DTMF signalling during a duplex telephone call. See DTMF Overdial on page 4-21. The call can be cleared down from either the mobile or the Test Set. See *Call Cleardown* on page 4-22.

Note

The mobile may be configured to only accept numbers within a certain limited range as valid ESN values for a telephone call. If unsure of what is acceptable, try a number which is valid in the PSTN of the country for which the TETRA mobile is configured, for example ten digits starting with zero. For some mobiles it may be necessary to prefix the ESN or make a menu selection to indicate that the number is a PSTN telephone number - refer to the mobile's operating instructions. For PABX calls, valid ESN values may be different, for example four digits not starting with zero.

Mobile Originated Emergency Call

This call type has the following characteristics:

- Call can be addressed to either an individual TETRA user or a group of TETRA users.
- May be addressed to special emergency SSI. e.g. 16777214.
- Can be simplex or duplex call.
- Can be direct set-up or hook signalling.
- Emergency priority level 15 i.e. Pre-emptive Priority level 4.

The ID of the called user(s) and the type of call which is set up are normally pre-programmed and are dependent on the configuration of the mobile for the requirements of a particular TETRA network.

The mobile may be configured to recognise an emergency call as a special case and behave differently, for example by displaying "EMERGENCY" and over-riding the user setting of the volume control, or by activating the mobile's microphone and transmitter without requiring the user to press the PTT.

The exact procedure will depend on the mobile under test, but normally the following operations are necessary to initiate an emergency call from the mobile to the Test Set:

- Ensure that the mobile has registered to the Test Set (SSI or full ITSI displayed).
- Press the mobile's red emergency button (may need to press and hold button briefly).

The Test Set will indicate the type of call which has been initiated and that it is in conversation on the traffic channel (MODE: CONVERSATION). The called user ID (SSI or GSSI) is shown. For a simplex call the mobile may initially be transmitting or receiving depending on the particular configuration. See *PTT operation* on page 4-20. It may not be possible to clear down an emergency call from the mobile, or this may be possible only by using the emergency button again; normally the call can be cleared from the Test Set. See *Call Cleardown* on page 4-22.

An emergency group call will be cleared down automatically by the Test Set after expiry of the Hang Timer. See *PTT operation* on page 4-20.

Note

Some mobiles may be configured to use the red emergency button to send an emergency status message and switch the mobile into 'emergency mode' without setting up a call; in this case it will then be necessary to use the PTT switch to initiate the emergency call. Some mobiles do not have a dedicated emergency button but may allow the '0' key or the '1' key to be used for this purpose, in which case it is necessary to press and hold the key until the mobile indicates that it is setting up an emergency call.

Mobile Terminated Calls

For mobile terminated calls, the relevant call type is selected on the Test Set after pressing the *[call mobile]* soft key. The relevant parameters can be configured for each call type. See *Call Types* in Chapter 3.

The following mobile-terminated call types are supported and are described below:

- Group Call.
- Private Call (Individual Call).
- Telephone Call (PSTN, PABX or ISDN Call).
- Emergency Call.
- User Defined Call

Mobile Terminated Group Call

This call type has the following characteristics:

- Call is addressed to a group of TETRA users (point-to-multipoint).
- Simplex call only.
- Direct set-up only.
- Automatic cleardown by network.

The exact procedure will depend on the mobile under test, but normally the following operations are necessary to initiate a group call from the Test Set to the mobile:

- Ensure that the mobile has registered to the Test Set (SSI or full ITSI displayed).
- Ensure that the mobile has attached a selected Group ID (GSSI) to the Test Set.
- Ensure that the mobile's selected Group ID (GSSI) is not 'NO GROUP'.
- Ensure that the mobile is in group mode, if it has a group selection.
- Do not alter the value of GSSI in the Test Set MOBILE PARAMETERS menu (see note 1 below).
- Set the Test Set group call type PRIORITY and CALLING PARTY SSI as required.
- Press the Test Set *[call mobile]* soft key followed by the Test Set *[group call]* soft key.

No action is required on the mobile to answer the call, as it is direct set-up. The Test Set will indicate that a Group Simplex Direct Call has been initiated and that it is in conversation on the traffic channel (MODE: CONVERSATION). The called Group ID (GSSI) is shown. The mobile will initially be receiving. See *PTT operation* on page 4-20. It is normally possible for the mobile to leave but not to cleardown the group call, as it is not the call owner; the call can be cleared down from the Test Set. See *Call Cleardown* on page 4-22..

The Test Set will cleardown the call automatically after expiry of the Hang Timer if neither the mobile nor the Test Set are transmitting. See *PTT operation* on page 4-20.

Notes

1. Mobile terminated group calls are addressed to the Group ID (GSSI) value shown in the MOBILE PARAMETERS menu, which is updated with the selected Group ID at registration or when a new group is selected on the mobile. If the mobile has attached other Group ID (GSSI) values in addition to the selected Group ID, it may be able to receive calls addressed to these groups. This may depend on whether the mobile has been configured to 'scan' for calls to other attached groups. In this case the GSSI value in the Test Set MOBILE PARAMETERS menu can be changed to one of the other Group ID values to test this behaviour. TETRA mobiles ignore group calls addressed to groups which they have not selected or attached.

2. Unlike an individual call, a group call is addressed to multiple mobiles that do not acknowledge the call set-up. Therefore as the Test Set is not expecting an acknowledgement from the mobile, the Test Set will always go into conversation on the TCH whether or not there is a mobile under test that recognises the Group ID.

Mobile Terminated Private Call (Individual Call)

This call type has the following characteristics:

- Call is addressed to an individual TETRA user (point-to-point).

- Can be simplex or duplex call.
- Can be direct set-up or hook signalling.

The ability to receive a simplex or duplex private call may depend on the mobile under test and the particular configuration programmed into it. The ability to accept direct set-up or hook signalling will normally depend on the mobile configuration, but may be affected by user selection of Loud Mode (speaker) or Discreet Mode (earpiece) on the mobile - a direct set-up call may be modified by the mobile under test to hook signalling. Use of duplex may automatically switch the mobile to use the earpiece instead of the speaker. The exact procedure will depend on the mobile under test, but normally the following operations are necessary to initiate a private call from the Test Set to the mobile:

- Ensure that the mobile has registered to the Test Set (SSI or full ITSI displayed).
- Do not alter the value of SSI in the Test Set MOBILE PARAMETERS menu (see note 1 below).
- Set the Test Set private call type PRIORITY to 00.
- Set the Test Set private call type to SIMPLEX or DUPLEX as required.
- Set the Test Set private call type to DIRECT or HOOK signalling as required.
- Set the Test Set private call type CALLING PARTY SSI as required. (See note 3 below.)
- Press the Test Set [*call mobile*] soft key, followed by the [*private call*] soft key.
- Simplex, direct set-up: no action is required on the mobile to answer the call.
- Simplex, hook signalling: press and hold the mobile's PTT switch to answer the call. (See Note2 below.)
- Duplex, hook signalling: press the mobile's key marked "SEND" (handset up) or "Y" (yes) or "OK" to answer the call. (See note 2 below.)

The Test Set will indicate the type of individual call which has been initiated and that it is either alerting or in conversation on the traffic channel (MODE: CONVERSATION). The called ID (SSI) is shown. For a simplex call, the mobile will initially be receiving (direct set-up) or transmitting (hook signalling). See *PTT operation* on page 4-20. For a duplex call, the mobile will be transmitting and receiving, and PTT operation is not required. The call can be cleared down from either the mobile or the Test Set.

Notes

1. Mobile terminated private (individual) calls are addressed to the user ID (SSI) shown in the MOBILE PARAMETERS menu, which is updated with the mobile's SSI at registration. TETRA mobiles ignore individual calls addressed to other users, i.e. different SSI values.
2. For some mobiles it may be necessary to select Private Mode before answering an individual call.
3. If the CALLING PARTY SSI matches an SSI entry stored in the mobile's directory, the mobile should display the name associated with the SSI as the calling party ID.

Mobile Terminated Telephone Call (PSTN, PABX or ISDN)

This call type has the following characteristics:

- Call is addressed to an individual TETRA user from a telephone (PSTN, PABX or ISDN) user (point-to-point).
- Called TETRA user is accessed via the gateway (PSTN, PABX or ISDN) from the external PSTN, PABX or ISDN.
- Normally duplex call for both users, but can be simplex call for TETRA user.
- Hook signalling only.
- DTMF over dial for end to end signalling.

Use of duplex may automatically switch the mobile to use the earpiece instead of the speaker. The exact procedure will depend on the mobile under test, but normally the following operations are necessary to initiate a phone call from the Test Set to the mobile:

- Ensure that the mobile has registered to the Test Set (SSI or full ITSI displayed).
- Do not alter the value of SSI in the Test Set MOBILE PARAMETERS menu (see note 1 below).

- Set the Test Set phone call type PRIORITY to 00.
- Set the Test Set phone call type CALLING PARTY SSI to the PSTN gateway address (16777184) or to the PABX gateway address (16777186) or to the ISDN gateway address (16777185) as required.
- Set the Test Set phone call type CALLING PARTY ESN as required (see note 3 below).
- Press the Test Set [*call mobile*] soft key, followed by the [*phone call*] soft key.
- Press the mobile's key marked "SEND" (handset up) or "Y" (yes) or "OK" to answer the call. (see note 4 below).

The Test Set will indicate the type of individual call which has been initiated and that it is either alerting or in conversation on the traffic channel (MODE: CONVERSATION). The called ID (SSI) is shown. As this is a duplex call, the mobile will be transmitting and receiving, and PTT operation is not required. It may be possible to send DTMF signalling during a duplex telephone call. See DTMF over dial on page 4-21. The call can be cleared down from either the mobile or the Test Set.

Notes

1. Mobile terminated telephone calls are addressed to the user ID (SSI) shown in the MOBILE PARAMETERS menu, which is updated with the mobile's SSI at registration. TETRA mobiles ignore individual calls addressed to other users, i.e. different SSI values.
2. For some mobiles it may be necessary to select Phone Mode before answering a telephone call.
3. If the CALLING PARTY ESN matches a phone number entry stored in the mobile's directory, the mobile should display the name associated with the phone number as the calling party ID.
4. Some mobiles are only capable of simplex operation and may either modify the duplex call to simplex or reject the duplex call. In the case of modification to simplex, press and hold the mobile's PTT switch to answer the call.

Mobile Terminated Emergency Call

This call type has the following characteristics:

- Call can be addressed to an individual TETRA user or a group of TETRA users.
- Can be simplex or duplex call.
- Can be direct set-up or hook signalling.
- Emergency priority level 15 i.e. Pre-emptive Priority level 4.

The mobile may be configured to recognise an emergency call as a special case and behave differently, for example by displaying "EMERGENCY" and over-riding the user setting of the volume control, or by activating the mobile's microphone and transmitter without requiring the user to press the PTT. The exact procedure will depend on the mobile under test, but normally the following operations are necessary to initiate an emergency call from the Test Set to the mobile:

- Ensure that the mobile has registered to the Test Set (SSI or full ITSI displayed).
- Group call: ensure that the mobile has attached a selected Group ID (GSSI) to the Test Set.
- Group call: ensure that the mobile's selected Group ID (GSSI) is not 'NO GROUP'.
- Do not alter the value of SSI or GSSI in the Test Set MOBILE PARAMETERS menu.
- Set the Test Set emergency call type to INDIVIDUAL or GROUP as required.
- Set the Test Set emergency call type to SIMPLEX or DUPLEX as required.
- Set the Test Set emergency call type to DIRECT or HOOK signalling as required.
- Set the Test Set emergency call type CALLING PARTY SSI as required.
- Press the Test Set [*call mobile*] soft key, followed by the [*emergency call*] soft key.
- Simplex, direct set-up: no action is required on the mobile to answer the call.
- Simplex, hook signalling: press and hold the mobile's PTT switch to answer the call.
- Duplex, hook signalling: press the mobile's key marked "SEND" (handset up) or "Y" (yes) or "OK" to answer the call.

The Test Set will indicate the type of emergency call which has been initiated and that it is either alerting or in conversation on the traffic channel (MODE: CONVERSATION). The called ID (SSI or GSSI) is shown. For a simplex call, the mobile will initially be receiving (direct set-up) or transmitting (hook signalling). See 'PTT operation' starting on page 4-20, for control of transmission and reception. For a duplex call, the mobile will be transmitting and receiving, and PTT operation is not required. The ability to clear down the emergency call from the mobile will depend on the mobile; in the case of an emergency group call the mobile may be able to leave the call but will not be able to clear it down as it is not the call owner. The call can be cleared down from the Test Set. See *Call Cleardown* on page 4-22.

An emergency group call will be cleared down automatically by the Test Set on expiry of the hang timer. See 'PTT operation' starting on page 4-20.

Note

For some mobiles it may be necessary to select Private Mode before answering an individual emergency call.

Mobile Terminated User Defined Call

This call type allows all mobile terminated call parameters to be set to the users requirements. It can be used to define call types which are not covered by the four pre-defined call types (for example a simplex telephone call). It can also be used to provide an additional call type (for example a private duplex call when the pre-defined private call type is set to simplex).

- Set the Test Set user defined call type to INDIVIDUAL or GROUP as required.
- Set the Test Set user defined call type to SIMPLEX or DUPLEX as required.
- Set the Test Set user defined call type to DIRECT or HOOK as required.
- Set the Test Set user defined call type PRIORITY as required.
- Set the Test Set user defined call type CALLING PARTY SSI as required.
- Set the Test Set user defined call type ESN as required.
- Press the Test Set *[call mobile]* soft key, followed by the *[user call]* soft key.
- Answer the call as required by the call type.

Mobile Terminated Call in Alerting State:

Caller ID, Call Modification, Call Reject and Call Abort

When a mobile terminated hook signalling call has been initiated from the Test Set to the mobile, the mobile will be in the **Alerting** state waiting for you to answer the call on the mobile; the Test Set indicates MODE: ALERTING. The Test Set supplies a calling user ID to the mobile, as configured for the call type, and the mobile may show this number if it has caller ID display capability. It is also possible to be in this state if a direct set-up call has been initiated from the Test Set to the mobile, and the mobile performs **Call Modification** to a hook signalling call by indicating to the Test Set that it is in the alerting state. In the alerting state there are six possible mobile and Test Set actions:

- You can answer the call; normally by using the mobile's PTT switch (for a simplex call) or the mobile's key marked "SEND" (handset up) or "Y" (yes) or "OK" (for a duplex call). The Test Set indicates MODE: CONVERSATION.
- The mobile can auto-answer the call; if it has this capability. This may depend on the mobile's setting of Loud Mode (speaker) or Discreet Mode (earpiece), if applicable.
- You can reject the call; normally by using the mobile's key marked "END" (handset down) or "C" (clear). The Test Set indicates MODE: MCCH / STATUS: CALL TO MOBILE REJECTED.
- The mobile can auto-reject the call if it has this capability, and you do not answer the call before expiry of the mobile's call set-up timer. (Typically 20 to 60 seconds duration.) The Test Set indicates MODE: MCCH/STATUS: CALL TO MOBILE NOT ANSWERED.
- You can abort the call; using the Test Set *[abort call]* soft key, simulating the calling user abandoning the call (hanging up) when the called user does not answer. This may cause the mobile to indicate that there has been a "missed call". The Test Set indicates MODE: MCCH / STATUS: CALL TO MOBILE ABORTED.
- The Test Set automatically aborts the call after 65 seconds if there is no answer or reject.

Mobile Terminated Call:

Call Type, Call Modification and Call Rejection by Mobile

It is possible to set any combination of CALL TYPE parameters in the Test Set SYSTEM PARAMETERS: CALL TYPE USER DEFINED CALL menu. These parameters are used to determine the type of mobile terminated call which will be set up from the Test Set to the mobile when the [*call mobile*] and [*user call*] soft keys are pressed. Some parameter combinations are invalid (for example GROUP with DUPLEX or HOOK), and a TETRA mobile will not necessarily support every valid combination of call type parameters. When a non-supported call type is set-up, the mobile may perform call modification to a type which it does support; in this case the Test Set will indicate the actual call type which is now taking place. Alternatively, the mobile may reject the call set-up, indicating that it does not support the service requested; in this case the Test Set indicates MODE: MCCH / STATUS: CALL TO MOBILE REJECTED.

PTT Operation

During a simplex call on a real TETRA network, only one user at a time may talk (transmit). Mobile users indicate their wish to talk by holding or releasing the PTT (Push To Talk) switch which is normally mounted on the side of a hand-held TETRA mobile, or is mounted on a separate microphone or handset for a vehicle-mounted TETRA mobile. The PTT switch does not directly control the mobile's transmitter - signalling is sent to the base station requesting or relinquishing permission to transmit.

When a simplex call has been set up on the Test Set, holding the mobile's PTT switch pressed will request transmit permission; the Test Set will grant transmit permission to the mobile, the mobile will turn on its transmitter, and the Test Set will indicate R6RX. Releasing the mobile's PTT switch will relinquish the transmit permission; the mobile will turn off its transmitter and the Test Set will indicate TxRx or TxP6. See also 'Selection of Trunking Mode' below. PTT operation does not apply to duplex calls, since the mobile's transmitter and receiver are both active during the call. (The Test Set indicates R6P6.)

Note

For simplex calls, mobiles will normally be configured with a transmit timer, typically of one minute duration. If the PTT switch is held pressed for longer than this period, the mobile will autonomously cease transmission as if the user had released the PTT switch. To avoid transmission time-out when testing a TETRA mobile, you are recommended to set up a duplex call (if possible) if you wish to examine the mobile transmitter measurements in detail. Using duplex also avoids the need to hold the mobile's PTT switch pressed during transmitter measurements.

Selection of Trunking Mode

During a simplex call on a real TETRA network, only one user at a time may talk (transmit). Other users wishing to talk (transmit) may make transmission requests either before or after the user with transmit permission has finished talking. Depending on the relative priorities of the users, a transmit request may either interrupt the transmitting user or be queued pending the end of transmission. When the transmitting user ceases transmission, there will either be another user granted transmit permission or the call will be silent (no user transmitting). The **Trunking Mode** defines the system behaviour when no user is transmitting. In the case of **Message Trunking**, the mobiles in the call remain assigned to the traffic channel until the call is cleared down. In the case of **Transmission Trunking**, the mobiles in the call are returned to the Main Control Channel (MCCH), with a traffic channel being assigned each time transmit permission is granted.

In the case of Quasi Transmission Trunking, the mobiles remain on the traffic channel for a short period and are changed to the MCCH if no PTT requests are made within this period (the Quasi Tx trunking hang timer).

For a simplex call on the Test Set, the SYSTEM PARAMETERS: TRUNKING TYPE parameter allows selection of MESSAGE TRUNKING, TRANSMISSION TRUNKING or QUASI TRANSMISSION TRUNKING. This selection is only applicable to simplex calls, and is only effective when the conversation mode is used in connection with the SYSTEM PARAMETERS: CALL TIMERS menu. The Test Set TRUNKING MODE selection applies to mobile originated calls as well as mobile terminated calls.

For a simplex call on the Test Set, the SYSTEM PARAMETERS: CALL TIMERS: TEST SET TRANSMIT MODE parameter allows selection of NONE, TIMED, or CONTINUOUS.

Note

The Test Set transmit mode parameter is only effective when the Test Set trunking type is set to Transmission Trunking. For Message Trunking and Quasi Transmission Trunking, the Test Set transmit mode is CONTINUOUS regardless of the setting of this parameter.

When the Test Set transmit mode is set to NONE, and the Test Set trunking mode is set to TRANSMISSION TRUNKING, the following Test Set behaviour applies to simplex calls.

When the mobile's PTT is released, the Test Set indicates MODE:MCCH CALL ACTIVE and TxRx and the mobile is returned to the MCCH. Further PTT requests can be made by the mobile, but the Test Set does not simulate another user talking. If no PTT requests are made, and the call type is a Group Call, the Test Set will automatically clear down the call on expiry of the CALL HANG TIMER.

When the Test Set transmit mode is set to TIMED, and the Test Set trunking mode is set to Transmission Trunking, the following Test Set behaviour applies to simplex calls.

When the mobile's PTT is released, the Test Set indicates MODE:MCCH CALL ACTIVE and TxRx and the mobile is returned to the MCCH. Further PTT requests can be made by the mobile, otherwise the mobile simulates another user talking on expiry of the TESTSET QUIET TIME. The Test Set indicates MODE:CONVERSATION and TxP6. The Test Set simulates the other user releasing PTT on expiry of the TEST SET TRANSMIT TIME.

Further PTT requests can be made by the mobile. If no PTT requests are made, and the call type is a group call, the Test Set will automatically clear down the call on expiry of the CALL HANG TIMER.

When the Test Set transmit mode is set to CONTINUOUS, the following Test Set behaviour applies to simplex calls, whether mobile originated or mobile terminated. When the mobile's PTT is released, the Test Set indicates TxP6. The Test Set immediately indicates to the mobile that another user has been granted transmit permission, and the mobile under test receives the audio on the traffic channel (silence, test tone or talk-back). The mobile under test will continue to receive the audio on the traffic channel until either the PTT is pressed or the call is cleared down. In the case of message trunking with conversation mode talk-back, the previously stored speech will be continuously replayed. When the mobile's PTT is pressed, the Test Set indicates R6Rx and the mobile's transmitter is turned on.

DTMF over dial

During a duplex telephone call, some mobiles are able to indicate DTMF (Dual Tone Multi Frequency) signalling to communicate with automated equipment connected to the PSTN, e.g. Auto Attendant, Voice Mail etc. This capability may depend on the mobile's configuration and may be a user settable option in the mobile's settings menu.

If enabled, pressing any of the keys 0 ~ 9, *, # on the mobile's keypad causes these digits to be sent to the Test Set which displays e.g. DTMF:123456789*0#. The mobile may be configured to indicate a DTMF tone for as long as the key is held pressed, in which case the Test Set shows the digit in inverse video until the key is released.

Call Cleardown

It is always possible to clear down a call from the Test Set, using the [*clear down*] soft key. It is often also possible to clear down a call from the mobile, depending on the mobile configuration and the type of call. The method for clearing down will depend on the mobile, but is typically an “END” (handset down) key, a “C” (clear) key or a soft key which indicates “END” only when a call is in progress and mobile clear down is possible. For a Private Mode (individual) Call or a Phone Mode (PSTN,PABX or ISDN) Call, it will normally be possible to clear down the call from the mobile. For a Mobile Originated Group Call set up from the mobile to the Test Set, it is possible with most mobiles to clear down from the mobile. For a Mobile Terminated Group Call set up from the Test Set to the mobile, it will normally be possible for the mobile to leave the call and return to the Main Control Channel (MCCH) by pressing ‘END’.

For a mobile originated or mobile terminated group call, the Test Set will automatically clear down the call on expiry of the call hang timer (transmission trunking, timed or no transmission).

Note

If you attempt to clear down a Mobile Terminated Group Call from the mobile, this will not clear down the call on the Test Set, as the mobile does not send any signalling to the Test Set in this case. The mobile will simply leave the Traffic Channel (TCH) and return to the Main Control Channel (MCCH). However, since the Test Set is still in conversation mode on the Traffic Channel, the mobile will not find the MCCH until you press the [*clear down*] soft key on the Test Set, which causes the Test Set to return to MODE: MCCH.

Mobile Originated Status and Short Data (SDS) Messages

For mobile-originated messages, the relevant message type is selected on the mobile; user operations on the mobile cause signalling to be sent to the Test Set to send the message.

The following mobile-originated message types are supported and are described below:

- Status
- SDS Type 1
- SDS Type 2
- SDS Type 3
- SDS Type 4

Mobile Originated Status Message

This message type has the following characteristics:

- 16-bit number with pre-defined meaning
- Message is addressed to an individual TETRA user or a group of TETRA users

The ability to originate a status message may depend on the mobile under test and the particular configuration programmed into it. The exact procedure will depend on the mobile under test, but normally the following operations are necessary to send a status message from the mobile to the Test Set:

- Ensure that the mobile has registered to the Test Set (SSI or full ITSI displayed)
- Ensure that the Test Set is in MCCH mode
- Select ‘STATUS’ using the mobile’s soft keys and menu functions
- Select a pre-coded status message or enter a status number on the mobile
- Enter or select an individual or group message destination address on the mobile
- Send the status message by briefly pressing the mobile’s PTT switch or ‘SEND’ (handset up) key or ‘OK’ key or ‘send status’ soft key

The mobile may indicate that the status message has been successfully sent. The Test Set will indicate the numerical value of the status message (in hexadecimal and decimal), and the type of status message which has been received, in the format:-

‘MSG IN: xxxxHEX (dddd) cccccccccccccccc’

With *x* being HEX digits, *d* being decimal digits and *c* being text characters making up the message type text. The destination SSI or GSSI of the status message is shown by pressing [show detail] then [status message].

Status messages in the range 8000 - EFFF HEX (32768 - 61439) are defined by network operators or user applications.

Other status message values are for system use. Some of these values have standard defined meanings, shown in the following table.

Status values displayed

HEX Value	Decimal Value	Text indicated by Test Set	Explanation
0000	0	EMERGENCY STATUS	May precede an emergency call
0001	1	RESERVED FOR TETRA	Base station fall back
0002 to 7BFF	2 to 31743	RESERVED FOR TETRA	Reserved for future definition in the TETRA standard.
7C00 to 7CFF	31744 to 31999	TEXT PROTOCOL 000 to TEXT PROTOCOL 255	SDS-TL message number <i>nnn</i> was not understood by the receiving MS
7D00 to 7DFF	32000 to 32255	TEXT MEM FULL 000 to TEXT MEM FULL 255	Text message number <i>nnn</i> could not be stored by the receiving MS
7E00 to 7EFF	32256 to 32511	TEXT RECEIVED 000 to TEXT RECEIVED 255	Text message number <i>nnn</i> received by the MS
7F00 to 7FFF	32512 to 32767	TEXT CONSUMED 000 to TEXT CONSUMED 255	Text message number <i>nnn</i> read by the MS user
8000 to EFFF	32768 to 61439	RESERVED FOR USER	Values defined by user applications
F000 to FDFF	61440 to 65023	RESERVED FOR SYSTEM	Reserved for future system definition
FE00	65024	GENERAL STATUS ACK	Status message received by MS
FE01	65025	GENERAL STATUS NO ACK	Negative acknowledgement. (SwMI error code)
FE02	65026	NOT AUTHORISED	SwMI error code
FE03	65027	UNKNOWN ADDRESS	SwMI error code
FE04	65028	NO DESTINATION	SwMI error code
FE05	65029	DEST NOT REACHABLE	SwMI error code
FE06	65030	UNAUTH DESTINATION	SwMI error code
FE07	65031	DESTINATION BUSY	SwMI error code
FE08	65032	VALUE OUT OF RANGE	Status message received by MS outside the range of defined values
FE09 to FE1F	65033 to 65055	RESERVED FOR SYSTEM	Reserved for future definition of system error codes
FE20 to FEFF	65056 to 65263	RESERVED FOR SYSTEM	Reserved for future definition

(continued)

Status values displayed (continued)

FEF0	65264	TX INHIBIT ON	RF sensitive area use
FEF1	65265	TX INHIBIT OFF	RF sensitive area ended
FEF2 to FEF5	65266 to 65269	RESERVD FOR SYSTEM	Reserved for future system definition
FEF6	65270	NORMAL AL CALL REQ	MS requests system to set up an ambience listening call
FEF7	65271	EMERG AL CALL REQ	MS requests system to set up an emergency ambience listening call
FEF8	65272	SCANNING OFF	MS is not scanning groups
FEF9	65273	SCANNING ON	MS is scanning groups
FEFA	65274	ENTRY REQUEST	MS wants to be a member of a group
FEFB	65375	RESERVD FOR SYSTEM	Reserved
FEFC	65276	URGENT CALLBACK	Urgent call-back request
FEFD	65277	SELECTIVE ALERT	Notification of important call
FEFE	65278	RESERVD FOR SYSTEM	reserved
FEFF	65279	CALLBACK REQUEST	Call-back request
FF00 to FFFF	65280 to 65535	RESERVD FOR SYSTEM	Reserved for future system definition

Mobile Originated Short Data (SDS) Message

This message type has the following characteristics:

- Type 1: 16-bit number
- Type 2: 32-bit number
- Type 3: 64-bit number
- Type 4: variable length text message or other SDS-TL application
- Message is addressed to an individual TETRA user or a group of TETRA users

The ability to originate a short data message may depend on the mobile under test and the particular configuration programmed into it. Type 4 is more commonly supported than types 1, 2 and 3. The exact procedure will depend on the mobile under test, but normally the following operations are necessary to send a short data message from the mobile to the Test Set:

- Ensure that the mobile has registered to the Test Set (SSI or full ITSI displayed)
- Ensure that the Test Set is in MCCH mode
- Select the 'send message' function using the mobile's soft keys and menu functions
- Enter a 16-bit, 32-bit or 64-bit number or a text message on the mobile
- Enter or select an individual or group message destination address on the mobile
- Send the short data message by briefly pressing the mobile's PTT switch or 'SEND' (handset up) key or 'OK' key or 'send message' soft key as appropriate

The mobile may indicate that the short data message has been successfully sent.

The Test Set reacts to Type 1, 2 and 3 messages differently to Type 4 messages.

Type 1, 2 and 3 messages

The Test Set will indicate the numerical value (in hexadecimal) and the type of short data message which has been received.

Short data message types are indicated as follows:

- “MSG IN: xxxx HEX (SDS TYPE 1)” (4 HEX characters.)
- “MSG IN: xxxxxxxx HEX (SDS TYPE 2)” (8 HEX characters.)
- “MSG IN: xxxxxxxxxxxxxxxx HEX (SDS TYPE 3)” (16 HEX characters.)

The destination SSI or GSSI of the SDS message is shown by pressing *[show detail]* then *[SDS message]*.

Type 4 messages

SDS Type 4 messages are of variable length and use the SDS Transport Layer (SDS-TL) protocol to identify their content and request delivery reports. All SDS Type 4 messages should contain at least a minimal 2 byte SDS-TL header as specified in ETSI EN 300 392-2 Clause 29.

The Test Set provides a minimal single line display of the form “MSG IN:XXXXXXX” when any SDS type 4 message is received. Press *[show detail]* then *[SDS message]* to display the full message content.

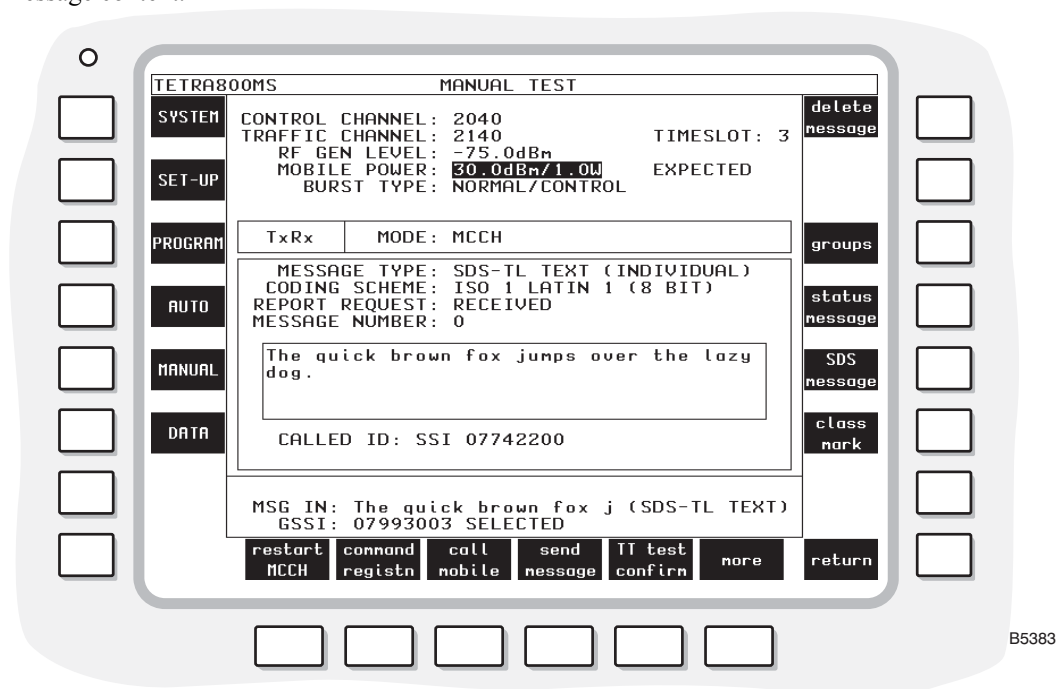


Fig. 4-8 TETRA Mobile Manual test screen showing SDS Message screen.

MESSAGE TYPE indicates whether the message is identified as an SDS-TL text message. (INDIVIDUAL) or (GROUP) indicate whether the message is sent to an SSI or a GSSI.

For other SDS-TL message types the SDS-TL protocol identifier is shown as a HEX value.

CODING SCHEME indicates whether the text message was sent using 8 bit TETRA standard coding or 7 bit GSM compatible coding.

REPORT REQUEST indicates the report type requested by the MS (received, consumed or none). The Test Set sends the requested report type as a short report.

MESSAGE NUMBER is the identification number of the message. The Test Set includes this number in the message report (if requested).

The message text box shows the text message sent by the mobile, up to 160 characters. For non-text messages this box shows up to 80 HEX bytes of data sent by the MS.

CALLED ID shows the SSI (individual) or GSSI (group) to which the message was sent.

Mobile Terminated Status and Short Data (SDS) Messages

The following mobile terminated message types are supported and are described below:-

- Status
- SDS Type 1
- SDS Type 2
- SDS Type 3
- SDS Type 4

For information about setting up the Messages parameters see Chapter 3, *Selecting and customizing the system*.

Mobile Terminated Status Message

This message type has the following characteristics:

- 16-bit number with pre-defined meaning (see page 4-23).
- Message is addressed to an individual TETRA user or a group of TETRA users

The exact procedure will depend on the mobile under test, but normally the following operations are necessary to send a status message from the Test Set to the mobile:

- Ensure that the mobile has registered to the Test Set (SSI or full ITSI displayed)
- Ensure that the Test Set is in MCCH mode
- On the Test Set SYSTEM PARAMETERS, MESSAGES MENU set the following parameters as required:
INDIVIDUAL MESSAGE or GROUP MESSAGE
CALLING PARTY SSI
STATUS MESSAGE value.
- Press the Test Set *[send message]* soft key followed by the Test Set *[status message]* soft key.

The mobile may indicate that the status message has been received and you should be able to read the message and see the calling party address. The Test Set will indicate that the message has been sent and the numerical value of the status message (in hexadecimal and decimal), in the format:-

‘MSG OUT: xxxxHEX (dddd) STATUS

Mobile Terminated Short Data (SDS) Message Types 1, 2 and 3

This message type has the following characteristics:

- Type 1: 16-bit number
- Type 2: 32-bit number
- Type 3: 64-bit number
- Message is addressed to an individual TETRA user or a group of TETRA users

The ability to receive a short data message may depend on the mobile under test and the particular configuration programmed into it. Type 4 is more commonly supported than types 1, 2 and 3. The exact procedure will depend on the mobile under test, but normally the following operations are necessary to send a short data message from the Test Set to the mobile:

- Ensure that the mobile has registered to the Test Set (SSI or full ITSI displayed)
- Ensure that the Test Set is in MCCH mode
- On the Test Set SYSTEM PARAMETERS, MESSAGES MENU set the following parameters as required:
INDIVIDUAL MESSAGE or GROUP MESSAGE

CALLING PARTY SSI
 SDS TYPE 1, 2 or 3 MESSAGE value.

- Press the Test Set *[send message]* soft key followed by the Test Set *[other SDS]* soft key followed by the Test Set *[SDS type 1]*, *[SDS type 2]* or *[SDS type 3]* soft key.

The mobile may indicate that the SDS message has been received and you should be able to read the message and see the calling party address. The Test Set will indicate that the message has been sent and the numerical value (in hexadecimal) and the type of SDS message, in the format:-

- 'MSG OUT: xxxx HEX (SDS TYPE 1) or
- 'MSG OUT: xxxxxxxx HEX (SDS TYPE 2) or
- 'MSG OUT: xxxxxxxxxxxxxxxx HEX (SDS TYPE 3)

Mobile Terminated Short Data (SDS) Message Type 4

This message type has the following characteristics:

- Variable length text message or user application data
- SDS-TL header identifies message content
- SDS-TL delivery reports may be requested
- Message is addressed to an individual TETRA user or a group of TETRA users

The ability to receive a type 4 short data message may depend on the mobile under test and the particular configuration programmed into it. Some mobiles may not be capable of receiving 7-bit coded messages.

The exact procedure will depend on the mobile under test, but normally the following operations are necessary to send a short data message from the Test Set to the mobile:

- Ensure that the mobile has registered to the Test Set (SSI or full ITSI displayed)
- Ensure that the Test Set is in MCCH mode
- On the Test Set SYSTEM PARAMETERS, MESSAGES MENU set the following parameters as required:

INDIVIDUAL MESSAGE or GROUP MESSAGE
 CALLING PARTY SSI

Press the Test Set *[SDS type 4]* soft key and select *[type4SDS-TL]*, *[type 4 simple]* or *[type 4 in hex]* to configure the required message type as follows:

Type 4 SDS-TL text message configuration:

Use this message type for normal text messages.

- Set REPORT TYPE to request a message received report from the MS if required
- Set REPORT SIZE to SHORT (short report) or FULL (standard report) as required.
- Set TEXT CODING to 8-BIT or 7-BIT as required
- Set TIME STAMP to REQUIRED or NOT REQUIRED

Note

The time stamp uses the time and date from the Test Set real time clock, not including the year. To set the real time clock select *[HELP/SET-UP]*, *[SET-UP]*, *[MISC CONFIG]*, *[time & date]*.

- The text message box shows a pre-defined text message of length 120 characters. Press the *[message]* soft key if you want to enter your own text message. See *Channel Plan Title* in Chapter 3 for details of entering text.

Type 4 simple text message configuration

Use this message type for terminals that do not support the SDS-TL protocol for text messages.

- Set TEXT CODING to 8 BIT or 7 BIT as required.

- The text message box shows a pre-defined text message of length 120 characters. Press the *[message]* soft key if you want to enter your own text message. See *Channel Plan Title* in Chapter 3 for details of entering text.

Type 4 HEX data message configuration

Use this message type for other types of messages, including user applications.

- The message box contains the HEX data equivalent of an 8-bit text message with no report request and no time stamp. Use the DATA keypad to enter your own data in HEX.

Note

The HEX data should be an even number of HEX digits, representing an integer number of bytes (octets) of data. The data you enter is the SDS type 4 message content starting with the SDS-TL protocol identifier as defined in ETSI EN 300 392-2 Clause 29. The Test Set calculates the SDS data length from the number of octets of your data, and generates the remaining protocol necessary to send your data to the MS.

- Press the Test Set *[send message]* soft key followed by the Test Set *[SDS-TL text]* or *[simple text]* soft key to send a text message.
- Press the Test Set *[send message]* soft key followed by the Test Set *[other SDS]* followed by *[type 4 in hex]* soft key to send a user application message.

The Test Set will indicate that the message has been sent and the type of SDS type 4 message in the format:-

- 'MSG OUT:XXXX (SDS-TL TEXT)' or
- 'MSG OUT:XXXX (SIMPLE TEXT)' or
- 'MSG OUT:XXXX HEX (SDS-TYPE4)'

Neighbour Cell Broadcast, Cell Selection, Cell Re-selection and Call Restoration

On a real TETRA network, a TETRA mobile continually monitors the signal strength of neighbouring base stations (cells) to determine whether it would obtain better service from another base station (cell); this operation is performed when idle on the control channel (MCCH) and when engaged in a call on a traffic channel (TCH). The mobile uses information transmitted (broadcast) by its current base station (serving cell) to identify the frequencies on which to monitor neighbour cells; this information (the **Neighbour Cell Broadcast**) also enables the mobile to make decisions about the signal level conditions under which it should attempt to obtain better service (perform **Cell Re-Selection**). The mobile may also use this information when first switched on to determine the most suitable base station on which to register (perform **Initial Cell Selection**). If the mobile is engaged in a call when it performs Cell Re-Selection, it will also perform **Call Restoration** signalling on the new base station (cell) in order to continue with the call. There are a number of different types of cell re-selection which are used according to the mobile's call involvement and its monitoring capabilities.

The Test Set supports the following cell re-selection operations, which are described below:

- Neighbour Cell Broadcast.
- Initial Cell Selection.
- Undeclared Cell Re-Selection.
- Unannounced Cell Re-Selection and Call Restoration.
- Announced Type 3 Cell Re-Selection and Call Restoration.
- Announced Type 2 Cell Re-Selection and Call Restoration.

Note

To test the mobile's ability to perform the above operations you need two Test Sets which each have Phase 3 or later software. Each Test Set simulates a separate TETRA base station (cell), using different carrier frequencies, as in a real TETRA network. The RF inputs and outputs of the two Test Sets are combined so that the mobile can receive signals generated by each Test Set, and so that each Test Set can receive signals transmitted by the mobile. See the section below 'Configuration for Cell Re-Selection Tests'.

Configuration for Cell Re-Selection Tests

The equipment required for these tests is as follows:

- Two IFR 2968 TETRA Test Sets (Phase 3, software version 11.00 or higher).
- 6dB Power Splitter (3-port, 2-way resistive divider, 50Ω) rated for mobile transmit power.
- 50Ω attenuator rated for mobile transmit power (only if necessary to protect splitter).
- Connecting cables.

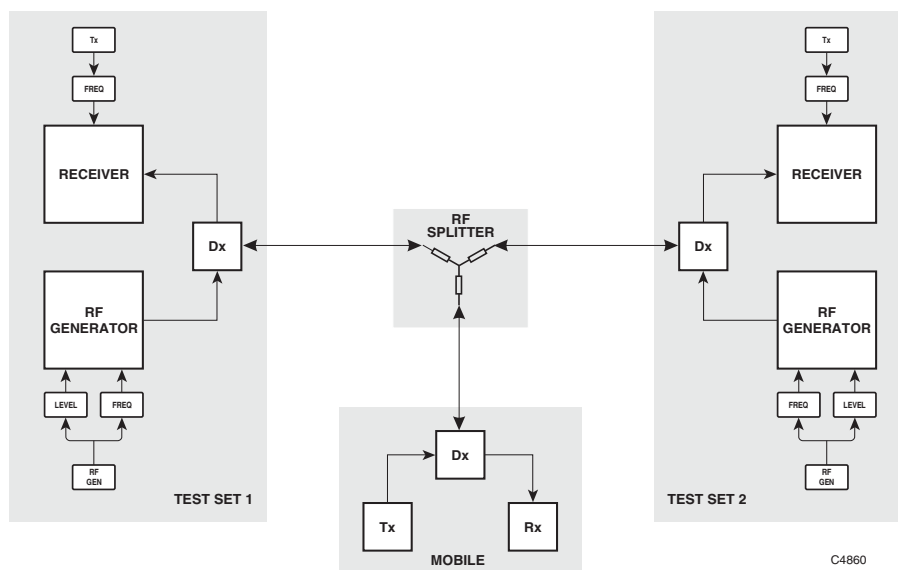


Fig. 4-9 Cell re-selection test set-up

Instead of connecting the mobile directly to a Test Set, use the power splitter and connecting cables to connect the mobile to both Test Sets via the power splitter. Connect the RF port of each Test Set (normally this will be N-type, single port duplex) to a port on the splitter, and connect the mobile under test to the third port on the splitter. The mobile will receive signals generated by each Test Set, subject to 6dB loss. Each Test Set will receive signals transmitted by the mobile, subject to 6dB loss. The Test Sets can compensate for this loss, so that the displayed values for signal generator level and mobile transmitter power correspond to the levels at the mobile's antenna connection. On each Test Set use the [HELP SET-UP] hard key, then the soft keys [SET-UP], [TEST OPTIONS], [RF port set-up], [set offset] to set offset values of -6.0 dB for both the Test Set RF generator and the Test Set receiver (negative values indicate a loss, positive values indicate a gain). If you need to use an attenuator to protect the splitter, this should be connected between the splitter and the mobile's antenna connection. If you have available calibrated loss figures for the connecting cables, mobile antenna connection adaptor, attenuator (if required) or other hardware, these should be included in the offset values. To return to TETRA operation either press the [return] soft key repeatedly or press the [HELP SET-UP] hard key twice.

On each Test Set, select TETRA Mobile as the system.

On each Test Set, ensure that the following SYSTEMS PARAMETERS are set up as follows:

- **CHANNEL PLAN:** On each Test Set, select the same CHANNEL PLAN, to suit the mobile under test. If this is a user defined channel plan, ensure that it is defined identically on each Test Set.
- **CONTROL CHANNEL:** Select different CONTROL CHANNEL numbers for each Test Set, preferably spaced at least four channels (100 kHz) apart. Depending on the mobile, you may need to ensure that at least one of the control channels corresponds to a channel that the mobile is configured to use for initial cell selection.
- **TRAFFIC CHANNEL:** Select different TRAFFIC CHANNEL numbers for each Test Set, which can be different from the CONTROL CHANNEL numbers and on any timeslots.
- **MOBILE COUNTRY CODE:** On each Test Set, select the same MCC value, which must match the MCC value expected by the mobile under test.
- **MOBILE NETWORK CODE:** On each Test Set, select the same MNC value, which must match the MNC value expected by the mobile under test.
- **BASE COLOUR CODE:** The BCC can be set to any value (although 00 is best avoided) and does not need to be the same on each Test Set.
- **LOCATION AREA:** Select different LA values for each Test Set. The actual values are arbitrary, but they must be different in order to simulate different cells.
- **MIN Rx LEVEL FOR ACCESS:** On each Test Set this must be set to the same value. The actual value to use will depend on the testing to be performed, as this parameter is used in conjunction with the cell re-selection parameters in the Neighbour Cell Broadcast. The suggested initial value for this parameter is -105 dBm.
- **MAX Tx LEVEL:** On each Test Set this must be set to the same value. The actual value to use will depend on the testing to be performed. If the 6dB power splitter (or attenuator) is not capable of handling the full transmit power class of the mobile, then you should ensure that the MAX Tx LEVEL: parameter is set to a safe value to avoid damage to the splitter. The suggested initial value for this parameter is 30 dBm.
- **ACCESS PARAMETER:** On each Test Set this must be set to the same value. The actual value to use will depend on the testing to be performed. The suggested initial value for this parameter is -33 dBm.
- **BASE SERVICES:** On each Test Set, set POWER ON REGISTRATION and POWER OFF DE-REGISTRATION to REQUIRED and ensure that the rest of the parameters are set to the same state on each Test Set.
- **NEIGHBOUR CELL INFO:** See below.
- **TRUNKING TYPE:** Set as required on each Test Set.
- **CALL TYPES:** Set as required on each Test Set.
- **CALL TIMERS:** Set as required on each Test Set.

Neighbour Cell Broadcast

Neighbour Cell Broadcast is a message which is generated periodically (every 5s ~ 30s) by each Test Set on both the control channel (MCCH) and the traffic channel (TCH). It contains the information about the other Test Set (cell) to enable the mobile to locate and monitor the signal level of the other Test Set (cell). It also contains the parameters necessary for the mobile to make cell re-selection decisions, in conjunction with the 'minimum Rx level for access' parameter, based on the signal levels of the two Test Sets (cells). On each Test Set, ensure that the Neighbour Cell Broadcast parameters are set as follows:

- **NEIGHBOUR CELL BROADCAST:** On each Test Set, select REQUIRED to allow the Test Set to generate the Neighbour Cell Broadcast Message.
- **BROADCAST INTERVAL:** This value is arbitrary, and controls how often the Test Set generates the Neighbour Cell Broadcast message. The suggested value for this parameter is 10s.

- **NEIGHBOUR CELL CHANNEL:** On each Test Set, select the channel number of the Main Control Channel (MCCH) which is set up on the *other* Test Set. This enables the mobile to tune its receiver to the correct frequency for monitoring and scanning the other Test Set.
- **NEIGHBOUR CELL LOCATION AREA:** On each Test Set, select the Location Area which is set up on the *other* Test Set. This enables the mobile to recognise the other Test Set as a valid neighbour cell to which it should register when performing cell re-selection.
- **NEIGHBOUR CELL IDENTIFIER:** This value is arbitrary, and is only used in Type 2 Cell Re-selection for the mobile to indicate which of a number of neighbour cells it has selected. The suggested value for this parameter is 1 on each Test Set.
- **SLOW RE-SELECT THRESHOLD:** This parameter sets the threshold level for judging the serving cell (the Test Set which the mobile is currently using) to be *improvable* (see 'C1 and C2 values' below). This parameter defines the additional threshold above the Fast Threshold. The suggested initial value for this parameter is 24 dB, setting the slow threshold level at $(-105 \text{ dBm} + 18 \text{ dB} + 24 \text{ dB}) = -63 \text{ dBm}$.
- **SLOW RE-SELECT HYSTERESIS:** This parameter sets the threshold level for judging the neighbouring cell (the other Test Set) to be better than the serving cell (the Test Set which the mobile is currently using) when the serving cell is *improvable* (see 'C1 and C2 values' below). The suggested initial value for this parameter is 6 dB.
- **FAST RE-SELECT THRESHOLD:** This parameter sets the threshold level for judging the serving cell (the Test Set which the mobile is currently using) to be *relinquishable* (see 'C1 and C2 values' below). The suggested initial value for this parameter is 18 dB, setting the fast threshold level at $(-105 \text{ dBm} + 18 \text{ dB}) = -87 \text{ dBm}$.
- **FAST RE-SELECT HYSTERESIS:** This parameter sets the threshold level for judging the neighbouring cell (the other Test Set) to be better than the serving cell (the Test Set which the mobile is currently using) when the serving cell is *relinquishable* (see 'C1 and C2 values' below). The suggested initial value for this parameter is 6 dB.

C1 and C2 values

In a real TETRA network, the mobile is required to *scan* the serving cell (the base station which it is currently using) and measure the signal level (the RSSI value). The difference between the RSSI value and the MIN Rx LEVEL FOR ACCESS parameter is known as 'C1'. The mobile is also required to *monitor* neighbouring cells and calculate for each neighbour cell the difference between the RSSI value and the MIN Rx LEVEL FOR ACCESS parameter, which is known as 'C2'. The mobile uses the C1 and C2 values in conjunction with the cell re-selection parameters (FAST THRESHOLD, FAST HYSTERESIS, SLOW THRESHOLD, SLOW HYSTERESIS) to make cell re-selection decisions. These decisions are only valid when the conditions have been established for a period of 5s and there have been no previous cell re-selections for a period of 15s; the decisions are categorised as follows:

- **Serving cell *relinquishable*:** $(C1 < \text{Fast Threshold}) \text{ AND } (C2 > (C1 + \text{Fast Hysteresis}))$. Using the suggested values, this would be satisfied with the currently used Test Set RF gen level below -87 dBm and the other Test Set RF gen level at least 6dB higher.
- **Serving cell *improvable*:** $(C1 < \text{Slow Threshold}) \text{ AND } (C2 > (C1 + \text{Slow Hysteresis}))$. Using the suggested values, this would be satisfied with the currently used Test Set RF gen level below -63 dBm and the other Test Set RF gen level at least 6dB higher.
- **Cell *usable*:** $(C1 > (\text{Fast Threshold} + \text{Fast Hysteresis}))$. Using the suggested values, this would be satisfied with the Test Set RF gen level above -81 dBm.
- **Initial cell selection:** $(C1 > 0)$. Using the suggested values, this would be satisfied with the test RF gen level above -105 dBm.
- **Radio link failure:** $(C1 < 0)$. Using the suggested values, this would be satisfied with the Test Set RF gen level below -105 dBm.

The mobile's cell re-selection decision also takes account of whether the neighbouring cells offer a better or worse level of service than the serving cell:

- If the service levels are the same in each cell, which is the case with the two Test Sets, the mobile should perform cell re-selection if the serving cell is *improvable*.
- If the service level in the neighbouring cell is worse, the mobile should only perform cell re-selection if the serving cell is *relinquishable*.
- If the service level in the neighbouring cell is better, the mobile should perform cell re-selection if the neighbouring cell is *usable*.

The criteria for cell re-selection are complex; for more detailed information you are referred to EN 300 392-2 subclauses 18.3.4 to 18.3.4.7.5, 18.5.4 and 23.7.1 to 23.7.5.2.3. During testing, the mobile may provide a diagnostic display indicating its calculated C1 and C2 values - you should refer to the mobile manufacturer or supplier to access this information.

Note

The values of C1 and C2 are further reduced if the value of the 'maximum Tx level' parameter is greater than the mobile's maximum power capability. The descriptions in this manual assume for the sake of simplicity that the 'maximum Tx level' parameter is set equal to or less than the mobile's maximum power capability, so that C1 and C2 are only dependent on the RSSI values and the 'minimum Rx level for access' parameter.

Initial Cell Selection

The mobile is allowed to initially select any suitable cell which has a positive C1 value, i.e. the received signal level is greater than the MINIMUM RX LEVEL FOR ACCESS parameter. If the mobile does not search the frequency band for suitable channels, it may select the Test Set generating the MCCH on one of its pre-defined channels and initially ignore the other Test Set, even if the other Test Set is generating the MCCH at a higher level. To test initial cell selection, ensure that both Test Sets are generating MCCH, set the RF gen levels on each Test Set as required, and switch on the mobile. The mobile will make its initial cell selection of one of the Test Sets, which will indicate STATUS: REGISTERED (ITSI ATTACH). If the mobile initially ignored the Test Set with the higher signal level, it may subsequently perform undeclared cell re-selection if the conditions are satisfied (see below).

Undeclared Cell Re-Selection

When the mobile is registered on one cell (Test Set) and is idle on the MCCH (i.e. not engaged in a call) it is required to monitor the neighbouring cells (the other Test Set) to determine whether cell re-selection conditions are satisfied. If this is the case, it will perform *undeclared* cell re-selection to the selected neighbour cell (the other Test Set). No signalling is sent to the serving cell (the currently used Test Set), i.e. it does not declare its intention of leaving the serving cell. The mobile performs *roaming location updating* signalling on the selected neighbour cell (the other Test Set), i.e. it is registering in the new location area which is indicated by the neighbour cell (the other Test Set). To test undeclared cell re-selection, ensure that both Test Sets are generating MCCH, and that the mobile is registered on one of the Test Sets. Set the RF gen levels on the Test Sets as required to satisfy the conditions for cell re-selection and wait for the mobile to decide to perform undeclared cell re-selection. The other Test Set will indicate STATUS: REGISTERED (ROAMING LOCATION UPDATE).

Unannounced Cell Re-Selection and Call Restoration

When the mobile is engaged in a call on one cell (Test Set) it is required to continue to monitor the neighbouring cells (the other Test Set) to determine whether cell re-selection conditions are satisfied. The type of cell re-selection depends on whether or not the serving cell (the currently used Test Set) needs to know that the mobile is leaving the serving cell. If the mobile is engaged in a group call and it is not transmitting (i.e. it is one of a number of mobiles which are receiving) then the serving cell does not need to know, i.e. the mobile does not need to *announce* that it is performing cell re-selection. To test unannounced cell re-selection, set up a group call on the currently used Test Set with the mobile receiving, and ensure that the other Test Set is generating MCCH. Set the RF gen levels on the Test Sets as required to satisfy the conditions for cell re-selection and wait for the mobile to decide to perform unannounced cell re-selection. The other Test Set will indicate CALL TYPE: RESTORED INDIVIDUAL CALL or CALL TYPE: RESTORED GROUP CALL. The mobile will be receiving the group call on the other Test Set and can continue with its normal operations. It is recommended to set both Test Sets to CONVERSATION MODE

TEST SOUND and TEST SET TRANSMIT MODE CONTINUOUS so that the duration of the break and re-establishment of the audio path can be observed.

Note

In the case of unannounced cell re-selection, the original Test Set does not know that the mobile has performed cell re-selection, and will remain in conversation on the traffic channel. You should press [clear down] on the original Test Set to return it to generating MCCH after the mobile has performed unannounced cell re-selection to the other Test Set.

Announced Type 3 Cell Re-Selection and Call Restoration

When the mobile is engaged in a call on one cell (Test Set) it is required to continue to monitor the neighbouring cells (the other Test Set) to determine whether cell re-selection conditions are satisfied. The type of cell re-selection depends on whether or not the serving cell (the currently used Test Set) needs to know that the mobile is leaving the serving cell. If the mobile is engaged in a private call or a phone call, or it is transmitting in a group call, then the serving cell needs to know, i.e. the mobile needs to *announce* that it is performing cell re-selection. To test announced cell re-selection, set up a private call or a phone call on the currently used Test Set, or a group call with the mobile transmitting, and ensure that the other Test Set is generating MCCH. Set the RF gen levels on the Test Sets as required to satisfy the conditions for cell re-selection and wait for the mobile to decide to perform announced cell re-selection. The currently used Test Set will indicate STATUS: MOBILE RE-SELECTED CELL (TYPE 3) / MODE: MCCH. The other Test Set will indicate CALL TYPE: RESTORED INDIVIDUAL CALL or CALL TYPE: RESTORED GROUP CALL. The mobile will continue with the call on the other Test Set and can continue with its normal operations. It is recommended to set both Test Sets to CONVERSATION MODE TEST SOUND so that the duration of the break and re-establishment of the audio path can be observed. When the mobile announces cell re-selection, the Test Set automatically returns to MCCH generation.

Announced Type 2 Cell Re-Selection and Call Restoration

The procedure for testing Announced Type 2 Cell Re-Selection is the same as for Type 3, the difference being due to the capability of the mobile under test, Type 2 being an optional requirement for TETRA mobiles. A Type 2 capable mobile will perform *background scanning* of the intended neighbour cell (the other Test Set) whilst it is engaged in the call on the serving cell (the currently used Test Set), and will have acquired the necessary synchronisation and system information from the neighbour cell (the other Test Set) before announcing cell re-selection of the identified neighbour cell on the serving cell. The currently used Test Set will indicate STATUS: MOBILE RE-SELECTED CELL (TYPE 2) in this case. The break in the audio path should be shorter for a type 2 re-selection, which can be observed by setting the CONVERSATION MODE on both Test Sets to TEST SOUND.

Analysis of Signalling messages

Analysis of the signalling messages exchanged between the mobile and the Test Set can be performed, using the *IFR TETRALOG MS Protocol Analyzer for 2968* application (IFR order code 81514).

Mobile Transmitter Testing

During registration, call set up or while transmitting during a call, the mobile will transmit bursts to the Test Set allowing the transmitter RF parameters to be measured. For a description of the measurements available see *Transmitter measurements* on page 4-36. These are displayed on the Manual Test screen, or may be examined in more detail in the [DUPLEX] mode of the Test Set. See the note on page 4-12. To select the burst type to be measured use the [burst type] soft key. Duplex call set-up is recommended, if supported by the mobile, as it avoids the need to hold the PTT switch pressed on the mobile. Simplex calls may also be subject to limited transmission periods due to autonomous call timers or transmit timers within the mobile.

Note

Measurements can only be made on bursts with a valid training sequence.

Mobile Receiver Testing

The receiver sensitivity of the mobile may be tested using a Bit Error Rate (BER) measurement in which PRBS frames sent to the mobile by the Test Set are returned to it by the RF transmission from the mobile. The Test Set then compares the data that it received with the data that it sent. This requires the mobile to be in a special mode in which it sends back all the traffic frames that it receives. This mode may be enabled with the Test Set and mobile in T1 Test mode, or with the Test Set and mobile in TT loopback mode. See *T1 Test mode* starting on page 4-37 and *TT LOOPBACK modes* starting on page 4-6.

Note

The T1 Test mode or TT loopback mode may not be supported by all mobiles.

If the mobile does not support the T1 loopback mode or TT loopback mode, receiver BER testing may still be possible using the standard signals generated by the Test Set. If the mobile is able to output its received data on a test connector it can be fed to and measured on a separate BER meter. Alternatively, some mobiles are able to measure their own BER internally.

T1 loopback mode is controlled by special parameters in the system information sent by the Test Set. To enable this mode in the Test Set press the *[T1 test]* soft key. The mode must also be enabled in the mobile. Refer to the manufacturer for instructions on how to do this. In this mode the BURST MODE and LOOPBACK parameters control the type of bursts transmitted by the mobile. With *[TCH/7.2]* selected and *[loop back]* ON the Test Set will perform a BER measurement on NORMAL bursts which the mobile has looped back.

Mobile Audio Testing

Audio performance testing allows for a subjective test of the audio performance of a mobile and may be used to confirm correct operation of the audio components within it. It is also a useful aid to fault diagnosis to audio components within the mobile.

Audio performance testing is achieved in the Test Set by storing speech received from the mobile and sending it back to the mobile after a 2 second delay (sometimes referred to as talkback or speech loopback).

In a duplex call, speech is continuously fed back to the mobile's loudspeaker; sounds entering the mobile's microphone should be heard from the mobile's speaker after a delay of 2 seconds.

In a simplex call sounds entering the mobile's microphone while the PTT is pressed should be heard from the mobile's speaker when the PTT is released; this is best achieved by speaking into the mobile's microphone briefly, then immediately releasing the PTT. The last 2 seconds stored speech will be sent to the mobile repeatedly, if the TEST SET TRANSMIT MODE is CONTINUOUS, to simulate a continuing transmission from another mobile.

The Test Set can send a test tone to the mobile, useful in ascertaining whether a fault is in the audio input or audio output section of the mobile if the talkback test has failed. The Test Set can also send silence to the mobile, useful in locating the source of noise in the audio system.

Selecting the Audio Signal

When a call has been established, the audio signal can be set to talkback, silence or test sound.

Pressing the *[talkback]* soft key selects the CONVERSATION (TALKBACK) mode. The speech frames produced by the signal from the microphone in the mobile are returned after a 2 second delay, to be output from the mobile's speaker.

Pressing the *[silence]* soft key selects the CONVERSATION (SILENCE) mode. *Silence* speech frames are sent to the mobile.

Pressing the *[test sound]* soft key selects the CONVERSATION (TEST SOUND) mode. *Test sound* speech frames are sent to the mobile. This encoded representation of a 1 kHz signal is provided as an aid to checking the audio output and speaker of the mobile, and represents an audio level approximately 10 dB below the theoretical maximum output level.

Note

Speech codecs in TETRA radios are not optimised for steady tones, and you may detect some fluctuation in level when played through the radio's earpiece or speaker.

Mobile Audio analysis

The Test Set audio generator and audio analysis facilities are available for performing audio testing while a TETRA radio is under test.

Press the [AF TEST] mode key followed by the [AF IN/OUT] soft key to access the AUDIO INPUT/OUTPUT screen. Refer to the Operating Manual for the Test Set 2968, Part Number 46882/274, for full details of AF TEST MODE operation.

The legend TETRA MS is shown in the title bar of this screen to indicate that the TETRA MOBILE TEST SYSTEM remains active. The signal generator continues to generate the signal set up in SYSTEMS mode.

Set up a call (preferably a duplex call) in SYSTEMS mode, select the required CONVERSATION mode and RF GEN LEVEL, then go AF TEST mode to perform audio analysis. Use a test connector, an accessory earpiece connector (or an acoustic coupler and microphone) to obtain an audio output signal from the mobile. Connect this to the Test Set AF input connector. The Test Set can now be used for objective measurements on the mobile's recovered audio signal. (Level, SINAD, distortion, oscilloscope and FFT audio spectrum analysis.)

If required, the mobile audio transmit performance can also be tested, in conjunction with the audio receive performance testing. In this case set up a duplex call in SYSTEMS mode with CONVERSATION MODE TALKBACK then select AF TEST MODE. In this configuration, audio input signals are TETRA speech encoded by the mobile, transmitted to the Test Set as digital data, re-transmitted by the Test Set to the mobile, and TETRA speech decoded by the mobile to produce audio output signals.

Connect the Test Set AF GEN OUTPUT connector to a suitable audio input test connector or accessory/microphone connector on the mobile (or an acoustic coupler and loudspeaker).

Note

The Test Set does not respond to signalling operations when in AF TEST mode (PTT press/release, cleardown, .call placement). Return to SYSTEMS mode to perform TETRA operations.

Mobile Receiver Testing using Audio Analysis

Some TETRA mobiles may not provide T1 loopback or TT loopback functions, or you may have difficulty in obtaining access to the mobile manufacturer's special test modes for performing internal or external BER testing. In these cases you will not be able to perform a BER test on the mobile's receiver. However, you may be able to achieve an approximate test of receiver sensitivity by using a conventional SINAD test.

In SYSTEMS mode, set up a call, preferably a duplex call. For a simplex call, set the Test Set TRANSMIT MODE to CONTINUOUS and do not hold the PTT so that the mobile is receiving.

Set CONVERSATION MODE TEST TONE and set the RF GEN LEVEL to the required level for sensitivity testing, e.g. -112 dBm.

Go to AF TEST mode and connect a suitable audio output signal from the mobile to the Test Set AF IN connector. If possible adjust the volume control (or accessory volume setting) of the mobile to achieve a measured AF LEVEL in the range 100 mV to 1 V.

Check that the audio distortion or SINAD measurement is acceptable, typically this might be $\leq 5\%$ distortion or $\geq 26\text{dB}$ SINAD. Using the CCITT psophometric band pass filter may produce more consistent measurements.

Note

The RF GEN LEVEL can only be adjusted in the SYSTEMS mode. Use [SYSTEMS] and [AF TEST] mode keys to toggle between the two modes if you need to examine distortion or SINAD measurements at different RF Signal levels.

Transmitter measurements

The measurements made on the transmitted signal from the mobile when in Manual Test mode are listed below, with information about them.

The following Tx measurements except power profile are measured over a number of bursts and the average or worst case result displayed. See *Tx Measurement Options* in Chapter 2.

Notes

1. The measurement averaging can be restarted by pressing the *[Restart measure]* soft key.
2. Measurements can only be made on bursts with a valid training sequence. If the mobile is believed to be transmitting but no measurements are indicated, the *spectrum analyzer* facility can be used to detect this. See Chapter 8, *Digital Duplex testing*.

Burst power

The burst power measurement shows the average power during the measured burst. This measurement is taken over the useful part of the burst (all modulation symbols $SN_0 \sim SN_{max}$) measured at the symbol points through a TETRA filter (Root Nyquist, $\alpha = 0.35$).

Burst type

The indicator next to the BURST POWER measurement shows whether the measured burst is a normal burst or a control burst.

Power profile

The power profile measurement gives an automatic pass/fail indication which shows whether or not the burst power profile is within the t1 and t3 power masks.

Timing error

The timing error or frame alignment measurement shows the symbol timing of Mobile's bursts measured with respect to the Test Set signal. A positive value indicates a burst which is early with respect to the Test Set signal. EARLY and LATE indications are also shown with the measurement.

Vector error RMS, peak and residual

The vector error measurement shows the vector error of the received symbols with respect to the ideal symbol points for the burst. This measurement is taken over the useful part of the burst measured at the symbol points through a TETRA filter. The RMS reading is the root mean squared of all the symbols, the peak reading is the vector error of the symbol with the highest error and the residual is the mean residual carrier magnitude.

Frequency error

The frequency error measurement shows the difference between the frequency of the received signal and the receiver frequency of the Test Set. This measurement is taken over the useful part of the burst measured at the symbol points through a TETRA filter.

Receiver measurements

To test the mobile's receiver, the Test Set generates specific test signals which are received and re-transmitted to the Test Set by the mobile (RF loopback). The measurements made on the mobile when in manual test mode are listed below, with information about them.

Bit Error Rate (T1 Test, TCH/7.2 DATA BER)

When the T1 Test mode is selected with TCH/7.2 and loopback on the result of the Bit Error Rate (BER) measurement is displayed. See *T1 Test mode* on page 4-37. While the measurement is being made the SETTLED: % figure increments to reach 100% when the required number of bits have been sampled. See *Mobile Receiver Testing* on page 4-34 and *Rx Measurement Options* in Chapter 2.

Bit Error Rate (TT Loopback BER) Class 0, Class 1 and Class 2

When the TT Loopback BER mode is selected the results of the Bit Error Rate measurements are displayed. While the measurement is being made the SETTLED: % figure increments to reach 100% when the required number of bits have been sampled. See *TT LOOPBACK modes* on page 4-6 *Mobile Receiver Testing* on page 4-34 and *Rx Measurement Options* in Chapter 2.

Residual Bit Error Rate (TT Loopback RBER Class 0, Class 1 and MER)

When the TT Loopback RBER mode is selected the results of the Bit Error Rate measurements are displayed. While the measurement is being made the SETTLED: % figure increments to reach 100% when the required number of bits have been sampled. See *TT LOOPBACK modes* on page 4-6, *Mobile Receiver Testing* on page 4-34 and *Rx Measurement Options* in Chapter 2.

Note

If it is not possible to use TT Test mode or T1 Test mode for measuring mobile BER performance, an approximate receiver sensitivity test using audio SINAD is possible if the mobile's audio output is accessible. See page 4-35

T1 Test mode

[more], [T1 test]

Selecting the T1 Test mode displays the screen shown in Fig. 4-10. In this mode the Test Set generates test signals - known as T1 Test signals - for conformance testing of TETRA mobile radios, as defined in ETSI EN 300 394-1.

Note

To use T1 Test mode on a mobile it may be necessary to perform additional procedures on the mobile and/or the Test Set. You may need to set the Test Set MCC value to 001 and the MNC value to 00001, so that the mobile recognises these special test values. You may need to select channel plan NO PLAN rather than a TETRA channel plan. You may need to enter an access code on the mobile or use an external configuration/control application to place the mobile into T1 Test mode.

Refer to the mobile manufacturer for information on using T1 Test mode and T1 loopback (if supported).

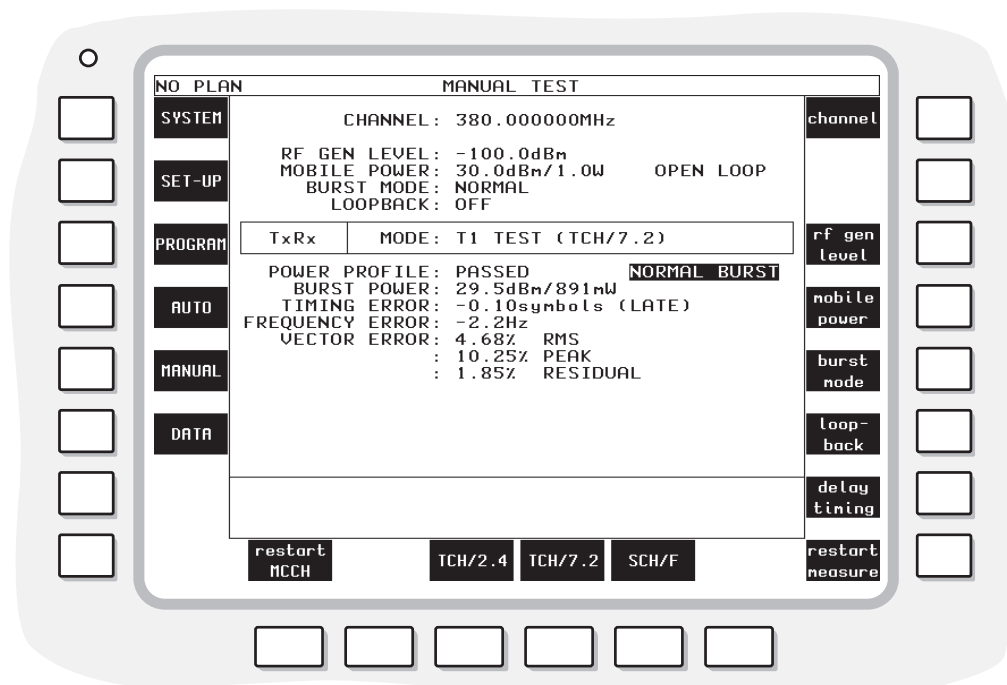


Fig. 4-10 Manual test T1 Test mode

Three T1 signal types are provided:-

TCH/7.2 TCH/2.4 SCH/F

The required signal type is selected by pressing the appropriate soft key at the bottom of the screen. Descriptions of the three signal types are given below, followed by details of the function of the other soft keys on the T1 Test screen.

[TCH/7.2] A T1 type 1 signal is generated. Frames 1 to 17 of the TN1 carry O. 153 PRBS data, coded as TCH/7.2, indicated in BNCH/T T1 burst type. If loopback is on (see below) a Bit Error Rate (BER) test is made on the signal received from the mobile.

[TCH/2.4] A T1 type 4 signal is generated. Frames 1 to 17 of TN1 carry O. 153 PRBS data, coded as TCH/2.4, indicated in BNCH/T T1 burst type.

[SCH/F] A T1 type 2 signal is generated. Frames 1 to 17 of TN1 carry O. 153 PRBS data, coded as SCH/F, indicated in BNCH/T T1 burst type.

These options affect the 'T1 burst type' parameter in the T1 BNCH/T information as shown in the following table.

	T1 Burst Type parameter
TCH/7.2	1 = TCH/7.2
SCH/F	2 = SCH/F
TCH/2.4	4 = TCH/2.4

[restart MCCH] Re-establishes the MCCH mode.

[channel] Controls the frequency of the receiver and generated T1 signal.

Note

If channel plan NO PLAN is selected, the CHANNEL frequency is that of the mobile's transmitter. The Test Set generates the T1 signal at the frequency of the mobile's receiver, which is offset from the mobile transmitter frequency by the value of the RX OFFSET on the SYSTEM PARAMETERS menu. For example, if RX OFFSET is 10 MHz and CHANNEL is 380MHz, the Test Set generates the T1 signal at 390 MHz and receives the MS transmissions at 380 MHz.

[rf gen level] Controls the level of the generated T1 signal.

[mobile power] Controls the way that the Test Set estimates the power that the mobile will transmit which is used to set the input sensitivity of the Test Set. See *Mobile Power* on page 4-8.

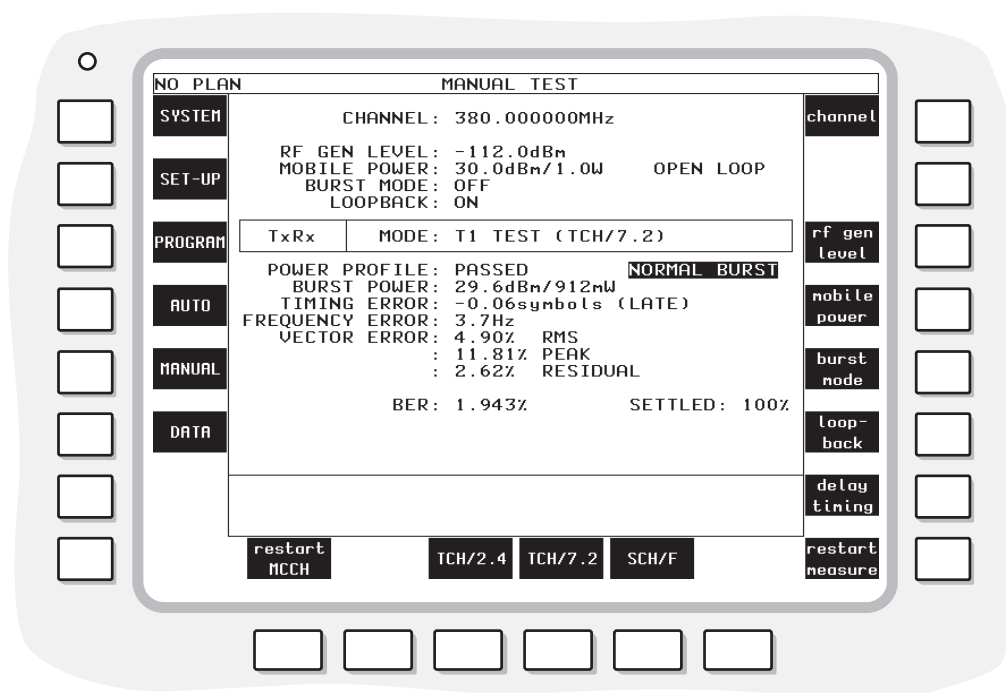
[burst mode] Allows selection of the type of burst generated by the mobile (if supported in the mobile) and measured by the Test Set in T1 Test mode. The options are; OFF, NORMAL, CONTROL. These options affect the Tx ON and Tx burst type parameters in the T1 BNCH/T information (ETSI EN 300 394-1 Annex A.3) as shown in the following table.

BURST MODE	Tx ON parameter	Tx Burst Type parameter
OFF	0 = off	0 = NUB
NORMAL	1 = on	0 = NUB
CONTROL	1 = on	1 = CB

[loopback]

Controls the loopback mode of the mobile in T1 Test mode (if supported in the mobile), which can be ON or OFF. With loopback mode 'ON' the mobile loops back the bursts received from the Test Set, allowing a BER measurement to be made if the TCH/7.2 signal is selected. This option affects the 'Loop back' parameter in the T1 BNCH/T information as shown in the following table.

LOOPBACK	Loop Back parameter
OFF	0 = off
ON	1 = on



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Fig. 4-11 T1 loopback BER Measurement screen

Note

If BER measurement does not start when the mobile transmits loopback data to the Test Set, press the *[channel]* soft key twice or select another screen and return to *[MANUAL]*. If the initial BER reading is very high, press *[restart measure]* to re-synchronize the PRBS sequence.

[delay timing]

Delays the timing of the T1 Test signal by 1 symbol each time it is pressed. This is used in conjunction with the timing error measurement to make the 'Mobile Frame alignment performance' test specified in ETSI EN 300 394-1.

Note

Generation of a faded T1 signal (TU50 or HT200) requires the use of an external propagation simulator.

[restart measure]

Restart the measurement averaging or worst case accumulation process.

This is used to make a number of the tests specified in ETSI EN 300 394-1 where it is required to wait for a period (e.g. 5 seconds) and then perform an average or worst case measurement on a number of bursts (e.g. 200).

TCH test mode

[more], [TCH test]

Selecting the TCH test mode displays the screen shown in Fig. 4-12. In this mode you are able to carry out tests on mobiles that have special test modes, e.g. mobiles that can be set into specific states via the keypad or via a PC. It forces the Test Set to generate a traffic channel frame structure in accordance with the system set-up parameters. It does not perform any signaling to put the mobile on a traffic channel.

The Test Set operates in *open receiver* (unsynchronized) mode and displays measurements for any bursts that it receives from the mobile.

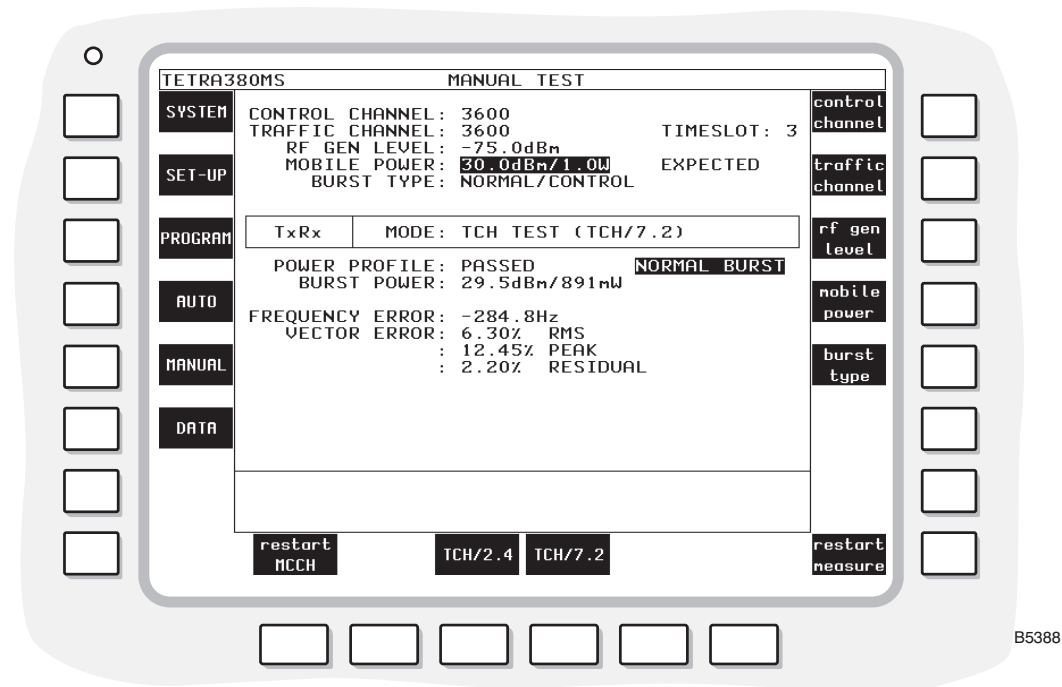


Fig. 4-12 Manual test TCH mode

Two signal types are provided:-

TCH/7.2 TCH/2.4

The required signal type is selected by pressing the appropriate soft key at the bottom of the screen. Descriptions of the signal types are given below, followed by details of the function of the other soft keys on the top level TCH test screen.

[TCH/7.2] A TCH/7.2 signal is generated. Frames 1 to 17 of the selected timeslot carry O. 153 PRBS data, coded as TCH/7.2.

[TCH/2.4] A TCH 2.4 signal is generated. Frames 1 to 17 of the selected timeslot carry O. 153 PRBS data, coded as TCH/2.4.

[restart MCCH] Re-establishes the MCCH mode.

[control channel] Controls the frequency of the receiver and generated MCCH signal.

[traffic channel] Controls the frequency of the receiver and generated TCH signal.

Note

If channel plan NO PLAN is selected, the CHANNEL frequency is that of the mobile's transmitter. The Test Set generates the traffic channel signal at the frequency of the mobile's receiver, which is offset from the mobile transmitter frequency by the value of the RX OFFSET on the SYSTEM PARAMETERS menu. For example, if RX OFFSET is 10 MHz and CHANNEL is 380MHz, the Test Set generates the traffic channel signal at 390 MHz and receives the MS transmissions at 380 MHz.

[rf gen level] Controls the level of the generated TCH signal.

[mobile power] Controls the way that the Test Set estimates the power that the mobile will transmit which is used to set the input sensitivity of the Test Set. See *Mobile Power* on page 4-8.

[burst type] Allows selection of the type of burst measured by the Test Set. The options are; NORMAL, CONTROL, NORMAL/CONTROL.

[Restart measure] Restart the measurement averaging or worst case accumulation process.

TT Test mode

To use this mode the mobile must have the *TT Test protocol* implemented.

Following an ITSI Attach registration, press the *[TT test confirm]* soft key to instruct the mobile to complete an extended (TT Test mode) registration. During this extended registration the mobile sends its TETRA Equipment Identity (TEI), Power Class and Receiver Class. These are then displayed on the manual screen.

During a duplex call the *[BER TT loopbk]* and *[RBER TT loopbk]* soft keys are available. These command the mobile to enter BER or RBER loopback on the speech traffic channel (TCH/S). For further information on TT Test Mode refer to ETSI EN 300 394-1 Annex D.

Note

To use TETRA Test (TT) Test mode on a mobile it may be necessary to perform additional procedures on the mobile and/or the Test Set to enable the use of TT mode on the mobile. You may need to set the Test Set MCC value to 001 and the MNC value to 00001, so that the mobile recognizes these as special test values. You may need to enter an access code on the mobile to enable it to register to the test values of MCC/MNC. You may need to complete ITSI Attach Registration and press *[TT test confirm]* within 30 seconds of entering the access code to prevent the mobile from reverting to normal TETRA operation. You may find that some of the normal functions and display indications are changed or absent while the mobile is in TT Test mode. Refer to the manufacturer for information on using TT Test mode and TT loopback (if supported).

Data displays

To enable the data display mode, press the *[DATA]* soft key at the left side of the screen. (See Chapter 9 *Data displays*).

Chapter 5

TETRA BASE STATION MANUAL TESTING

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Introduction

A typical TETRA Base Station test arrangement is shown below. A two port set-up will be required for base stations that require separate receiver and transmitter connections. See *RF connections* in Chapter 2.

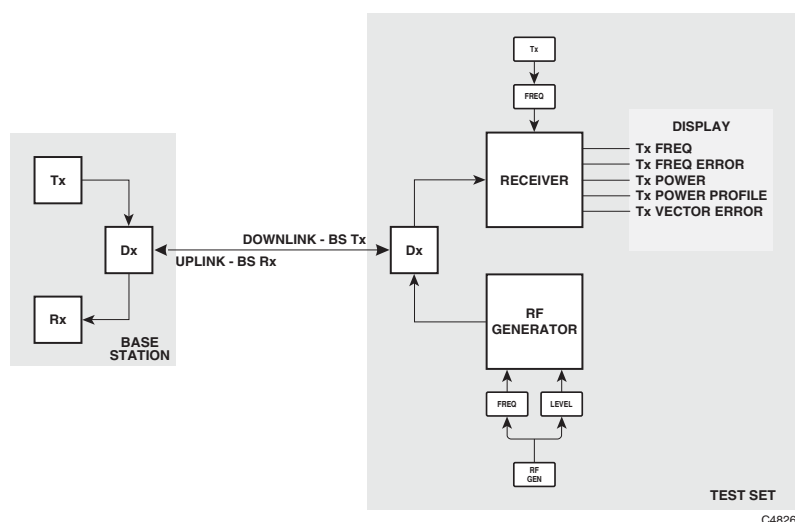


Fig. 5-1 TETRA Base Station test arrangement

Initial Configuration

The TETRA Base Station (BS) system is selected as described under *System selection* in Chapter 3.

The parameters referred to below must be set correctly in the Test Set to allow a base station to be tested.

They are set on the SYSTEM SET-UP SYSTEM PARAMETERS menu, accessed using the following key sequence:-

[SYSTEMS], [SET-UP]

For further information see Chapter 3, *Customizing the system*.

Channel plans

The correct channel plan configuration must be set up such that the Rx and Tx frequencies and parameters of the Test Set correspond to those recognised by the base station. You may need to contact the manufacturer or TETRA network operator supplying the base station to obtain this information.

Alternatively you can set the Test Set to the BS Tx frequency directly by setting the Test Set channel plan to NO PLAN.

Note

If the frequency of the BS transmitter is not known, you can use the Spectrum Analyzer to locate it. See *Spectrum Analyzer* in Chapter 8 *Digital Duplex testing*.

Predefined Channel plans

For a base station which is designed to use one of the standard TETRA channel plans, set the CHANNEL PLAN parameter on the SYSTEM PARAMETERS menu to the corresponding channel plan. To do this, select the [channel plan] soft key, then the appropriate soft key. The Channel Plans are defined in Chapter 11 *Performance data*.

The CHANNEL parameter set on the SYSTEM PARAMETERS menu will be used as the default for the Manual test mode.

User Defined Channel plans

For a base station which does not conform to one of the standard TETRA channel plans supported by the Test Set, precise channel plan information can be set up by setting the CHANNEL PLAN to USER DEFINED on the SYSTEM PARAMETERS menu (see Chapter 11). The parameters for the user defined channel plan can then be set on the USER DEFINED CHANNEL PLAN menu accessed by the [user defined] soft key.

Selecting the [channel block 1] sub-menu allows you to define the relationship between the channel number parameter and the base station's uplink/transmitter frequency. The channel block must be set to INCLUDED for it to take effect. See *Channel Block* in Chapter 3.

For further information see *Setting up a user-defined channel plan* in Chapter 3.

No channel plan

For testing base stations with unknown channel plans but where the frequency of operation is known, the base station Tx FREQUENCY and Rx OFFSET can be set directly on the Test Set by setting the CHANNEL PLAN to NO PLAN. The Tx FREQUENCY and Rx OFFSET set on the SYSTEM PARAMETERS menu will be used as the default for the Manual Test mode.

Note

The Tx FREQUENCY on the Test Set is set to the frequency of the BS transmitter. The T1 signal generated by the Test Set is set to (Tx FREQUENCY – Rx OFFSET).

System Identity

The correct MOBILE COUNTRY CODE, MOBILE NETWORK CODE and BASE COLOUR CODE must be set on the SYSTEM PARAMETERS menu to allow the Test Set to decode the downlink signalling from the base station and to encode the uplink T1 signal to the base station. These values are entered and displayed in *decimal* not *hexadecimal*. You may need to contact the manufacturer or TETRA network operator supplying the base station to obtain this information.

The Test Set can set these parameters automatically. See the note under *Auto Synchronization mode* on page 5-6.

The Test Set can make transmitter measurements on the base station regardless of these parameters.

Selecting manual test modes

You select the Manual test modes using the soft keys at to bottom of the Manual test screen. The MODE field in the middle of the screen indicates the current mode.

MCCH mode

[restart MCCH]

This is the normal mode for testing base stations and is the default mode when the manual test mode is first selected.

In this mode the Test Set is expecting to receive a Main Control Channel signal at a frequency determined by the CHANNEL or Tx FREQUENCY setting.

The MCCH mode screen is shown in Fig. 5-2.

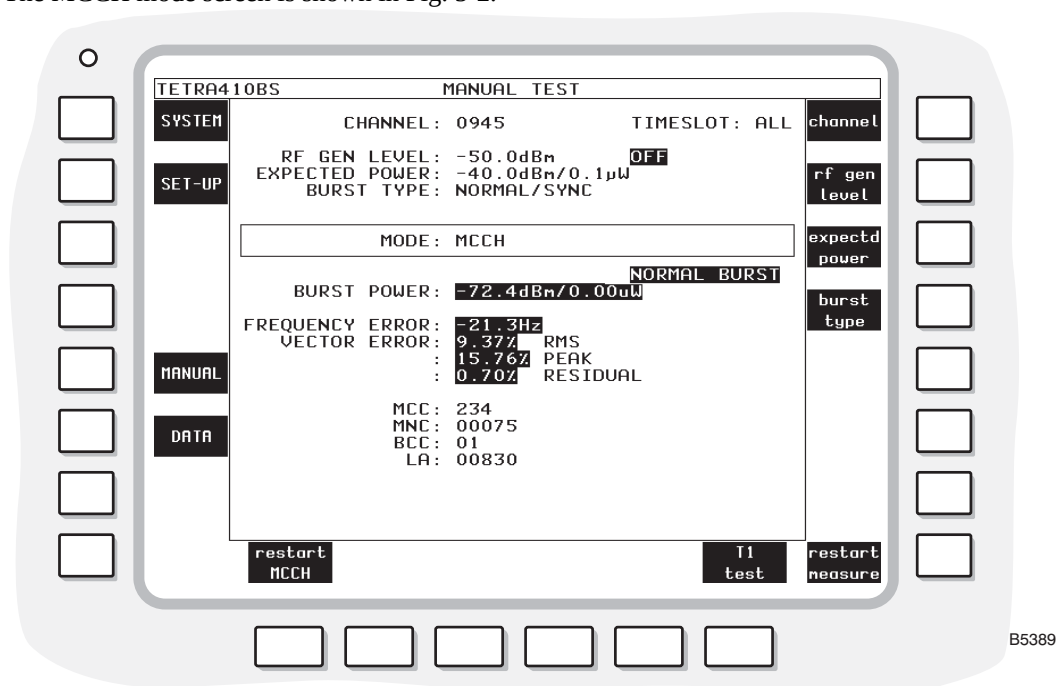


Fig. 5-2 TETRA Base Station, Manual test screen (MCCH mode)

From this screen you can:-

- Change manual test settings to suit your requirements.
- View the results of manual test measurements.
- View the identity information of the base station.
- Select the T1 Test mode.

T1 TEST mode

[T1 test]

In this mode the Test Set is expecting to receive a T1 Test signal from the base station at a frequency determined by the CHANNEL or Tx FREQUENCY setting. When the Test Set has synchronized to the base station frame structure, it generates a T1 signal with a Pseudo Random Binary Sequence (PRBS) signal that allows a Bit Error Rate (BER) test to be carried out on the receiver of the base station under test.

The T1 Test mode is described starting on page 5-6.

Changing Manual test settings

Channel or **Tx Frequency** (if channel plan is set to **NO PLAN** on the **SYSTEM PARAMETERS** menu)

[channel] [Tx freq]

Sets the frequency of the RF receiver within the Test Set. The RF generator is offset from this frequency by the value of RF OFFSET

The CHANNEL can be set to any value permitted by the selected channel plan (specified under *Test parameters* in Chapter 11). The TX FREQUENCY can be set to any frequency in the range 100 MHz to 999.999999 MHz. Initially the CHANNEL or TX FREQUENCY defaults to that set on the SYSTEM PARAMETERS menu. See Chapter 3, *Selecting and customizing the system*.

Timeslot

Press the [channel] or [Tx freq] soft key twice to change the timeslot value.

Sets the timeslot to be received (and measured) by the test.

The TIMESLOT can be set to ALL, 1, 2, 3 or 4. Initially, the TIMESLOT defaults to ALL.

RF generator level

The RF generator is enabled when the T1 Test mode is selected and the Test Set has synchronized to the base station. The Test Set produces a T1 PRBS test signal that is used to perform BER tests on the base station receiver. (See page 5-6.) The default setting is -50 dBm. The RF generator level range depends on the setting of the RF input/output connector selection as shown in the following table:-

RF output	RF input	RF generator level range
N type	N type	-50.0 to -135.0 dBm
N type	TNC	-40.0 to -135.0 dBm
TNC	TNC	-30.0 to -135.0 dBm
TNC	N type	-20.0 to -135.0 dBm

Note

If an RF generator level offset has been set via the HELP AND SET-UP menu it will be included in the RF GEN LEVEL value displayed and may therefore restrict its range. See *RF connections* in Chapter 2.

Expected Power

[expected power]

This field displays the power level that the Test Set estimates the base station will transmit at and is used to set the input sensitivity of the Test Set.

The level is set by pressing the [expected power] soft key, then pressing the [+10dB step] or [-10dB step] soft key as required.

Note

The range over which the expected level can be adjusted is between -40 dBm and 70 dBm but depends on the RF input port selected and any input offset introduced.

Burst type

The BURST TYPE: parameter can be set to:-

- SYNC
- NORMAL/SYNC
- NORMAL TS1/TS2
- NORMAL TS1 ONLY

The Test Set will measure the selected burst type(s) and disregard bursts of the alternative type(s).

If NORMAL/SYNC is selected, all types of burst will be measured. Setting NORMAL TS1 ONLY will exclude any linearisation bursts from the measurements.

Restart measurements

[restart measure]

Pressing this key restarts the measurement averaging or worst case accumulation process. See *Tx Measurement options* in Chapter 2 for information on setting up measurement averaging or worst case accumulation.

Measurements

The measurements made on the transmitted signal from the base station when in Manual Test mode are listed below, with information about them.

The measurements (except power profile) are measured over a number of bursts and the average or worst case result displayed. See *Tx Measurement Options* in Chapter 2.

Burst power

The burst power measurement shows the average power during the measured burst. This measurement is taken over the useful part of the burst (all modulation symbols $SN_0 \sim SN_{max}$) measured at the symbol points through a TETRA filter (Root Nyquist, $\alpha = 0.35$).

Power profile

The POWER PROFILE is displayed only if Base Station Tx mode is set to Discontinuous on the System Parameters menu.

The power profile measurement gives an automatic pass/fail indication which shows whether or not the burst power profile is within the t1 and t3 power masks.

Vector error RMS, peak and residual

The vector error measurement shows the vector error of the received symbols with respect to the ideal symbol points for the burst. This measurement is taken over the useful part of the burst measured at the symbol points through a TETRA filter. The RMS reading is the root mean squared of all the symbols, the peak reading is the vector error of the symbol with the highest error and the residual is the mean residual carrier magnitude.

Frequency error

The frequency error measurement shows the difference between the frequency of the signal transmitted by the BS and the frequency of the Test Set signal analyzer. This measurement is taken over the useful part of the burst measured at the symbol points through a TETRA filter.

Note

The accuracy of the 2968 internal frequency standard is the same as that specified in ETSI EN 300 394-1 for the base station. i.e. 0.2 ppm. In order to make frequency error measurements on a base station, the 2968 should be connected to an external frequency standard of greater accuracy than the base station under test. Base stations may have very low long term frequency error but higher short term fluctuations. Averaging of a number of bursts should be used for measuring long term accuracy; worst case of a number of bursts can be used for measuring short term fluctuations.

Base station identity information

The identity details of the base station, that are transmitted within the base station signal, are displayed on the screen. These are:-

- MCC Mobile Country Code.
- MNC Mobile Network Code.
- BCC Base Color Code.
- LA Location Area.

Note

The Location Area parameter can only be decoded if the Test Set values of MCC, MNC and BCC are set correctly (normally by using AUTO UPDATE).

T1 test mode

The T1 Test mode provides a Pseudo Random Binary Sequence (PRBS) signal (T1 type 7, TCH/7.2 uplink) that allows a Bit Error Rate (BER) test to be carried out on the receiver of the base station under test. In order to do this, the Test Set must be synchronized to the base station frame structure, so that the transmitted signal is sent in the correct timeslot and frames and within the specified alignment limits.

The Test Set has two synchronization modes, AUTO and PULSE. Selection between the two modes is made by setting the SYNC TO BASE STATION parameter in the SET-UP SYSTEM PARAMETERS menu. A description of the two modes is given below and a block diagram is provided in Fig. 2- 3

Auto Synchronization mode

The downlink signal from the base station under test is fed to the RF input of the Test Set. The SYNC bursts from this signal are used to establish the current position in the downlink frame structure. The signal generated by the Test Set is automatically timed to provide correct synchronization with the base station.

Note

If MCC-MNC-BCC UPDATE: AUTO is selected in system parameters, the T1 uplink signal generated by the Test Set will automatically use the MCC-MNC-BCC values captured in the sync burst for scrambling.

The AUTO SYNC PATH DELAY parameter controls the time offset between the downlink frame structure and the T1 uplink signal from the Test Set within the range -45.00 to $+45.00$ bits. Positive values increase the time delay between the downlink frame structure and the Test Set signal. For example a value of 10 bits will delay the Test Set signal by 10 bits relative to the base station downlink frame structure.

Pulse Synchronization mode

A synchronization pulse is fed from the base station by physical connection to the Test Set rear panel Accessory Port, pin 2, with ground return to pin 4. The base station does not need to be transmitting to carry out testing using the PULSE mode. The signal generated by the Test Set is timed to provide correct synchronization with the base station synchronization pulse.

Note

The T1 signal generated by the Test Set must use the same MCC-MNC-BCC values for scrambling as the base station under test. If there is no base station Tx signal, these values must be entered manually in the base station SYSTEM PARAMETERS menu.

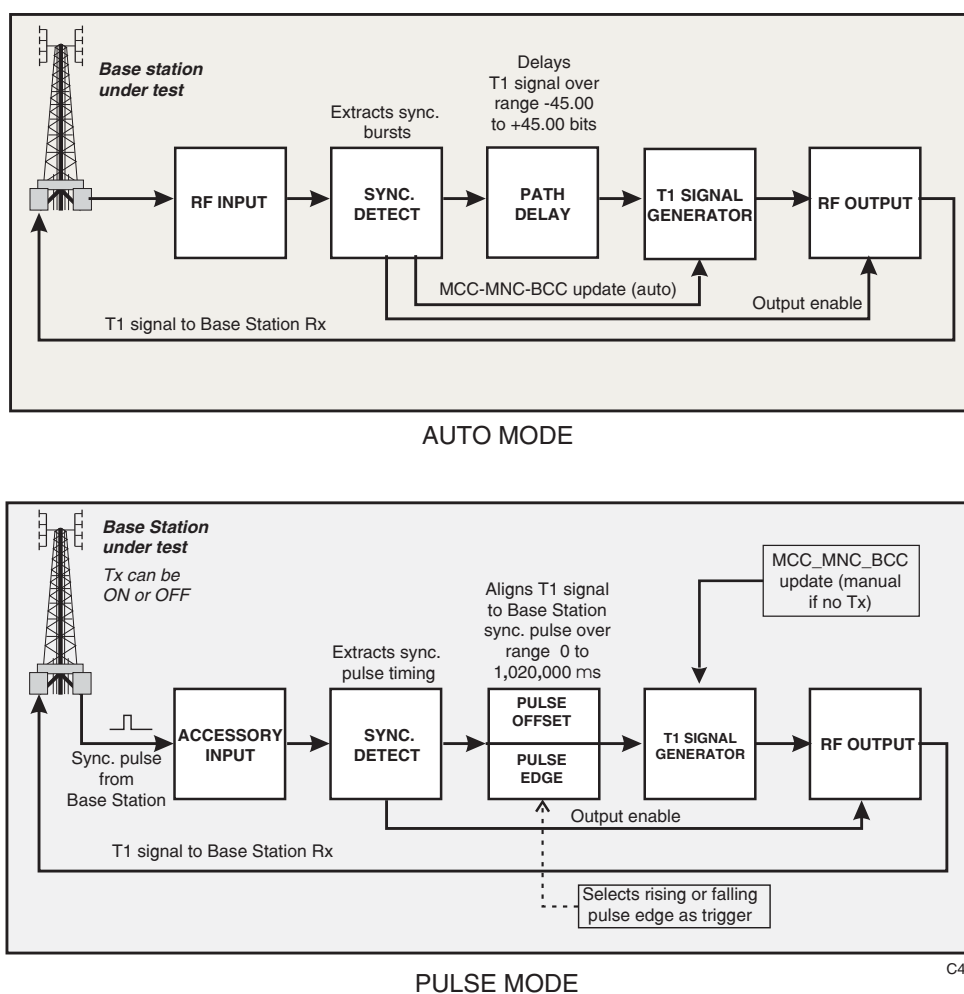


Fig. 2-3 Block diagram of SYNC TO BASE STATION, AUTO and PULSE modes

The SYNC PULSE OFFSET parameter represents the offset between the start of frame 1 timeslot 1 of the base station's downlink frame structure and the synchronization pulse, and may take values within the range 0 to 1,020,000 μ s. For example a value of 10 μ s indicates that the base station's synchronization pulse occurs 10 μ s after the start of frame 1 timeslot 1 of the base station's downlink frame structure.

The DATA keys [kHz / mV] and [MHz / V] can be used to enter values in milliseconds (ms) or seconds (s) respectively, otherwise the entered value is taken as being in microseconds (μ s).

The SYNC PULSE EDGE parameter selects whether timing is referenced to the rising or falling edge of the base station's synchronization pulse.

The Pulse Specification

The signal applied to the input of pin 2 of the accessory connector should conform to the specification shown below.

Logic 1 level	+3.85 to +5.00 V (max) [†]
Logic 0 level	0 (min) [†] to 1.35 V
Pulse width	> 1.0 μ s
Pulse polarity	Rising or Falling
Input resistance	10k Ω $\pm$ 5%

[†] The voltage on pin 2 should at no time exceed +5.0 V or be allowed to go negative.

RF generator control

The RF signal generator is normally off in base station mode to prevent any non-synchronized signals from being transmitted. This is shown by the OFF indicator adjacent to the RF GEN LEVEL setting on the MANUAL TEST screen. To start the T1 transmission, select the T1 Test mode. The RF generator remains off until synchronization is established, at which point it is turned on automatically. The synchronization process can take several seconds.

Deselecting T1 Test mode turns off the signal generator.

Data displays

To enable the data display mode, press the *[DATA]* soft key at the left side of the screen. The use of this facility is described in Chapter 9 *Data displays*.

Off-Air Base Station testing

The Test Set can be used to receive signals off air from a nearby TETRA Base Station.

Connect a suitable antenna to the Test Set TNC RF port, and select TNC input/N type output.

Set the expected power level to -40 dBm.

Set the CHANNEL or TX FREQUENCY to the base station transmitter frequency. (The spectrum analyzer can be used to locate the frequencies of off-air signals.)

Set SYNC TO BASE STATION to AUTO and MCC-MNC-BCC UPDATE to AUTO.

Note

Reception of off-air signals is unlikely to be reliable at signal levels below -75 dBm, or where other strong base station signals are on nearby frequencies.

Chapter 6

TETRA DIRECT MODE MANUAL TESTING

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Introduction

The TETRA Direct Mode test arrangement is shown below.

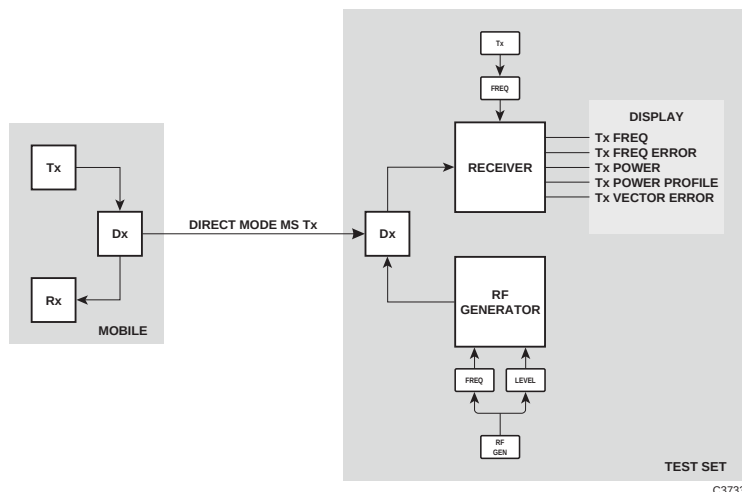


Fig. 6-1 TETRA Direct Mode test arrangement

The TETRA Direct Mode test system allows you to test the operation of TETRA mobile radios operating in the direct (mobile to mobile) mode. The Test Set simulates the receiver of a mobile radio to allow you to test the operation of the transmitter of the mobile radio under test.

The Test Set does not generate an RF signal as only the transmitter of the mobile under test is being tested. The MANUAL TEST screen is shown in Fig. 6-2.

Initial Configuration

The TETRA Direct Mode (DM) system is selected as described under *System selection* in Chapter 2.

The parameters referred to below must be set correctly in the Test Set to allow a mobile to be tested.

They are set on the SYSTEM SET-UP SYSTEM PARAMETERS menu, accessed using the following key sequence:-

[SYSTEMS], [SET-UP]

For further information see *Customizing the system* in Chapter 3.

Channel plans

The correct channel plan configuration must be set up such that the receiver frequency of the Test Set corresponds to the Direct Mode frequency of the mobile. You may need to contact the manufacturer or TETRA network operator supplying the mobile to obtain this information.

Alternatively you can set the frequency directly by setting the Test Set channel plan to NO PLAN.

No channel plan

For testing mobiles with unknown channel plans but where the frequency of operation is known, the mobile Tx FREQUENCY can be set directly by setting the CHANNEL PLAN to NO PLAN. The Tx FREQUENCY set on the SYSTEM PARAMETERS menu will be used as the default for the Manual Test mode.

Note

A mobile that supports Direct Mode will have one or more Direct Mode talk groups programmed in it, which will be configured with particular Direct Mode frequencies. You can locate these frequencies using the Test Set spectrum analyzer. Initially set a wide span to cover the possible frequency range (e.g. REF FREQ 405 MHz span 5 MHz/div will cover the range 380 to 430 MHz) and press the PTT with direct mode selected. Use the expanded spectrum *[max hold]* function to hold the peaks of the bursty signal, and use the *[peak find]* and *[set ref to M1]* functions to home in on the signal. Repeat this process with reduced span widths until the characteristic TETRA modulation signal is centered on a span of 5 kHz/div. Make a note of the REF FREQ, adjusting it if necessary to be a multiple of 6.25 kHz. Repeat this process for other Direct Mode talk groups if required. Return to SYSTEMS mode, which overrides the DUPLEX mode settings, and enter the appropriate REF FREQ value as the TX FREQUENCY for the selected Direct Mode talk group.

Predefined Channel plans

For a mobile which is designed to use one of the standard TETRA channel plans, set the CHANNEL PLAN parameter on the SYSTEM PARAMETERS menu to the corresponding channel plan. To do this, select the *[channel plan]* soft key, then the appropriate soft key. The Channel Plans are defined in Chapter 11, *Performance data*.

The CHANNEL parameter set on the SYSTEM PARAMETERS menu will be used as the default for the Manual test mode.

User Defined Channel plans

For a mobile which does not conform to one of the standard TETRA channel plans supported by the Test Set, precise channel plan information can be set up by setting the CHANNEL PLAN to USER DEFINED on the SYSTEM PARAMETERS menu. The parameters for the user defined system can then be set on the USER DEFINED CHANNEL PLAN menu accessed by the *[user defined]* soft key.

Selecting the *[channel block 1]* sub-menu allows you to define the relationship between the TETRA channel number parameter and the mobile's Direct Mode frequency. The channel block must be set to INCLUDED for it to take effect. See *Channel Block* in Chapter 3.

For further information see *Setting up a user-defined channel plan* in Chapter 3.

Direct Mode Transmit Mode

Direct Mode mobiles in normal mobile-to-mobile operation transmit discontinuously, so the Test Set [DM Tx mode] should be set to DISCONTINUOUS. Do not set this parameter to CONTINUOUS unless you are measuring a continuous signal otherwise measurements will be invalid.

Using the manual test mode

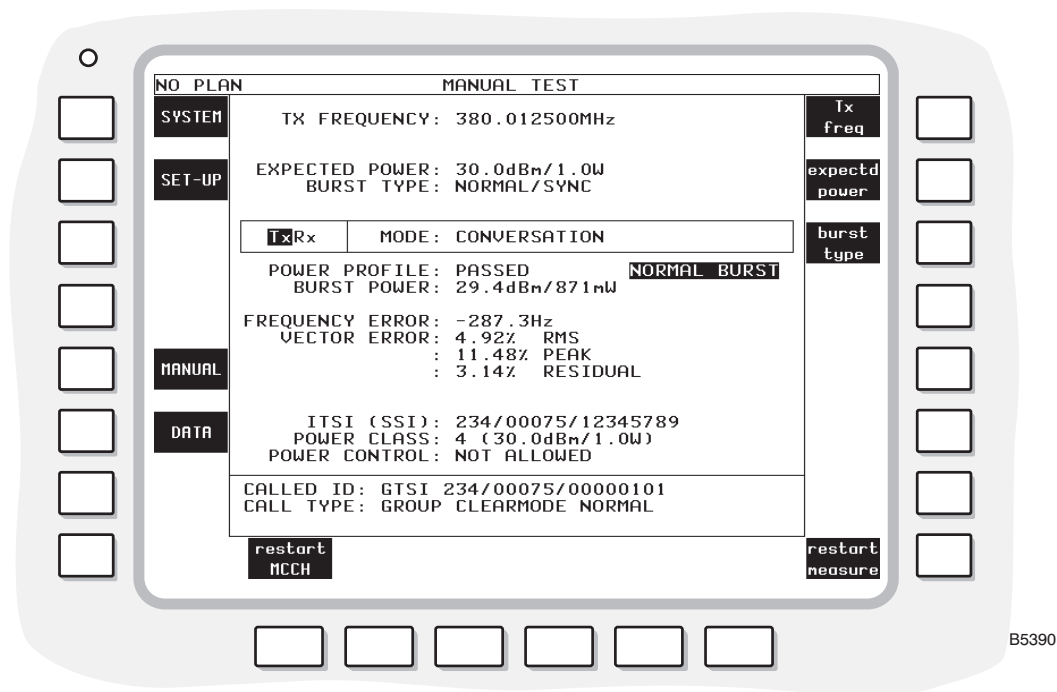


Fig. 6-2 TETRA Direct Mode Manual test screen

From this screen you can:-

- Change manual test settings to suit your requirements.
- View the results of manual test measurements.
- Set up a group call from the mobile to the Test Set
- Cleardown a group call from the mobile
- view the mobiles direct mode parameters
- View the call set-up information

Changing Manual test settings

Channel or **Tx Frequency** (if channel plan is set to **NO PLAN** on the **SYSTEM PARAMETERS** menu)

[channel]

[Tx freq]

Sets the frequency of the RF generator and receiver within the test.

The CHANNEL can be set to any value permitted by the selected channel plan (specified under *Test parameters* in Chapter 11). The TX FREQUENCY can be set to any frequency in the range 100 MHz to 999 MHz. Initially the CHANNEL or TX FREQUENCY defaults to that set on the SYSTEM PARAMETERS menu.

Expected Power

[expected power]

This field displays the power level that the Test Set estimates the mobile will transmit at and is used to set the input sensitivity of the Test Set.

The level is set by pressing the *[expected power]* soft key, then pressing the *[+10 dB step]* or *[-10 dB step]* soft key as required.

Note

The range over which the expected level can be adjusted is between -40 dBm and 70 dBm but depends on the RF input port selected and any input offset introduced.

Burst type

The burst type can be set to:-

- NORMAL
- SYNC
- NORMAL/SYNC

The Test Set will measure the selected burst type and disregard bursts of the alternative type. If NORMAL/SYNC is selected, both types of burst will be measured.

The measurements (except power profile) are measured over a number of bursts and the average or worst case result displayed. See *Tx Measurement Options* in Chapter 2.

Restart measurements

[restart measure]

Pressing this key restarts the measurement averaging or worst case accumulation process. For information on setting up measurement averaging or worst case accumulation, see *Tx Measurement options* in Chapter 2.

Setting up Direct Mode calls from the mobile

Calls may be set up from a Direct Mode mobile to the Test Set (Mobile Originated or MO calls). The Test Set does not support Direct Mode call setup from the Test Set to a Direct Mode mobile (Mobile Terminated or MT calls).

Note

MT call set up can be tested with two similarly configured Direct Mode mobiles. Test each mobile for correct MO call setup and transmitter performance, then use the known good DM transmitter to test the DM reception and MT call setup of the other DM mobile (either off air or using a direct RF connection and an attenuator).

The Test Set is compatible with mobiles conforming to TTR002 DMO TIP version 1.0.3.

Mobile Originated Calls

For mobile originated calls, user operations on the mobile cause signalling to be sent to the Test Set to set up a call.

Mobile originated group call

This call type has the following characteristics:-

- Call is addressed to a group of TETRA Direct Mode users. (point to multi-point).
- Simplex calls only.
- Direct setup only.

The exact procedure will depend on the mobile under test, but normally the following operations are necessary to initiate a group call from the mobile to the Test Set:-

- Switch the mobile to Direct Mode, typically by a menu selection or the mobile may have a programmable key assigned to this function.
- Set the mobile's group selector to the required Direct Mode talk group.
- Set the Test Set Tx Frequency or CHANNEL to the correct frequency for the selected Direct Mode talk group.
- Press and hold the mobile's PTT (Push To Talk) switch.

The Test Set will indicate that a group call has been initiated and that it is in conversation on the Direct Mode channel (MODE: CONVERSATION). The called group ID (GTSI) is shown. The Test Set also indicates the type of group call set up, CLEAR MODE or ENCRYPTED, and the call priority (NORMAL, HIGH, PRE-EMPTIVE or EMERGENCY).

The call is normally cleared down each time the PTT is released. (STATUS: CLEARDOWN COMPLETE).

Note

The Direct Mode MANUAL TEST Screen uses the term "MCCH" common to the TETRA MS Test Mode which is a misnomer for Direct Mode Operation. When no call is in progress, the Test Set indicates MODE: MCCH. This should be taken to mean 'no call activity'. The Test Set does not generate a Main Control Channel (MCCH) or any other signalling when the TETRA Direct Mode system is selected. The soft key *[restart MCCH]*, also common to TETRA MS test mode, does not cause generation of a MCCH but merely serves to clear the call set-up information from the display.

Direct mode mobile information

The following information, obtained during Direct Mode call set-up, is displayed on the manual screen, and provides details of the Direct Mode mobile under test.

ITSI

The Individual TETRA Subscriber Identity, displayed in *decimal* not *hexadecimal*, consisting of the Mobile Country Code (MCC), Mobile Network Code (MNC) and Short Subscriber Identity (SSI).

Note

The MCC, MNC and SSI values may not necessarily be the same as the mobile's normal MCC, MNC and SSI values for trunked mode operation. The SSI value may be a 'PSEUDO SSI' which changes its value on each call setup.

Power class

This defines the mobile's Direct Mode power level.

Note

This may be different from the mobile's normal trunked mode power level. The mobile is not able to indicate the lower power 'L' classes. e.g. a direct mode Power Class 4L (1/2 Watt) is indicated as Class 4 (1 Watt). Some mobiles may not indicate a valid power class value, in which case the Test Set indicates INVALID. The power class information may be INVALID when the mobile clears down the call.

Power control

Indicates whether other Direct Mode mobiles are ALLOWED or NOT ALLOWED to apply open loop power control to reduce the power of their transmissions in response to this call setup.

Note

The power control information may be NOT ALLOWED when the mobile clears down the call.

Measurements

The measurements made on the transmitted signal from the direct mode terminal when in Manual Test mode are listed below, with information about them.

The measurements (except power profile) are measured over a number of bursts and the average or worst case result displayed. See *Tx Measurement Options* Chapter 2.

Burst power

The BURST POWER measurement shows the average power during the measured burst. This measurement is taken over the useful part of the burst (all modulation symbols $SN_0 \sim SN_{max}$) measured at the symbol points through a TETRA filter (Root Nyquist, $\alpha = 0.35$).

Power profile

The POWER PROFILE is displayed only if the Direct Mode Tx Mode is set to Discontinuous on the System Parameters menu.

The power profile measurement gives an automatic pass/fail indication which shows whether or not the burst power profile is within the t1 and t3 power masks.

Vector error RMS, peak and residual

The VECTOR ERROR measurement shows the vector error of the received symbols with respect to the ideal symbol points for the burst. This measurement is taken over the useful part of the burst measured at the symbol points through a TETRA filter. The RMS reading is the root mean squared of all the symbols, the peak reading is the vector error of the symbol with the highest error and the residual is the mean residual carrier magnitude.

Note

If the vector error measurements appear to be very high, check that you have not selected DM Tx mode CONTINUOUS on the System Parameters menu.

Frequency error

The FREQ ERROR measurement shows the difference between the frequency of the signal transmitted by the direct mode mobile and the frequency of the Test Set signal analyzer. This measurement is taken over the useful part of the burst measured at the symbol points through a TETRA filter.

Note

If the FREQ ERROR is greater than ± 1 kHz, the Test Set may not be able to capture and measure the DM signal. Ensure that TX FREQUENCY is set to an integer multiple of 6.25 kHz closest to the centre frequency of the DM signal.

Data displays

To enable the data display mode, press the [DATA] soft key at the left side of the screen. The use of this facility is described in Chapter 9, *Data displays*.

Chapter 7

TETRA MOBILE AUTOMATIC TESTING

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Introduction

The SYSTEM SET-UP, AUTORUN CONTROL menu allows you to select how the Autorun program will respond to the results of tests. For details, see Chapter 3, *Selecting and customizing the system*.

Test results

The results of a test program are displayed and placed in a results store. Five stores are available, each of which you can select by repeatedly pressing the *[next store]* soft key on the AUTORUN screen shown in Fig. 7-2 on page 7-3, which is accessed via the *[AUTO]* soft key as described below. Any results in the current store will be overwritten if you start a new test run, so if you want to keep them you must either select the next store or make a permanent record of them. (See below.)

When all five stores have been used, or you run out of store capacity, you can make a permanent record of results by:-

- Printing out hard copies using the *[print store]* soft key on the AUTORUN screen. Note that this key only prints out the *current* store.
- Saving the results to a memory card, as explained in Chapter 10 *Store and recall facility*.

It is possible for the results to exceed the capacity of the stores, particularly during comprehensive testing. Before this happens, a warning message showing the percentage used appears. To maximize space, make a permanent record of any results that you want to keep, and then clear all stores before starting the test.

All test results in stores are lost when the Test Set is switched off.

Initial configuration

Key parameters must be set correctly in the Test Set to allow a mobile to be tested.

They are set on the SYSTEM SET-UP SYSTEM PARAMETERS menu, accessed using the following key sequence:-

[SYSTEMS], [SET-UP], [System Params]

These parameters and how to set them are described in the *Initial configuration* section of Chapter 4, *TETRA Mobile manual testing*, with further information under *Customizing the system* in Chapter 3.

Selecting an Autorun test program

[PROGRAM]

The [PROGRAM] soft key allows you to select one of three test programs from the PROGRAM menu, shown in Fig. 7-1 on page 2-2. The selected program will then be available when you select Auto test mode (see below).

The first two test programs are supplied with the system; the third option is for a user-defined test program, which can be downloaded into the Test Set via the GPIB or RS232 interface or from a memory card. (See Chapter 10, *Store and recall facility*.) Programming information is provided in the Programming Manual.

Note

You cannot select a user-defined test program until such a program has been downloaded.

The test programs are made up of selected tests, which are described on pages 7-7 to 7-30; the built-in programs are described under *Summary of Autotest programs* starting on page 7-5.

Selecting Auto test mode

[AUTO]

With a mobile connected to the Test Set, and with settings configured to suit the mobile, you can carry out automatic testing.

First, select the program that you want to run using the PROGRAM menu, as described above. For example, if you want to run CALL PROCESSING ONLY, press the [PROGRAM] soft key followed by [call process]. Then bring up the AUTORUN screen by pressing [AUTO]. (See Fig. 7-2.) The selected program is shown at the top of the display.

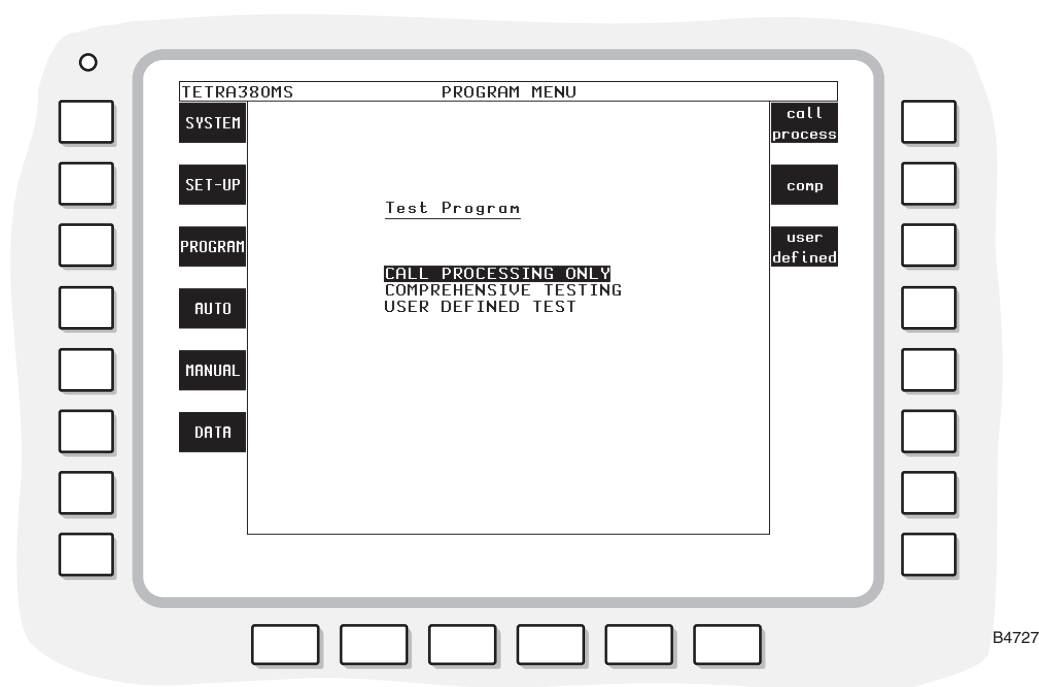


Fig. 7-1 Program menu

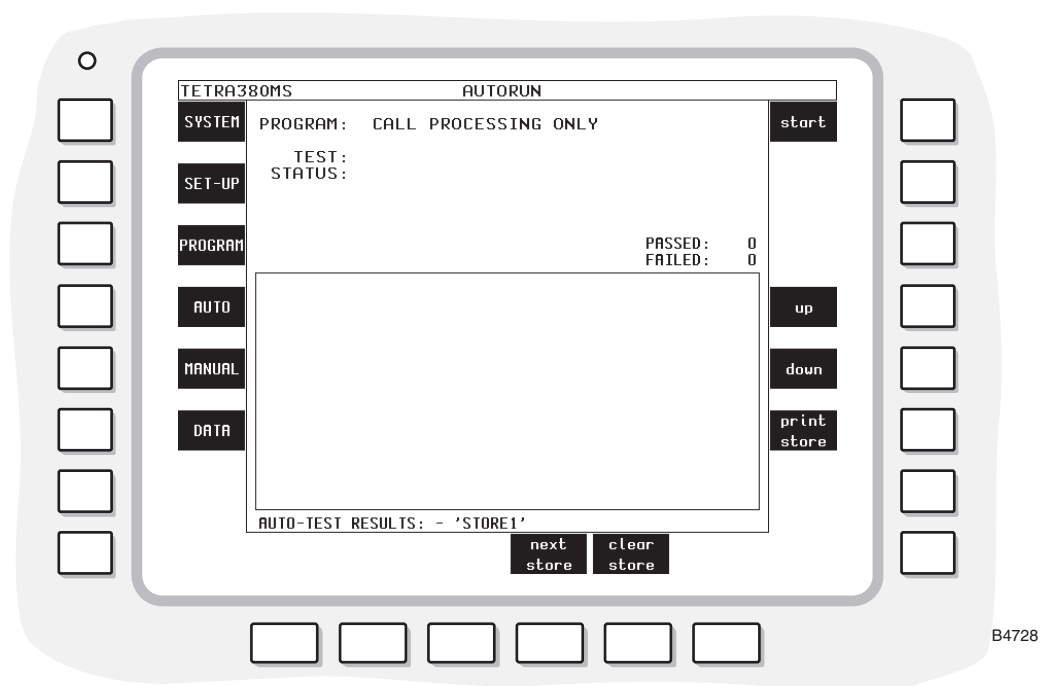


Fig. 7-2 Autorun screen

Running the test program

To run the program, press the *[start]* soft key.

When an Autorun program is running, soft keys *[stop]* and *[pause]* are displayed.

Pressing the *[stop]* soft key stops the program immediately and aborts the current test. The message **★FAIL★ ABORTED BY USER** will be displayed and a *[cont]* soft key will appear. Pressing this key continues the Autorun program, commencing at the next test.

Pressing the *[stop]* soft key rather than the *[cont]* soft key at this point stops the program completely.

Pressing the *[pause]* soft key while a program is running makes the program pause at the end of the current test. The program is then in the same condition as when it is paused via the Pause Mode setting on the AUTORUN CONTROL menu. See Chapter 3, *Selecting and customizing the system*.

If you select one of the standard test modes, such as DUPLEX TEST, by pressing the associated hard key while an automatic test is paused, the mobile remains active and all the settings of the basic test modes relate to the paused system test. This allows measurements to be made on the mobile receiver or transmitter. You can then return directly to the automatic test by pressing the *[SYSTEMS]* hard key. Further information about duplex testing is given in Chapter 9.

In a similar manner, you can use the MANUAL system mode, by pressing the *[MANUAL]* soft key; to return to AUTO mode press the *[AUTO]* soft key.

General information about the automatic tests

This section gives some general information on how to enable or disable particular tests and also how to modify the channels, timeslots, mobile power levels and RF generator levels that are used in the tests by changing parameter values.

Enabling or disabling tests

You can enable or disable each test by setting the test status parameter associated with the test on the SYSTEM SET-UP, AUTORUN PARAMETERS menu to ON or OFF (e.g. the Registration test on page 7-8).

Refer to *Autorun Parameter Settings* in Chapter 11 for details of which tests are disabled by default.

Channels tested

The control channel, traffic channel and timeslot used by the autotests are set on the SYSTEM SET-UP, SYSTEM PARAMETERS menu. See *Customizing the system* in Chapter 3.

Mobile Power Levels for the tests

The **Registration**, **TT Test Mode** and **Call** tests assume the mobile will transmit at its maximum power level during system access. They use the MOBILE POWER FOR ACCESS parameter on the AUTORUN PARAMETERS, TEST CONFIGURATION menu (default 30 dBm) to set the Test Set input attenuator to suit the expected mobile power level.

All other tests except the **Tx Power Level** and **Tx Power Profile** tests assume the mobile will use open loop power control (Link Control) to transmit at a power level determined by the following:

- The ACCESS PARAMETER as set on the SYSTEM PARAMETERS menu (default –45 dBm).
- The MAX TX LEVEL as set on the SYSTEM PARAMETERS menu (default 30 dBm).
- The POWER CLASS of the mobile as set on the MOBILE PARAMETERS menu (default Class 1, 45 dBm Max) or as received from the mobile if it is in the TETRA test mode (TT).
- The RF Generator Level (RF GEN LEVEL) as set on the AUTORUN PARAMETERS, TEST CONFIGURATION menu (default –75 dBm).

The **Tx Power Level** and **Tx Power Profile** tests assume the mobile will use open loop power control to transmit at its minimum and maximum power levels determined by the following:

- The ACCESS PARAMETER as set on the SYSTEM PARAMETERS menu (default –45 dBm).
- The MAX TX LEVEL as set on the SYSTEM PARAMETERS menu (default 30 dBm).
- The POWER CLASS of the mobile as set on the MOBILE PARAMETERS menu (default Class 1, 45 dBm Max) or as received from the mobile if it is in the TETRA test mode (TT).
- The RF generator level set during the test to 5dB above the level required to ensure the mobile transmits at its minimum power and set to 5dB below the level required to ensure the mobile transmits at its maximum power as determined by the above parameters.

At the start of a test if the mobile power level is expected to change (because the RF generator level has changed) the Test Set will wait for a period as set on the AUTORUN PARAMETERS, TEST CONFIGURATION menu (MOBILE POWER SETTling), to allow the mobile's power level to settle.

RF Generator Levels for the tests

All tests except for those tests that specifically change it (**Speech Quality**, **Tx Timing**, **Tx Power Level** and **Tx Power Profile** and the **BER**, **RBER** and **MER** tests) are run with the RF generator level as set on the AUTORUN PARAMETERS, TEST CONFIGURATION menu (RF GEN LEVEL).

The **Speech Quality**, **Tx Timing**, **BER**, **RBER** and **MER** tests are run with an RF generator level as set on the AUTORUN PARAMETERS menu for the specific test (RF GEN LEVEL).

The RF generator level for the **Tx Power Level** and **Tx Power Profile** tests is determined by requirements explained under *Mobile power levels for the tests* starting on page 7-4.

Bursts tested

The **Tx Timing**, **Tx Power Level**, **Tx Frequency**, **Tx RMS Vector Error**, **Tx Peak Vector Error** and **Tx Residual Carrier** tests all have a BURSTS TESTED parameter on the AUTORUN PARAMETERS menu for the test. This parameter specifies the number of bursts on which measurements are to be performed. The returned value is the worst case or average result as specified by the MEASUREMENT TYPE on the AUTORUN PARAMETERS menu for the test.

Types of Calls tested

The autorun program can include up to six call setup and cleardown tests. This enables the functionality and configuration of the mobile to be tested as well as the parametric RF measurements. The parametric tests are performed after the final call type has been set up, so that operations requiring user intervention are completed first, leaving the remainder of the test to run unattended. The final call cleardown test is performed after the parametric tests have been completed, for unattended operation this should be a cleardown from the Test Set.

For each CALL test the following parameters can be configured:-

- TEST: ON or OFF
- ORIGIN: TEST SET or MOBILE
- CALL TYPE: GROUP, PRIVATE, PHONE or EMERGENCY
- TIMEOUT: 1 to 600 s.
- ACCESSORY PORT: LOGIC 0, LOGIC 1, LOGIC 2, LOGIC 3, or OFF.

For mobile terminated (MT) calls (ORIGIN: TESTSET) the Test Set sets up a call of the specified type, according to the parameters for the particular call type defined in the SYSTEM PARAMETERS:CALL TYPES menu.

For Mobile Originated (MO) calls (ORIGIN: MOBILE), the Test Set prompts the user to set up a call of the specified type. If the MOBILE CALL TYPE CHECK parameter is ON, the Test Set will determine whether or not the user has set up the type of call requested.

For automatic testing including simplex calls, you are recommended to set the SYSTEM PARAMETERS: CALL TIMERS: TEST SET TRANSMIT MODE parameter to CONTINUOUS or NONE. Use of TIMED mode in automatic tests may interfere with the END CALL tests.

For each END CALL test the following parameters can be configured:-

- TEST: ON or OFF
- ORIGIN: TEST SET or MOBILE
- TIMEOUT: 1 to 600 s
- ACCESSORY PORT: LOGIC 0, LOGIC 1, LOGIC 2, LOGIC 3, or OFF.

Accessory port, logic lines

The **Call** and **End Call** tests provide the possibility of controlling external equipment via the Test Set accessory port logic lines 0, 1, 2, and 3. If the USE ACCESSORY PORT parameter for a test is set to ON, the associated logic line is raised to +5 V at appropriate points during the test. See *Accessory port* in Chapter 2 of the Test Set 2968 Operating Manual (46882/274) and *Logic lines within the accessory port* in Chapter 3 of the same manual.

Summary of Autotest programs

The built-in test programs are summarized below.

For a more detailed description, refer to the section *Autorun test programs* in Chapter 11. You can control whether or not a test is run during an automatic program by enabling or disabling it.

Call processing only program

This program performs the *Registration* test, or the *TT Test Mode* test depending on which is enabled. See *TT Test mode* in Chapter 4. If the mobile passes the registration test, the ITSI is

displayed. If it passes the TT Test Mode test, then the TEI, the Power Class and the Receiver Class are also displayed.

One or more call setup and cleardown tests will be performed according to the AUTORUN PARAMETERS configuration. For mobile originated CALL tests, you are prompted to place a call of a particular type on the mobile; if the CALL test is passed, the Call Type and Called ID are displayed.

For mobile terminated CALL tests, you are prompted to answer the call if it is a hook signalling call (i.e. the mobile is alerting).

For mobile originated END CALL tests, you are prompted to clear the call from the mobile

If enabled, the speech quality test is performed, during which you are prompted to judge the speech quality by performing a talk-back test.

The test program ends with the final END CALL test that is enabled.

Comprehensive testing program

This program performs all the tests contained in the Call processing only program and also does the following:-

After the *Speech Quality* test, the *Tx Timing* test is performed.

Next the *Power Level* and *Tx Power Profile* tests are carried out at the at the minimum and maximum power levels of the mobile.

Then the *Tx Frequency* test, the *Tx RMS Vector Error* test, the *Tx Peak Vector Error* test and the *Tx Residual Carrier* tests are performed.

Any required Bit Error Rate tests are then carried out. These tests are disabled by default, but can be enabled individually. Those enabled will run in the following order:-

Rx BER Class 0, Rx BER Class 1, Rx BER Class 2,
Rx RBER Class 0, Rx RBER Class 1, Rx MER

The test program ends with the final END CALL test that is enabled.

Autotest *Pause* facility

A feature of the Auto test mode is the pause facility. This allows you to select the conditions under which the test program will pause. The options are:-

- At the end of any test, following a request from you
- Following an individual test that fails
- After every individual test in the program

When in the paused condition, the current parameters and results are displayed. Conversation is maintained between the Test Set and the mobile if applicable.

You can select other operating modes of the Test Set to carry out a detailed analysis of the mobile's performance.

The test program can be resumed at any time.

Automatic Test descriptions

The following descriptions of each test explain the system function that is being tested and the signaling between the Test Set and the mobile under test. The descriptions are arranged in the same order as that in which they appear on the AUTORUN PARAMETERS menu.

The descriptions apply to the tests when performed in the built-in test programs or user-defined programs using the MI BASIC TEST command.

Tests can be omitted from the built-in test programs by switching them off - refer to Autorun Parameters in Chapter 3.

When a test gives a numerical result and the result is outside the test limits, a message of the form ***FAIL*** *nn* is displayed, where *nn* is the test result. Other possible failure messages associated with the tests are explained in the test descriptions.

Registration (Basic: TEST REGISTER)

This test checks the ability of the mobile to Register (ITSI Attach) at Power on.

If the mobile supports an extended "TT Test mode" registration you should enable the TT Test mode test and disable the Registration test.

The message **** POWER ON MOBILE **** prompts you to power up the mobile.

The RF generator level (default -75 dBm) and Mobile power level expected for access (default 30 dBm) are as specified under TEST CONFIGURATION on the AUTORUN PARAMETERS MENU.

If the mobile does not register within the specified Timeout period (default 30 seconds), the test exits.

Following ITSI attach registration, the Test Set waits for the specified period for the mobile to complete its group attachment process.

If a successful registration is completed, the message PASSED is displayed along with the ITSI/SSI and selected group identity (GSSI) if attached.

Failure messages:

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON CONTROL CHANNEL	Test set not generating a control channel at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL NO RESPONSE	Mobile failed to register in the user defined timeout period. (Default 30sec)
FAIL WRONG RESPONSE	Mobile failed to complete registration because it started something else
FAIL ITSI: INVALID	No ITSI received from mobile

TT Test Mode

This test is only available for use in the built-in automatic test programs
It is not available for the MI-BASIC facility.

This test checks the ability of the mobile to perform an extended “TT Test Mode” registration. The main purpose of the TT TEST Mode test is to obtain information about the mobile, such as the ITSI/SSI, TEI, Power Class, and Receiver Class for use in later tests. The TT Test Mode test also enables the use of “TT” RF Loopback for performing the later receiver sensitivity tests (Rx BER, Rx RBER, Rx MER).

You should refer to the mobile manufacturer’s documentation for the method required to place a particular mobile into “TT Test Mode”. You may need to set the System Parameters MCC and MNC parameters to the test values of MCC = 1, MNC = 1.

The “TT Test Mode” may not be supported by some TETRA mobiles. If the mobile does not support “TT Test Mode” registration, you should enable the Registration test and disable the TT Test Mode test.

The message ***** START TEST MODE ***** prompts you to place the mobile into test mode. You should power up the mobile if you haven’t already done so and perform the operation required to place the mobile into test mode.

The RF generator level (default –75 dBm) and Mobile power level expected for access (default 30 dBm) are as specified under TEST CONFIGURATION on the AUTORUN PARAMETERS MENU.

If the mobile does not register within the specified Timeout period (default 30 seconds), the test exits.

Following ITSI Attach registration, the Test Set waits for the specified period for the mobile to complete its group attachment process. Mobiles may refrain from attaching groups when in TT Test Mode.

If a successful registration is completed, the Test Set asks the mobile to confirm that TT Test Mode is set and to supply its TEI, Power Class and Receiver Class.

If a successful TT Test Mode registration is completed, the message PASSED is displayed along with the ITSI/SSI, TEI, Power Class and Receiver Class.

Failure messages:

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON CONTROL CHANNEL	Test set not generating a control channel at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL NO RESPONSE	Mobile failed to register in the user defined timeout period. (Default 30sec)
FAIL WRONG RESPONSE	Mobile failed to complete registration because it started something else
FAIL ITSI: INVALID	No ITSI received from mobile
FAIL NOT IN TEST MODE	Mobile failed to complete extended (TT Test Mode) registration

Call *n*

This test is only available for use in the built-in automatic test programs
It is not available for the MI-BASIC facility.

There are six Call tests, *Call 1* to *Call 6*. These tests check the ability of the mobile to place calls to a Base Station (ORIGIN:MOBILE) or receive calls from a base station (ORIGIN:TEST SET).

The RF generator level (default -75 dBm), and the mobile power level expected for access (default 30 dBm) are as specified under TEST CONFIGURATION on the AUTORUN PARAMETERS menu.

A traffic channel and timeslot are assigned as specified by the TRAFFIC CHANNEL and TIMESLOT parameters on the SYSTEM PARAMETERS menu.

For calls that are ORIGIN:MOBILE, the following actions occur:

You are prompted to place a call of the requested type from the mobile. The message options are:-

*** PLACE GROUP CALL ***

** PLACE PRIVATE CALL **

*** PLACE PHONE CALL ***

* PLACE EMERGENCY CALL *

If the AUTORUN PARAMETERS USE ACCESSORY PORT option is ON (default ON), the selected ACCESSORY PORT logic output (default LOGIC 1) is raised to +5V.

The test times out if a call is not placed within the specified Timeout period (default 30 seconds).

The call can be of any type e.g. group/private/phone/emergency, simplex or duplex; however, the test will fail if the call type does not match that requested and the autorun parameters MOBILE CALLING TYPE CHECK option is ON.

For calls that are ORIGIN:TEST SET, the following actions occur:

The type of call will be as configured on the AUTORUN PARAMETERS menu for the test. The parameters for each call type are as configured on the SYSTEM PARAMETERS menu (CALL TYPES, TRUNKING TYPE, CALL TIMERS). For the final Call Test you are recommended to use the PHONE CALL type (if the mobile supports this type) to avoid the need to hold the mobile's PTT switch during the following parametric tests.

When the mobile is alerting, during a hook set-up call, you are prompted by the message ***** ANSWER CALL *****. If the AUTORUN PARAMETERS USE ACCESSORY PORT option is ON (default ON), the selected ACCESSORY PORT logic output (default LOGIC 3) is raised to +5V.

The test times out if the call is not answered within the specified Timeout period (default 30 seconds).

If the test is successful, the mobile passes into conversation, and the message PASSED is displayed, together with the called ID. The Call Type (Group, Individual, Duplex/Simplex etc.) and the External subscriber Number (ESN) (if available) are also displayed.

Failure messages:

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the Test Configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON CONTROL CHANNEL	Test set not generating a control channel at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL NO RESPONSE	Mobile failed to start call placement in the user defined timeout period (default 30sec)
FAIL CALL ABORTED	Mobile started call placement but it was aborted by user with mobile "end" key
FAIL CALL REJECTED	Call rejected before alerting by the mobile or during alerting by the user
FAIL NO ANSWER	User failed to answer call during alerting within the user defined timeout period (default 30 seconds)
FAIL NOT A GROUP CALL	Mobile originated call type did not match the requested call type.
FAIL NOT A PRIVATE CALL	
FAIL NOT A PHONE CALL	
FAIL NOT AN EMERGENCY CALL	

Called ID formats

PASSED TSI 000/00000/00000000	Call placed with a TETRA Subscriber ID
PASSED SNA 000	Call placed with a Short Number Address
PASSED SSI 00000000	Call placed with a Short Subscriber ID
PASSED GSSI 00000000	Call placed with a Group Short Subscriber ID

Place Call

(Basic: TEST PLACECALL)

This test is only available for the MI-BASIC facility.
It is not available for use in the built-in automatic test programs

This test checks the ability of the mobile to place a call to a Base Station.

The RF generator level (default -75 dBm), and the mobile power level expected for access (default 30 dBm) are as specified under TEST CONFIGURATION on the AUTORUN PARAMETERS menu.

The message ***** PLACE CALL ***** prompts you to place a call from the mobile. If the AUTORUN PARAMETERS USE ACCESSORY PORT option is ON (default ON), the selected ACCESSORY PORT logic output (default LOGIC 1) is raised to +5V.

The test times out if a call is not placed within the specified Timeout period (default 30 seconds).

The call can be of any type e.g. group/private/phone/emergency, simplex or duplex.

A traffic channel and timeslot are assigned as specified by the TRAFFIC CHANNEL and TIMESLOT parameters on the SYSTEM PARAMETERS menu.

If the test is successful, the mobile passes into conversation, and the message PASSED is displayed, together with the called ID. The Call Type (Group, Individual, Duplex/Simplex etc.) and the External subscriber Number (ESN) (if available) are also displayed.

Failure messages:

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the Test Configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON CONTROL CHANNEL	Test set not generating a control channel at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL NO RESPONSE	Mobile failed to start call placement in the user defined timeout period (default 30sec)
FAIL CALL ABORTED	Mobile started call placement but it was aborted by user with mobile "end" key

Called ID formats

PASSED TSI 000/00000/00000000	Call placed with a TETRA Subscriber ID
PASSED SNA 000	Call placed with a Short Number Address
PASSED SSI 00000000	Call placed with a Short Subscriber ID
PASSED GSSI 00000000	Call placed with a Group Short Subscriber ID

End Call *n*

This test is only available for use in the built-in automatic test programs
It is not available for the MI-BASIC facility.

There are six ENDCALL tests, ENDCALL1 to ENDCALL6, corresponding to the six CALL tests. The final ENDCALL test which is enabled is performed after the parametric tests have been completed. Normally this should be ORIGIN:TESTSET to avoid the need for operator intervention at the end of the test program.

These tests check the ability of the mobile to end calls (ORIGIN:MOBILE) or to end calls on an instruction from a base station (ORIGIN:TESTSET).

The RF generator level (default -75 dBm) is as specified under TEST CONFIGURATION on the AUTORUN PARAMETERS menu.

For ORIGIN:MOBILE the message ***** END CALL ***** prompts you to clear down the call from the mobile. If the AUTORUN PARAMETERS USE ACCESSORY PORT option is ON (default ON), the selected ACCESSORY PORT logic output (default LOGIC 2) is raised to +5V.

If the call ends, the result PASSED CLEARED DOWN is displayed and a control channel is generated.

The test times out if the call has not ended within the specified Timeout period (default 30 seconds).

Failure messages:

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the Test Configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set not generating a Traffic Channel at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL NO RESPONSE	Mobile failed to complete clear down in the user defined timeout period (default 30sec)

Clear From Mobile**(Basic: TEST MOBILECLEAR)**

This test is only available for the MI-BASIC facility.
It is not available for use in the built-in automatic test programs

This test checks the ability of the mobile to end a call.

The RF generator level (default –75 dBm) is as specified under TEST CONFIGURATION on the AUTORUN PARAMETERS menu.

The message ***** END CALL ***** prompts you to clear down the call from the mobile. If the AUTORUN PARAMETERS USE ACCESSORY PORT option is ON (default ON), the selected ACCESSORY PORT logic output (default LOGIC 2) is raised to +5V.

If the call ends, the result PASSED CLEARED DOWN is displayed and a control channel is generated.

The test times out if the call has not ended within the specified Timeout period (default 30 seconds).

Failure messages:

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the Test Configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set not generating a Traffic Channel at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL NO RESPONSE	Mobile failed to complete clear down in the user defined timeout period (default 30sec)

Call Mobile**(Basic: TEST CALLMOBILE)**

This test is only available for the MI-BASIC facility.
It is not available for use in the built-in automatic test programs

This test checks the ability of the mobile to receive a call from a Base Station.

The type of call will be as configured on the AUTORUN PARAMETERS menu for the Call Mobile test. You are recommended to use the PHONE CALL type (if the mobile supports this type) to avoid the need to hold the mobile's PTT switch during the following tests.

The RF generator level (default -75 dBm) and the mobile power level expected for access (default 30 dBm) are as specified under TEST CONFIGURATION on the AUTORUN PARAMETERS menu.

A traffic channel and timeslot are assigned as specified by the TRAFFIC CHANNEL and TIMESLOT parameters on the SYSTEM PARAMETERS menu.

When the mobile is alerting, during a hook set-up call, you are prompted by the message ***** ANSWER CALL *****. If the AUTORUN PARAMETERS USE ACCESSORY PORT option is ON (default ON), the selected ACCESSORY PORT logic output (default LOGIC 3) is raised to +5V.

The test times out if the call is not answered within the specified Timeout period (default 30 seconds). If the call is answered, the result PASSED is displayed, together with the called ID, and the mobile passes into conversation. The Call Type (Group, Individual, Duplex/Simplex etc.) is also displayed.

Failure messages:

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON CONTROL CHANNEL	Test set not generating a control channel at the start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL NO RESPONSE	Mobile failed to respond to call setup in 30sec
FAIL CALL REJECTED	Call rejected before alerting by the mobile or during alerting by the user
FAIL NO ANSWER	User failed to answer call during alerting within the user defined timeout period (default 30 seconds)

Clear From Land

(Basic: TEST LANDCLEAR)

This test is only available for the MI-BASIC facility.
It is not available for use in the built-in automatic test programs

This test checks the ending of a call on an instruction from a Base Station.

The RF generator level (default –75 dBm) is as specified under TEST CONFIGURATION on the AUTORUN PARAMETERS MENU.

If the call ends, the result PASSED CLEARED DOWN is displayed and a control channel is generated.

Failure messages:-

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set not generating a Traffic Channel at the start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL NO RESPONSE	Mobile failed to complete cleardown

Speech Quality (Basic: TEST SPEECHQUAL)

This is a subjective test in which you are asked to judge the quality of the complete audio/radio/audio path by talking into the mobile's microphone and listening to the result from the earpiece or speaker (delay approx. 2 seconds).

Some mobiles may not support speech when TT TEST Mode has been selected, in which case this test should be disabled.

The default RF generator level for the test is -75 dBm, but you can change this under the AUTORUN PARAMETERS menu for the test.

You are prompted by the message ****** PRESS PASS/FAIL****** to do a talkback test and accept or reject the result by pressing the *[pass test]* or *[fail test]* soft key.

If a simplex call has been established, use the PTT key on the mobile to speak and listen as required.

If accepted, the result PASSED is displayed, together with the RF generator level for the test.

The test times out if *[pass test]* or *[fail test]* is not pressed within the specified timeout period (default 30 s).

Failure messages:-

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for this test is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set not generating TCH at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL PASS/FAIL NOT SELECTED	User failed to press "pass test" or "fail test" key in user defined timeout (30sec)
FAIL AT -xxx.x dBm	User pressed "fail test" key

Tx Timing Error (Basic: TEST TIMING)

The timing error or frame alignment test measures the symbol timing of the bursts from the Mobile, measured with respect to the signal transmitted by the Test Set.

The Tx timing error (worst case or average as set by the MEASUREMENT TYPE parameter) is measured over the number of bursts specified for this test by the BURSTS TESTED parameter on the AUTORUN PARAMETERS menu for this test (default 5).

The default RF generator level for the test is -75 dBm, but you can change this under the AUTORUN PARAMETERS menu for the test.

If the error is less than the value specified by the ERROR TOLERANCE parameter on the AUTORUN PARAMETERS menu (default 0.25 symbols) the result PASSED is displayed, together with the RF generator level for the test.

Failure messages:-

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set not generating a traffic channel at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL PTT NOT PRESSED	Mobile's PTT key has not been pressed when requested by the Test Set
FAIL NO BURSTS	No bursts received from mobile
FAIL -xxx.xxsym at xxx.x dBm	Failed user defined limits at user defined RF generator level

Power Level (Basic: TEST POWERLEVEL)

The power level test measures the average power during a burst. The measurement is taken over the useful part of the burst measured at the symbol points through a TETRA filter (Root Nyquist, $\alpha = 0.35$).

The power level (maximum or minimum or average as set by the MEASUREMENT TYPE parameter) is measured over the number of bursts specified for this test by the BURSTS TESTED parameter on the AUTORUN PARAMETERS menu for this test (default 5).

The test is repeated at the mobile's minimum and maximum power level assuming that the mobile is using open loop power control. See *Mobile Power Levels for the tests* on page 7-4.

The displayed result shows the power measured and the difference from the nominal level (error). The test passes if the error is less than the upper and lower limits specified on the AUTORUN PARAMETERS menu for this test. There are two sets of limits :-

- HI PWR UPPER ERROR and HI PWR LOWER ERROR for the maximum mobile power as indicated by the Power Class of the mobile (defaults ± 2 dB).
- UPPER ERROR and LOWER ERROR for all other mobile power levels (defaults ± 2.5 dB).

Failure messages:-

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set not generating a traffic channel at the start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL PTT NOT PRESSED	Mobile's PTT key has not been pressed when requested by the Test Set
FAIL NO BURSTS	No bursts received from mobile
FAIL xx.xdBm +xx.xdB	Failed user defined limits

Tx Power Profile (Basic: TEST PROFILE)

This test checks the conformance of the mobile's burst power profile to the TETRA Specification mask limits.

The test is repeated at the mobile's minimum and maximum power level assuming that the mobile is using open loop power control. See *Mobile Power Levels for the tests* on page 7-4.

The test passes if the power profile is within the Specification mask limits at the chosen power level.

Failure messages:-

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set not generating a traffic channel at the start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL PTT NOT PRESSED	Mobile's PTT key has not been pressed when requested by the Test Set
FAIL NO BURSTS	No bursts received from mobile
FAIL MASK AT xxdBm	Profile exceeded the TETRA mask at xxdBm expected mobile power

Tx Frequency (Basic: TEST TXFREQ)

The Tx Frequency test determines the accuracy of the radio frequency transmitted by the mobile on the current traffic channel . The measurement made is the difference between the frequency of the signal received from the mobile and the receiver frequency of the Test Set. This measurement is taken over the useful part of the burst measured at the symbol points through a TETRA filter.

The RF generator level (default –75 dBm) is as specified under TEST CONFIGURATION on the AUTORUN PARAMETERS menu.

The frequency error (worst case or average as set by the MEASUREMENT TYPE parameter) is measured over the number of bursts specified for this test by the BURSTS TESTED parameter on the AUTORUN PARAMETERS menu for this test (default 5).

The test result is the actual frequency and frequency error. The mobile passes if the error is within the specified ERROR TOLERANCE parameter for this test (default 100 Hz).

Failure messages:-

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set not generating a traffic channel at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set “stop” key
FAIL PTT NOT PRESSED	Mobile's PTT key has not been pressed when requested by the Test Set
FAIL NO BURSTS	No bursts received from mobile
FAIL xxxx.xxxxxxMHz -xxxx.xHz	Failed user defined limits

Tx RMS Vector Error (Basic: TEST RMSVECTOR)

This test checks the modulation accuracy of the mobile's transmitter.

The RMS vector error test measures the vector error of the received symbols with respect to the ideal symbol points for the burst. This measurement is taken over the useful part of the burst measured at the symbol points through a TETRA filter. The RMS value is the root mean square of all the symbols.

The RF generator level (default -75 dBm) is as specified under TEST CONFIGURATION on the AUTORUN PARAMETERS menu.

The RMS Vector Error (worst case or average as set by the MEASUREMENT TYPE parameter) is measured over the number of bursts specified for this test by the BURSTS TESTED parameter on the AUTORUN PARAMETERS menu for this test (default 5).

The test passes if the error is below the figure specified by the UPPER Limit parameter for this test (default 10%)

Failure messages:-

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set not generating a traffic channel at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL PTT NOT PRESSED	Mobile's PTT key has not been pressed when requested by the Test Set
FAIL NO BURSTS	No bursts received from mobile
FAIL xx.xx%	Failed user defined limits

Tx Peak Vector Error (Basic: TEST PKVECTOR)

The Peak Vector Error test measures the vector error of the received symbols with respect to the ideal symbol points for the burst. This measurement is taken over the useful part of the burst measured at the symbol points through a TETRA filter. The peak value is the vector error of the symbol with the highest error.

The RF generator level (default –75 dBm) is as specified under TEST CONFIGURATION on the AUTORUN PARAMETERS menu.

The Peak Vector Error (worst case or average as set by the MEASUREMENT TYPE parameter) is measured over the number of bursts specified for this test by the BURSTS TESTED parameter on the AUTORUN PARAMETERS menu for this test (default 5).

The test passes if the error is below the figure specified by the UPPER Limit parameter for this test (default 30%).

Failure messages:-

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set not generating a traffic channel at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL PTT NOT PRESSED	Mobile's PTT key has not been pressed when requested by the Test Set
FAIL NO BURSTS	No bursts received from mobile
FAIL xx.xx%	Failed user defined limits

Tx Residual Carrier (Basic: TEST RESIDUAL)

The Residual Carrier test measures the vector error of the received symbols with respect to the ideal symbol points for the burst. This measurement is taken over the useful part of the burst measured at the symbol points through a TETRA filter. The residual carrier value is the mean residual carrier magnitude.

The RF generator level (default -75 dBm) is as specified under TEST CONFIGURATION on the AUTORUN PARAMETERS menu.

The Residual Carrier error (worst case or average as set by the MEASUREMENT TYPE parameter) is measured over the number of bursts specified for this test by the BURSTS TESTED parameter on the AUTORUN PARAMETERS menu for this test (default 5).

The test passes if the error is below the figure specified by the UPPER Limit parameter for this test (default 5%).

Failure messages:-

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set not generating a traffic channel at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL PTT NOT PRESSED	Mobile's PTT key has not been pressed when requested by the Test Set
FAIL NO BURSTS	No bursts received from mobile
FAIL xx.xx%	Failed user defined limits

Rx BER Class 0**(Basic: TEST RXBERO)**

The character O is used in this MI-BASIC command as numerals are not permitted.

This test checks the mobile receiver's bit error rate for Class 0 bits.

The default RF generator level for the test is –112 dBm, under the AUTORUN PARAMETERS menu for this test.

The mobile is commanded to go into Loopback (BER) mode. (This may fail if the TT loopback protocol is not implemented in the mobile or if TT Test mode has not been enabled.)

The result is averaged over the number of samples specified for this test by the Number of Samples parameter on the AUTORUN PARAMETERS menu (default 15000, equivalent to 9 seconds approx.).

The test passes if the result is less than the upper limits specified on the AUTORUN PARAMETERS menu for this test. There are 3 possible limits which are used, depending on the Receiver Class of the mobile as set on the MOBILE PARAMETERS menu (default Class A) or as received from the mobile if it has completed an Extended (TETRA Test Mode) registration. See *TT Test mode* in Chapter 4.

Failure messages:-

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set not generating a traffic channel at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL NO LOOPBACK	Unable to establish mobile TT Loopback (BER) mode.
FAIL NO BURSTS	No bursts received from mobile
FAIL xx.xxx% AT xxx.x dBm	Failed user-defined limits.

Rx BER Class 1**(Basic: TEST RXBERI)**

The character I is used in this MI-BASIC command as numerals are not permitted.

This test checks the mobile receiver's bit error rate for Class 1 bits.

The default RF generator level for the test is –112 dBm, under the AUTORUN PARAMETERS menu for this test.

The mobile is commanded to go into Loopback (BER) mode (this may fail if the TT loopback protocol is not implemented in the mobile or if TT Test mode has not been enabled.)

The result is averaged over the number of samples specified for this test by the Number of Samples parameter on the AUTORUN PARAMETERS menu (default 15000, equivalent to 8 seconds approx.).

The test passes if the result is less than the upper limits specified on the AUTORUN PARAMETERS menu for this test. There are 3 possible limits which are used, depending on the Receiver Class of the mobile as set on the MOBILE PARAMETERS menu (default Class A) or as received from the mobile if it has completed an Extended (TETRA Test Mode) registration. See *TT Test mode* in Chapter 4.

Failure messages:-

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set is not generating a traffic channel.
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key.
FAIL NO LOOPBACK	Unable to establish mobile TT Loopback (BER) mode.
FAIL NO BURSTS	No bursts received from mobile
FAIL xx.xxx% AT xxx.x dBm	Failed user-defined limits.

Rx BER Class 2 (Basic: TEST RXBERII)

The characters II are used in this MI-BASIC command as numerals are not permitted.

This test checks the mobile receiver's bit error rate for Class 2 bits.

The default RF generator level for the test is –112 dBm, under the AUTORUN PARAMETERS menu for this test.

The mobile is commanded to go into Loopback (BER) mode (this may fail if the TT loopback protocol is not implemented in the mobile or if TT Test mode has not been enabled.).

The result is averaged over the number of samples specified for this test by the Number of Samples parameter on the AUTORUN PARAMETERS menu (default 15000, equivalent to 15 seconds approx.).

The test passes if the result is less than the upper limits specified on the AUTORUN PARAMETERS menu for this test. There are 3 possible limits which are used, depending on the Receiver Class of the mobile as set on the MOBILE PARAMETERS menu (default Class A) or as received from the mobile if it has completed an Extended (TETRA Test Mode) registration. See *TT Test mode* in Chapter 4.

Failure messages:-

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set not generating a traffic channel at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL NO LOOPBACK	Unable to establish mobile TT Loopback (BER) mode.
FAIL NO BURSTS	No bursts received from mobile
FAIL xx.xxx% AT xxx.x dBm	Failed user-defined limits.

Rx RBER Class 0 (Basic: TEST RXRBERO)

The character O is used in this MI-BASIC command as numerals are not permitted.

This test checks the mobile receiver's residual bit error rate for Class 0 bits.

The default RF generator level for the test is –112 dBm, under the AUTORUN PARAMETERS menu for this test.

The mobile is commanded to go into Loopback (RBER) mode (this may fail if the TT loopback protocol is not implemented in the mobile or if TT Test mode has not been enabled.)

The result is averaged over the number of samples specified for this test by the Number of Samples parameter on the AUTORUN PARAMETERS menu (default 15000, equivalent to 9 seconds approx.).

The test passes if the result is less than the upper limits specified on the AUTORUN PARAMETERS menu for this test. There are 3 possible limits which are used, depending on the Receiver Class of the mobile as set on the MOBILE PARAMETERS menu (default Class A) or as received from the mobile if it has completed an Extended (TETRA Test Mode) registration. See *TT Test mode* in Chapter 4.

Failure messages:-

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set not generating a traffic channel at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL NO LOOPBACK	Unable to establish mobile TT Loopback (RBER) mode.
FAIL NO BURSTS	No bursts received from mobile
FAIL xx.xxx% AT xxx.x dBm	Failed user-defined limits.

Rx RBER Class 1 (Basic: TEST RXRBERI)

The character I is used in this MI-BASIC command as numerals are not permitted.

This test checks the mobile receiver's residual bit error rate for Class 1 bits.

The default RF generator level for the test is –112 dBm, under the AUTORUN PARAMETERS menu for this test.

The mobile is commanded to go into Loopback (RBER) mode (this may fail if the TT loopback protocol is not implemented in the mobile or if TT Test mode has not been enabled.)

The result is averaged over the number of samples specified for this test by the Number of Samples parameter on the AUTORUN PARAMETERS menu (default 15000, equivalent to 8 seconds approx.).

The test passes if the result is less than the upper limits specified on the AUTORUN PARAMETERS menu for this test. There are 3 possible limits which are used, depending on the Receiver Class of the mobile as set on the MOBILE PARAMETERS menu (default Class A) or as received from the mobile if it has completed an Extended (TETRA Test Mode) registration. See *TT Test mode* in Chapter 4.

Failure messages:-

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set not generating a traffic channel at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL NO LOOPBACK	Unable to establish mobile TT Loopback (RBER) mode.
FAIL NO BURSTS	No bursts received from mobile
FAIL xx.xxx% AT xxx.x dBm	Failed user-defined limits.

Rx MER (Basic: TEST RXMER)

This test checks the mobile receiver's message error rate.

The default RF generator level for the test is –112 dBm, under the AUTORUN PARAMETERS menu for this test.

The mobile is commanded to go into Loopback (BER) mode (this may fail if the TT loopback protocol is not implemented in the mobile or if TT Test mode has not been enabled.)

The result is averaged over the number of samples specified for this test by the Number of Samples parameter on the AUTORUN PARAMETERS menu (default 300, equivalent to approx. 18 seconds).

The test passes if the result is less than the upper limits specified on the AUTORUN PARAMETERS menu for this test. There are 3 possible limits which are used, depending on the Receiver Class of the mobile as set on the MOBILE PARAMETERS menu (default Class A) or as received from the mobile if it has completed an Extended (TETRA Test Mode) registration. See *TT Test mode* in Chapter 6.

Failure messages:-

FAIL RFGEN LEVEL RANGE ERROR	RF generator level autorun parameter for the test configuration is not within the range allowed for the RF port selected & any offset entered
FAIL NOT ON TRAFFIC CHANNEL	Test set not generating a traffic channel at start of the test
FAIL ABORTED BY USER	Test aborted by user with Test Set "stop" key
FAIL NO LOOPBACK	Unable to establish mobile TT Loopback (RBER) mode.
FAIL NO BURSTS	No bursts received from mobile
FAIL xx.xxx% AT xxx.x dBm	Failed user-defined limits.

Chapter 8

DIGITAL DUPLEX TESTING

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Introduction

When you are operating the Test Set in one of the TETRA systems, you can examine the operation of the mobile or base station under test in greater detail by entering the DIGITAL DUPLEX TEST mode. You access this by pressing the [DUPLEX TEST] hard key. The top level DIGITAL DUPLEX TEST screen (Rx/Tx Test screen) is shown in Fig. 8-1. From this you can select Rx or Tx options that provide enhanced facilities. These are:-

Rx Digital Duplex Test

- Incremental adjustment to the RF generator frequency and level.
- BER barchart max. hold and range hold.

Tx Digital Duplex Test

- Incremental adjustment to the Test Set receiver frequency setting.
- Power, frequency error and vector error barcharts max. hold and range hold.
- Power analyzer displays.
- Spectrum analyzer displays.
- Vector analyzer displays.
- Vector diagram displays.

The use of these facilities is explained later in this chapter.

If, when you select DIGITAL DUPLEX TEST from the SYSTEMS test mode, the Test Set signal generator is already enabled, it remains active. The signal generator and receiver in the Test Set can be set independently to any frequency (in Hz. rather than by channel number).

If the PRBS signal is already enabled to perform BER tests on the receiver of the unit under test using loopback, this will remain active.

The DIGITAL DUPLEX TEST mode is similar in concept to the analog DUPLEX TEST mode. However, the signal generator continues to produce TETRA system modulation, and TETRA transmitter measurements are performed on the TETRA transmissions of the unit under test.

You can return to the automatic or manual SYSTEMS mode at any time by pressing the [SYSTEMS] hard key; any changes to settings that you have made while in DUPLEX TEST mode will be reset to the SYSTEMS mode values.

The DIGITAL DUPLEX TEST mode can be entered at any time from the SYSTEMS mode. In most instances you will re-enter the facility that you last used.

Note

System protocol

You must return to SYSTEMS mode before you can use the system protocol to carry out functions such as PLACE CALL, CLEARDOWN etc.

Tx and Rx defined

When in DIGITAL DUPLEX TEST mode, the term Tx applies to the transmitter section of the Unit Under Test. In TETRA Mobile this is the mobile transmitter; in TETRA Base Station it is the base station transmitter; in TETRA Direct Mode, the transmitter of the mobile under test.

Similarly the term Rx applies to the receiver section of the Unit Under Test. In TETRA Mobile this is the mobile receiver; in TETRA Base Station it is the base station receiver; in TETRA Direct Mode the receiver of the mobile under test.

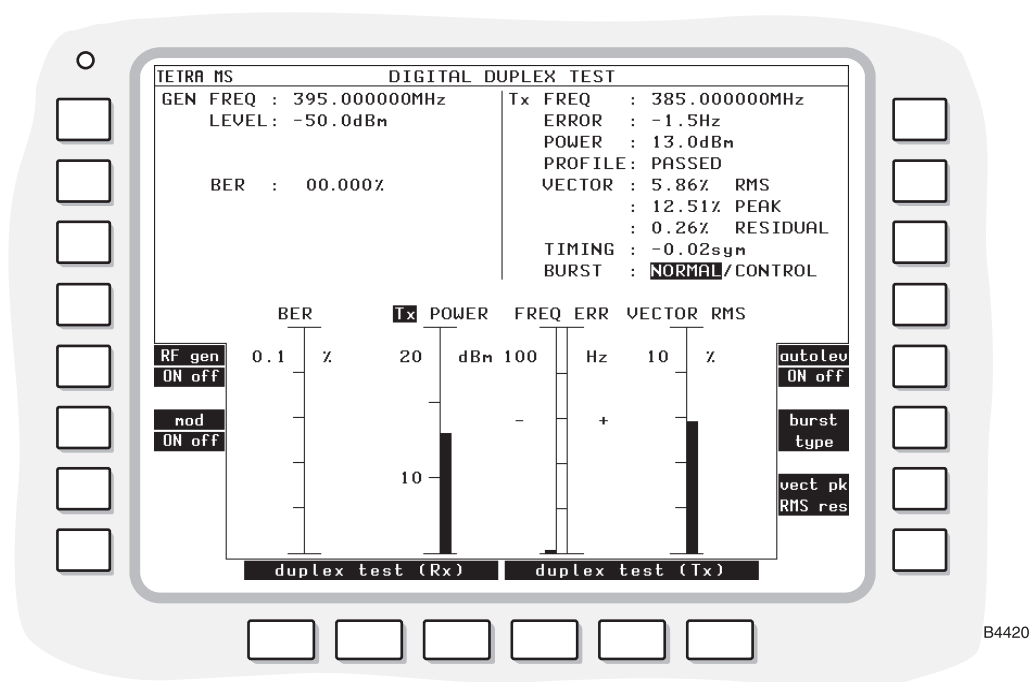


Fig. 8-1 Digital duplex test screen (top-level)

Rx/Tx Test screen

The top-level DIGITAL DUPLEX TEST screen is shown in Fig. 2.1. The settings and measurement results shown are:-

GEN FREQ

The Test Set RF generator output frequency setting.

(Gen) LEVEL

The Test Set RF generator output level setting.

BER (mobile mode only)

The bit error rate (BER) measurement is made on a mobile which is looping back (i.e. re-transmitting) a PRBS signal transmitted by the Test Set, and shows the percentage of bits in the received signal which are in error, i.e. do not match those in the transmitted signal.

Tx FREQ

The Test Set receiver frequency setting.

(Tx) ERROR

The difference between the measured frequency of the transmitter of the unit under test and the Test Set receiver frequency.

(Tx) POWER

The average power during the measured burst. This measurement is taken over the useful part of the burst (all modulation symbols $SN_0 \sim SN_{max}$) measured at the symbol points through a TETRA filter (Root Nyquist, $\alpha = 0.35$).

(Tx) PROFILE

The power profile measurement gives an automatic pass/fail indication which shows whether or not the burst power profile is within the t1 and t3 power masks.

Note

When in Base or Direct Mode the (POWER) PROFILE measurement is displayed only if Tx mode is set to Discontinuous on the System Parameters menu.

(Tx) VECTOR

The vector error measurement shows the vector error of the received symbols with respect to the ideal symbol points for the burst. This measurement is taken over the useful part of the burst measured at the symbol points through a TETRA filter. The RMS reading is the root mean squared of all the symbols, the peak reading is the vector error of the symbol with the highest error and the residual is the mean residual carrier magnitude.

(Tx) TIMING (Mobile only)

The timing error or frame alignment measurement shows the symbol timing of the mobile's bursts measured with respect to the signal from the Test Set.

Some of these measurements are also shown in bar chart form.

Automatic reference level setting

The *[autolev ON off]* soft key enables you to select whether the reference level is set automatically (this is the default) or manually. Note that auto-leveling is switched off if you use the spectrum analyzer *[▼ ref level ▲]* soft key on the Tx test screen. If auto-leveling is switched off, the receiver in the Test Set may not be optimally ranged, so the accuracy of transmitter measurements may be reduced. When this occurs the results are shown in inverse video.

Measured burst type selection

The *[burst type]* soft key allows you to select the type of signal burst that will be measured. The types of burst that can be selected depend on the TETRA system selected, as explained under *Burst type* in TETRA Mobile, TETRA Base Station and TETRA Direct Mode operation.

The burst type options are tabled below:-

TETRA Mobile	TETRA Base Station	TETRA Direct Mode
NORMAL CONTROL NORMAL/CONTROL	NORMAL NORMAL TS1 SYNC NORMAL/SYNC	NORMAL SYNC NORMAL/SYNC

When NORMAL/CONTROL or NORMAL/SYNC are selected, the burst type of the currently measured burst is highlighted.

Vector readings

The results of VECTOR measurements are displayed as RMS, peak and residual values.

You can select which of these values is shown by the VECTOR bar chart. Pressing the *[vect pk RMS res]* soft key will select each in rotation.

Highlighted measurements

Certain measurements may be highlighted in inverse video. For details refer to *Highlighted measurements* in Chapter 2, *Testing mobiles and base stations*.

Rx Test screen

You select the DIGITAL DUPLEX TEST (Rx) screen, by pressing any one of the three [duplex test (Rx)] soft keys at the bottom of the DIGITAL DUPLEX TEST screen or by pressing the [duplex Rx Tx] soft key from the DIGITAL DUPLEX TEST (Tx) screen. Three examples of digital duplex test (Rx) screens are shown in Fig. 8-2, Fig. 8-3 and Fig. 8-4.

If you need to return to the top level DIGITAL DUPLEX TEST display, press the [DUPLEX TEST] key.

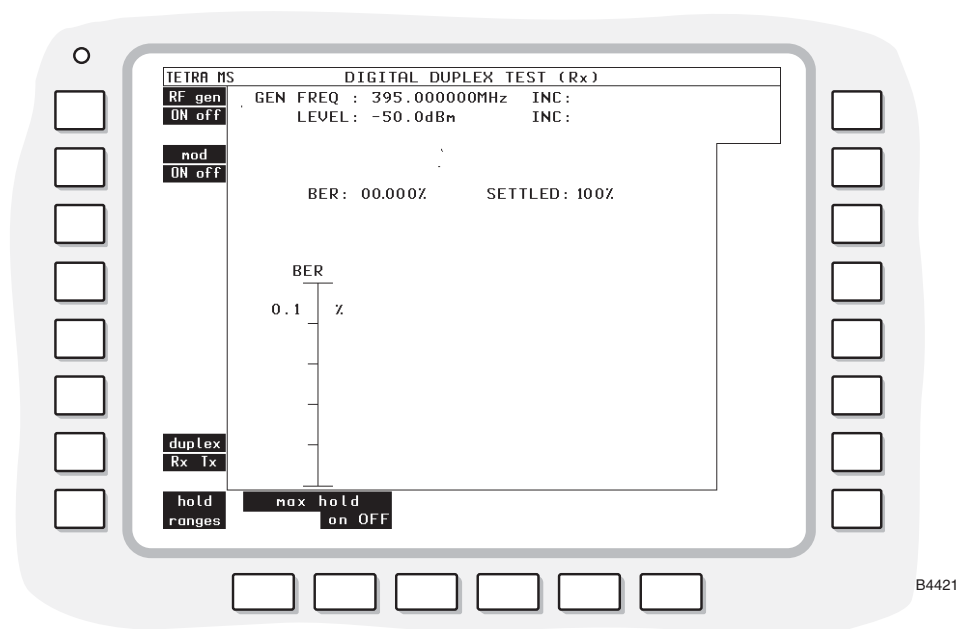


Fig. 8-2 Digital duplex test (Rx) screen (T1 loopback BER)

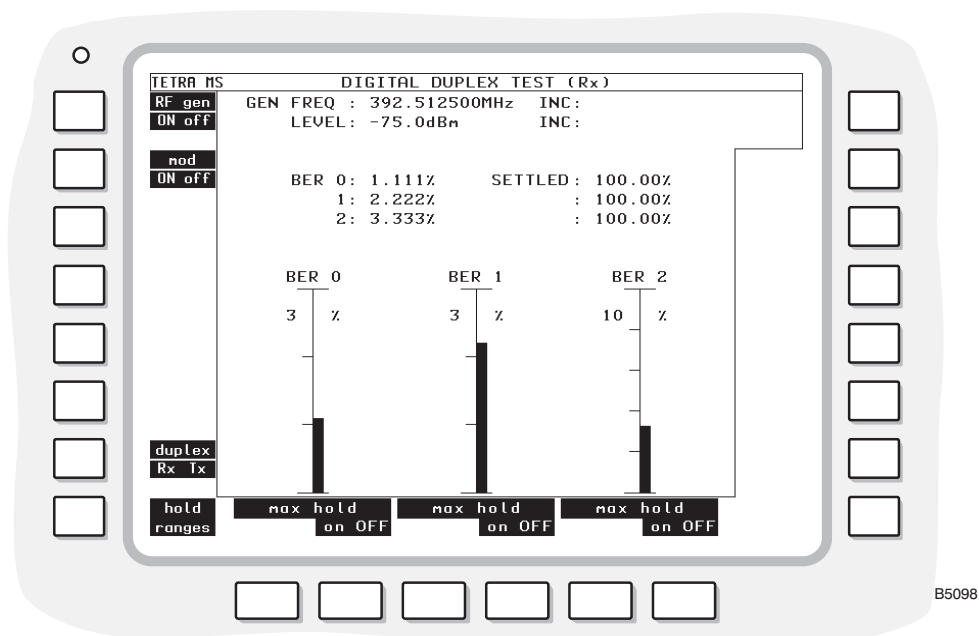


Fig. 8-3 Digital duplex test (Rx) screen (TT loopback BER)

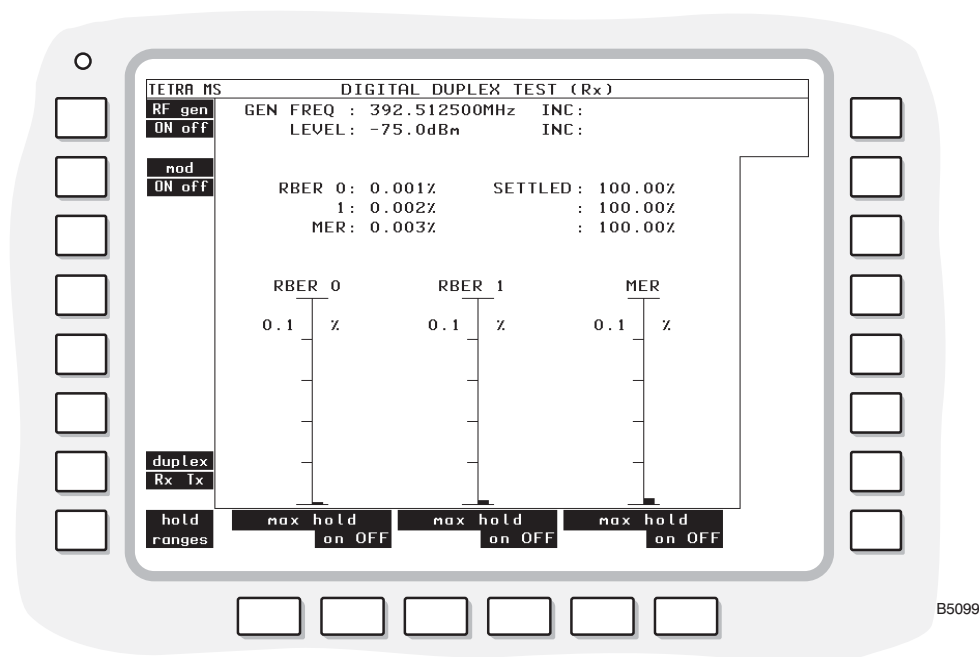


Fig. 8-4 Digital duplex test (Rx) screen (TT loopback RBER and MER)

The generator frequency and level (GEN FREQ, LEVEL) are inherited from the Systems configuration on entry to DIGITAL DUPLEX TEST mode.

You can set these values at will, using the [RF GEN], [FREQ], and [LEVEL] keys, to any value permitted by the normal operating range of the Test Set. As well as entering discrete values via the data keys, you can also use the VARIABLE control or INC (incremental) keys. See *Duplex transmitter testing* and *Duplex receiver testing* in Chapter 3 of the Test Set 2968 Operating Manual (46882/274).

This applies to any of the DIGITAL DUPLEX TEST screens on which these parameters are displayed. The increment values can only be set from the DIGITAL DUPLEX TEST (Rx) or DIGITAL DUPLEX TEST (Tx) screens, but once set can be used in any DIGITAL DUPLEX TEST screen showing Δ or δ . This includes the top level Digital DUPLEX TEST screen.

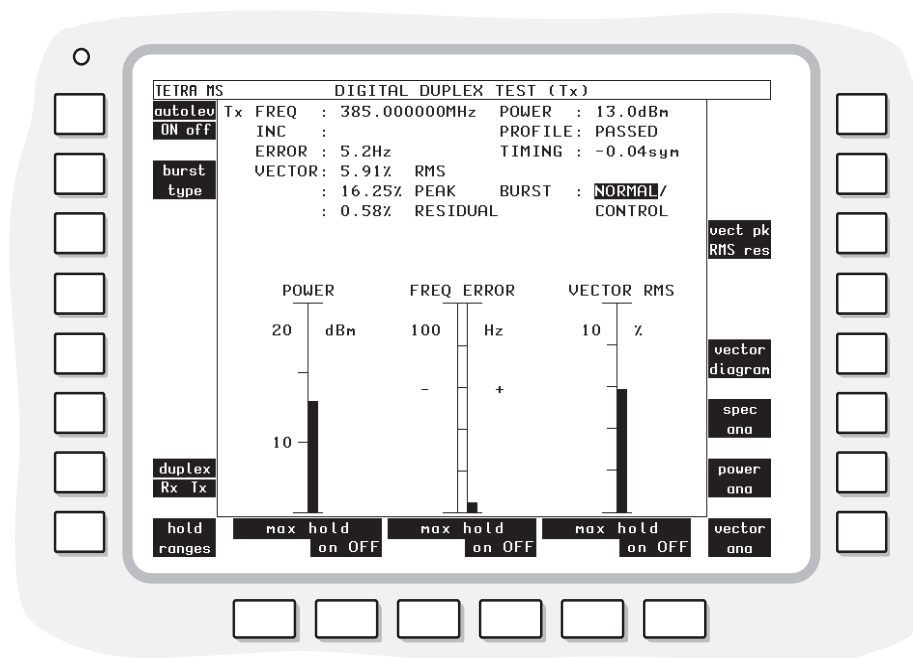
Highlighted measurements

Certain measurements may be highlighted in inverse video. For details refer to *Highlighted measurements* in Chapter 2, *Testing mobiles and base stations*.

Tx Test screen

You select the DIGITAL DUPLEX TEST (Tx) screen, shown in Fig. 8-5 by pressing any one of the three *[duplex test (Tx)]* soft keys at the bottom of the DIGITAL DUPLEX TEST screen or by pressing the *[duplex Rx Tx]* soft key from the DIGITAL DUPLEX TEST (Rx) screen.

If you need to return to the top level DIGITAL DUPLEX TEST display, press the *[DUPLEX TEST]* hard key.



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Fig. 8-5 Digital duplex test (Tx) screen

The DIGITAL DUPLEX TEST (Tx) screen provides access to measurement bar charts, power analyzer, spectrum analyzer, vector analyzer and vector diagram displays.

Highlighted measurements

Certain measurements may be highlighted in inverse video. For details refer to *Highlighted measurements* in Chapter 2, *Testing mobiles and base stations*.

Bar charts

Bar charts on the Tx screen give analogue indications of power, frequency error, and vector error measurements. The *[vect pk RMS res]* soft key selects the RMS or peak vector error bar chart or the residual carrier power bar chart. The bar charts have a *max hold* facility controlled via the *[max hold]* soft keys at the bottom of the screen. When the *max hold* is active on a particular bar chart, the corresponding digital readout is also in *max hold*. See *Maximum hold* in Chapter 3 of the Test Set 2968 Operating Manual (46882/274).

Spectrum analyzer

The *[spec ana]* soft key selects the spectrum analyzer (Fig. 8-6 on page 8-8).

If you are displaying a discontinuous signal (e.g. in TETRA Mobile mode), this may have an intermittent appearance on display because of the TDMA operation during the spectrum analyzer sweep. The use of the *max hold* function described below will allow a contiguous display to be produced.

Selecting the expanded (full screen) spectrum analyzer (by pressing the *[expand on OFF]* soft key) gives you access to the max hold function. Pressing the *[maxhold on OFF]* soft key captures the peak spectrum during active transmissions. This builds up during a number of sweeps to give a continuous display. (Fig. 8-7 on page 8-8.)

The expanded display also gives access to the trace markers.

See *Transmitter testing;; non-expanded: Spectrum analyzer* and *Transmitter testing;; Expanded: Spectrum analyzer* in Chapter 3 of the Test Set 2968 Operating Manual (46882/274).

These sections describe the uses of the spectrum analyzer facility and its operating controls.

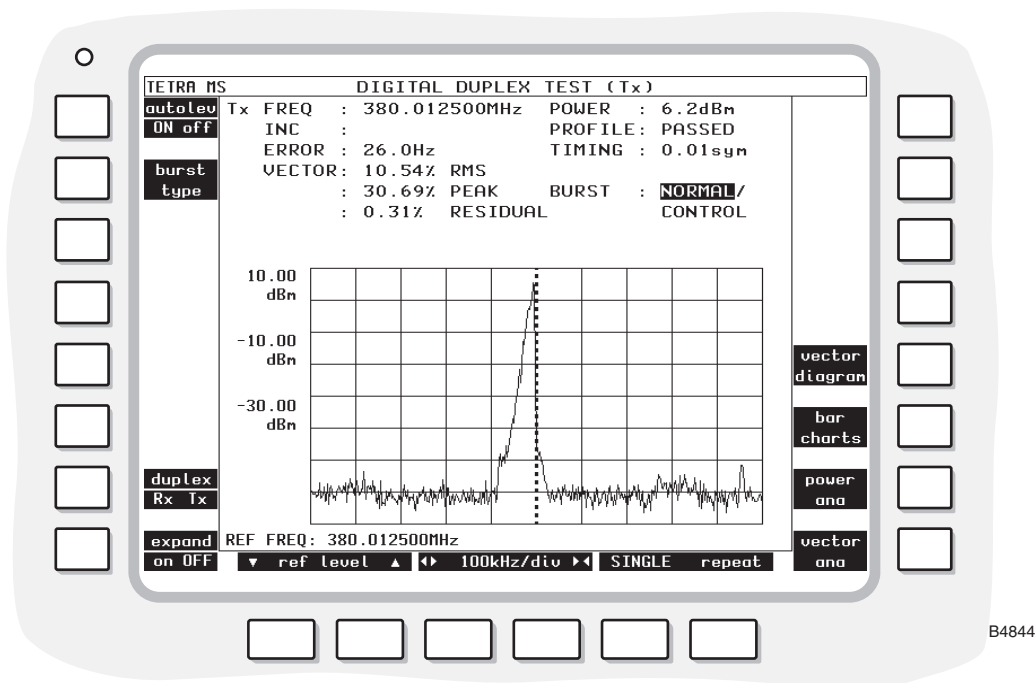


Fig. 8-6 Spectrum analyzer

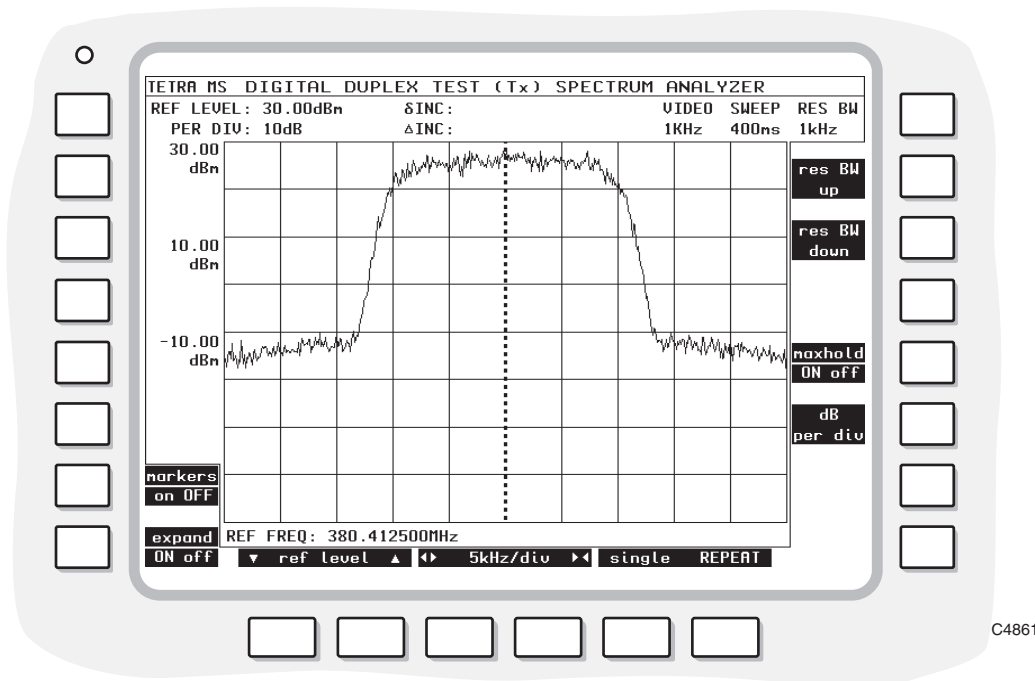


Fig. 8-7 Spectrum analyzer - expanded, with max. hold ON

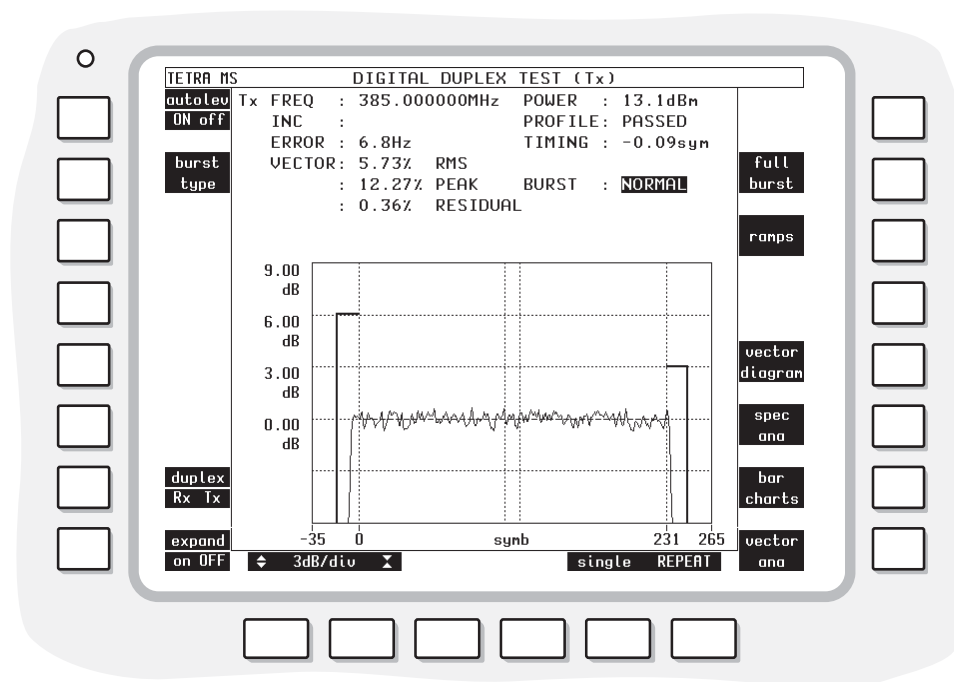
Power analyzer

Use the *[power ana]* soft key to select the power analyzer (Fig. 8-8). Two additional soft keys appear; *[full burst]* and *[ramps]*.

Selecting *[full burst]* displays the entire burst profile including the rising and falling edges (Fig. 8-8). The vertical resolution can be set to 3 dB/division or 10 dB/division using the *[↕]* and *[↗]* soft keys.

To avoid large variations due to modulation, the full burst and the ramp displays show power only at the symbol points.

Note that all the power analyzer examples in this chapter are for normal bursts but other burst types can also be analyzed.



B4404

Fig. 8-8 Power analyzer (full burst)

Fig. 8-9 shows the full burst in expanded mode.

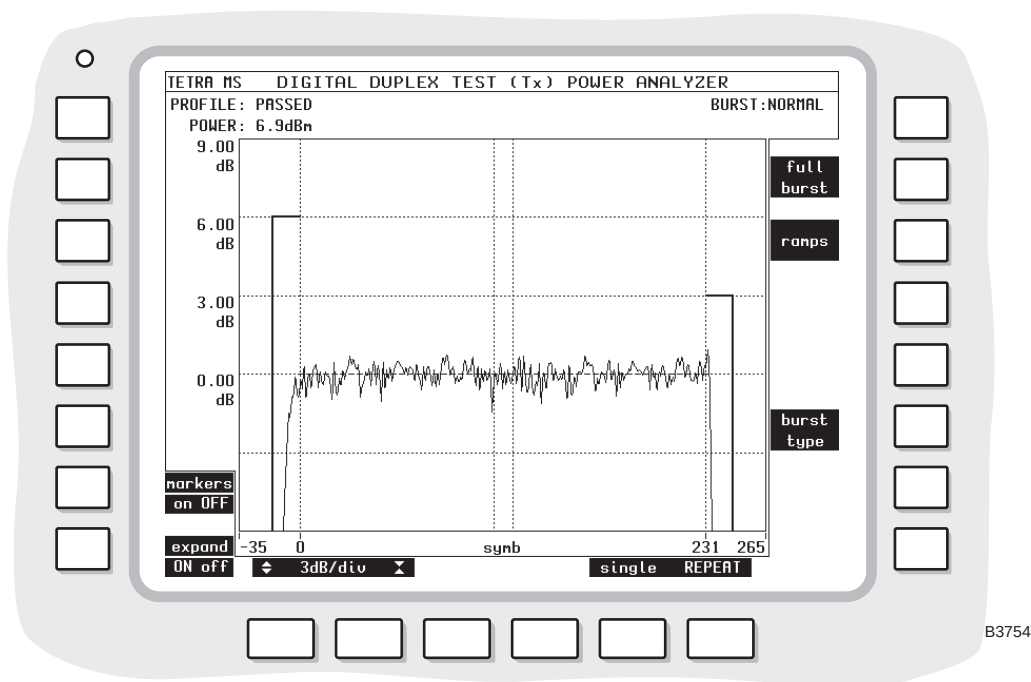


Fig. 8-9 Power analyzer (full burst) - expanded

Selecting *[ramps]* displays only the rising and falling edges of the burst, with greater horizontal resolution than in the full burst display.

The vertical resolution can be set to 3 dB/division or 10 dB/division using the *[◆]* and *[▼]* soft keys.

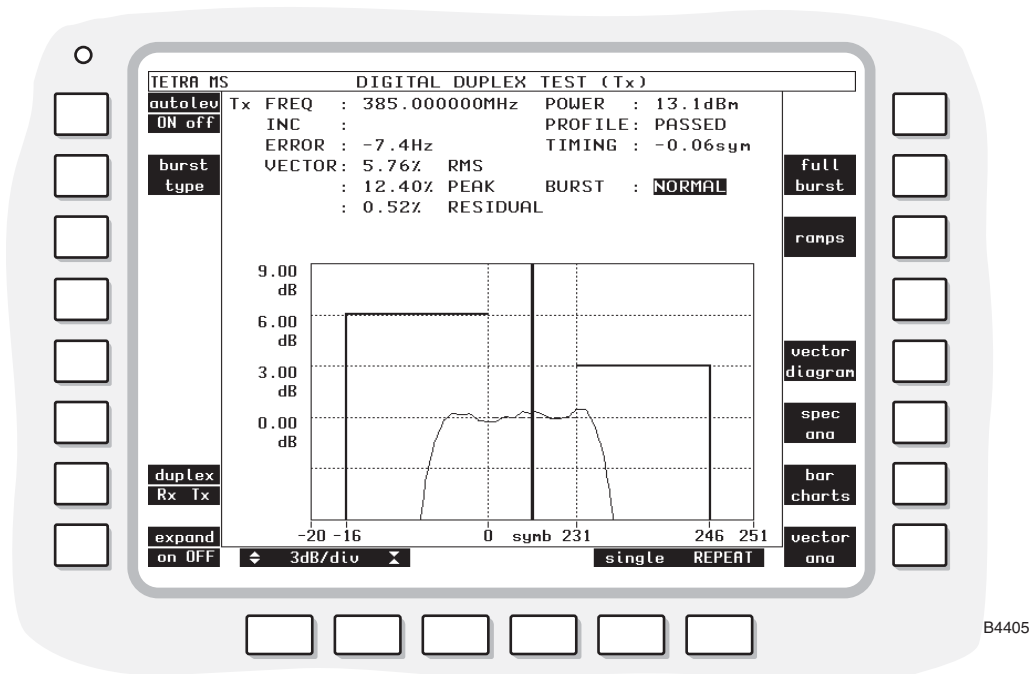


Fig. 8-10 Power analyzer (ramps)

The *[expand on OFF]* soft key provides access to a full screen display (Fig. 8-11 on page 8-11).

The condition of the displayed burst is shown as **PASSED** or **FAILED** against the **PROFILE**: indicator at the top left of the display. Below this the **POWER** indicator shows the RF level.

Markers can be selected, with the readings from them shown at the top of the display (Fig. 8-12). Intelligent markers accommodate the missing section of the burst when performing 'delta marker' measurements.

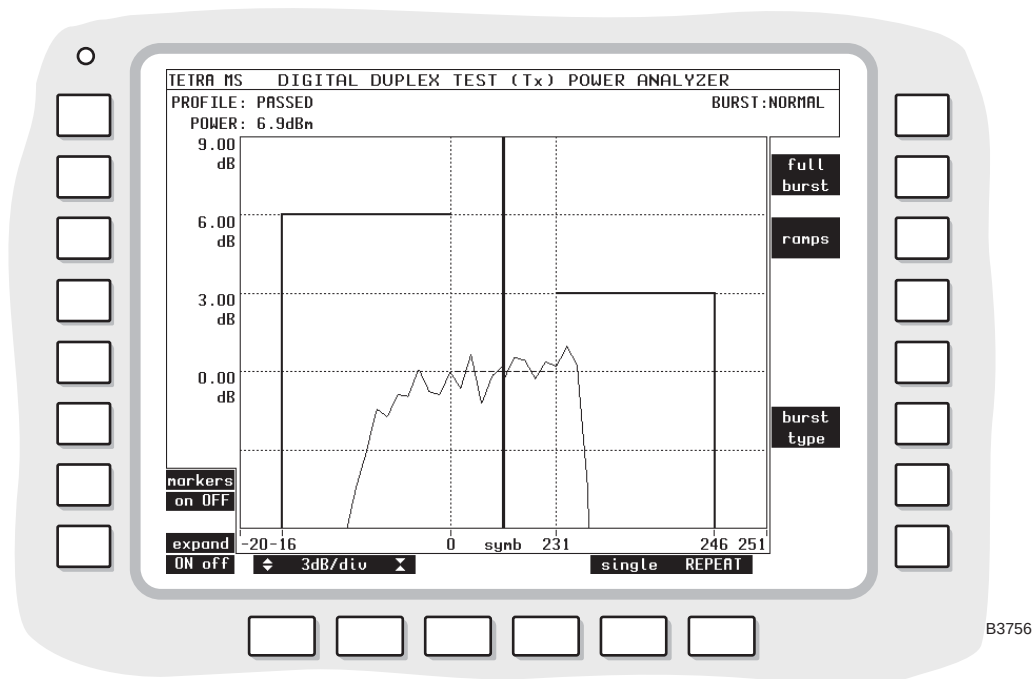


Fig. 8-11 Power analyzer (ramps) - expanded

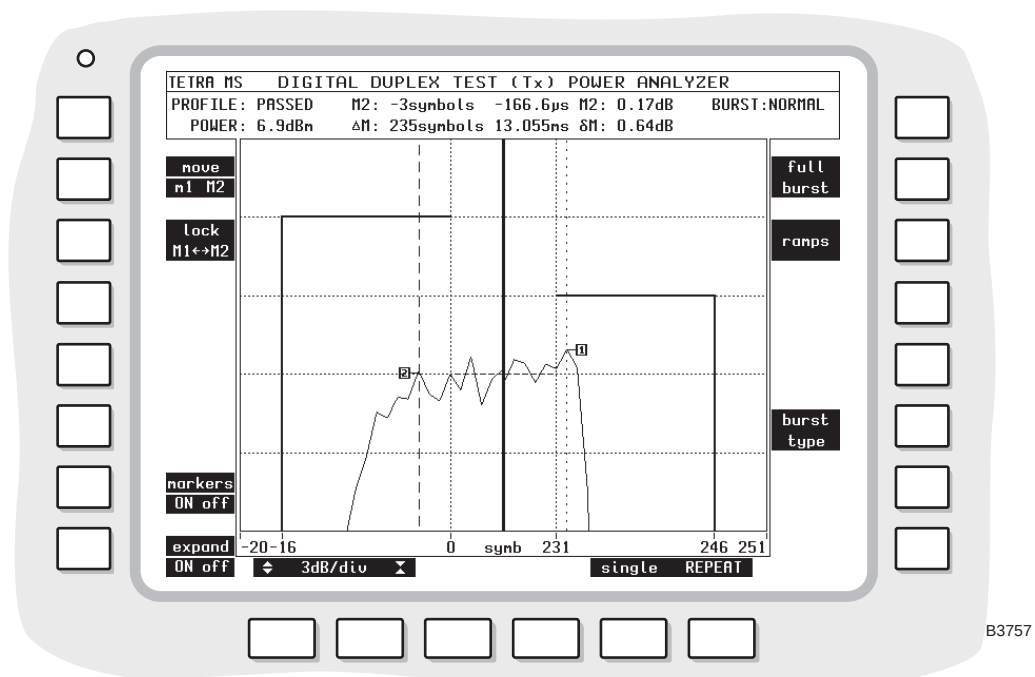


Fig. 8-12 Power analyzer (ramps) - expanded with markers

Vector analyzer

The vector analyzer facility gives displays of the vector error of each symbol in the burst against time, similar to the power profile displays, to allow vector error values for individual symbols to be examined using markers. There are three versions of this display:-

- The error vector for each symbol *against time*.
- The magnitude component of the error vector for each symbol *against time*.
- The phase component of the error vector for each symbol *against time*.

You can access the vector analyzer facility from any of the normal (not expanded) DIGITAL DUPLEX TEST (Tx) screens.

Use the *[vector ana]* soft key to select the vector analyzer. The last used vector analyzer screen is displayed and three additional soft keys appear; *[vector error]*, *[magnitude error]* and *[phase error]*. The soft key relating to the current selection is highlighted.

Examples of displays of each version of the vector analyzer displays are shown below in Figs. 8-13 to 8-17 starting on page 8-13.

Burst types

The *[burst type]* soft key allows you to select the type of burst that will be captured. Pressing this key will step through the available burst types.

The options available for each system are shown below.

TETRA Mobile	TETRA Base Station	TETRA Direct Mode
NORMAL CONTROL	NORMAL NORMAL TS1 SYNC	NORMAL SYNC

When the vector analyzer is accessed from the bar charts display, the burst type selected there will be carried over. However, if NORMAL/SYNC or NORMAL/CONTROL was selected, the burst type set on the vector analyzer will be NORMAL.

Expanded displays

The *[expand on OFF]* soft key provides access to a full screen display.

Markers are available with the expanded display of all vector analyzer versions.

Single or Repeat display

The *[Single]* and *[Repeat]* soft keys select display of a single burst which you can update when required or continual updating from each burst of the selected type.

Vertical resolution

The vertical resolution of the vector analyzer displays can be selected by use of the *[♦]* and *[✕]* soft keys. The vertical resolution range applicable is shown for each display version.

Vector error

The vertical resolution can be set to 5, 10 or 20 %/division giving full scale ranges of 20, 40 or 80 %.

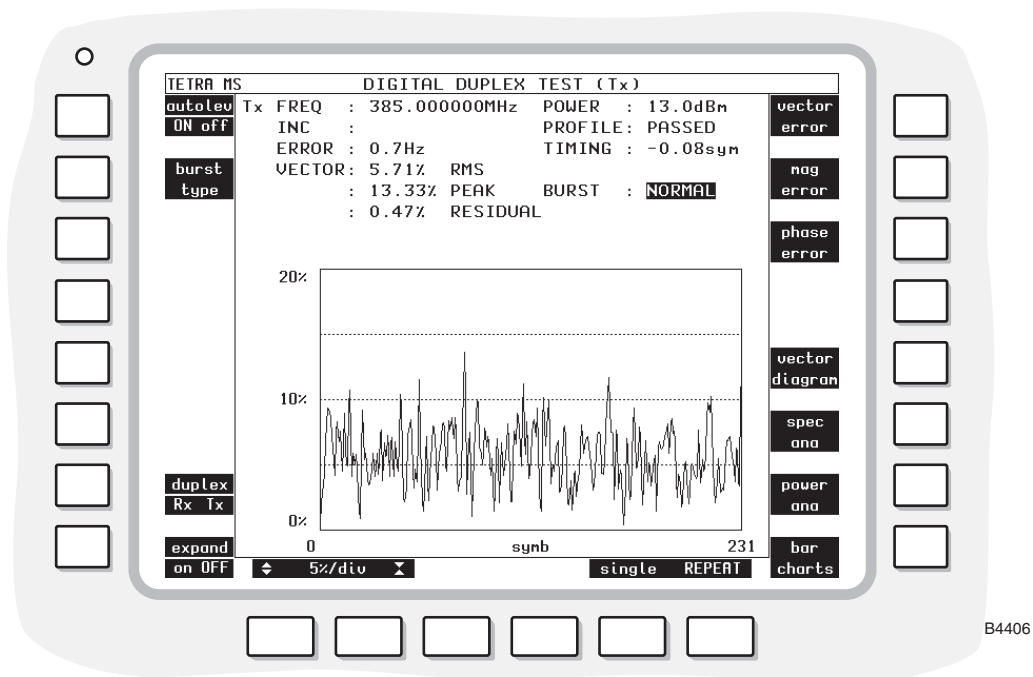


Fig. 8-13 Vector analyzer (vector error) display

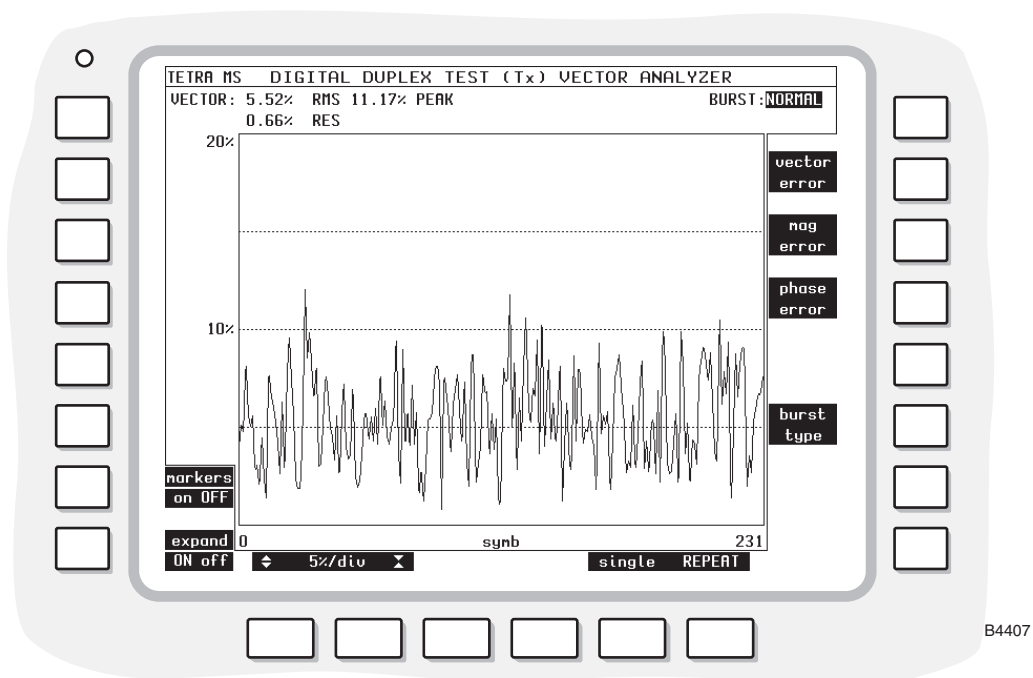


Fig. 8-14 Vector analyzer (vector error) - expanded

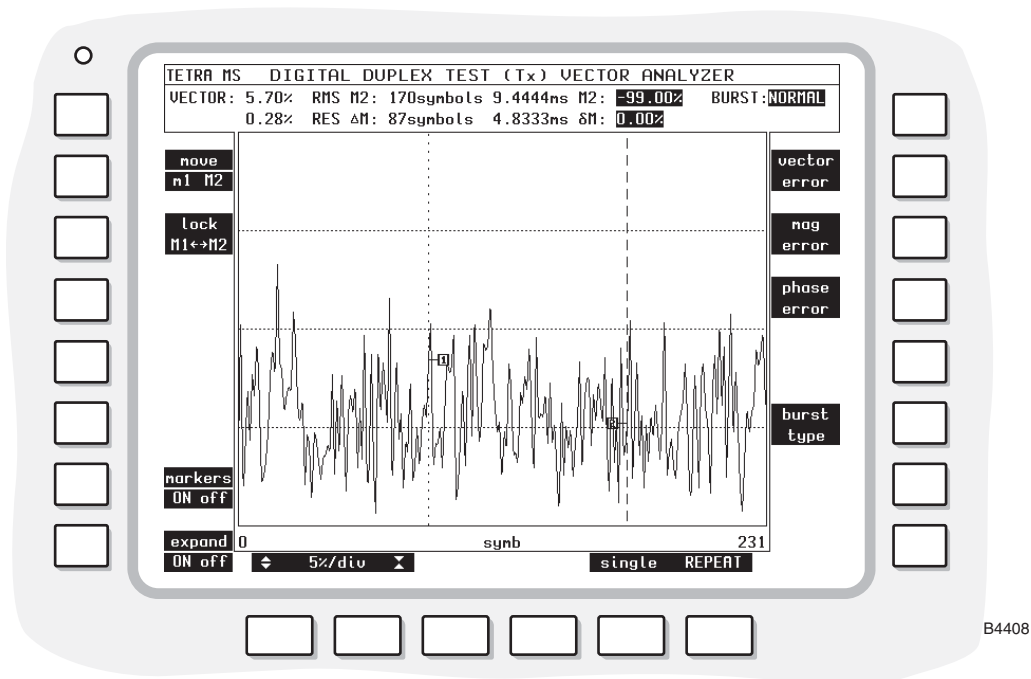


Fig. 8-15 Vector analyzer (vector error) - expanded with markers

Magnitude error

The vertical resolution can be set to 0.1, 0.2 or 0.4/division giving full scale ranges of ± 0.2 , 0.4 or 0.8.

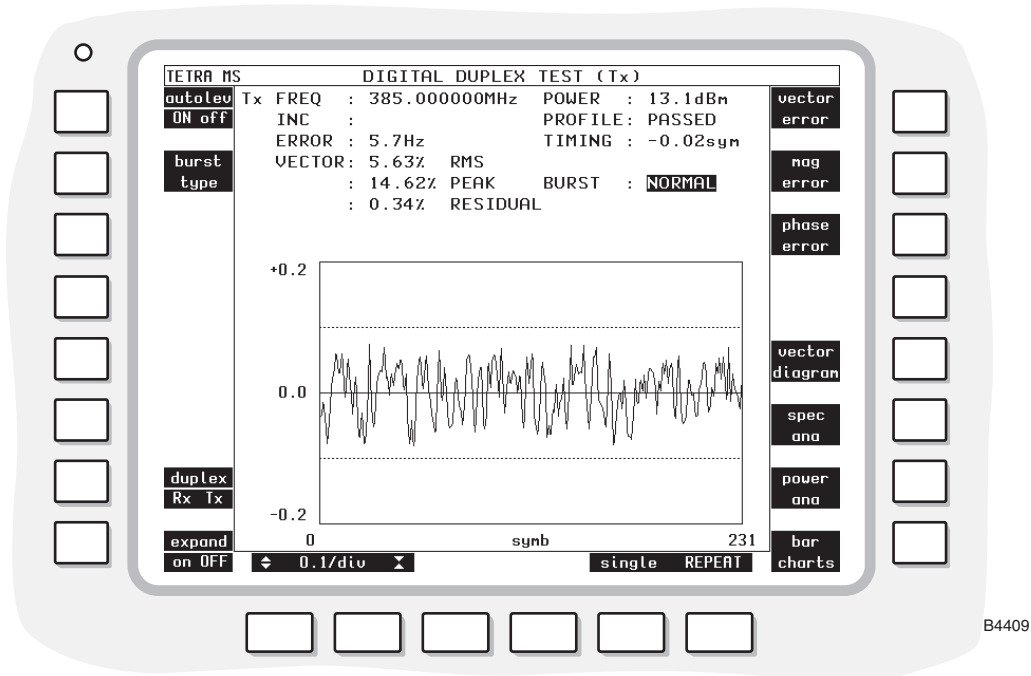


Fig. 8-16 Vector analyzer (magnitude error)

Phase error

The vertical resolution can be set to 5°, 10° or 20°/division giving full scale ranges of $\pm 10^\circ$, 20° or 40° .

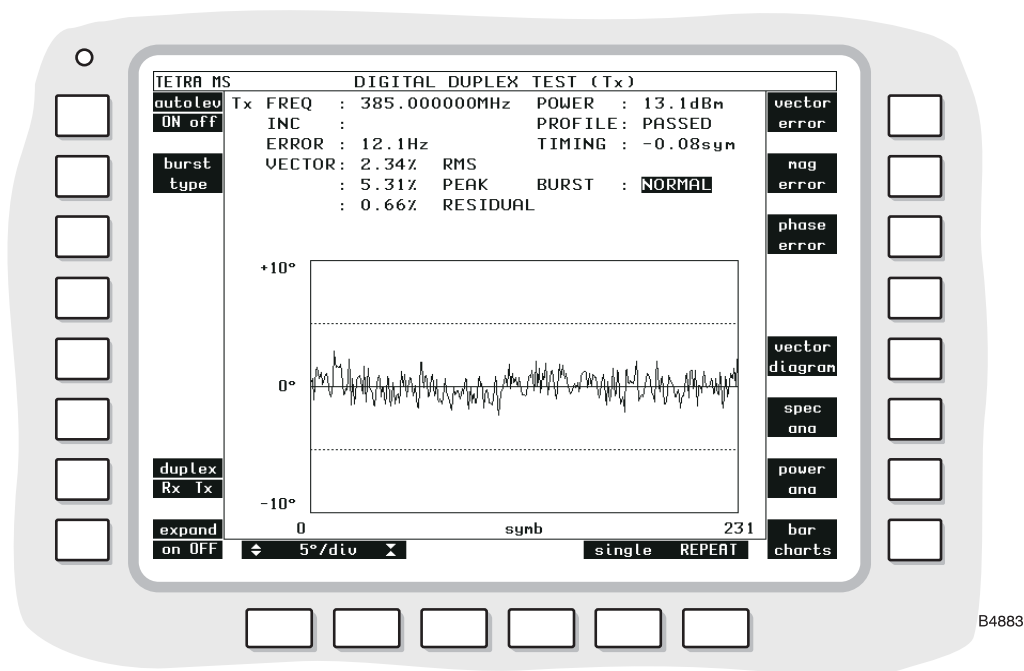


Fig. 8-17 Vector analyzer (phase error)

Vector diagrams

The vector diagram function shows modulation errors in graphic form. Measurements relating to the displayed data are shown on the display. Three aspects of the modulation are provided:-

- Symbol constellation, normal or expanded display.
- Phase trajectory, normal or expanded display.
- Rotated vector, normal or expanded display.

Display updating

At the bottom left of the display are soft keys that allow you to select the way in which the vector diagrams are updated. The selected key is shown in UPPER CASE.

[REFRESH]

Pressing this key sets the *REFRESH* display option. As each burst of the selected type is received, the display relating to the previous burst is erased and that relating to the current burst is displayed.

[PERSIST]

Pressing this key sets the *PERSISTENCE* display option. A vector display relating to several previous bursts is displayed. As each burst of the selected type is received, the data is added to the display but that of the earliest of the previous bursts is erased.

[ACCUM]

Pressing this key sets the *ACCUMULATE* display option. As each burst of the selected type is received, the data is added to the vector display.

Symbol constellation

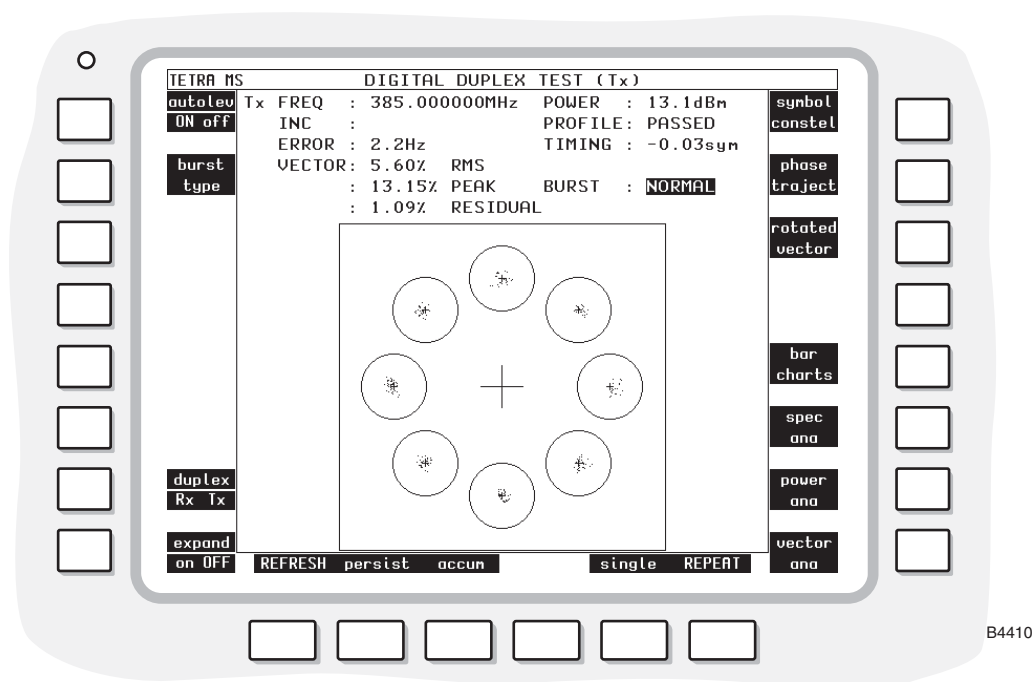


Fig. 8-18 TETRA symbol constellation vector diagram

The symbol constellation display shows the modulation at the symbol points. The normal display, shown in Fig. 8-18, includes vector measurement results of % RMS, % PEAK and % RESIDUAL. Transmission measurement results are also included.

The circles represent the 30% peak vector error limits for each constellation point.

Expanded display

Selecting the expanded display increases the area of the symbol constellation display but removes some readings and keys. The selected burst type is indicated and readings of vector % RMS, % peak and % residual remain.

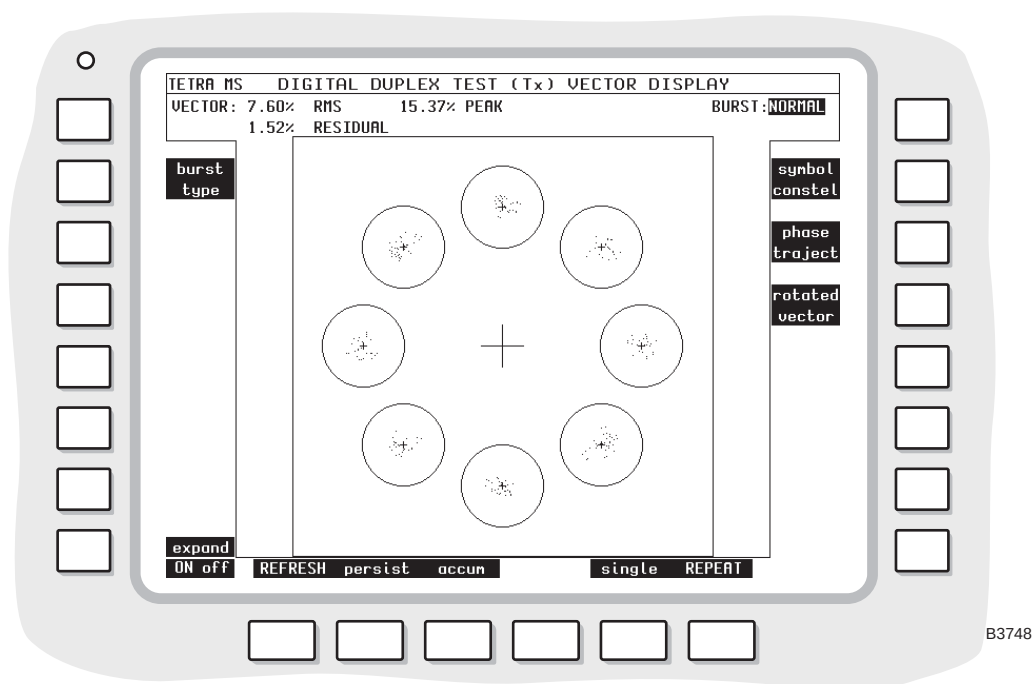


Fig. 8-19 TETRA symbol constellation vector diagram (expanded)

Phase trajectory

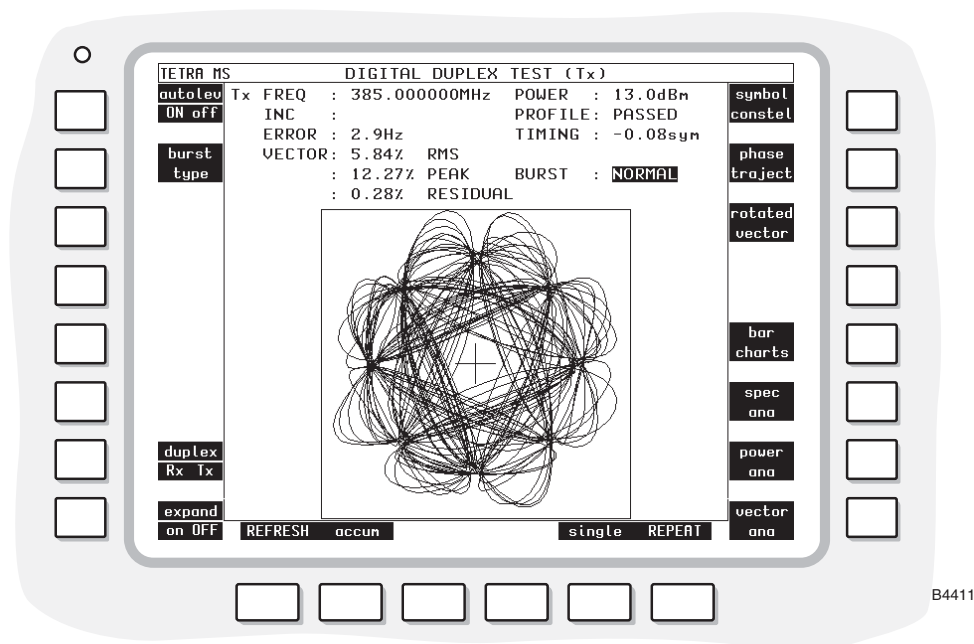


Fig. 8-20 TETRA phase trajectory vector diagram

The phase trajectory vector diagram shows the modulation at the symbol points and the transition between points (phase trajectory). The normal display, shown in Fig. 8-20, includes vector measurement results of % RMS, % PEAK and % RESIDUAL. Transmission measurement results are also included.

Expanded display

Selecting the expanded display increases the area of the phase trajectory display but removes some readings and keys. The selected burst type is indicated and readings of vector % RMS % peak and % residual remain.

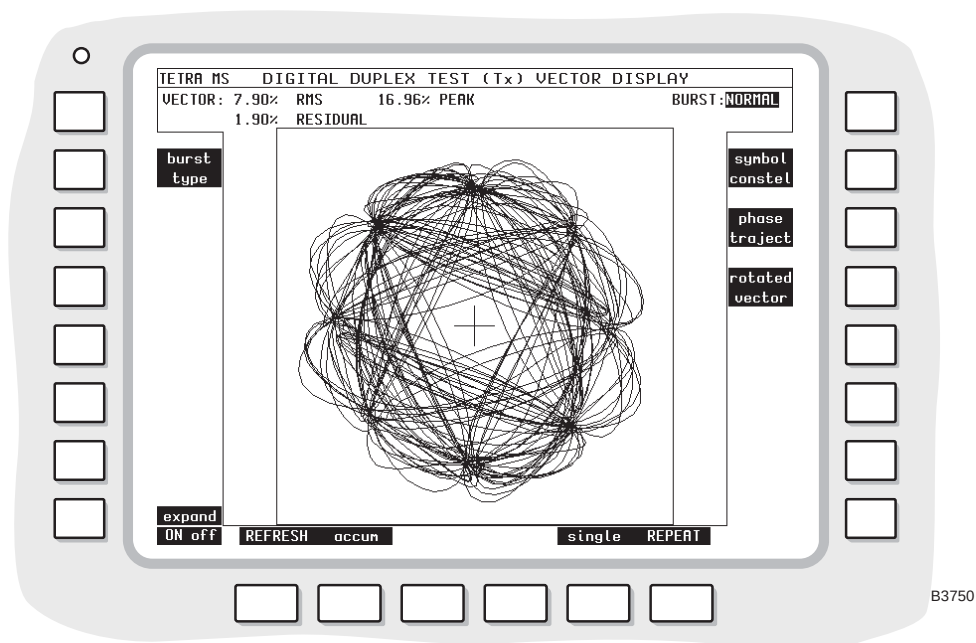


Fig. 8-21 TETRA phase trajectory vector diagram (expanded)

Rotated vector

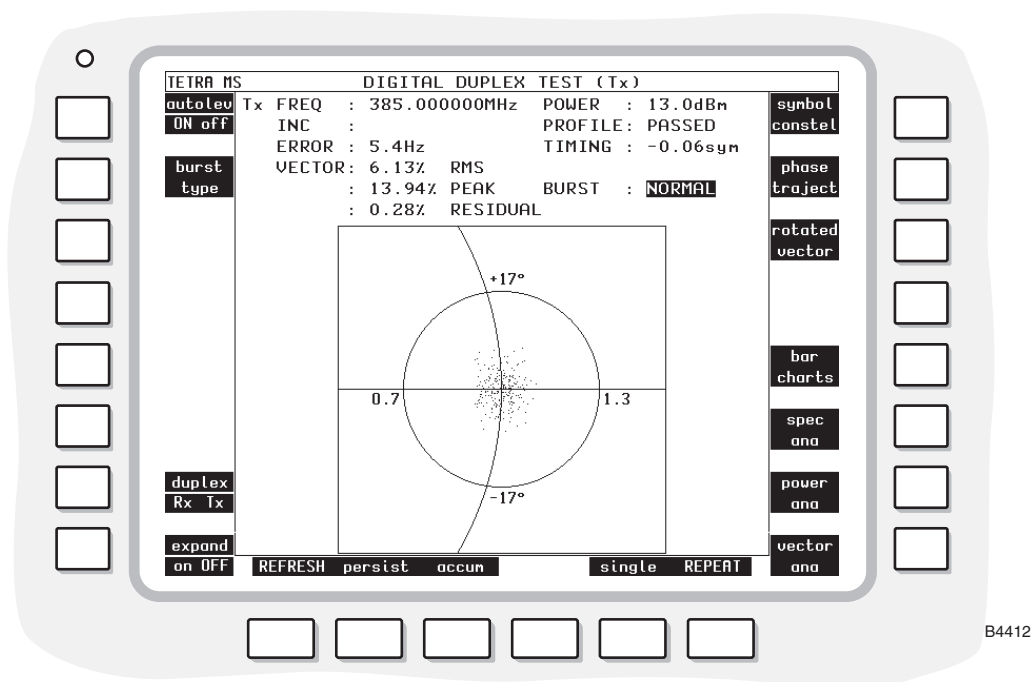


Fig. 8-22 TETRA rotated vector diagram

The rotated vector diagram shows all eight constellation diagram symbol points together on a single point. The display is obtained by rotating all points, other than the EAST point, about the common axis until they are superimposed on the EAST point. The display then shows positive phase errors upward, negative phase errors downward, negative magnitude errors to the left and positive magnitude errors to the right.

The normal display, shown in Fig. 8-22, includes vector measurement results of % RMS, % PEAK and % RESIDUAL. Transmission measurement results are also included.

Expanded display

Selecting the expanded display increases the area of the rotated vector display but removes some readings and keys. The selected burst type is indicated and readings of vector % RMS % peak and % residual remain.

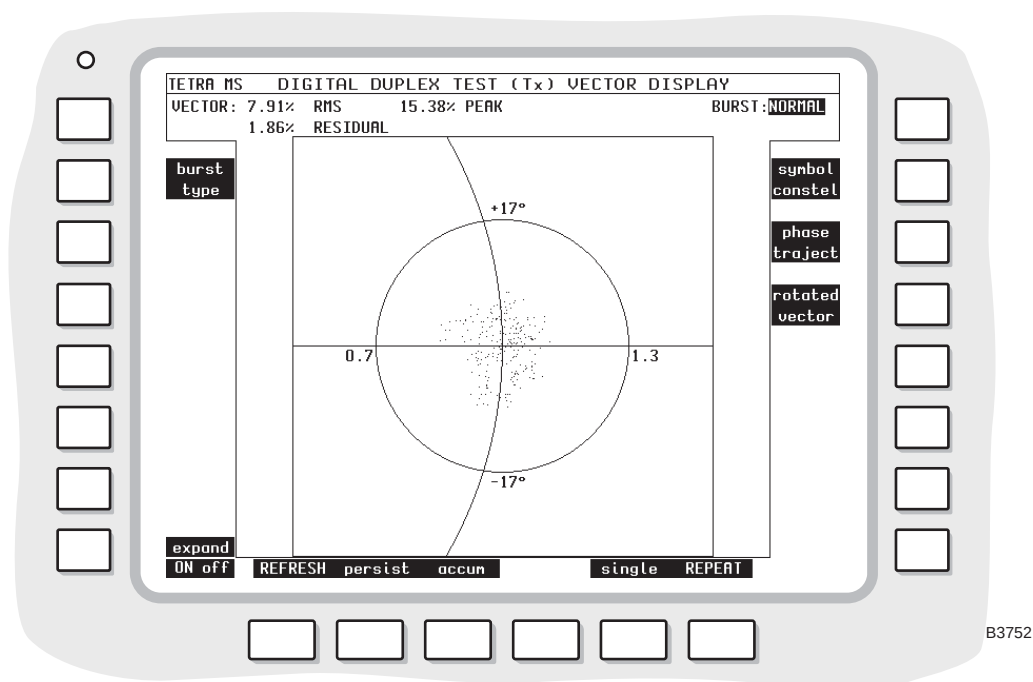


Fig. 8-23 TETRA rotated vector diagram (expanded)

Chapter 9

DATA DISPLAYS

Data displays

The DATA DISPLAY screen is opened by the *[DATA]* soft key that is on all of the TETRA System screens.

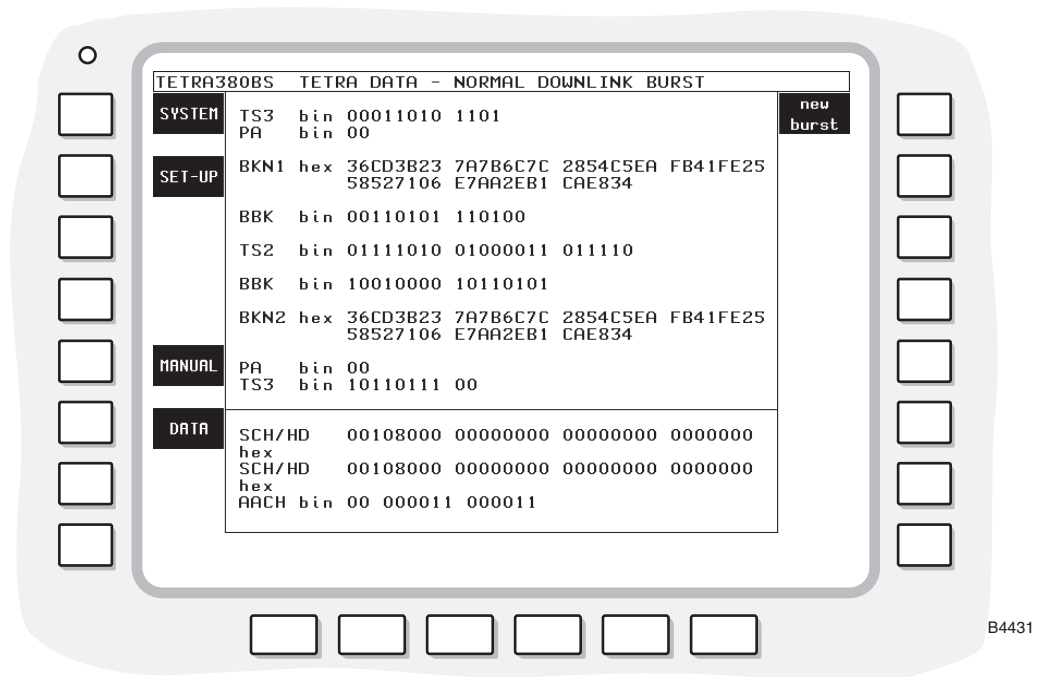


Fig. 9-1 Typical TETRA data display screen

The data display shows the data contained in a single burst. The burst type is shown at the top of the display. This will be of the type selected in the MANUAL TEST screen.

The components of the demodulated burst data are indicated; these are tabled below, and are indicated as either **bin** to identify binary data or **hex** to identify hexadecimal data.

The data is displayed in groups of eight digits, starting with the most significant digit.

The top part of the screen displays demodulated burst data; the lower part displays the same data following de-scrambling and channel decoding.

Demodulated burst data types displayed	
Indication	Full title and comments
Blocks and Bits	
BBK	B roadcast B lock
BKN1	B lock N umber 1
BKN2	B lock N umber 2
CBK	C ontrol B lock (in Control Burst)
FC	F requency C orrection bits
P1	P re-amble 1 (in Direct Mode Normal Burst)
P2	P re-amble 2 (in Direct Mode Normal Burst)
P3	P re-amble 3 (in Direct Mode Synchronization Burst)
PA	P hase A justment bits
SBK	S ynchronization B lock (in Synchronization Burst)
TAIL	T AIL bits
Training sequences	
TS1	T raining S equences 1 Normal training sequence, full slot usage
TS2	T raining S equences 2 Normal training sequence, half slot usage
TS3	T raining S equences 3 Inter-slot training sequence
TSex	e xtended T raining S equences (Control Burst)
TSSync or TSsy	s ynchronization T raining S equences (Synchronization Burst)

Decoded logical channel types displayed	
Indication	Full title and comments
Channels	
AACH	A ccess A ssignment C hannel
BNCH	B roadcast N etwork C hannel
BSCH	B roadcast S ynchronization C hannel
SCH/F	F ull slot S ignaling C hannel
SCH/H	H alf slot S ignaling C hannel (Direct Mode)
SCH/HD	H alf slot S ignaling C hannel (D own link)
SCH/HU	H alf slot S ignaling C hannel (U p link)
SCH/S	S ynchronisation S ignaling C hannel (Direct Mode)
STCH	S Tealing C hannel
TCH/2.4	2.4 kbit/s T raffic C hannel
TCH/4.8	4.8 kbit/s T raffic C hannel
TCH/7.2	7.2 kbit/s T raffic C hannel
TCH/S	S peech T raffic C hannel

Chapter 10

STORE AND RECALL FACILITY

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Introduction

You can store instrument settings internally within the Test Set. Instrument settings, instrument results, systems settings, systems test results, captured screen images (screen dumps) and user-defined MI-BASIC programs can be stored externally on a memory card inserted in the memory card slot. A block diagram of the store and recall facility is shown in Fig. 10-1 on page 10-3.

This section describes how to use the store and recall facility. This information is also given in Chapter 3 of the Test Set 2968 Operating Manual (46882/274).

Systems settings

When you store systems settings, the information on the SYSTEM PARAMETERS, MOBILE PARAMETERS, AUTORUN CONTROL and AUTORUN PARAMETERS menus for *all* systems are stored, not just the currently selected system.

Systems results

When you are using Autotest mode, the results of the tests carried out by the program are placed in one of the five test result stores, as explained under *Test results* in Chapter 7, *TETRA Mobile automatic testing*. The memory used for this is volatile, so the results are lost when the Test Set is switched off, but the contents of the current test results store can be saved to a memory card and recalled later.

User defined MI-BASIC programs

The current user-defined MI-BASIC program can be saved on a memory card, or you can overwrite the current program by a program stored on a memory card. (See *Selecting an Autorun test program* in Chapter 7.)

Screen dumps

Captured screen images (screen dumps) can be made of any display which is visible on the Test Set. Pressing the [COPY] key freezes the display in its current state, with all data and graphics intact. A message box is displayed over the screen but this is not printed or stored. The screen dump can be stored on a memory card or reproduced on a suitable printer connected to the Test Set. Screen dumps stored on a memory card can be recalled to the display and then reproduced on a printer in the same manner as directly captured screens.

Suitable memory cards

Memory cards, suitable for use in the Test Set, are available from IFR Ltd. as part number 59000/189.

Accessing the Store and recall facility

Access to the STORE FACILITY or RECALL FACILITY menu is via the [STORE] key or [RECALL] key respectively.

Formatting a memory card

The memory card to be used must be formatted for use with the Test Set. When you insert a memory card in the memory card slot, the Test Set checks the card format. Provided that one of the STORE FACILITY or RECALL FACILITY menus is on the screen, and the card format is correct (i.e. MI DOS Format), the card format, title and size, and the number of bytes used and free, are displayed. If the card format is not correct, the message Memory Card Format is incorrect is displayed.

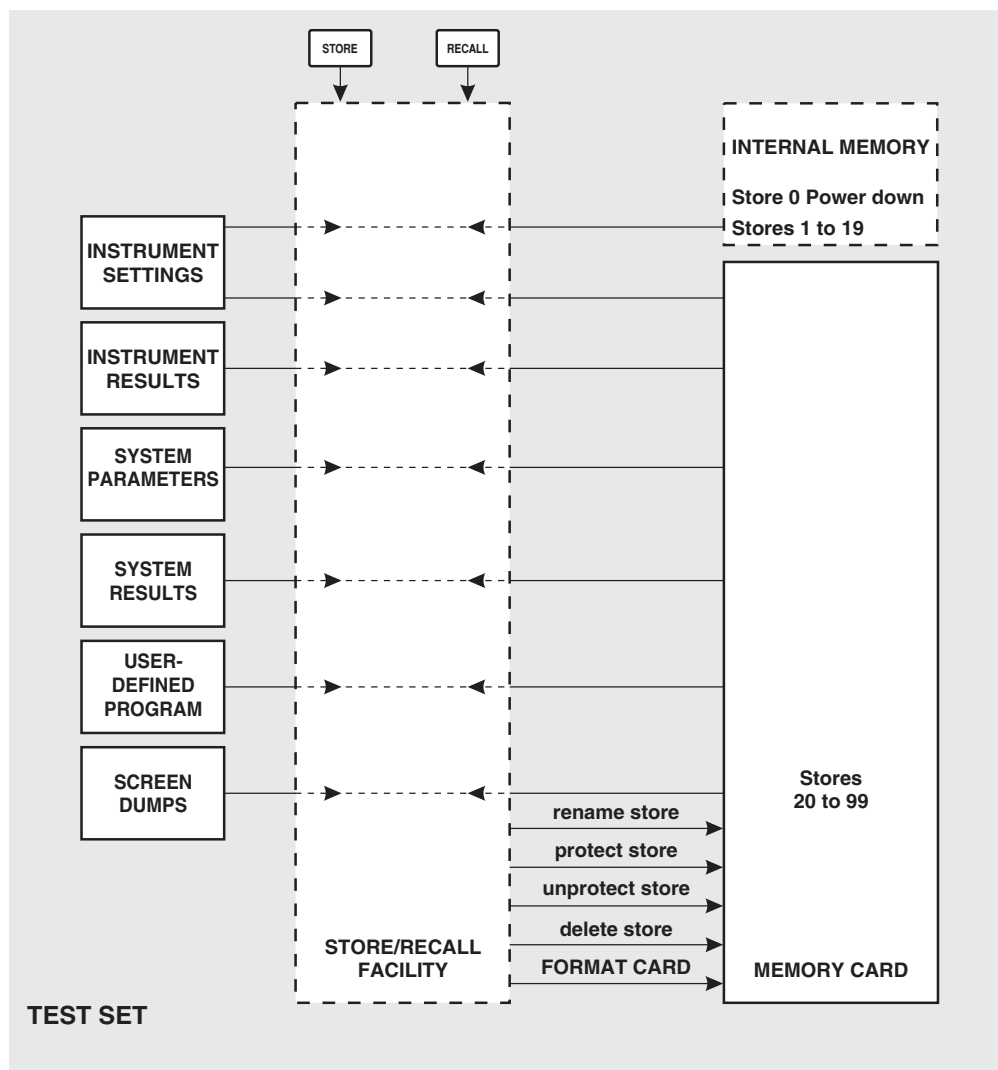
Before formatting (or reformatting) a memory card, read the **CAUTION** note below. To format a card, select any STORE FACILITY menu by pressing the [STORE] function key, and then press [FORMAT CARD]. When formatted, the card has 80 store locations (20-99).

CAUTION

Formatting a memory card results in the loss of all data previously stored on the card. Memory cards should not be subjected to extremes of temperature.

Protection

You can prevent the unintentional deletion or overwriting of a specific store location on a memory card by protecting it. (See page 10-7.) However, there is no protection against formatting, other than the physical *Read Only* switch on the card. See the **CAUTION** above.



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Fig. 10-1 Store and recall facility - block diagram

Store

System settings, test results or User Defined MIBASIC programs

To store the systems settings, system test results, or the current user defined MIBASIC program, to a memory card:-

1. Access the store facility by pressing the [STORE] function key. One of the STORE FACILITY menus appears. This will either be the factory default (the STORE FACILITY - SETTINGS menu; see Fig. 10-2) or the STORE FACILITY menu that was last selected in the current test session.
2. Insert a memory card, and format it if necessary. See page 10-2.
3. Use the *[next page]* or *[prev page]* soft key to display one of the pages showing memory card stores, i.e. stores from 20 to 99. A typical memory card page is shown in Fig. 10-3 on page 10-5. Note that stores 1 to 19 are internal and cannot be used for systems data. Information adjacent to each store number indicates that the store is either empty or, if not, its name, the date and time at which data was stored, and the type of data stored. Store locations can be re-used if their contents are no longer required, and renamed if necessary.
4. Select the type of data that you want to store by pressing *[systems setting]*, *[systems results]* or *[BASIC]* as appropriate.
5. Decide at which store location the data is to be stored and enter the store number using the appropriate numerical DATA keys. Press either the *[store]* soft key or the ENTER data key. The screen changes to show soft keys that can be used to enter or change the name of the store location. See Fig. 10-4 on page 10-6.
6. If required, enter or change the store location name using the procedure described on page 10-6. If you do not do this, the default name No Name Stored will be assigned to it. The *[rename store]* soft key can be used to rename the store location subsequently.
7. Press the ENTER data key. The message Storing <data type> to Memory Card. Please wait. is displayed. When this message disappears from the screen, the data has been stored on the memory card in the chosen store location.

Screen dumps

To capture and store screen dumps to a memory card:-

1. Press the [COPY] key to freeze the display. A message prompts you to select store to memory card or send to printer.
2. Press the [STORE] key. One of the STORE FACILITY menus appears. This will either be the factory default (the STORE FACILITY - SETTINGS menu; see Fig. 10-2) or the STORE FACILITY menu that was last selected in the current test session.
3. Insert a memory card, and format it if necessary. (See page 10-2.)
4. Use the *[next page]* or *[prev page]* soft key to display one of the pages showing memory card stores, i.e. stores from 20 to 99. A typical memory card page is shown in Fig. 10-3 on page 10-5. Note that stores 1 to 19 are internal and cannot be used for systems data. Information adjacent to each store number indicates that the store is either empty or, if not, its name, the date and time at which data was stored, and the type of data stored. Store locations can be re-used if their contents are no longer required, and renamed if necessary.
5. Decide at which store location the data is to be stored and enter the store number using the appropriate numerical DATA keys. Press either the *[store]* soft key or the ENTER data key. The screen changes to show soft keys that can be used to enter or change the name of the store location. See Fig. 10-4 on page 10-6.
6. If required, enter or change the store location name using the procedure described on page 10-6. If you do not do this, the default name No Name Stored will be assigned to it. The *[rename store]* soft key can be used to rename the store location subsequently.
7. Press the ENTER data key. The message Storing <data type> to Memory Card. Please wait. is displayed. When this message disappears from the screen, the data has been stored on the memory card in the chosen store location.

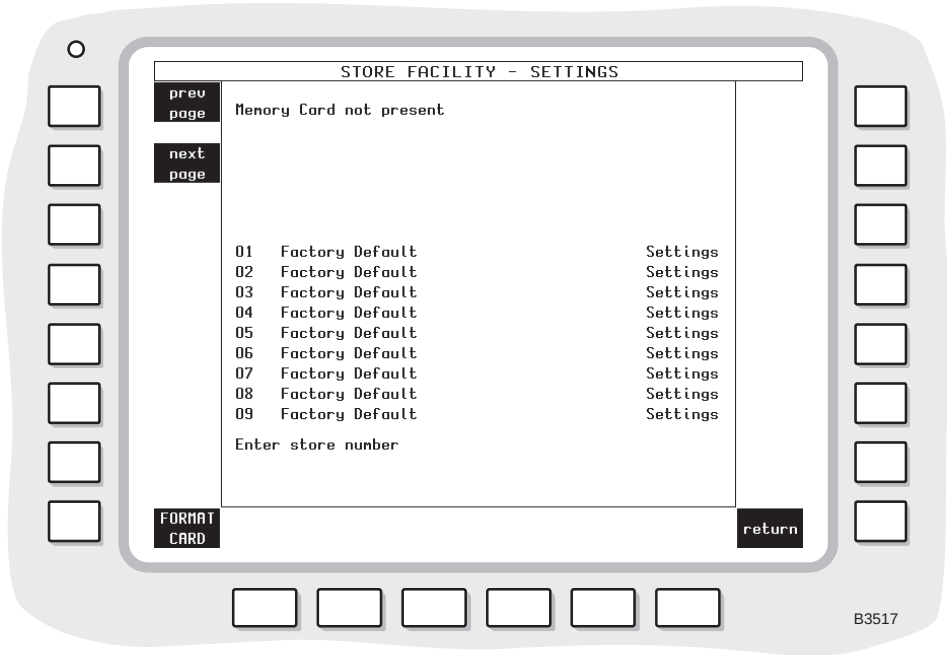


Fig. 10-2 STORE FACILITY - SETTINGS menu

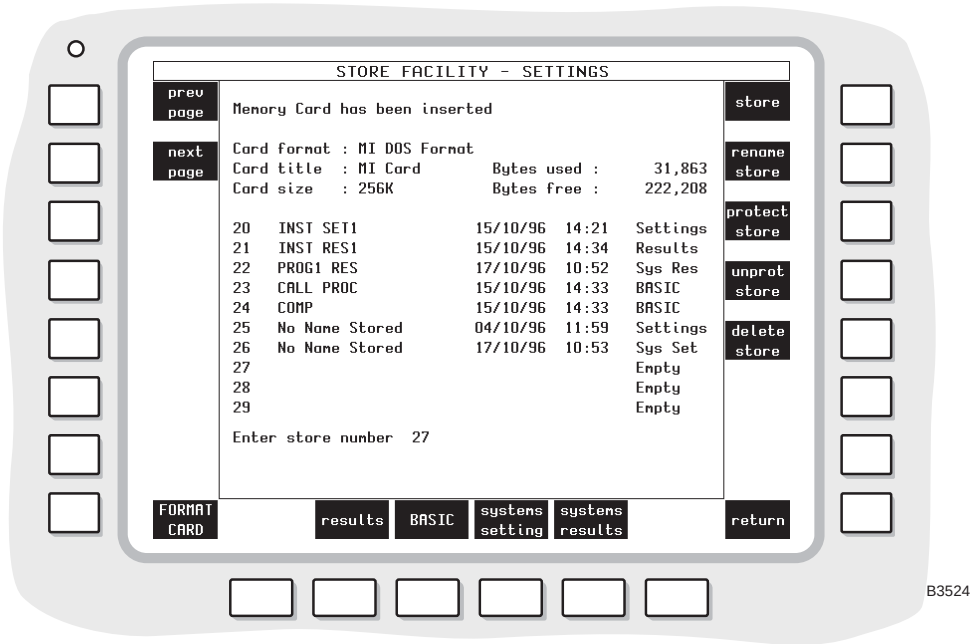


Fig. 10-3 STORE FACILITY - memory card store locations 20 to 29

Naming a memory card store location

The name of a store location can contain up to 19 characters, including spaces.

The following procedure describes how to name, or rename, a memory card store location as part of a storage operation - see steps 1 to 6 in the *Store* section on page 10-4:-

1. Use the character set provided by the soft keys [A-B], [C-D], [E-F] etc (see Fig. 10-4 on page 10-6), or one of the numerical DATA keys, to enter the first character of the name. To select the alphabetic characters B, D, F, ... to Z, press the associated key twice in quick succession. Similarly, for the remaining two character keys press once or several times in quick succession until the character that you need appears. You can also use the DATA keys to enter the full-stop (.) or dash (-) characters.
2. Enter the next character of the name in the same way.
3. You can delete characters while the naming procedure is active by pressing the [DELETE] data key. The character to the left of the current cursor position will be deleted. Repeated pressing deletes further characters.
4. The functions of the other soft keys are described below. Note that you can use the VARIABLE rotary control to reposition the cursor.
5. When you have set up the required name, press the ENTER data key to complete the storage procedure.

Other soft key functions

The functions of the soft keys at the bottom of the screen are:-

[insert/exchange] Toggles between insert or exchange modes; the current mode is shown on the screen. In insert mode, characters that you enter are *added* to the name at the current cursor position. In exchange mode, characters that you enter *replace* the character at the current cursor position.

[caps lock] Allows you to select uppercase (caps) or lowercase alphabetic characters. In caps mode, the message CAPS on is displayed.

[home] Moves the cursor to the start of the name.

[end] Moves the cursor to the end of the name.

[clear] Deletes all characters.

[abort] Aborts the storage operation.

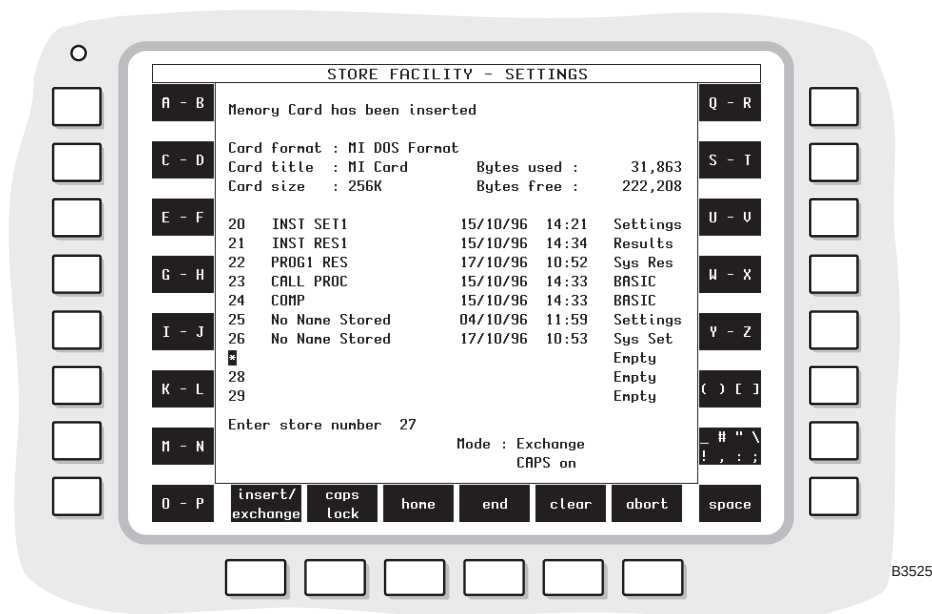


Fig. 10-4 Entering a store location name

Renaming a memory card store location

The storage procedure, as described in the *Store* section (page 10-4), always gives you the option of renaming a store location that has already been used that you want to overwrite. However, you may want simply to change the name of a store location *without* changing its contents.

You can do this by accessing the appropriate STORE FACILITY screen as described previously, entering the store number, and then pressing the *[rename store]* soft key. This brings up the naming screen (Fig. 10-4). The procedure is then the same as that described in the section *Naming a memory card store location* (page 10-6).

Protecting or unprotecting a memory card store location

To protect a memory card store location from accidental erasure, select the store location as described previously, either as part of data storage or as a separate operation, and press the *[protect store]* soft key. Each protected store location has the letter P beside the store number. You can remove this protection by selecting the store location and pressing the *[unprot store]* soft key.

Deleting a memory card store location

You can delete the contents of a memory card store location, provided that it is not protected, by selecting it as described previously, either as part of data storage or as a separate operation, and then pressing the *[delete store]* soft key. The title and other text beside the store number is replaced by the word Empty.

Recall

System settings

Systems test results

User-defined MI-BASIC programs

To recall systems settings, systems test results or user-defined MI-BASIC programs from a memory card, insert the card and press the [RECALL] function key.

Use the *[next page]* or *[prev page]* soft key to display the page showing the store that you want to recall (theoretically, provided that you are sure that you know the required store number, you do not need to do this, but it is usually safer to at least check the store title and data type before recalling it). Enter the store number using the numerical DATA keys and then press the ENTER data key. The data recalled from the memory card will overwrite the existing data (see the note below).

The Test Set mode and display then reverts to that which was active before pressing the [RECALL] key.

Note

If you recall systems settings, remember that settings for *all* systems are recalled, not just the currently selected system.

Screen dumps

To recall screen dumps from a memory card, insert the card and press the [RECALL] function key.

Use the *[next page]* or *[prev page]* soft key to display the page showing the store that you want to recall. Enter the store number using the numerical DATA keys and then press the ENTER data key. The screen dump recalled from the memory card will be displayed on the Test Set and a message will prompt you to print the screen dump, clear the message from the screen dump or clear the screen dump from the display.

Error, warning and information messages

There are a number of error, warning and information messages associated with the store facility that will be displayed under certain conditions. These are listed below:-

Error Messages

Message	Explanation
Memory Card Error Store is empty	The Test Set could not read data from the selected store because no information was stored in it. Data must be stored to a memory card store before it may be recalled.
Memory Card Error Store is protected	The selected store is protected to prevent it being overwritten. Use the <i>[unprot store]</i> soft key to unprotect the store. Note that any new data saved to the store will overwrite the existing data.
Memory Card Error No Card present	The Test Set was unable to detect a memory card in the memory card slot. Remove and reinsert the memory card. If this does not cure the problem the memory card may be faulty.
Memory Card Error Wrong format	The memory card in the memory card slot has the wrong format and may contain data from a different instrument or system. To use the memory card with this instrument it can be reformatted using the <i>[FORMAT CARD]</i> soft key. Note that all data previously stored on the card will be lost.
Memory Card Error Card directory is full	The directory on the memory card is full therefore no more data can be stored on the card. Delete unwanted data from the card using the <i>[delete store]</i> soft key.
Memory Card Error Card FAT is full	The File Allocation Table on the memory card is full therefore no more data can be stored on the card. Delete unwanted data from the card using the <i>[delete store]</i> soft key.
Memory Card Error File not found	The Test Set could not find a file on the memory card and is unable therefore to read back data. The memory card data may be corrupted or the memory card may be faulty. It will not be possible to retrieve the data.
Memory Card Error Store is too big to fit on Memory Card	There was insufficient blank memory on the card to store the data. Delete unwanted data from the card using the <i>[delete store]</i> soft key.
Memory Card Error User aborted store	The user hit the <i>[ABORT STORE]</i> soft key while the Test Set was in the process of saving data to the memory card.
Memory Card Error Unknown Error	An unknown error occurred with the memory card preventing the current operation from being carried out. Remove and reinsert the memory card. If this does not cure the problem the memory card may be faulty or the data stored in it may be corrupted.

Warning Messages

Message	Explanation
Memory Card Formatting in progress. Please wait	The Test Set is formatting the memory card because the <i>[FORMAT CARD]</i> soft key has been pressed. Memory card formatting takes a few seconds during which the Test Set may not be used.
Memory Card Formatting is finished	The Test Set has finished formatting the memory card and the Test Set may now be used.
Memory Card not present	The Test Set has detected that the memory card is not present in the memory card socket.
Memory Card is present Please wait while reading format	The Test Set requires a few seconds to read the format of a memory card when it is inserted into the memory card socket and the Test Set may not be used during this time.
Memory Card not inserted properly	The Test Set has detected that the memory card is not inserted in the socket correctly. Check that the memory card is pushed fully into the socket.

Information Messages

Storing data to or recalling data from the internal stores or a data card takes a few seconds, during which time the Test Set may not be used. An information message from the list below is displayed during this period.

Storing Instrument Settings to internal memory. Please wait.
Storing Instrument Settings to Memory Card. Please wait.
Storing Instrument Results to Memory Card. Please wait.
Storing BASIC Program to Memory Card. Please wait.
Storing Systems Settings to Memory Card. Please wait.
Storing Systems Results to Memory Card. Please wait.

Recalling Instrument Settings from internal memory. Please wait.
Recalling Instrument Settings from Memory Card. Please wait.
Recalling Instrument Results from Memory Card. Please wait.
Recalling BASIC Program from Memory Card. Please wait.
Recalling Systems Settings from Memory Card. Please wait.
Recalling Systems Results from Memory Card. Please wait.

Chapter 11

PERFORMANCE DATA

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Performance data

The performance data shown below is provided in addition to that contained in the operating manual for the 2968 (IFR part number 46882/274).

System options

Each of the three system modes of TETRA operation are individual *Options*. These are:-

TETRA Mobile	Option 30
TETRA Base Station	Option 31
TETRA Direct Mode	Option 32

TETRA Direct Mode is possible only if TETRA Mobile is fitted.

TETRA Mobile (MS)

Channel plans

Channel plan	Mobile Rx frequency of minimum channel number (MHz)	Mobile Tx frequency of minimum channel number (MHz)	Minimum channel number	Maximum channel number	Channel spacing (kHz)	Duplex spacing (Rx - Tx) (MHz)	Offset (kHz)
TETRA 380 MS	390.0125	380.0125	3600	3999	25	10	12.5
TETRA 380 +0 MS	390.0000	380.0000	3600	4000	25	10	0
TETRA 410 MS	420.0125	410.0125	800	1199	25	10	12.5
TETRA 410 +0 MS	420.0000	410.0000	800	1200	25	10	0
TETRA 410 -6 MS	420.01875	410.01875	801	1200	25	10	-6.25
TETRA 450 MS	460.0125	450.0125	2400	2799	25	10	12.5
TETRA 450 +0 MS	460.0000	450.0000	2400	2800	25	10	0
TETRA 800 MS	805.0125	850.0125	2000	2799	25	45	12.5
TETRA 800 +0 MS	805.0000	850.0000	2000	2800	25	45	0
TETRA 870 MS	915.0125	870.0125	600	839	25	45	12.5
TETRA 870 +0 MS	915.0000	870.0000	600	840	25	45	0
USER DEFINED	390.0000	380.0000	0	4095	25	10	0

The values shown for USER DEFINED in the above table are the default setting of the user-defined channel plan.

TETRA Mobile (MS) (continued)**Note****Channel plans**

The pre-defined channel plans are based on the channel numbering scheme defined in ETSI TS 100 392-15 v1.1.2. These standardized channel plans may not necessarily be implemented or configured in the mobile under test, which may instead, employ a proprietary channel numbering scheme. A TETRA mobile will normally use the 2968 SYSTEM INFORMATION parameters *Channel number*, *Band*, *Offset*, *Duplex spacing* and *Duplex reverse* to set its receive and transmit frequencies.

The TETRA 380, 410, 450, 800 and 870 channel plans assume an offset value of 12.5 kHz.

The TETRA 380 +0, 410 +0, 450 +0 800 +0 and 870 +0 channel plans assume an offset value of 0 kHz.

The TETRA 410 –6 channel plan assumes an offset value of –6.25 kHz.

If a different offset value is required, use the USER DEFINED channel plan and set the appropriate values. Similarly if the mobile under test employs proprietary channel numbering, or ignores some of the network parameters, use the USER DEFINED channel plan, with the appropriate values set up.

If the channel plan is unknown, but the frequency of operation is known, the frequency of the MS transmitter can be set directly on the Test Set by setting the channel plan to NO PLAN.

Mobile transmitter power levels

Control level	Nominal output power		Comments
MS PL1	45 dBm	31.6 W	Power Class 1 Max Power
MS PL1L	42.5 dBm	17.8 W	Power Class 1L Max Power
MS PL2	40 dBm	10 W	Power Class 2 Max Power
MS PL2L	37.5 dBm	5.62 W	Power Class 2L Max Power
MS PL3	35 dBm	3.16 W	Power Class 3 Max Power
MS PL3L	32.5 dBm	1.78 W	Power Class 3L Max Power
MS PL4	30 dBm	1 W	Power Class 4 Max Power
MS PL4L	27.5 dBm	562 mW	Power Class 4L Max Power
MS PL5	25 dBm	316 mW	
MS PL6	20 dBm	100 mW	
MS PL7	15 dBm	31.6 mW	

Test parameters

This section lists the TETRA Mobile (MS) System parameter default settings. Where appropriate, it shows alternative settings. Details of how to change the settings are given in Chapter 3, *Selecting and customizing the system*.

If the parameters are modified, you can restore the defaults by pressing the following keys:-

[HELP SET-UP], [SET-UP], [MISC CONFIG], [default values], [init current]

System parameter settings**Defaults****Range**

Channel Plan

TETRA380MS

NO PLAN
USER DEFINED
TETRA380MS
TETRA380 +0 MS
TETRA410MS
TETRA410 +0 MS
TETRA410 –6 MS
TETRA450MS
TETRA450 +0 MS
TETRA800 MS
TETRA800 +0 MS
TETRA870MS
TETRA870 +0 MS

System parameter settings	Defaults	Range
User Defined Plan	(Only available with USER DEFINED channel plan)	
Channel Plan Title	USER DEFINED	up to 12 characters as defined by user
Frequency Band (Sys Info)	3 (300 MHz)	0 to 15
Offset (Sys Info)	0 (0 kHz)	0 to 3
Duplex Spacing (Sys Info)	0 (10 MHz)	0 to 7
Reverse Operation (Sys Info)	0 (normal)	0 (normal) or 1 (reverse)
Lowest Channel	0000	0000 to 4095
Highest Channel	4095	0000 to 4095
Lowest Channel Tx Freq	380 MHz	10 MHz to 999 MHz
Duplex Offset	10 MHz	-100 MHz to +100 MHz
Channel Spacing	25 kHz	-5 kHz to -500 kHz and 5 kHz to 500 kHz
Control Channel	(Control Channel MS Tx Frequency for NO PLAN Channel Plan)	
NO PLAN	380 MHz	100 to 999.999999 MHz
USER DEFINED	0	0 to 4095
TETRA380MS	3600	3600 to 3999
TETRA380 +0 MS	3600	3600 to 4000
TETRA410MS	800	800 to 1199
TETRA410 +0 MS	800	800 to 1200
TETRA410 -6 MS	801	801 to 1200
TETRA450MS	2400	2400 to 2799
TETRA450 +0 MS	2400	2400 to 2800
TETRA800 MS	2040	2000 to 2799
TETRA800 +0 MS	2040	2000 to 2800
TETRA870MS	600	600 to 839
TETRA870 +0 MS	600	600 to 840
Traffic Channel	(Traffic Channel MS Tx Frequency for NO PLAN Channel Plan)	
NO PLAN	380 MHz	100 to 999.999999 MHz
USER DEFINED	0	0 to 4095
TETRA380MS	3700	3600 to 3999
TETRA380 +0 MS	3700	3600 to 4000
TETRA410MS	900	800 to 1199
TETRA410 +0 MS	900	800 to 1200
TETRA410 -6 MS	901	801 to 1200
TETRA450MS	2500	2400 to 2799
TETRA450 +0 MS	2500	2400 to 2800
TETRA800 MS	2140	2000 to 2799
TETRA800 +0 MS	2140	2000 to 2800
TETRA870MS	700	600 to 839
TETRA870 +0 MS	700	600 to 840
Rx Offset	(Available only with NO PLAN Channel Plan)	
	10 MHz	-100 to +100 MHz
Timeslot	3	1 to 4
Mobile Country Code (MCC)	001	000 to 999
Mobile Network Code (MNC)	00001	00000 to 16383
Base Color Code (BCC)	01	00 to 63
Location Area Code (LA)	00001	00000 to 16383
Min Rx Level for Access	-125 dBm	-125 dBm to -50 dBm (5 dB steps)
Max Tx Level	+30 dBm	15 dBm to 45 dBm (5 dB steps)
Access Parameter	-45 dBm	-23 dBm to -53 dBm (2 dB steps)

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System parameter settings	Defaults	Range
Base Services		
Power on registration	Required	Not required, Required
Power off de-registration	Required	Not required, Required
Priority call	Yes	No, Yes
Minimum mode service	Never used	Never used, May be used
Migration	Supported	Not supported, Supported
System wide services	Normal mode	Not supported, Normal mode
TETRA voice service	Supported	Not supported, Supported
Circuit mode data service (Reserved)	Not supported	Not supported, Supported
SNDP service	Not available	Not available, Available
Air interface encryption	Not available	Not available, Available
Advanced link	Not supported	Not supported, Supported
Neighbour Cell Broadcast		
Neighbour Cell Broadcast	Not supported	Not supported, Supported
Broadcast Interval	10 s	5 s to 30 s
Neighbour Cell Channel	0000	0000 to 4095
Neighbour Cell Location Area	00001	00001 to 16383
Neighbour Cell Identifier	01	01 to 30
Slow Re-select Threshold	10 dB	0 dB to 30 dB (2 dB steps)
Slow Re-select Hysteresis	10 dB	0 dB to 30 dB (2 dB steps)
Fast Re-select Threshold	10 dB	0 dB to 30 dB (2 dB steps)
Fast Re-select Hysteresis	10 dB	0 dB to 30 dB (2 dB steps)
Trunking Type	Transmission	Message, Transmission, Quasi-transmission
Call Type		
Group Call		
Priority	00	00 to 15
Calling Party SSI	00000001	00000000 to 16777215
Private Call		
Simplex/Duplex Signalling Type	Simplex Call Hook	Simplex Call, Duplex Call Direct, Hook
Priority	00	00 to 15
Calling Party SSI	00000001	00000000 to 16777215
Phone Call		
Priority	00	00 to 15
Calling Party SSI	16777184 (PSTN Gateway)	00000000 to 16777215
Calling Party ESN	01438742200	0 to 9999999999999999999999 Up to 24 digits
ESN Required	Required	Required, Not required
Emergency Call		
Group/Individual Simplex/Duplex Signalling Type	Individual Call Simplex Call Direct	Individual Call, Group Simplex Call, Duplex Call Direct, Hook
Calling Party SSI	00000001	00000000 to 16777215
User Defined Call		
Group/Individual Simplex/Duplex Signalling Type	Individual Call Duplex Call Hook	Individual Call, Group Simplex Call, Duplex Call Direct, Hook
Priority	00	00 to 15
Calling Party SSI	16777186 (PABX Gateway)	00000000 to 16777215
Calling Party ESN	742200	0 to 9999999999999999999999 Up to 24 digits
ESN Required	Required	Required, Not required

System parameter settings	Defaults	Range
Call Timers		
Testset Transmit Mode	Timed	None, Timed, Continuous
Testset Quiet Time	2 s	0 to 30 s
Testset Transmit Time	2 s	1 to 30 s
Quasi Tx Trunking Hang Timer	5 s	1 to 30 s
Call Hang Timer	15 s	1 to 30 s
Messages		
Group/Individual	Individual Message	Individual Message, Group Message
Calling Party SSI	00000001	00000000 to 16777215
Status Message	FEFF HEX (65279)	
	Callback Request	0000 to FFFF HEX
SDS Type 1 Message	5431 HEX ("T1")	0000 to FFFF HEX
SDS Type 2 Message	54595032 HEX ("TYP2")	00000000 to FFFFFFFF HEX
SDS Type 3 Message	54595045 33534453 HEX ("TYPE3SDS")	00000000 to FFFFFFFF HEX (word 1) 00000000 to FFFFFFFF HEX (word 2)
SDS Type 4 Messages		
Type 4 SDS-TL Text Message		
Report Type	None	None, Received, Consumed, Received and Consumed
Report Size	Short	Short, Full (standard)
Text Coding	8 bit (Latin)	7 bit (GSM) 8 bit (Latin)
Time Stamp	Not Required	Not Required, Required
Message	"This SDS type 4 SDS-TL text message was sent by the IFR 2968 and is one hundred and twenty characters long and ends here"	
	Up to 120 characters A-Z, a-z, 0-9, , . : ; + - = < > ! ? \$ % & # * () { } [] @ _ / "	
Type 4 Simple Text Message		
Text Coding	8 bit (Latin)	7 bit (GSM) 8 bit (Latin)
Message	"This SDS type 4 SDS-TL text message was sent by the IFR 2968 and is one hundred and twenty characters long and ends here"	
	Up to 120 characters A-Z, a-z, 0-9, , . : ; + - = < > ! ? \$ % & # * () { } [] @ _ / "	
Type 4 Message in HEX		
Message	82020101 HEX followed by	Up to 120 bytes HEX data
	"This SDS type 4 message in hex, was sent by the IFR 2968 and is one hundred and twenty characters long and ends here"	

TETRA Mobile (MS) (continued)

Mobile parameter settings	Defaults	Range
Short Subscriber ID	00000001	00000001 to 16777215
Group Short Subscriber ID	00000001	00000001 to 16777215
Power Class	1 (45 dBm / 32 W)	1 (45 dBm / 31.6 W) 1L (42.5 dBm / 17.8 W) 2 (40 dBm / 10 W) 2L (37.5 dBm / 5.62 W) 3 (35 dBm / 3.16 W) 3L (32.5 dBm / 1.78 W) 4 (30 dBm / 1 W) 4L (27.5 dBm / 562 mW)
Receiver Class	A	A, B or E
Autorun control settings	Defaults	Range
Display Format	Summary	Summary or Full
Pause Mode	Manual Only	Manual Only, On Failure, or Always
Print	Off	On or Off
Program settings	Default	Range
Test Programs	Call Processing Only Testing,	Call Processing Only, Comprehensive User Defined Test

Autorun parameter settings

This section shows (in the first two columns) the default parameter values for all of the available automatic tests for TETRA Mobile. These parameters are accessed by the high-level test routines (called from MI-BASIC), or they can be accessed directly by the MI-BASIC commands shown in the right hand columns. See chapter 3 of the programming manual 46882/280.

If the parameters are modified, you can restore the defaults by pressing the following keys:-

[HELP SET-UP], [SET-UP], [MISC CONFIG], [default values], [init current]

Test Configuration

READPARAM CONFIGURATION WRITEPARAM CONFIGURATION

(This is *not* a test, but is used for general test parameters)

RF Gen Level	-75 dBm	RFGENLEVEL	<expr>[<units>]
Mobile Power for Access	30 dBm	MOBILELEVEL	<expr>[<units>]
Mobile Power Settling	10 s	TIMEOUT	<expr>[<units>]

Registration

READPARAM REGISTER WRITEPARAM REGISTER TEST REGISTER

Registration Test	On	STATUS	{ON/OFF}
Timeout	30 s	TIMEOUT	<expr>[<units>]
Wait for Group Attachment	2 s		

TT Test Mode

This test is only available for use in the built-in automatic test programs
It is not available for the MI-BASIC facility.

TT Test Mode Test	Off
Timeout	30 s
Wait for Group Attachment	2 s

Place Call

This test is only available for the MI-BASIC facility. It is not available for use in the built-in automatic test programs

READPARAM PLACECALL WRITEPARAM PLACECALL TEST PLACECALL

Place Call Test	On	STATUS	{ON/OFF}
Timeout	30 s	TIMEOUT	<expr>[<units>]
Use Accessory Port	On	USEACCPORT	{ON/OFF}
Accessory Port	Logic1	ACCPORT	{LOGIC0/1/2/3}

Clear From Mobile

This test is only available for the MI-BASIC facility. It is not available for use in the built-in automatic test programs

READPARAM MOBILECLEAR WRITEPARAM MOBILECLEAR TEST MOBILECLEAR

Clear From Mobile Test	On	STATUS	{ON/OFF}
Timeout	30 s	TIMEOUT	<expr>[<units>]
Use Accessory Port	On	USEACCPORT	{ON/OFF}
Accessory Port	Logic2	ACCPORT	{LOGIC0/1/2/3}

TETRA Mobile (MS) (continued)**Call Mobile**

This test is only available for the MI-BASIC facility. It is not available for use in the built-in automatic test programs

**READPARAM CALLMOBILE
WRITEPARAM CALLMOBILE
TEST CALLMOBILE**

Call Mobile Test	On	STATUS	{ON/OFF}
Call Type	Phone Call	CALLTYPE	{GROUP/PRIVATE/ PHONE/EMERGENCY}
Timeout	30 s	TIMEOUT	<expr>[<units>]
Use Accessory Port	On	USEACCPORT	{ON/OFF}
Accessory Port	Logic3	ACCPORT	{LOGIC0/1/2/3}

Clear From Land

This test is only available for the MI-BASIC facility. It is not available for use in the built-in automatic test programs

**READPARAM LANDCLEAR
WRITEPARAM LANDCLEAR
TEST LANDCLEAR**

Clear From Land Test	On	STATUS	{ON/OFF}
----------------------	----	--------	----------

Call 1

This test is only available for use in the built-in automatic test programs
It is not available for the MI-BASIC facility.

Call 1 Test	On
Origin	Mobile
Call Type	Group Call
Mobile Call Type Check	On
Timeout	30 s
Use Accessory Port	On
Accessory Port	Logic 1

End Call 1

This test is only available for use in the built-in automatic test programs
It is not available for the MI-BASIC facility.

End Call 1 Test	On
Origin	Test Set
Timeout	30 s
Use Accessory Port	Off
Accessory Port	Logic 2

Call 2

This test is only available for use in the built-in automatic test programs
It is not available for the MI-BASIC facility.

Call 2 Test	Off
Origin	Mobile
Call Type	Private Call
Mobile Call Type Check	On
Timeout	30 s
Use Accessory Port	On
Accessory Port	Logic 1

End Call 2

This test is only available for use in the built-in automatic test programs
It is not available for the MI-BASIC facility.

End Call 2 Test	Off
Origin	Mobile
Timeout	30 s
Use Accessory Port	On
Accessory Port	Logic 2

Call 3

This test is only available for use in the built-in automatic test programs
It is not available for the MI-BASIC facility.

Call 3 Test	Off
Origin	Mobile
Call Type	Phone Call
Mobile Call Type Check	On
Timeout	30 s
Use Accessory Port	On
Accessory Port	Logic 1

End Call 3

This test is only available for use in the built-in automatic test programs
It is not available for the MI-BASIC facility.

End Call 3 Test	Off
Origin	Mobile
Timeout	30 s
Use Accessory Port	On
Accessory Port	Logic 2

Call 4

This test is only available for use in the built-in automatic test programs
It is not available for the MI-BASIC facility.

Call 4 Test	Off
Origin	Test Set
Call Type	Group Call
Mobile Call Type Check	On
Timeout	30 s
Use Accessory Port	Off
Accessory Port	Logic 3

End Call 4

This test is only available for use in the built-in automatic test programs
It is not available for the MI-BASIC facility.

End Call 4 Test	Off
Origin	Test Set
Timeout	30 s
Use Accessory Port	Off
Accessory Port	Logic 2

TETRA Mobile (MS) (continued)**Call 5**

This test is only available for use in the built-in automatic test programs
It is not available for the MI-BASIC facility.

Call 5 Test	Off
Origin	Test Set
Call Type	Private Call
Mobile Call Type Check	On
Timeout	30 s
Use Accessory Port	On
Accessory Port	Logic 3

End Call 5

This test is only available for use in the built-in automatic test programs
It is not available for the MI-BASIC facility.

End Call 5 Test	Off
Origin	Test Set
Timeout	30 s
Use Accessory Port	Off
Accessory Port	Logic 2

Call 6

This test is only available for use in the built-in automatic test programs
It is not available for the MI-BASIC facility.

Call 6 Test	On
Origin	Test Set
Call Type	Phone Call
Mobile Call Type Check	On
Timeout	30 s
Use Accessory Port	On
Accessory Port	Logic 3

End Call 6

This test is only available for use in the built-in automatic test programs
It is not available for the MI-BASIC facility.

End Call 6 Test	On
Origin	Test Set
Timeout	30 s
Use Accessory Port	Off
Accessory Port	Logic 2

Speech Quality
**READPARAM SPEECHQUAL
WRITEPARAM SPEECHQUAL
TEST SPEECHQUAL**

Speech Quality Test	On	STATUS	{ON/OFF}
RF Gen Level	−75 dBm	RFGENLEVEL	<expr>[<units>]
Timeout	30 s	TIMEOUT	<expr>[<units>]

Tx Timing		READPARAM TIMING WRITEPARAM TIMING TEST TIMING	
Tx Timing Test	On	STATUS	{ON/OFF}
RF Gen Level	–75 dBm	RFGENLEVEL	<expr>[<units>]
Measurement Mode	Worst Case	MEASTYPE	{WORSTCASE/AVERAGE}
Bursts Tested	5	BURSTTST	<expr>
Error Tolerance	0.25 Symbols	ERROR	<expr>[<units>]
Power Level		READPARAM POWERLEVEL WRITEPARAM POWERLEVEL TEST POWERLEVEL	
Power Level Test	On	STATUS	{ON/OFF}
Measurement Mode	Average	POWMEASTYPE	{MAXIMUM/ MINIMUM/ AVERAGE}
Bursts Tested	5	BURSTTST	<expr>
Hi Pwr Upper Error	2.0 dB	HIRUPPER	<expr>[<units>]
Hi Pwr Lower Error	2.0 dB	HIRLOWER	<expr>[<units>]
Upper Error	2.5 dB	RUPPER	<expr>[<units>]
Lower Error	2.5 dB	RLOWER	<expr>[<units>]
Tx Power Profile		READPARAM PROFILE WRITEPARAM PROFILE TEST PROFILE	
Tx Power Profile Test	On	STATUS	{ON/OFF}
Tx Frequency		READPARAM TXFREQ WRITEPARAM TXFREQ TEST TXFREQ	
Tx Frequency Test	On	STATUS	{ON/OFF}
Measurement Mode	Worst Case	MEASTYPE	{WORSTCASE/AVERAGE}
Bursts Tested	5	BURSTTST	<expr>
Error Tolerance	100 Hz	ERROR	<expr>[<units>]
Tx RMS Vector Error		READPARAM RMSVECTOR WRITEPARAM RMSVECTOR TEST RMSVECTOR	
Tx RMS Vector Error Test	On	STATUS	{ON/OFF}
Measurement Mode	Worst Case	MEASTYPE	{WORSTCASE/AVERAGE}
Bursts Tested	5	BURSTTST	<expr>
Upper Limit	10.00 %	ERROR	<expr>[<units>]
Tx Peak Vector Error		READPARAM PKVECTOR WRITEPARAM PKVECTOR TEST PKVECTOR	
Tx RMS Vector Error Test	On	STATUS	{ON/OFF}
Measurement Mode	Worst Case	MEASTYPE	{WORSTCASE/AVERAGE}
Bursts Tested	5	BURSTTST	<expr>
Upper Limit	30.00 %	ERROR	<expr>[<units>]

TETRA Mobile (MS) (continued)

Tx Residual Carrier		READPARAM RESIDUAL WRITEPARAM RESIDUAL TEST RESIDUAL	
Tx Residual Carrier Test	On	STATUS	{ON/OFF}
Measurement Mode	Worst Case	MEASTYPE	{WORSTCASE/AVERAGE}
Bursts Tested	5	BURSTTST	<expr>
Upper Limit	5.00 %	ERROR	<expr>[<units>]

Rx BER Class 0		READPARAM RXBERO WRITEPARAM RXBERO TEST RXBERO	
		The character O is used in these MI-BASIC commands as numerals are not permitted.	
Rx BER Class 0 Test	OFF	STATUS	{ON/OFF}
RF Gen Level	-112dBm	RFGENLEVEL	<expr>[<units>]
Number of Samples	15000	SAMPLES	<expr>
Upper Limit for Rx Class A	4.270%	UPPER	<expr>[<units>]
Upper Limit for Rx Class B	4.880%	BUPPER	<expr>[<units>]
Upper Limit for Rx Class E	4.270%	EUPPER	<expr>[<units>]

Rx BER Class 1		READPARAM RXBERI WRITEPARAM RXBERI TEST RXBERI	
		The character I is used in these MI-BASIC commands as numerals are not permitted.	
Rx BER Class 1 Test	OFF	STATUS	{ON/OFF}
RF Gen Level	-112dBm	RFGENLEVEL	<expr>[<units>]
Number of Samples	15000	SAMPLES	<expr>
Upper Limit for Rx Class A	0.230%	UPPER	<expr>[<units>]
Upper Limit for Rx Class B	0.230%	BUPPER	<expr>[<units>]
Upper Limit for Rx Class E	0.230%	EUPPER	<expr>[<units>]

Rx BER Class 2		READPARAM RXBERII WRITEPARAM RXBERII TEST RXBERII	
		The characters II are used in these MI-BASIC commands as numerals are not permitted.	
Rx BER Class 2 Test	OFF	STATUS	{ON/OFF}
RF Gen Level	-112dBm	RFGENLEVEL	<expr>[<units>]
Number of Samples	15000	SAMPLES	<expr>
Upper Limit for Rx Class A	0.045%	UPPER	<expr>[<units>]
Upper Limit for Rx Class B	0.045%	BUPPER	<expr>[<units>]
Upper Limit for Rx Class E	0.045%	EUPPER	<expr>[<units>]

Rx RBER Class 0**READPARAM RXRBERO
WRITEPARAM RXRBERO
TEST RXRBERO**

The character O is used in these MI-BASIC commands as numerals are not permitted.

Rx RBER Class 0 Test	OFF	STATUS	{ON/OFF}
RF Gen Level	−112dBm	RFGENLEVEL	<expr>[<units>]
Number of Samples	15000	SAMPLES	<expr>
Upper Limit for Rx Class A	4.270%	UPPER	<expr>[<units>]
Upper Limit for Rx Class B	4.880%	BUPPER	<expr>[<units>]
Upper Limit for Rx Class E	4.270%	EUPPER	<expr>[<units>]

Rx RBER Class 1**READPARAM RXRBERI
WRITEPARAM RXRBERI
TEST RXRBERI**

The character I is used in these MI-BASIC commands as numerals are not permitted.

Rx BER Class 1 Test	OFF	STATUS	{ON/OFF}
RF Gen Level	−112dBm	RFGENLEVEL	<expr>[<units>]
Number of Samples	15000	SAMPLES	<expr>
Upper Limit for Rx Class A	0.230%	UPPER	<expr>[<units>]
Upper Limit for Rx Class B	0.230%	BUPPER	<expr>[<units>]
Upper Limit for Rx Class E	0.230%	EUPPER	<expr>[<units>]

Rx MER**READPARAM RXMER
WRITEPARAM RXMER
TEST RXMER**

Rx MER Test	OFF	STATUS	{ON/OFF}
RF Gen Level	−112dBm	RFGENLEVEL	<expr>[<units>]
Number of Samples	300	SAMPLES	<expr>
Upper Limit for Rx Class A	0.045%	UPPER	<expr>[<units>]
Upper Limit for Rx Class B	0.045%	BUPPER	<expr>[<units>]
Upper Limit for Rx Class E	0.045%	EUPPER	<expr>[<units>]

Autorun test programs

This section defines what tests are run in each of the built-in test programs. For information about each test, see *TETRA Mobile automatic testing, Automatic Test descriptions*, in Chapter 7, *TETRA mobile automatic testing*.

Enabling or disabling tests

You can enable or disable each test by setting the test status parameter associated with the test on the AUTORUN PARAMETERS menu to ON or OFF (e.g. the Registration test on page 11-7).

By default, certain tests are disabled. See *Autorun parameter settings* on page 11-7

TETRA Mobile (MS) (*continued*)

Note

In the following program descriptions, a start brace { and a finish brace } indicate the start and finish of tests that are run as a *group*. Each group of tests is repeated in accordance with the condition preceding the group (e.g., For *Minimum and Maximum Power*).

Call processing only program

If enabled
 TEST REGISTER
 if passed
 Display **ITSI**

If enabled
 TEST TT TEST MODE
 if passed
 Display **TEI, Power Class and Receiver Class**

If enabled
 TEST CALL 1
 if passed
 Display **Call Type and Called ID**

If enabled
 TEST ENDCALL 1

If enabled
 TEST CALL 2
 if passed
 Display **Call Type and Called ID**

If enabled
 TEST ENDCALL

If enabled
 TEST CALL 3
 if passed
 Display **Call Type and Called ID**

If enabled
 TEST ENDCALL 3

If enabled
 TEST CALL 4
 if passed
 Display **Call Type and Called ID**

If enabled
 TEST ENDCALL 4

If enabled
 TEST CALL 5
 if passed
 Display **Call Type and Called ID**

If enabled
 TEST ENDCALL 5

If enabled
 TEST CALL 6
 if passed
 Display **Call Type and Called ID**

If enabled*
 TEST SPEECH QUAL (Talkback)*

If enabled
 TEST ENDCALL 6

* This test takes place after the last enabled call in the program, assumed to be CALL 6 in this case.

TETRA Mobile (MS) (*continued*)

Comprehensive testing program

```
If enabled
    TEST REGISTER
    if passed
        Display ITSI

If enabled
    TEST TT TEST MODE
if passed
    Display TEI, Power Class and Receiver Class

If enabled
    TEST CALL 1
    if passed
        Display Call Type and Called ID

If enabled
    TEST ENDCALL 1

If enabled
    TEST CALL 2
    if passed
        Display Call Type and Called ID

If enabled
    TEST ENDCALL 2

If enabled
    TEST CALL 3
    if passed
        Display Call Type and Called ID

If enabled
    TEST ENDCALL 3

If enabled
    TEST CALL 4
    if passed
        Display Call Type and Called ID

If enabled
    TEST ENDCALL 4

If enabled
    TEST CALL 5
    if passed
        Display Call Type and Called ID

If enabled
    TEST ENDCALL 5

If enabled
    TEST CALL 6
    if passed
        Display Call Type and Called ID

If enabled*
    TEST SPEECHQUAL (Talkback)*

If enabled
    TEST TIMING

For Minimum and Maximum Power
{
    If enabled
        TEST POWERLEVEL
    If enabled
        TEST PROFILE
}
If enabled
    TEST TXFREQ
```

If enabled
TEST RMSVECTOR

If enabled
TEST PKVECTOR

If enabled
TEST RESIDUAL

If enabled
TEST RXBERO

If enabled
TEST RXBERI

If enabled
TEST RXBERII

If enabled
TEST RXBERO

If enabled
TEST RXRBERO

If enabled
TEST RXRBERI

If enabled
TEST RXMER

If enabled
TEST ENDCALL 6

NOTE:-The characters O, I and II are used in these
MI-BASIC commands as numerals are not permitted.

*** This test and the following parametric tests take place after the last enabled call in the
program, assumed to be CALL 6 in this case.**

TETRA Base Station (BS)

Channel plans

Channel plan	Base station Rx frequency of minimum channel number (MHz)	Base station Tx frequency of minimum channel number (MHz)	Minimum channel number	Maximum channel number	Channel spacing (kHz)	Duplex spacing (Tx - Rx) (MHz)	Offset (kHz)
TETRA 380 BS	380.0125	390.0125	3600	3999	25	10	12.5
TETRA 380 +0 BS	380.0000	390.0000	3600	4000	25	10	0
TETRA 410 BS	410.0125	420.0125	800	1199	25	10	12.5
TETRA 410 +0 BS	410.0000	420.0000	800	1200	25	10	0
TETRA 410 -6 BS	410.01875	420.01875	801	1200	25	10	-6.25
TETRA 450 BS	450.0125	460.0125	2400	2799	25	10	12.5
TETRA 450 +0 BS	450.0000	460.0000	2400	2800	25	10	0
TETRA 800 BS	805.0125	850.0125	2000	2799	25	45	12.5
TETRA 800 +0 BS	805.0000	850.0000	2000	2800	25	45	0
TETRA 870 BS	870.0125	915.0125	600	839	25	45	12.5
TETRA 870 +0 BS	870.0000	915.0000	600	840	25	45	0
USER DEFINED	380.0000	390.0000	0	4095	25	10	N/A

The values shown for USER DEFINED in the above table are the default setting of the user-defined channel plan.

Note

See the note regarding channel plans on page 11-2.

Base station transmitter power classes

Power class	Nominal output power	
BS Class 1	46 dBm	40 W
BS Class 2	44 dBm	25 W
BS Class 3	42 dBm	16 W
BS Class 4	40 dBm	10 W
BS Class 5	38 dBm	6 W
BS Class 6	36 dBm	4 W
BS Class 7	34 dBm	2.5 W
BS Class 8	32 dBm	1.6 W
BS Class 9	30 dBm	1.0 W
BS Class 10	28 dBm	600 mW

Test parameters

This section lists the TETRA Base Station (BS) System parameter default settings. Where appropriate, it shows alternative settings. Details of how to change the settings is given in Chapter 3, *Selecting and customizing the system*.

If the parameters are modified, you can restore the defaults by pressing the following keys:-

[HELP SET-UP], [SET-UP], [MISC CONFIG], [default values], [init current]

System parameter settings	Defaults	Range
Channel Plan	TETRA380BS	NO PLAN USER DEFINED TETRA380BS TETRA380 +0 BS TETRA410BS TETRA410 +0 BS TETRA410 -6 BS TETRA450BS TETRA450 +0 BS TETRA800 BS TETRA800 +0 BS TETRA870BS TETRA 870 +0 BS
User Defined Plan	(Only available with USER DEFINED channel plan)	
Channel Plan Title	USER DEFINED	up to 12 characters as defined by user
Lowest Channel	0000	0000 to 4095
Highest Channel	4095	0000 to 4095
Lowest Channel Tx Freq	390 MHz	10 MHz to 999 MHz
Duplex Offset	10 MHz	-100 MHz to +100 MHz
Channel Spacing	25 kHz	-5 kHz to -500 kHz and 5 kHz to 500 kHz
Channel	(BS Tx Frequency for No Plan channel plan)	
NO PLAN	380 MHz	100 to 999.999999 MHz
USER	0	0 to 4095
DEFINEDTETRA380BS	3600	3600 to 3999
TETRA380 +0 BS	3600	3600 to 4000
TETRA410BS	800	800 to 1199
TETRA410 +0 BS	800	800 to 1200
TETRA410 -6 BS	801	801 to 1200
TETRA450BS	2400	2400 to 2799
TETRA450 +0 BS	2400	2400 to 2800
TETRA800 BS	2040	2000 to 2799
TETRA800 +0 BS	2040	2000 to 2800
TETRA870BS	600	600 to 839
TETRA870 +0 BS	600	600 to 840
Rx Offset	10 MHz	-100 to +100 MHz (only available with NO PLAN channel plan)
Base Station Tx Mode	Continuous All Slots	Continuous All Slots Continuous 1 Slot Discontinuous
Mobile Country Code	001	000 to 999
Mobile Network Code	00001	00000 to 16383
Base Colour Code	01	00 to 63
MCC-MNC-BCC Update	Auto	Auto, Manual
Sync to Base Station	Auto	Auto, Pulse
Auto Sync Path Delay	0.00 bits	-45.00 to +45.00 bits
Sync Pulse Offset	0.0 s	0.000000 to +1.020000 s
Sync Pulse Edge	Rising	Rising or Falling

TETRA Direct Mode (DM)

Channel plans

Channel plan	Mobile Rx frequency of minimum channel number (MHz)	Mobile Tx frequency of minimum channel number (MHz)	Minimum channel number	Maximum channel number	Channel spacing (kHz)	Offset (kHz)
TETRA 380 DM	380.0125	380.0125	3600	3999	25	12.5
TETRA 380 +0 DM	380.0000	380.0000	3600	4000	25	0
TETRA 410 DM	410.0125	410.0125	800	1199	25	12.5
TETRA 410 +0 DM	410.0000	410.0000	800	1200	25	0
TETRA 410 –6 DM	410.01875	410.01875	801	1200	25	–6.25
TETRA 450 DM	450.0125	450.0125	2400	2799	25	12.5
TETRA 450 +0 DM	450.0000	450.0000	2400	2800	25	0
TETRA 800 DM	805.0125	805.0125	2000	2799	25	12.5
TETRA 800 +0 DM	805.0000	805.0000	2000	2800	25	0
TETRA 870 DM	870.0125	870.0125	600	839	25	12.5
TETRA 870 +0 DM	870.0000	870.0000	600	840	25	0
USER DEFINED	380.0000	380.0000	0	4095	25	N/A

USER DEFINED is the default setting of the user-defined channel plan.

Note

See the note regarding channel plans on page 11-2.

Test parameters

This section lists the TETRA Direct Mode (DM) System parameter default settings. Where appropriate, it shows alternative settings. Details of how to change the settings is given in Chapter 3, *Selecting and customizing the system*.

System parameter settings	Defaults	Range
ChannelPlan	TETRA380DM	NO PLAN USER DEFINED TETRA380DM TETRA 380 +0 DM TETRA410DM TETRA 410 +0 DM TETRA 410 –6 DM TETRA450DM TETRA 450 +0 DM TETRA 800 DM TETRA 800 +0 DM TETRA870DM TETRA 870 +0 DM
User Defined Plan	(Only available with USER DEFINED channel plan)	
Channel Plan Title	USER DEFINED	up to 12 characters as defined by user
Lowest Channel	0000	0000 to 4095
Highest Channel	4095	0000 to 4095
Lowest Channel Tx Freq	380 MHz	10 MHz to 999 MHz
Channel Spacing	25 kHz	-5 kHz to –500 kHz and 5 kHz to 500 kHz

System parameter settings	Defaults	Range
Channel	(DM Tx Frequency for NO PLAN channel plan)	
NO PLAN	380 MHz	100 to 999.999999 MHz
USER DEFINED	0	0 to 4095
TETRA380DM	3600	3600 to 3999
TETRA380 +0 DM	3600	3600 to 4000
TETRA410DM	800	800 to 1199
TETRA410 +0 DM	800	800 to 1200
TETRA410 -6 DM	801	801 to 1200
TETRA450DM	2400	2400 to 2799
TETRA450 +0 DM	2400	2400 to 2800
TETRA800 DM	2040	2000 to 2799
TETRA800 +0 DM	2040	2000 to 2800
TETRA870DM	600	600 to 839
TETRA870 +0 DM	600	600 to 840
Direct mode Tx Mode	Discontinuous	Continuous Discontinuous

Appendix A

GLOSSARY

This glossary provides additional information about TETRA terms and abbreviations used in this manual.

System Types

TETRA	Terrestrial Trunked Radio
DMO	Direct Mode Operation
V+D	Voice plus Data

TETRA Equipment Types

BS	Base Station
DM	Direct Mode
MS	Mobile Station

Basic Channel Types

BNCH/T	Broadcast Network Channel (in T1 signal)
BSCH	Broadcast Synchronization Channel
MCCH	Main Control Channel
TCH	Traffic Channel (generic term)
TCH/S	Traffic Channel for Speech
TCH/7.2	Traffic Channel for 7.2 kbit/s data
TCH/4.8	Traffic Channel for 4.8 kbit/s data
TCH/2.4	Traffic Channel for 2.4 kbit/s data
TN	Timeslot Number

Basic System Parameters

BCC	Base station Color Code 0 to 63
ECC	Extended Color Code = MCC + MNC + BCC, generates scrambling sequence.
DCC	Direct Mode Color Code (30 bits substitute for ECC in direct mode)
LA	Location Area 00000 to 16383
MCC	Mobile Country Code (E.212, 10 bit binary) 000 to 999
MNC	Mobile Network Code (14 bit binary) 00000 to 16383

Basic Identities

ESN	External Subscriber Number
GSSI	Group Short Subscriber Identity
ITSI	Individual Tetra Subscriber Identity
SNA	Short Number Address
SSI	Short Subscriber Identity
TEI	TETRA Equipment Identity

Measurement Terms

BER	Bit Error Rate
RBER	Residual Bit Error Rate
PL	Power Level
PRBS 0.153	Pseudo Random Binary Sequence 511 bit PRBS
RSSI	Received Signal Strength Indication
SN	Symbol Number
TDMA	Time Division Multiple Access
T1	Test signal (wanted signal).
t1	Power Profile ramp-up period
t3	Power Profile ramp-down period
TS1	Normal Training Sequence 1
TS2	Normal Training Sequence 2
TT	TETRA Test Protocol

Burst Types

CB	Control Burst (Uplink SCH/HU burst)
LB	Linearization Burst
NDB	Normal Downlink Burst (all downlink slots except BSCH slot)
NUB	Normal Uplink Burst (uplink TCH & SCH/F bursts)
SB	Synchronization Burst (BSCH)

Call Processing Terms

ISDN	Integrated Services Digital Network
PSTN	Public Switched Telephone Network
PABX	Private Automatic Branch eXchange
PTT	Push To Talk
SDS	Short Data Service
SDS-TL	Short Data Service Transport Layer

Specification Terms

EN 300 392-2	TETRA V+D Air Interface
EN 300 394-1	TETRA Radio Conformance Specification
MoU	Memorandum of Understanding
TIPv2	TETRA Interoperability Profile version 2

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